

The

GEOGRAPHY & ECOLOGY COMPENDIUM

**for CSAT Paper 1, State PSC, NDA, CDS
& other Competitive Exams**

Covers :

- Nature And Scope
- Universe And Our Planet Earth
- Hydrosphere
- Atmosphere
- World Geography (Physical, Social and Economics Aspect)
- Physiography And Drainage Pattern Of India
- Soil, Nature Vegetation, Wildlife And Agriculture Of India
- Mineral Resource, Industries Population
- Multipurpose Projects of India
- Ecology and Environment Bio-Diversity
- Environmental Issues

- Latest Pattern
- INFOGRAPHICS
- 13 Chapters
- 26 Exercises
- 1300+ MCQs
(Direct & Statement Based)
- 100% Syllabus Coverage

The
**Geography
and Ecology
Compendium**

- **Head Office** : B-32, Shivalik Main Road, Malviya Nagar, New Delhi-110017
- **Sales Office** : B-48, Shivalik Main Road, Malviya Nagar, New Delhi-110017
Tel. : 011-26691021 / 26691713

Typeset by Disha DTP Team



DISHA PUBLICATION

ALL RIGHTS RESERVED

© Copyright Publisher

No part of this publication may be reproduced in any form without prior permission of the publisher. The author and the publisher do not take any legal responsibility for any errors or misrepresentations that might have crept in. We have tried and made our best efforts to provide accurate up-to-date information in this book.

For further information about the books from DISHA,

Log on to www.dishapublication.com or email to info@dishapublication.com

CONTENTS

GEOGRAPHY

- 1. Nature and Scope** **G-1-G-6**
Introduction; Modern Geography; Branches of Geography;
- 2. Universe and Our Planet Earth** **G-7-G-32**
Introduction; Galaxy; Stars; Planets; Moon; Asteroids; Earth
- 3. Hydrosphere** **G-33-G-54**
Introduction; Hydrological cycle; Components of Hydrosphere
- 4. Atmosphere** **G-55-G-76**
Introduction; Composition; Structure; Atmospheric Pressure; Winds; Air mass
- 5. World Geography (Physical Aspect)** **G-77-G-106**
Introduction; Asia; Africa; North America; South America; Australia And Oceania; Europe; Antarctica
- 6. World Geography (Social Aspect)** **G-107-G-112**
Introduction; Race; Population; Migration
- 7. World Geography (Economic Aspect)** **G-113-G-124**
Introduction; Primary; Activities; Secondary; Activities; Tertiary Activities;
- 8. Physiography and Drainage Pattern of India** **G-125-G-154**
Introduction; Location - Area – Extent; Physical Features; Climate; Drainage
- 9. Soil, Natural Vegetation, Wildlife and Agriculture of India** **G-155-G-188**
Introduction; Description and Composition; Soil Formation; Major Soils
- 10. Mineral Resource, Industries Population Multipurpose Projects of India** **G-189-G-208**
Introduction; Mineral Resource of India; Agro-Based; Cotton Textiles; Woolen Textile; Jute Textiles; Sugar; Silk; Metallurgical; Iron And Steel; Aluminium Smelting; Engineering; Automobile; Fertilizer; Aircraft; Cement; Glass Industry

ECOLOGY AND ENVIRONMENT

1. Ecology and Environment

EE-1-EE-22

Ecosystem; Abiotic Components; Biotic Components Environment; Population

2. Biodiversity

EE-23-EE-40

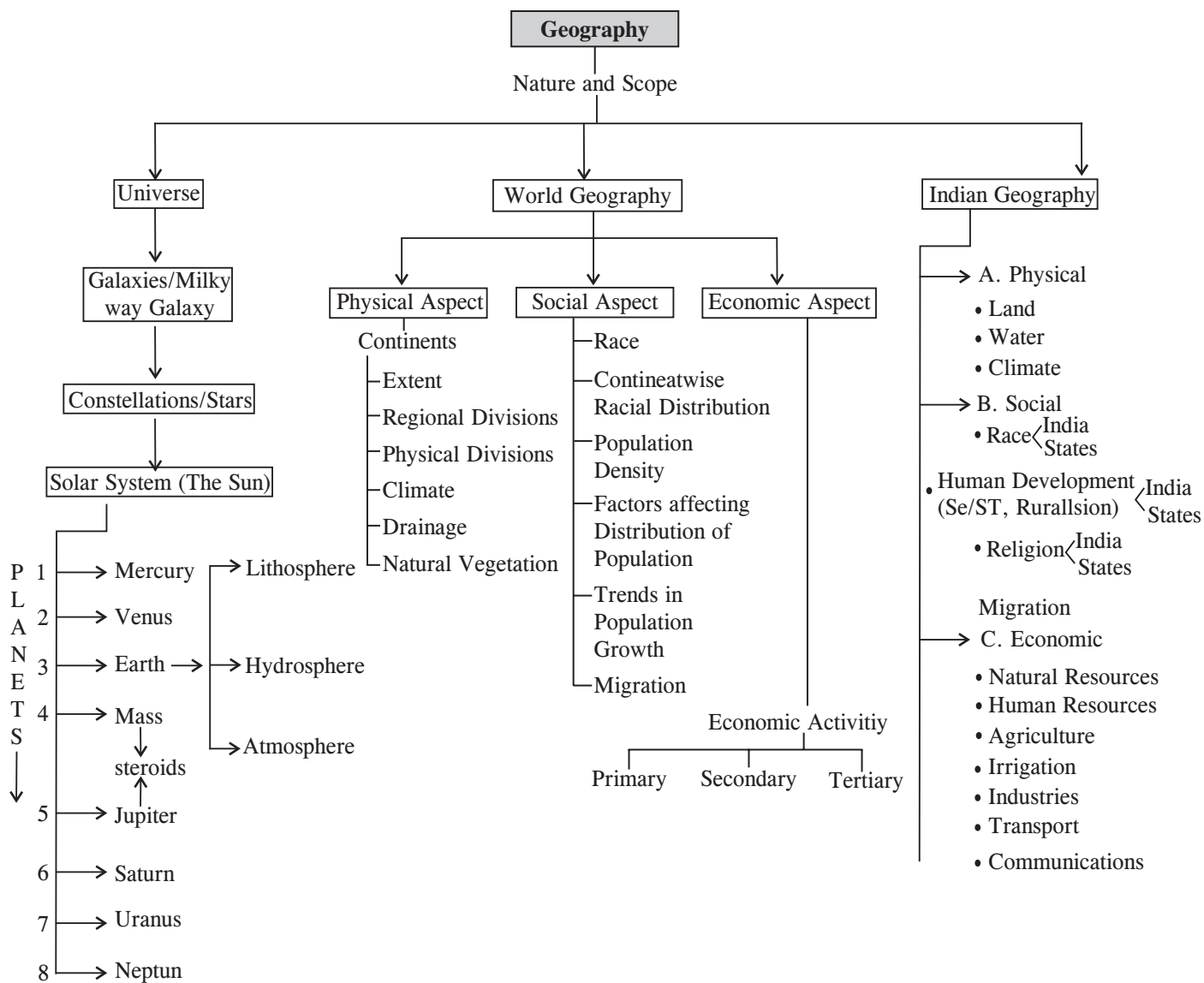
Levels of Biodiversity; Biodiversity of India; Loss & Conservation

3. Environmental Issues

EE-41-EE-72

Global Warming; Green House Effect; Ozone Depletion; Deforestation; Air; Water; Sound
Pollutions; Agro-chemical effects; Acid Rain; Solid Waste Management; Radioactive Waste
Management; Plastic Waste Management; Mercury Pollution; ECO Sensitive Zones; COP;
Environmental Acts

GEOGRAPHY



GEOGRAPHY

Geography is that branch of knowledge which links the study of natural world to that of society. It is the frame work of various fields of study which cover origin and development of universe, land scapes and its natural environment and pressure created over them, the global and local changes in the world and how and upto what extent the society is contributing in these changes. In this section we are going to cover all these above mentioned aspects in details.

Greek Scholar Eratosthenes is first man to discuss about Geography. It's the combination of two greek words and Geo means earth and graphy" means write. Thus geography means to write about the Earth or "earth descriptions".

Various other geographers defined Geography as follows:

In Classical Period

- (i) Pythagoras, a great Greek philosopher and mathematician, was the first to say that the Earth is spherical and revolves around the sun.
- (ii) Herodotus Greek historian is known as the "Father of history". He explained the deposition of silt in the Nile delta.
- (iii) Aristotle (384 - 322 B.C.) another Greek philosopher explained the eclipses.
- (iv) Eratosthenes (276 - 194 B.C) a Greek poet, mathematician observing the angle of the noon day Sun at Syene and at Alexandria.
- (v) Hipparchus (190 - 120 B.C.) was perhaps the greatest of the Greek astronomers.
- (vi) Ptolemy (2nd century A.D.). He wrote the book 'Geography' and he described system of latitudes and longitudes. He was a cartographer and he evolved the science of map-making.

Modern Geography

1. Bernhardus Varenius (1622-1650) first recognised the need for organisation of geographical knowledge. He wrote Geographia Generalis, the most highly regarded treatise on geography for more than a century into two - part - general (now called systematic) and special (also called regional) which identified regions according to the interactions between human and environmental processes.
2. Immanuel Kant (1724-1804) was also a German philosopher, is considered by many to be one of most influential thinkers of modern times. He tried to find a foundation for geography within the framework of other science. According to him, all knowledge can be organised into three groups according to the object of study.
3. Baron Alexander Von Humboldt (1769-1859) a German naturalist and explorer, moulded the substance of geography into a scientific form. He was interested in all aspects of natural history and has been described by Charles Darwin as "the greatest scientific traveller that ever lived".

* He invented the "isotherms" to compare temperatures.

4. Carl Ritter (1779-1859) A German geographer is considered the founder of modern geographic study. He stressed the importance of using all the sciences in the study of geography.

His most important work Die Erdkunde (Earth Science) emphasized the influence of physical environment on human activity. He divided the Earth into natural regions and showed each unit as a whole interrelated complex of elements. His plan of study became the model for regional study and presentation.

5. Ellsworth Huntington (1876-1947), an American geographer and explorer, was noted particularly for his study of the effects of climate on human heredity and civilization. The human aspect was neglected to a great extent. His approach to geographic study is known as determinism in which humans are passive agents while the physical environment is active.
6. Vidal de la Blache advocated the opposite theory of determinism, which is sometimes known as possibilism, in which humans are active agents, at liberty to choose between a wide range in environmental possibilities.

Branches of Geography

During the 20th century, more attention has been paid to systematic studies of geographical data. Phenomena have combined to make geography one of the first disciplines to fulfil a bridging function between the natural and social science. As a result, the branches of geographical inquiry fall into three major areas, studies associated with the regional concept, the major systematic branches of human geography, and physical geography.

Physical Geography

This is the most important branch of geography. The superstructure of the discipline of geography is built upon it. It studies relief, soil and structure of the Earth.

It is divided into a number of branches making the subject matter of geography more comprehensive.

- (i) **Geomorphology** studies the Earth's structure, the rocks that make up the Earth, relief features like mountains and plains and their evolution.
- (ii) **Geology** is the science of the study of rocks, and is helpful in the study of glaciers.
- (iii) **Glaciology** is concerned with the study of glaciers.
- (iv) **Seismology** is the study of earthquakes and their bearings on the internal structure of the Earth.
- (v) **Hydrology** is the study of the characteristics of rivers, lakes fluvial morphology, fluctuation of water table and underground water resources, development and change of coastal features.
- (vi) **Oceanography** is the branch of hydrology. It is the study of the ocean, tides waves and the ocean floor.
- (vii) **Climatology** studies the causes and distribution of temperature and winds, rainfall and runoff weather and climate.

- (viii) **Pedology** is the study of soil science. Soil, which is formed due to the result of complex physical and chemical reaction of parent rock materials, needs the support of other sciences like chemistry, geology and biology.
- (ix) **Biogeography** studies the distribution of flora (plant) and fauna (animal) in different parts of the world.
- (x) **Paleogeography** is the study of physical geography in the past geological ages. It studies the distribution of land and sea through successive geological times.

Human Geography

Human geography, also known as a branch of the modern geography, deals with the evolution of human beings i.e, changing distribution and spacial organisation of a variety of human characteristics, ranging from great urban centres built by man to geographical diffusion of specific technical innovations in agriculture. By applying geographical methods and techniques of analysis to such feature the knowledge accumulated in other social studies - history, politics, economics, and sociology - may be seen in a new light and often extended considerably. It is accepted that it is not the physical environment alone that determines human ability to make the best use of the natural sources.

It is divided into following sub fields to make the study more comprehensive :

- (i) **Economic Geography** : It deals with man's activities in improving his material through economic production, exchange distribution and consumption of useful goods and services, that human groups and their members need.
- (ii) **Cultural Geography** : This is also known as social geography. It deals with the cultural aspects of human's habitat, clothing, food, shelter, skills, tools, language, religions, social organisation and his outlook.
- (iii) **Historical Geography** : It describes geographical picture of a region or area. It gives us important clues in understanding of the regions as it is viewed at present.
- (iv) **Anthrogeography** : Discusses the distribution of human communities on the Earth in contence of their geographical environment.
- (v) **Demography** : Studies the different aspects of population like birth rate, death rate, age composition. It also deals with the socio-economic composition of the population and sex composition.
- (vi) **Settlement Geography** : It studies about the size, form and function of settlement of human beings and define their historic growth.
- (vii) **Agricultural Geography** : Studies about farms and farming systems have developed in particular areas and how they are different or similar too the farms and farming systems of other areas.
- (viii) **Urban Geography** : Study of urban Geography is based on the concepts of location interaction and accessibility as well as distribution and movements of populations.
- (ix) **Political Geography** : In study about the government states and countries. It studies human social activities which is concerned to the location and boundaries of cities, nations and groups of nations.

Systematic and Regional Geography

Systematic Geography

Systematic Geography, we choose any geographical factor and study its distribution for the whole world. Relief, drainage, climate, vegetation, soil, minerals wealth, agriculture, industry, transport, trade and commerce and population are some of the important geographical elements. These elements are studied separately with reference to a particular area. This method of studying geography is also known as 'Topical approach'. This is known as General Geography also.

Regional Geography

Regional Geography area as a whole first aims at identifying geographical factors or components of any area. Regional geography helps us in identifying the region. For example, if we take Ganga plain, Chhota Nagpur Platiau. or Assam Velley, rather than taking whole of India and study their location, relief, drainage, climate, soils, vegetation mineral wealth, agriculture, industry, transport, trade, population. In regional geography, the main emphasis is on the region.

Advantages of Regional Geography

- (i) Study becomes easy and effective.
- (ii) Basic principle becomes easy to understand.
- (iii) It describes relation between man and environment.
- (iv) We come to know the economic disparities among various regions.
- (v) The study is intense because the area is limited.

The above discussion makes it clear that both systematic and regional geography are essential to comprehend the geographical knowledge.

Features of Systematic Geography

- Systematic geography studies a particular element over the earth or a part thereof.
- It presents an the integrated form of the area. It is based on political units.
- The study presents the facts about any particular region on the earth surface.
- In the systematic, study types and sub-types are determined on one particular factor.
- It presents isolated form of the area, and it is based on geographical units.
- This study examines the man-environment relationship.

Cartography

It is the science and art of drawing maps and charts on selected scale.

Mathematical Geography

It is almost similar of maps and interpretation and also describes statical data.

Remote Sensing and Geographic Information System (GIS)

The techniques of remote sensing and GIS plays an important role in the study of geographic problems.

Exercise -1

1. Who was the first man to discuss about geography?
(a) Pythagoras (b) Eratosthenes
(c) Aristotle (d) None of these
2. Who was the first man to say that the Earth is spherical and revolves around the sun?
(a) Aristotle (b) Hipparchus
(c) Pythagoras (d) None of these
3. Which of these was great astronomer?
(a) Hipparchus (b) Herodotus
(c) Ptolemy (d) Carl Ritter
4. Who wrote 'Geographia Generalis'?
(a) Ptolemy (b) Bernhardus Varenius
(c) Herodotus (d) None of these
5. Who is considered the founder of modern geographic study?
(a) Humboldt (b) Hipparchus
(c) Carl Ritter (d) None of these
6. Who invented the Isotherms?
(a) Pythagoras (b) Homboldt
(c) Aristotle (d) Herodotus
7. Who advocated the opposite theory of determinism?
(a) Vidal de la Blache (b) Huntington
(c) Ptolemy (d) None of these
8. The study of earth's structure and the rocks that make up the earth is called
(a) Geology (b) Glaciology
(c) Hydrology (d) Geomorphology
9. The study of earthquakes and their bearings on the internal structure of the Earth is called
(a) Glaciology (b) Seismology
(c) Geology (d) None of these
10. The study of soil science is called
(a) Climatology (b) Geology
(c) Pedology (d) Biogeography
11. Oceanography is the branch of
(a) Geomorphology (b) Geology
(c) Hydrology (d) Pedology
12. Human geography is also known as
(a) Economic geography
(b) Demography
(c) Urban geography
(d) Cultural geography
13. The study of physical geography in the past geological ages is called
(a) Biogeography (b) Paleography
(c) Climatology (d) Geology
14. The science and art of drawing maps and charts on selected scale is called
(a) Hydrology (b) Geology
(c) Pedology (d) Cartography
15. The approach of 'Determinism' is related with
(a) Huntington (b) Aristotle
(c) Herodotus (d) Hipparchus
16. The study of the distribution of human communities on the Earth in contence of their geographical environment is called
(a) Biogeography (b) Paleography
(c) Anthropogeography (d) Climatology
17. Who wrote the five volume book 'Kosmos'?
(a) Pythagoras (b) Humboldt
(c) Carl Ritter (d) Aristotle
18. Who evolved the science of map making?
(a) Ptolemy (b) Hipparchus
(c) Herodotus (d) Huntington
19. Who wrote the book 'Geography'?
(a) Carl Ritter (b) Blanche
(c) Ptolemy (d) Humboldt
20. The study of the atmosphere that focuses on weather process and forcasting is called
(a) Biogeography (b) Glaciology
(c) Meteorology (d) Hydrology
21. Chain of roughly adjacent metropolitan areas are called
(a) Global city (b) Megalopolis
(c) Metropolis (d) None of these
22. General geography was written by
(a) Bernhardus Varenius (b) Ptolemy
(c) Blanche (d) None of these
23. Select the correct sequence in the chronological order of the following Greeks who have contributed vastly to the evolution of geographical thought during the ancient period:
(a) Aristotle-Eratosthenes-Anaximander-Ptolemy
(b) Eratosthenes-Anaximander-Ptolemy-Aristotle
(c) Anaximander-Aristotle-Eratosthenes-Ptolemy
(d) Ptolemy-Anaximander-Aristotle-Eratosthenes
24. Which one of the following is a key concept in humanistic geography?
(a) Development and quality of life
(b) Spatial location
(c) Spatial concentration
(d) Geographical distribution
25. Which one of the following scholars was the first to divide the world landmass into three continents : Europe, Asia and Libya (Africa) ? :
(a) Anaximander (b) Hecataeus
(c) Herodotus (d) Eratosthenes
26. An early statement of Geography as Chorology was provided by
(a) Kant (b) Humboldt
(c) Ritter (d) Varenius
27. Ratzel's work was based on the concept
(a) There is a dichotomy between physical and cultural aspects of geography
(b) Geography is a scientific discipline
(c) Physical environment controlled human activities
(d) Geography is necessarily descriptive

28. Huntington is noted for describing
- man as a product of the earth's surface
 - the evolution of landforms
 - the effects of climate on human life
 - the historical-ecological study of the cultural landscape
29. The behavioural environment is meant by
- reality as it exists in nature
 - the decisions based on reasoned thought
 - man's behaviour as a function of the environment
 - reality as is perceived by individuals]
30. Which one of the following concepts is associated with Vidal de La Blache?
- Determinism
 - Possibilism
 - Environmentalism
 - Probabilism

Exercise -2

Statement Based MCQ

1. With reference to the sub-tropical high pressure belt, consider the following statements
- It is affected by Earth's rotation and descent of winds from higher altitudes
 - It dynamically induced and characterized with anti-cyclonic conditions. Which of the statement/statements given above is/are correct
- 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
2. Consider the following rivers :
- Barak
 - Lohit
 - Subansiri
- Which of the above flows/flow through Arunachal Pradesh?
- 1 only
 - 2 and 3 only
 - 1 and 3 only
 - 1, 2 and 3
3. With reference to 'Changpa' community of India, consider the following statements :
- They live mainly in the State of Uttarakhand.
 - They rear the Pashmina goats that yield a fine wool.
 - They are kept in the category of Scheduled Tribes.
- Which of the statements given above is/are correct?
- 1 only
 - 2 and 3 only
 - 3 only
 - 1, 2 and 3
4. Assertion (A): In contemporary geography, there is an increasing awareness of the need for an interdisciplinary approach to the regional problems.
Reason (R): Groups of scholars with diverse backgrounds and skills focus on specific sets of regional problems more comprehensively and scientifically.
- Both A and R are individually true and R is the correct explanation of A
 - Both A and R are individually true but R is not the correct explanation of A
 - A is true but R is false
 - A is false but R is true
5. Consider the following statements:
- Human Geography covers all those aspects of Geography which are not directly concerned with Physical Geography.
 - Human Geography is the study of interrelationship between human beings and their environment.
 - Human Geography deals with the description and explanation of human phenomena around the variable earth surface.
 - Human Geography does not cover technical matters of Cartography.
- Which of the above statements are correct?
- 1 and 2
 - 2 and 3
 - Only 3
 - 1,2,3 & 4

Hints and Explanations

EXERCISE-1

- | | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1 | (b) | 2 | (c) | 3 | (b) | 4 | (b) | 5 | (c) |
| 6 | (b) | 7 | (a) | 8 | (d) | 9 | (b) | 10 | (c) |
| 11 | (c) | 12 | (d) | 13 | (b) | 14 | (d) | 15 | (a) |
| 16 | (c) | 17 | (b) | 18 | (a) | 19 | (c) | 20 | (c) |
| 21 | (b) | 22 | (a) | 23. | (b) | 24. | (a) | 25. | (c) |
| 26. | (b) | 27. | (c) | 28. | (c) | 29. | (a) | 30. | (b) |

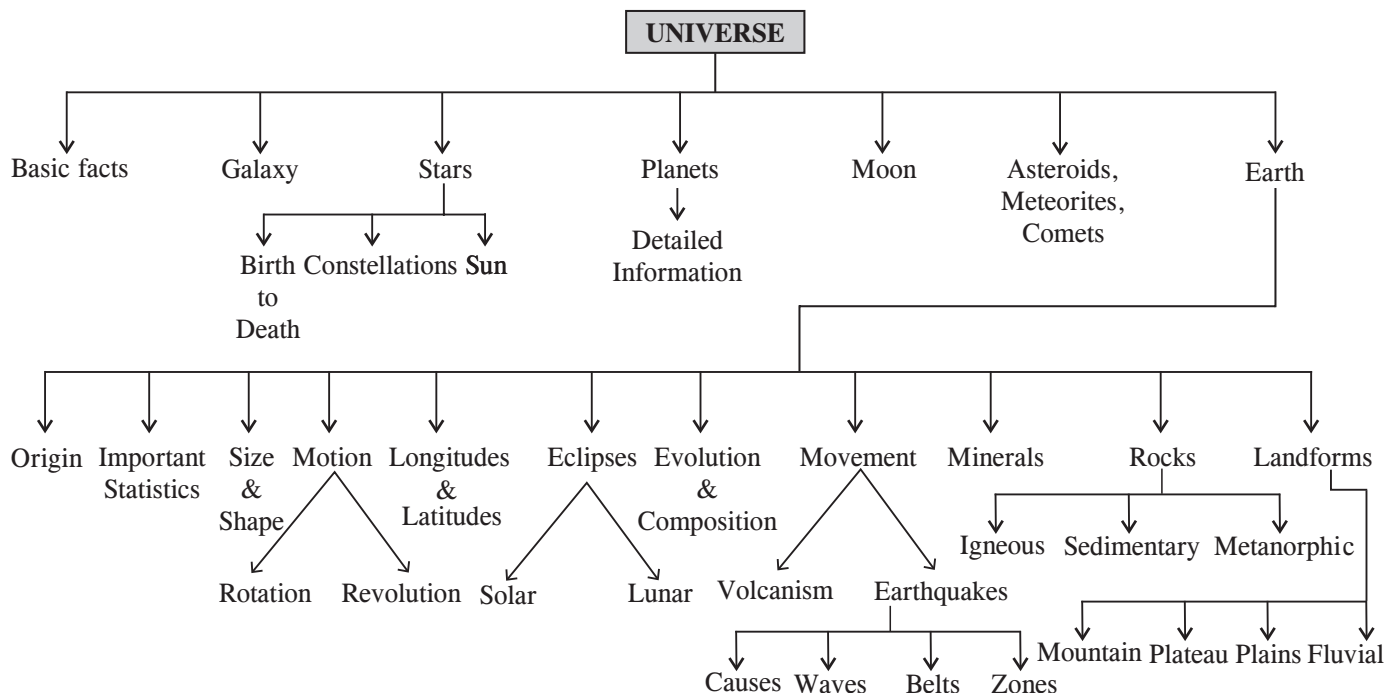
EXERCISE-2

- | | | |
|----|-----|---|
| 1. | (c) | |
| 2. | (b) | Rivers Lohit and Subansiri flow through Arunachal Pradesh. River Barak flows in south Assam. |
| 3. | (b) | The Changpa are a semi-nomadic Tibetan ethnic group found mainly in Zaskar region of Jammu and Kashmir. They rear the Pashmina goats that yield a fine wool. They are kept in the category of Scheduled Tribes. |
| 4. | (a) | |
| 5. | (c) | |



Introduction

Astronomy is a science that asks fundamental questions about the very basic of things, the universe. The **Universe** is all of time and space and its contents. The Universe includes planets, stars galaxies, the contents of intergalactic space, the smallest subatomic particles, and all matter and energy. The observable universe is about 28 billion parsecs (91 billion light-years) in diameter at the present time. The size of the whole Universe is not known and may be either finite or infinite. Observations and the development of physical theories have led to inferences about the composition and evolution of the Universe.



UNIVERSE

- Everything that exists, from the Galactic Megaclusters to the tiniest subatomic particles, comprises the **Universe**.
- As for the **age of Universe**, scientists agree that it is about 13.79 billion years old as 2015.
- The universe comprises of a number of galaxies.
- Optical and radio telescope studies indicate the existence of about 100 billion galaxies in the visible universe.
- The Big Bang Theory is most accepted for the origin of Universe in comparison to the Steady State and the Pulsating Universe Theory.

GALAXY

- Galaxy is a collection of millions or billions of stars and planets that are held together by gravitational pull.
- Milky Way is one such galaxy. The earth lies in this galaxy. It is called Milky Way, because it looks like a river of milky light flowing from one corner to another of the sky.
- It is spiral in shape.
- We call it Akash Ganga.
- The nearest galaxy to Milky Way is *Andromeda*. Andromeda is a *spiral galaxy* and approximately 2.5 million light-years from the earth.

STARS

- Luminous heavenly bodies which have their own light and other radiant energy are called star.
- They are made of extremely hot burning gases.
- **Star reflects Looks** - *Red* with low temperature, *Yellow* with higher and *blue* with very high temperature.

Star (Birth to Death)

- Star starts its life as clouds of dust and gas known as *Nebula*.
- The gaseous matter of Nebula further contracts to make dense region named *Proto Star*.
- The Proto Star further condenses to a critical stage of mass where *nuclear fusion* begins and *star* finally comes into existence.
- When all the *hydrogens* of a star are used up then its *helium* begins fusing into *carbon*. At a stage helium's fusion and energy production inside the star stops. As a result stars core contracts under its own weight to a very high density to make a white Dwarf star.
- White Dwarf star becomes *dark balls of matter* on cooling to make *Black Dwarf Star*.
- The mass of white Dwarf Star is less than 1.44 times the mass of the Sun named as *Chandrasekhar Limiting Mass*.
- White Dwarf Star is a dead star because of the end of fusion reaction and energy production.
- It shines by radiating its stored heat.
- Giant star expands into Red Supergiant after consuming its fuel (H & He). At a stage, it explodes as *Supernova* or changes into *Neutron* or *Black Holes*.
- The nearest star to the Earth is the Sun followed by Proxima and Alpha Centauri and radiant energy. (4.35 light years)

Constellations

- A *Constellation* is a group of stars that makes an imaginary shape in the sky at night.
- It helps in navigation of sea vessel during night as they are seen in a fixed direction at a particular period of time in a year.
- Orian, Big Dipper, Great Bear, Cassiopeia are some examples of constellations.
- Orion or Mriga can be seen in the late evening during winter; Cassiopeia in the Northern Sky is seen during winter.
- Great Bear consists of Ussa Minor (Laghu Saptarishi) and Ussa Major (Vrihat Saptarishi), and can be seen in early night during summer.

Sun

- It is a star made up of extremely hot gases, particularly by hydrogen (70%), Helium (26.5%) and others (3.5%) gases.
- It is 109 times bigger than the earth and weighs 2×10^{27} tonnes, and accounts for 99.83% of mass of the solar system.
- It is 150 million km away from the earth. The sun light takes 8 minutes to reach the earth's surface.
- It has immense gravitational pull which keeps the planets fixed in their orbit, revolving round the sun.
- It continuously gives off energy in the form of visible light, infra red, ultra violet, X-rays, gamma rays, radio waves and plasma gas.
- The period of revolution of the sun around the galactic centre is 250 million year. This period is called as *cosmic or galactic year*.
- Sudden flash of brightness observed near the sun's surface which is a collection of magnetic energy including electrons, protons and nuclei is called as *solar flares*. They are consized particles and are harmful for satellite communication.
- The layers of sun are divided according to their brightness level which is represented in the write features of each layer shown in the figure.

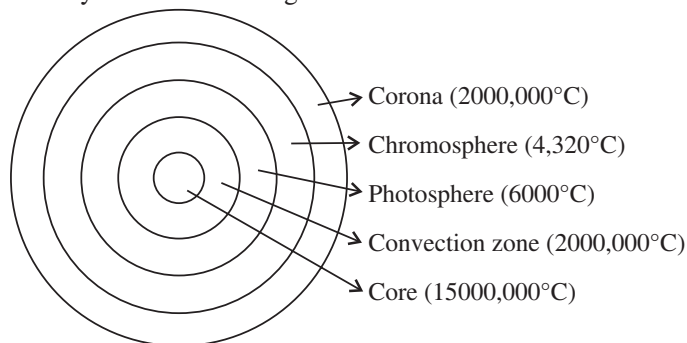


Fig. Layers of the Sun

- The core of the sun consists of hydrogen atoms which fuse together due to compression and creates helium. This is called as *nuclear fusion*.
- Nuclear fusion produces huge amount of energy. It is radiated outward to the surface, atmosphere and beyond.
- Convection zone is the next to the core of the sun. Here the temperature drops to 2 million degree C.
- Photosphere's temperature is 6,000°C.
- *Atmosphere* of the sun consists of chromosphere and corona.

- Corona is seen in a form of spectral lines emitted by iron, calcium and nickel ions. Ionization of these elements increases temperature of corona.
- Recently coronal heating puzzle has been related to magnetic carpet found in corona.
- The solar flare (wind) is a stream of charged particles released from upper atmosphere of the sun. These charged particles when get trapped by earth's magnetic field while entering in the upper atmosphere of the earth results in *auroral* (light) display.
- These auroral display in the northern hemisphere is called as *Aurora Borealis* (the Northern light) and when occurs in southern hemisphere is called as *Aurora Australis* (the Southern lights)
- *Sun-spots* are dark appearing areas present in photosphere from where solar flares originate. They are relatively a region cooler than its surrounding. It appears and disappears after every 11 years. This period is called Sun-spot Cycle.
- Plage is a brighter region in the chromosphere near to sunspot.

PLANETS

- Planets means 'wanderers'. There are eight planets in our solar system (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune).
- All these planets move around the sun in a fix orbit. Which is elongated in shape (elliptical).
- A new planet 2003 *UB 313* has been included recently in solar system. It is bigger than Pluto and farthest from the Sun.
- International Astronomical Union (IAU) recognized five dwarf planets such as : Ceres, Pluto, Haumea, Makemake, and Seden.
- The planets are grouped into two :
 - (1) **Terrestrial planets:** These are dense *rocky bodies* and are called as earth like planets. Mercury, Venus, Earth, and Mars are included in it. They are also called as inner planets.
 - (2) **Jovian Planets :** The outer planets which are gigantic in size and are *gaseous* in composition with large satellite are called as Jovian planets. These have similar features to that of Jupiter, thus called as Jupiter like planets.

Pluto, the Dwarf Planet

- Pluto was known as the smallest planet in the solar system and the ninth planet from the sun.
- Today Pluto is called a "dwarf planet".
- On average, Pluto is more than 3.6 billion miles away from the sun.
- Pluto is in a region called the Kuiper Belt. One day on Pluto is about 61/2 days on Earth.
- It has five moons. Its largest moon is named Charon. Pluto's four other moons are named Kerberos, Styx, Nix and Hydra.

Detail Information About The Planets

Planets/Features	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune
Equatorial & Diameter (km)	4,878	12,104	12,756	6,792	142,984	120,536	51,118	49,528
Mass (kg)	3.28×10^{23}	4.867×10^{24}	5.972×10^{24}	6.39×10^{23}	1.8913×10^{27}	5.683×10^{26}	8.681×10^{25}	1.024×10^{26}
Orbital Period (in days)	88	225	365	687	4,333	10,756	30,687	60,190
Inclination to Sun's Equator	3.38°	3.86°	7.25°	5.65°	6.07°	5.51°	6.48°	6.43°
Rotational Period (in days) and Direction	58.64 East to West	-243.02 West to East	1 East to West	1.03 East to West	0.41 East to West	0.43 East to West	-0.72 West to East	0.67 East to West
Distance from Sun (km)	57.91×10^6	10.82×10^7	14.96×10^7	22.79×10^7	77.85×10^7	14.33×10^8	28.77×10^8	44.98×10^8
Gravity (m/s ²)	3.7	8.89	9.78	3.71	24.79	10.44	8.69	11.5
No. of Moons	0	0	1	2	67	62	27	14
Axis Tilt	0.04°	177.36°	23.44°	25.19°	3.13°	26.73°	97.77°	28.32°
Mean density (g/cm ³)	5.43	5.25	5.52	3.93	1.33	0.71	1.27	1.67
Rings	no	no	no	no	yes	yes	yes	yes

Mercury

- Mercury is the smallest planet in our solar system - only slightly larger than the Earth's moon.
- It is the closest planet to the sun at a distance of about 58 million km (36 million miles) or 0.39 AU.
- One day on Mercury takes 59 Earth days.
- Mercury is a rocky planet, also known as a terrestrial planet.
- Mercury's thin atmosphere, or exosphere, is composed mostly of oxygen (O₂), sodium (Na), hydrogen (H₂), helium (He), and potassium (K). Atoms that are blasted off the surface by the solar wind and micrometeoroid impacts create Mercury's exosphere.
- Only two missions have visited this rocky planet: *Mariner 10* in 1974-5 and *MESSENGER*, which flew past Mercury three times before going into orbit around Mercury in 2011.
- Daytime Temperatures can reach 430° Celsius (800° Fahrenheit) and drop to -180° Celsius (-290° Fahrenheit) at night.

Venus

- Venus is only a little smaller than the Earth.
- Venus is the second closest planet to the sun at a distance of about 108 million km (67 million miles) or 0.72 AU.
- One day on Venus lasts as long as 243 Earth days (the time it takes for Venus to rotate or spin once).
- Venus is a rocky planet, also known as a terrestrial planet. Venus' solid surface is a created and volcanic landscape.
- Venus' thick and toxic atmosphere is made up mostly of carbon dioxide (CO₂) and nitrogen (N₂), with clouds of sulfuric acid (H₂SO₄) droplets.
- More than 40 spacecraft have explored Venus. The *Magellan* mission in the early 1990s mapped 98% of the planet's surface.
- The planet's extreme high temperatures of almost 480° Celsius (900° Fahrenheit) made it seem an unlikely place for life as we know it.
- Venus *spins backwards* (retrograde rotation) when compared to the other planets. This means that the *sun rises in the west and sets in the east on Venus*.

Earth

- Earth is the third planet from the sun at a distance of about 150 million km (93 million miles). That's one Astronomical Unit (AU).
- A day on Earth is 24 hours (the time it takes the Earth to rotate or spin once).
- Earth's atmosphere is 78% nitrogen (N₂), 21% oxygen (O₂) and 1% other ingredients - the perfect balance for Earthlings to breathe and live. Many planets in our solar system have atmospheres, but only Earth is breathable.
- Earth has one moon. Another name for a moon is *natural satellite*.
- Earth is the perfect place for life as we know it.
- Our atmosphere protects us from incoming meteoroids, most of which break up in our atmosphere before they can strike the surface as meteorites.

Mars

- Mars is the fourth planet from the sun at a distance of about 228 million km (142 million miles) or 1.52 AU.
- One day on Mars takes just a little over 24 hours (the time it takes for Mars to rotate or spin once).
- Mars is a rocky planet, also known as a terrestrial planet. Mars' solid surface has been altered by volcanoes, impacts, crustal movement and movement and atmospheric effects such as dust storms.
- Mars has a thin atmosphere made up mostly of carbon dioxide (CO₂), nitrogen (N₂) and argon (Ar).
- Mars has two moons named *Phobos* and *Deimos*.
- Several missions have visited this planet, from flybys and orbiters to rovers on the surface of the *Red Planet*. The first true Mars mission success was *Mariner 4* in 1965. At this time in the planet's history, Mars' surface cannot support life as we know it. Current missions exploring Mars on the surface and from orbit are determining Mars' past and future potential for life.
- Mars is known as the *Red Planet* because *iron minerals* in the Martian soil oxidize, or rust, causing the soil and the dusty atmosphere to look red.

Jupiter

- About 1,300 Earths could fit inside Jupiter.
- Jupiter is the fifth planet from the sun at a distance of about 778 million km (484 million miles) or 5.2 Astronomical Units (AU). Earth is one AU from the sun.
- One day on Jupiter takes about 10 hours (the time it takes for Jupiter to rotate or spin once).
- Jupiter is a gas-giant planet and therefore does not have a solid surface. Jupiter may have a solid, inner core about the size of the Earth.
- Jupiter's atmosphere is made up mostly of hydrogen (H₂) and helium (He).
- Jupiter has 50 known *moons*, with an additional 17 moons awaiting confirmation of their discovery, that is a total of *67 moons*.
- Jupiter has a faint *ring system* that was discovered in 1979 by the *Voyager-1 mission*. All four giant planets in our solar system have ring systems.
- Many missions have visited Jupiter and its system of moons. *The Juno mission* will arrive at Jupiter in 2016.
- Jupiter cannot support life as we know it. However, some of Jupiter's moons have oceans underneath their crusts that might support life.

Saturn

- Saturn is the sixth planet from the sun at a distance of about 1.4 billion km (886 million miles) or 9.5 AU.
- One day on Saturn takes 10.7 hours (the time it takes for Saturn to rotate or spin once).
- Saturn is a gas-giant planet and therefore does not have a solid surface.

- Saturn's atmosphere is made up mostly of hydrogen (H₂) and helium (He).
- Saturn has 53 known moons with an additional nine moons awaiting confirmation of their discovery, that is a total of 62 moons.
- Saturn has the most spectacular ring system, which is made up of seven rings with several gaps and divisions between them.
- Only a few missions have visited Saturn: Pioneer 11, Voyager 1 and 2 and Cassini-Huygens. Since 2004, Cassini has been exploring Saturn, its moons and rings.
- Fact: When Galileo Galilei was observing the planet Saturn in the 1600s, he noticed strange objects on each side of the planet and drew in his notes a triple-bodied planet system and later a planet with arms or handles. These "handles" were in fact the *rings of Saturn*.

Uranus

- Uranus is the seventh planet from the sun at a distance of about 2.9 billion km (1.8 billion miles) or 19.19 AU.
- One day on Uranus takes about 17 hours (the time it takes for Uranus to rotate or spin once).
- Uranus is an *ice giant*. Most (80 % or more) of the planet's mass is made up of a hot dense fluid of "icy" materials – water (H₂O), methane (CH₄), and ammonia (NH₃) – above a small rocky core.
- Uranus has an atmosphere which is mostly made up of hydrogen (H₂) and helium (He), with a small amount of methane (CH₄).
- Uranus has 27 *moons*. Uranus' moons are named after characters from the works of *William Shakespeare* and *Alexander Pope*.
- Uranus has 13 *known rings*. The inner rings are narrow and dark and the outer rings are brightly colored.
- *Voyager 2* is the only spacecraft to have visited Uranus.
- Uranus cannot support life as we know it.
- Unlike any of the other planets, Uranus *rotates on its side*, which means it *spins horizontally*.

Neptune

- Neptune is the eighth and farthest planet from the sun at a distance of about 4.5 billion km (2.8 billion miles) or 30.07 AU.
- One day on Neptune takes about 16 hours (the time it takes for Neptune to rotate or spin once).
- Neptune is a *sister ice giant to Uranus*.
- Neptune's atmosphere is made up mostly of hydrogen (H₂), helium (He) and methane (CH₄).
- Neptune has 13 *moons*. Neptune's moons are named after various *sea gods* and *nymphs* in Greek mythology.
- Neptune has *six rings*.
- *Voyager 2* is the only spacecraft to have visited Neptune.
- Neptune cannot support life as we know it.

Moon

- The moon is Earth's natural satellite and orbits the Earth at a distance of about 384 thousand km (239 thousand miles) or 0.00257 AU.
- The moon makes a complete orbit around Earth in 27 Earth days and rotates or spins at that same rate, or in that same amount of time. This causes the moon to keep the same side or face towards Earth during the course of its orbit.
- The moon is a rocky, solid-surface body, with much of its surface cratered and pitted from impacts.
- The moon has a very thin and tenuous (weak) atmosphere, called an *exosphere*.
- More than 100 spacecrafts have been launched to explore the moon. It is the only celestial body beyond Earth that has been visited by human beings.
- Twelve human beings have walked on the surface of the moon.

Asteroids

- Asteroids are minor planets especially those of the inner solar system.
- Asteroids orbit our sun in a region of space between the orbits of Mars and Jupiter known as the Asteroid Belt.
- Asteroids are solid, rocky and irregular bodies.
- Asteroids do not have atmospheres.
- More than 150 asteroids are known to have a small companion moon (some have two moons). The first discovery of an asteroid-moon system was of asteroid Ida and its moon *Dactyl* in 1993.
- Asteroids do not have rings.
- NASA space missions have flown by and observed asteroids, and one spacecraft (NEAR Shoemaker) even landed on an asteroid (*Eros*). The *Dawn* mission is the first mission to orbit (2011) a main belt asteroid (*Vesta*).
- Asteroids cannot support life as we know it.
- *Ceres*, the first and largest asteroid to be discovered (1801 by Giuseppe Piazzi), encompasses over one-third of the estimated total mass of all the asteroids in the asteroid belt.

Meteorites

- Meteorites may vary in size from tiny grains to large boulders. One of the largest meteorite found on Earth is the *Hoba* meteorite from southwest Africa, which weighs roughly 54,000 kg (119,000 pounds).
- Meteor showers are usually named after a star or constellation which is close to the radiant. Meteors and meteorites begin as meteoroids, which are little chunks of rock and debris in space.
- Most meteorites are either *iron*, *stony* or *stony-iron*.
- Leonid MAC (an airborne mission that took flight during the years 1998 - 2002) studied the interaction of meteoroids with the Earth's atmosphere.

- Meteoroids, meteors and meteorites cannot support life. However, they may have provided the Earth with a source of amino acids: the building blocks of life.
- Meteoroids become meteors or *shooting stars* when they interact with a planet's atmosphere and cause a streak of light in the sky. Debris that makes it to the surface of a planet from meteoroids are called *meteorites*.
- Meteorites may look very much like Earth rocks, or they may have a burned appearance. Some may have depression (thumbprint-like), roughened or smooth exteriors.
- Many of the meteor showers are associated with comets. The Leonids are associated with comet Tempel-Tuttle; Aquarids and Orionids with comet *Halley*, and the Taurids with comet Encke.

Comets

- Comets are cosmic snowballs of frozen gases, rock and dust.
- A comet warms up as it comes near the sun and develops an atmosphere, or coma. The coma may be hundreds of thousands of kilometers in diameter.
- Comets do not have moons.
- Comets do not have rings.
- Several missions have visited, impacted and even collected samples from comets. Two recent missions are Stardust-NExT and Deep Impact EPOXI.
- When *comets* come around the sun, they leave a dusty tail. Every year the Earth passes through the comet tails, which allows the debris to enter our atmosphere where it burns up and creates fiery and colorful streaks (meteors) in the sky.
- Comets may not be able to support life themselves, but they may have brought water and organic compounds, i.e., the building blocks of life -- through collisions with the Earth and other bodies in our solar system.
- *Comet Halley* makes an appearance in the Bayeux tapestry from the year 1066, which chronicles the overthrow of King Harold by William the Conqueror at the Battle of Hastings.

It is a short period comet visible from Earth every 75-76 years. Halley last appeared in the inner part of the solar system in 1986 and will next appear in mid - 2061.

Who is Called What	
Brightest Planet	Venus
Evening Star	Venus
Morning Star	Venus
Hottest Planet	Venus
Nearest Planet to Earth	Venus
Earth's twin	Venus
Fastest rotation in solar system	Jupiter
Slowest rotation in solar system	Venus
Green Planet	Uranus
Blue Planet	Earth
Red Planet	Mars
Smallest Planet	Mercury
Biggest Planet	Jupiter
Fastest revolution in Solar System	Mercury
Slowest revolution in Solar System	Neptune
Coldest Planet	Neptune
Closest star to the Sun	Proxima Centauri
Densest Planet	Earth
Least dense Planet	Saturn
Only satellite with an atmosphere like Earth	Titan
Smallest satellite	Deimos
Biggest Satellite	Gannymede

Venus is the hottest planet even though Mercury is the closest planet to the Sun. The reason behind it is that Venus has atmosphere made of carbon dioxide.

BLUE PLANET : THE EARTH

The Earth is the only known planet where life exists. Its surface area is covered with two-third of water that is why we call it blue planet.

Earth is the third planet from the sun, the densest planet in the solar system, the largest of the solar system's four terrestrial planets.

Origin of the Earth

- A number of theories have been proposed by different philosophers.
- *Immanuel Kant* gave gaseous hypothesis based on Newtonian law related to gravitation and rotatory motion in 1755.

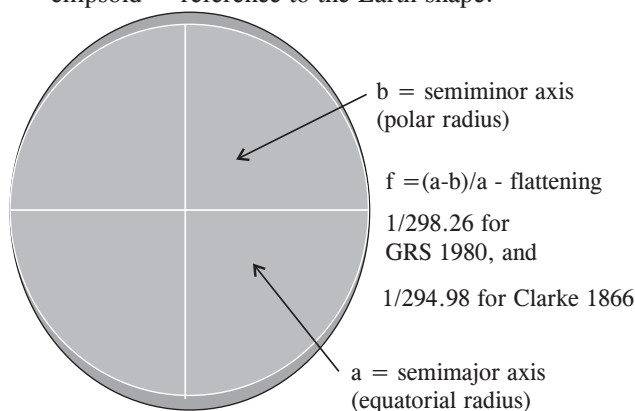
- Laplace gave Nebular hypothesis in 1786.
- Tidal hypothesis was given by James Jeans in 1929 and it was modified by Jeffrey and was called as Collision hypothesis.
- Big Bang theory was proposed by Georges Lemaitre (1927).
- According to *Big Bang theory* billion years ago cosmic matters were in highly compressed state. The expansion started with premordial explosion. It resulted in formation of superdense balls which travelled at a speed of thousands mile per second and gave rise to galaxies.
- The expansion of universe means increase in space between galaxies and formation of new galaxies.

Earth Statistics

• Age of The Earth	: 4.5 to 4.6 billion years
• Water-land Ratio	: 70.8 % Water & 29.2 % Land
• Earth's Circumference at the Equator	: 24,901.55 miles (40,075.16 km)
• Earth's Circumference between the North and South Poles	: 24,859.82 miles (40,008 km)
• Earth's Diameter at the Equator	: 7,926.41 miles (12,756.32 km)
• Average Distance from the Earth to the Sun	: 92,955,802 miles (149,579.870 km)
• Average Distance from the Earth to the Moon	: 238,855 miles (384,400.1 km)
• Highest Elevation on Earth	: Mt. Everest, Asia : 29,035 feet (8848 m)
• Tallest Mountain on Earth from Base to Peak	: Mauna Kea. Hawaii: 33,480 feet, i.e 10204 m (rising to 13796 feet above sea level) (10204 m, 4205 m)
• Point Farthest from the Center of the Earth	: The peak of the volcano chimborazo in Ecuador at 20561 feet (6269 m) is farthest from the center of the Earth due to its location near the equator and the oblateness of the Earth.
• Lowest Elevation on Land	: Dead Sea: 1369 feet below sea level (417 m)
• Deepest Point in the Ocean	: Challenger Deep (Mariana Trench) in Western Pacific Ocean: 36,740 feet (11022 m)
• Highest Temperature Recorded	: 136.8°F - AL Aziziyah, Libya, Sep. 13, 1922 (58.4°C)
• Lowest Temperature Recorded	: -128.6°F - Vostok, Antarctica. July 21, 1983 (-89.2°C)
• Atmosphere content	: 78% nitrogen, 21% oxygen and 1% traces of argon, carbon dioxide and water.
• Rotation on Axis	: 23 hours and 56 minutes and 04.09053 second. But, it takes an additional four minutes for the earth to revolve to the same position as the day before relative to the sun (i.e. 24 hours)
• Revolution Around Sun	: 365 . 2425 day
• Chemical Composition of the Earth	: 32.1 % Iron, 30.1 % Oxygen, 15.1 % Silicon, 13.9 %, Magnesium, 2.9 % Sulfur, 1.8 % Nickel, 1.5 % Calcium and 2.6 % other.
• Standard Time zones	: 24

Size and Shape of the Earth

- Shape of the Earth is called “geoid”
- The sciences of earth measurement is called “Geodesy”
- “ellipsoid” - reference to the Earth shape.



The geoid bulges at the North Pole and is depressed at the South Pole

- Earth shape is affected by two main facts:
- It bulges in midriff, because of pliability of Earth's lithosphere;
- Its shape is therefore an oblate spheroid.
- It has topographical irregularities.

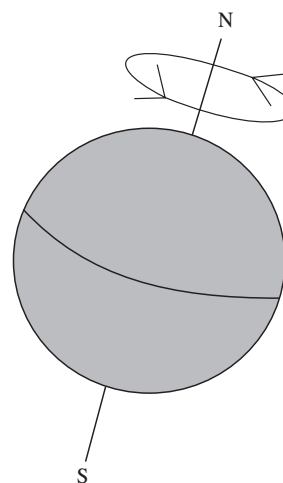
Motions of the Earth

The Earth is constantly in motion, revolving around the Sun and rotating on its axis. These motions account for many of

the phenomenon we see as normal occurrences: night and day, changing of the seasons, and different climates in different regions. With a globe ball properly mounted and rotating on its axis, the movements of the Earth around the Sun may be illustrated accurately.

Rotation

The Earth spins on its axis from West to East (counter-clockwise). It takes the Earth 23 hours, 56 minutes, and 4.09 seconds to complete one full turn. Day and night are produced by the rotation of the Earth. The speed of rotation at any point upon the equator is at the rate of approximately 1,038 miles per hour, decreasing to zero at the poles.

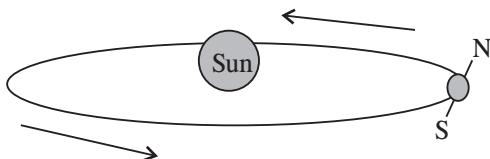


- **How rotation periods are calculated:** The period of rotation is calculated with reference to a star and with reference to the sun. When it is calculated with reference to a star, it is called a sidereal day and when it is calculated with reference to the sun, it is called a solar day.
- **Solar days and sidereal days:** The solar day is a time period of 24 hours, and the duration of a sidereal day is 23 hours 56 minutes. This difference of four minutes between a solar day and a sidereal day is due to the fact that the position of the Earth keeps changing with reference to the sun due to the revolution around it; while with reference to a star at infinity, it will remain unchanged. Thus, a sidereal day is the actual time taken by the planet for a rotation of exactly 360 degrees on its axis.

Revolution

While the Earth is spinning on its axis, it is revolving around the Sun in a counter-clockwise direction. It takes the Earth one full year to complete one full revolution around the Sun. This path is known as the Earth's orbit. It is very near a circle. The mean distance of the Earth from the Sun is about 93 million miles and the distance varies by 3 million miles, forming a slightly oval path.

The revolution of the Earth around the Sun traverse a distance of 595 million miles in 365 days, 6 hours, 9 minutes and 9.5 seconds. This means a speed of 18 miles a second (or 66,000 miles per hour) while at the same time rotating once each twenty-four hours.



Earth rotates in an elliptical orbit around the Sun

The orbit of the Earth around the sun is elliptical and not circular. Due to this, the distance between the Earth and the sun keeps changing.

- When this distance is minimum, the Earth is said to be in perihelion (around January 3).
- When the distance is the maximum, it is said to be in aphelion (around July 4).

The Earth's axis points constantly to the same point (the polar star) in the celestial sphere. As a consequence the latitude on the surface of the earth at which the sun's rays fall vertically keeps changing as the earth moves in its orbit around the sun. Due to this the earth attains four critical positions with reference to the sun.

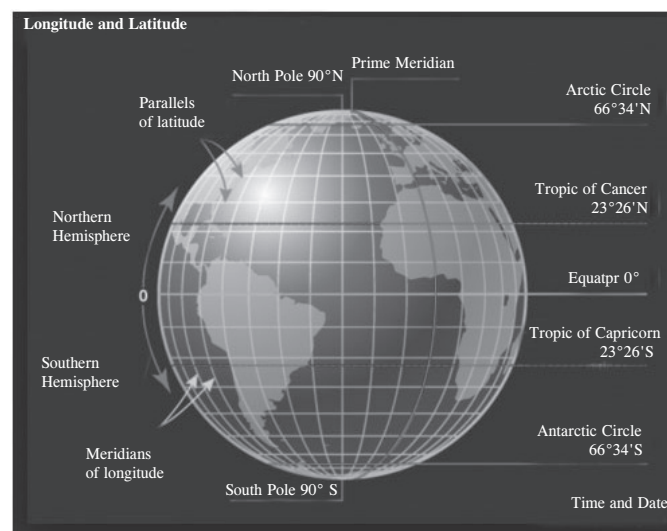
1. **The equinoxes:** On 21st March, the Earth is so positioned with reference to the sun that the sun's rays are vertical at the equator and the entire world experiences equal day and night.
2. **The autumnal equinox:** A similar situation occurs on September 23.
3. **Summer solstice:** On 21st of June the sun's rays are vertical over the Tropic of Cancer as the north pole of the Earth is inclined at its maximum towards the sun. At this time, the north pole experiences a long continuous day and the south pole a long continuous night (ergo, what we know as summer solstice). The northern hemisphere has

the summer season at this time and the southern hemisphere experiences winter now. Also the days are longer than the nights in the northern hemisphere at this time.

4. **Winter solstice:** On December 22, the position of the earth with respect to the sun is such that the south pole is inclined at its maximum towards the sun and the Tropic of Cancer receives the vertical rays of the sun. This position is called the winter solstice when the sun shines continuously in the south polar region and it is a long continuous night at the north pole. This is the winter season in the northern hemisphere and the summer in the southern hemisphere. During the winter solstice, the days are longer than the nights in the southern hemisphere.
- Thus, the variation in the duration of day and night and the change of seasons are due to the earth's revolution and the inclination of the axis of the earth. Also the seasons are reversed from the northern to the southern hemisphere.

Longitudes and Latitudes

Together, longitudes and latitudes form the Earth's geographical coordinates, and represent the angular distance of any location on the Earth from the Earth's Equator. Both latitudes and longitudes are measured in degrees.



The Equator

The Earth is (almost, but not quite) a sphere that rotates around its axis. If we draw a line passing through the center of the Earth along its rotational axis, the line would pass through the North and the South Pole.

The **Equator** is an imaginary line perpendicular to this axis. It is equidistant from the two poles, and divides the globe into the Northern Hemisphere and the Southern Hemisphere.

Most locations on the Equator experience consistently high temperatures throughout the year. They also experience almost 12 hours of daylight every day during the year. On the Equinoxes –50 autumn and spring – the sun is directly overhead the Equator, resulting in exactly 12 hour days and 12 hour nights.

Latitudes

- Latitudes are imaginary circles drawn parallel to the Equator.

- They are defined by the angle created by a line connecting the latitude and the center of the Earth, and the line connecting the Equator and the center of the Earth, and are named by the angle. Latitudes specify the North-South position of a location on the globe.
- When looking at a map, latitude lines run horizontally. Latitude lines are also known as parallels since they are parallel and are an equal distant from each other

Longitudes

- Longitudes are geographical positioning markers that run from the geographical North to the geographical South Pole, intersecting the Equator. They meet at both Poles, and specify the East-West position of a location.
- Longitudes are therefore imaginary circles that intersect the North and South Poles, and the Equator. Half of a longitudinal circle is known as a **Meridian**. Meridians are perpendicular to every latitude.
- Unlike, latitudes, there is no obvious central longitude. However, in order to measure the position of a location based on the longitude, cartographers and geographers over the course of history have designated different locations as the main longitudinal reference point. Today, the meridian line through Greenwich, England is considered as the reference point for longitudes. This line is also known as the **Prime Meridian**
- The Prime Meridian is set as 0° longitude and it divides the Earth into the Eastern and the Western Hemisphere. All the other longitudes are measured, and named after the angle they make with respect to the center of the Earth from the intersection of the Meridian and the Equator.
- Since a sphere has 360 degrees, the Earth is divided into 360 longitudes. The meridian right opposite the Prime Meridian (on the other side of the Earth) is the 180° longitude.
- Modern timekeeping systems use longitudes as references to keep time. Time zones are defined by the Prime Meridian and the longitudes.

Time and Longitudes

Local Time

Local time of any place is 12 noon when the Sun is exactly overhead.

Standard Time

It is the uniform time fixed by each country. It is fixed in the relation to mean time of a certain meridian which generally passes through it.

Greenwich Mean Time (GMT)

The line at 0° longitude is called Greenwich Mean Time. It is based on local time of the meridian passing through Greenwich near London.

Indian Standard Time

It is fixed on the mean of 82°30'E. Meridian, a place near Allahabad. It is 5½ hours ahead of Greenwich Mean Time.

Solstice and Equinoxes

Equinox

It is a day of the year when the duration of day and night is equal and the position of the sun is in its zenith. In year equinoxes occurs twice.

Solstice

Like equinoxes solstice also occur twice a year. This is the time when the sun reaches either its highest or lowest point at noon. It leads to the result of longest and shortest day of the year.

- **June solstice (approximately June 20-21):** This day begins summer in the Northern hemisphere and winter in the Southern hemisphere. This day is the Longest in the year. In the Northern Hemisphere and shortest. In the Southern hemisphere. On this day, the Sun is directly overhead the tropic of Cancer (23.5 degree at moon).
- **September Equinox (approximately September 22-23):** This day begins fall in the Northern hemisphere and spring in the Southern hemisphere. There are 12 hours of day light and 12 hours of darkness at all points on the Earth's surface on the two equinoxes. Sunrise is at 6 a.m. and sunset is at 6 pm.
- **December Solstice (Approximately December 21-22):** This day begins summer in the Southern hemisphere and is the longest day in the Southern hemisphere. It begins winter day in Northern hemisphere and is the shortest day of the year in the Northern hemisphere. The Sun is directly overhead the Tropic of capricorn on the December.
- **March Equinox (Approximately March 21-21):** Spring solstice equinox this day begins spring in the Southern hemisphere and spring in the Northern hemisphere. There are 12 hours of day light and 12 hours of darkness at all the points on the Earth's surface on the two equinoxes. Sunrise is at 6 a.m. and Sunset is at 6 pm. local (so far) time zone most points on the Earth's surface.

ECLIPSES

When the light of the Sun or the Moon is blocked by another body, the sun or Moon is said to be in eclipse. The Sun, Earth and Moon are in a straight line. Eclipses of the Sun is a solar eclipse and the Moon is a lunar eclipse.

Solar Eclipse

It is caused when the moon revolving around the Earth comes in between the Earth and the Sun, thus making a part or whole of the sun invisible from a particular part of the Earth thus, the eclipse can be partial or complete. It occurs during the day.

Type of Solar Eclipses

- **Total Eclipse :** Occurs when the Sun is completely obscured from view. Instead, the Sun's intense light is replaced by dark silhouette of the Moon that is outlined by the Sun's corona (the super heated plasma extending out from the Sun).
- **Annular Eclipse:** Occurs when the Sun and Moon are exactly in line but Moon appears smaller than the Sun. During an annular eclipse, the Sun appears as a bright ring around the Moon.

- **Partial Eclipse:** Occurs when the Sun and Moon are not completely aligned and the Sun is partially obscured.
- **Hybrid Eclipse:** is a combination of total and annular eclipse that takes place when a total eclipse changes to an annular eclipse or vice-versa along different sections of the eclipse's path.

Lunar Eclipse

- When the Earth comes between the Moon and the Sun, the shadow cast by the Earth on the Moon results in a lunar eclipse. It occurs at night.

Earth Theory

- A large number of hypothesis were put forth by different philosophers and scientists regarding the origin of the earth. One of the earlier and popular arguments was by German philosopher Immanuel Kant. Mathematician Laplace revised it in 1796. It is known as **Nebular Hypothesis**. The hypothesis considered that the planets were formed out of a cloud of material associated with a youthful sun, which was slowly rotating. Later in 1980, Chamberlin and Moulton considered that a wandering star approached the sun. As a result, a cigar-shaped extension of material was separated from the solar surface, as the passing star moved away, the material separated from the solar surface continued to revolve around the sun and it slowly condensed into planets. **Sir James Jeans** and later **Sir Harold Jeffrey** supported this argument. At a later date, the arguments considered of a companion to the sun to have been consisting. These arguments are called binary hypothesis. In 1950, Otto Schmidt in Russia and Carl Weizsäcker in Germany somewhat revised the 'nebular' hypothesis.
- **Modern Theories :** The most popular argument regarding the origin of the universe is the **Big Bang theory**. It is also called expanding universe hypothesis. Edwin Hubble, in 1920, provided evidence that the universe is expanding. As time passes, galaxies move further and further apart.

EVOLUTION OF THE EARTH

Continental Drift theory

- It was proposed by Wegener in 1912.
- According to the theory billion years ago all continents were joined together into one big landmass called as Pangaea. On the other hand all ocean joined together were called as Panthalassa.
- Pangaea floated over Panthalassa.
- Pangaea broke into two parts due to gravitational force and buoyancy.
- The northern part was known as Laurasia and southern part was known as Gondwanaland. These landmass were further splitted into smaller landmass forming different continents.
- Tethys sea was formed when Pangaea first splitted into Laurasia and Gondwanaland.
- Jig-saw fit conventional current theory and plate tectonic theory supports it.
- Lithosphere lies the **asthenosphere**, which is the inner part of the upper mantle. Asthenosphere is in semi fluid state. Lithosphere floats on the asthenosphere and is broken up into lithospheric plates.
- Plate Tectonics theory defined a tectonic plate or lithospheric plate as a massive, irregularly shaped slab of solid rock, composed of both continental and oceanic lithosphere. The plates move horizontally over the asthenosphere as rigid units. Lithosphere is divided into seven major and some minor plates. The 7 major plates are: (i) Antarctica plate, (ii) North American plate, (iii) South American plate, (iv) Pacific plate, (v) India-Australia-New Zealand plate, (vi) Africa plate and (vii) Eurasian plate. Also there are minor plates like (i) Cocos plate, (ii) Nazca plate, (iii) Arabian plate, (iv) Philippine plate, (v) Caroline plate and (vi) Fuji plate.
- Three types of plate boundaries are found :
 1. Transform boundaries occur where the plates slide or grind with each other (San Andreas fault).
 2. Divergent boundaries occur where two plates slide apart from each other (Mid-Atlantic Ridge).
 3. Convergent boundaries occur where two plates slide towards each other. Found near Pacific boundary. North and South America.

Plate Tectonic theory

- It was argued that the outermost part of the earth's interior consists of two layers: the **lithosphere** consists of crust, and the solidified uppermost part of the mantle. Below the

Geological Time Scale

Eons	Era	Period	Epoch	Age/years Before Present	Life/Major Events
	Cainozoic (From 65 million years to the present times)	Quaternary	Holocene Pleistocene	0 - 10,000 10,000 - 2 million	Modern Man Homo Sapiens
		Tertiary	Pliocene Miocene	2 - 5 million 5 - 24 million	Early Human Ancestor Ape : Flowering Plants and Trees Anthropoid Ape
			Oligocene Eocene Palaeocene	24-37 Million 37 - 58 Million 57 - 65 Million	Rabbits and Hare Small Mammals : Rats - Mice
	Mesozoic 65 - 245 Million Mammals	Cretaceous Jurassic Triassic		65 - 144 Million 144 - 208 Million 208 - 245 Million	Extinction of Dinosaurs Age of Dinosaurs Frogs and turtles

	Palaeozoic 245-570 Million	Permian Carboniferous Devonian Silurian Ordovician Cambrian		245-286 Million 286-360 Million 360-408 Million 408-438 Million 438-505 Million 505-570 Million	Reptile dominate-replace amphibians First Reptiles: Vertebrates : Coal beds Amphibians First trace of life on land: plants First Fish No terrestrial Life: Marine Invertebrate
Proterozoic Archean Hadean	Pre-Cambrian 570 Million 4,800 Million			570-2,500 Million 2,500-3,800 Million 3,800 - 4,800 Million	Soft-bodied arthropods Blue green Algae; Unicellular Bacteria Oceans and Continents form - Ocean and Atmosphere are rich in Carbon dioxide
Origin of Stars Supernova Big Bang	5,000-13,700 Million			5,000 Million 12,000 Million 13,700 Million	Origin of the sun Origin of the universe

Composition of the earth

Earth's Interior

- Interior of the earth is composed of non-uniform material. Velocity of seismic waves proves that the earth's interior consists of three different layers; (i) crust (ii) mantle and (iii) core. These layers are distinguished on the basis of their (i) physical and chemical properties, (ii) thickness, (iii) density, (iv) temperature, (v) metallic content and (vi) rocks.

Crust

The Outermost layer of the earth is called crust. Thickness of earth's crust varies from 8 to 60 km. It is divided into upper and lower crust.

- Upper crust has the density of 2.8.
- Lower crust has the density of 3.0.
- The upper crust has lower density due to the presence of lower density minerals than in lower crust.
- It composed of variety of igneous, metamorphic and sedimentary rocks.
- Most common rock found in earth's crust is igneous rocks. It covers 90% of the earth crust's volume.
- The temperature of crust increases with depth. Near mantle its temperature is recorded between 200°C to 400°C.
- It consists of different minerals like oxygen (46.6%), silicon (27.7%), aluminium (8.1%), iron (5%), others (12.6 %).
- The continent crust is thicker than the oceanic crust.
- The boundary between crust and mantle shows sudden increase in density and velocity of seismic wave which is called as moho-discontinuity.

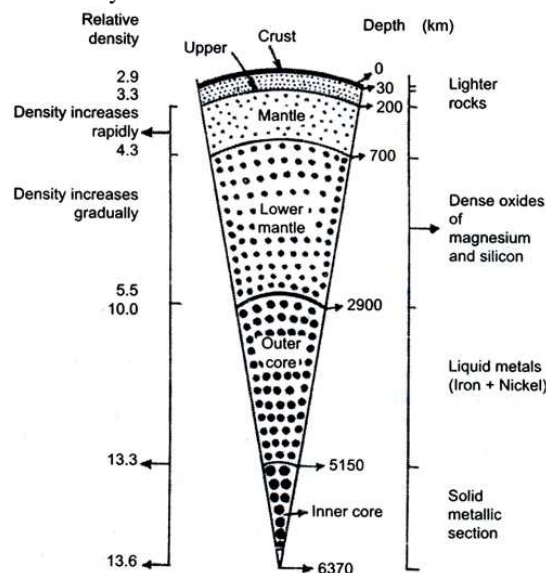
Mantle

- The second of the interior of the earth.
- It extends from 60 km to 2900 km. depth.
- Divided into two layers : upper mantle (60 – 1000 km) and lower mantle (1000 – 2900 km)
- Silica and magnesium are the dominant minerals found along with iron.

- The upper part of the mantle is in solid state as compared to its lower part.
- There is sudden change in the composition and structure of the interior of the earth at mantle core boundary. This zone of change in average density (5.5 to 10) is called as Wichter-Gutenberg discontinuity.

Core

- Core is the inner most layer of the earth's interior.
- Its density at the boundary of mantle is 10.3 and increases with increase in depth.
- It contains 32 percent of the total mass of the earth.
- It occupies 16 percent of the total volume of the earth.
- It is divided into two sub-zones. Outer core and inner core.
- The line dividing core into inner and outer core is at 5150 km of depth from earth's crust.
- The outer case is mainly a liquid layer of nickel and iron. The inner core is in solid state mainly consisting of iron nickel alloy.



Section through Earth
(Relation between Density, Section and Depth)

EARTH MOVEMENT

Movement in earth causes vast changes. The forces affecting earth, crust is divided into two on the basis of their origin:

1. Endogenetic forces
2. Exogenetic forces
 - Endogenetic forces result in sudden movement or diastrophic movement.
 - Sudden movement results in volcanic eruption and earthquake.
 - Diastrophic movement results in tensional and compressional forces.
 - Cracks and faulting is the result of tensional force.
 - Warping and folding is associated to compressional force.
 - Rift valley is formed due to tensional and compressional force.
 - Wave like bends in crustal rocks due to tangential compression force resulting in horizontal movement is called as fold.
 - There are five major types of folds : symmetrical, asymmetrical, monoclinical, isoclinal and recumbent folds.
 - Fracture in the crustal rock causing displacement of rock is called as fault.
 - There are four types of faults : Normal faults, Reverse faults, Lateral or strike-slip faults and Step faults.
 - Exogenetic forces are generated from the atmosphere due to varying combination of temperature and moisture.
 - Exogenetic forces result in denudational processes (denudation = weathering + erosion)

Volcanism

Movement of the heated material from the interior of the earth to a towards the surface is called as volcanism. Volcano is a vent opening through which these heated material like gases, water, liquid lava and fragment of rocks are ejected causing widespread of devastation.

Distribution of Volcanic activity in world

- (1) Circum-Pacific belt
- (2) Mid-continental belt
- (3) Mid-Atlantic belt
- (4) Intra-plate volcanoes
- (1) **Circum-Pacific belt:** The 'Ring of Fire', the belt starts from Erebus Mountain of Antarctica, running through Andes and Rockies mountains reaches Alaska from where it enters Asiatic coast turning eastward spreading over volcanoes of Island arc (Sakhaline, Kamchatka, Japan, Hawaii and Aloutin etc.) It is the subduction zone.
- (2) **Mid-Continental belt :** Convergent zone of continental plates. This belt includes the volcanoes of the Alpine mountain chain alongwith Mediterranean sea and reaches the fault zone of Africa in the east. It is formed due to collision of Eurasian plates with African and Indian plates.
- (3) **Mid-Atlantic belts :** It is a zone of divergent where two plates split and form new landmass. Here volcanoes of fissure eruption are found. It runs across whole of mid-

Atlantic stretching 11,300 km. The Hekla volcanic mountain of Iceland makes Iceland world's largest volcanic Island.

- (4) **Intra-Plate volcanoes :** The massive eruption that occurs in north-western part of North America, Indian Peninsula, Parana, of Barazil and Paraguay sometimes in past are part of intra-plate volcanoes.

Types of Volcanoes

- (1) **Active Volcanoes :** The volcanoes which have at least erupted one's in past 10,000 years. Etna (Sicily), Hekla (Iceland), Erta Ale (Ethiopia) Bromo (Java), Nevado del Ruiz (Colombia) etc. are in active state.
- (2) **Dormant Volcanoes :** They are active volcanoes but are not erupting at present Mt. Fuji (Japan) and Mt. St. Helen (Washington) are such examples.
- (3) **Extinct Volcanoes :** They are those volcanoes which have not erupted at least for 10,000 years and are not expected to erupt in future. Mt. Kilimanjaro (Tanzania) and Mt. Buninyong Australia are examples of extinct volcanoes.
2. **Classification based on periodicity of eruptions**
 - (a) **Active volcano:** Such volcanoes erupt frequently or at least have erupted in recent periods like barren islands in India and Mt Stromboli in Italy (Lighthouse of mediterranean sea).
 - (b) **Dormant volcano:** They are known to have erupted and show signs of possible eruption in future like Mt Kilimanjaro.
 - (c) **Extinct volcano:** They have stopped erupting but retain the features of volcanoes.

Earthquakes

- An earthquake is basically the vibration of earth produced by the rapid release of energy. This energy radiates in all directions from the source, in the form of waves. It is accompanied by a rumbling sound and tremors. The place of origin of an earthquake is its **focus**. The point on the earth's surface vertically above the focus is the **epicenter**. Intensity of vibrations is the maximum near the epicenter.
- Seismic sensors, located throughout the world can record the event.
- The magnitude or intensity of energy released by an earthquake is measured by the **Richter scale**. It ranges between 0 and 9.
- Elastic Rebound theory explains the mode and cause of earthquake.

Causes of Earthquake

- Most of the earthquakes are related with compressional or tensional stresses built up at margins of the moving lithospheric plates.
- Plates are dynamic and often push into one another or are pulled apart at times they even slide over each other. When the stress is too much these rocks break and the beings shake causing earthquake. The process of volcanicity,

faulting and elastic rebounding, hydrostatic pressure along with anthropogenic and plate tectonic results in earthquake occurrence. Sometimes, the tremendous energy released by atomic explosions or by volcanic eruptions can also produce earthquakes, but these events are usually too weak and infrequent.

Earthquake Waves

Earthquakes generate pulses of energy which are called **seismic waves** that can pass through the entire Earth. There are three major types seismic waves.

- **Primary (P) waves**, of short wavelength and high frequency, are longitudinal waves which travel not only through the solid crust and mantle but also through the liquid part of the earth's core. This is compression wave. Its speed is highest so it reaches first at earth surface.
- **Secondary (S) waves**, of short wavelength and high frequency, are transverse waves which travel through all the solid parts of the Earth but not the liquid part of the core. It is also called distortional wave and shear wave.
- **Long (L) Waves**, of long wavelength and low frequency, are confined to the skin of the earth's crust, thereby, causing most of the earthquakes structural damage. Long (L) Wave is also called Long period wave. These waves cover the longest distance of all seismic waves.

Distribution of Earthquakes belts in World

- Circum Pacific Belt** accounts for 68% of total earthquakes. Circum Pacific belt runs through the west coasts of North and South America, Aleutian Islands and the island group along the eastern coast of Asia.
 - Mid-Atlantic Belt** causes 21% of total earthquakes. This range causes mid-atlantic Ridge and several islands nearer the ridge. Earthquake even here are of moderate to shallow magnitude.
 - Mid Continental Belt** causes 11% of total earthquakes. Mid-continental mountain belt runs through the middle of Asia from east to west and goes beyond the Mediterranean Sea. Its axis lies along the mountain belt of the Himalayas, Caucasus, and the Alps.
- The Himalayas, making the great mountain wall of the north, also happen to form the northern margin of the Indian plate and hence are marked by frequent and severe earthquakes.
- Others**
 - Seismicity may also occur away from such regions. Such seismic activities, located within the plates away from the plate margins, are called in-plate seismicity.

- They are generally confined to the weaker zones of the earth's crust.
- These weaker zones are represented by faults or fractures within the earth's crust and are generally less intense than the ones found near the plate margins.
- One such zone of in-plate seismicity is the Narmada-Son lineament, cutting across the northern plane.
- The earthquake of January 26, 2001 of Gujarat was also caused by the growing 'in-plate stress' : whole Kutch is a fault. Bhuj, lies close to the Allah bund fault.
- Hence, in-plate seismicity is due to the reactivation of concealed shields (stable parts of the Earth) and release of energy.

The Earthquake Zones in India

On the basis of intensity of the earthquakes a map of India has been published by the Meteorological Department in collaboration of the Indian Standard Institution. The map shows the five seismic zones based on modified Mercalli scale.

Zone I - Intensity V or below (feeble, slight, moderate rather strong)

Zone II - Intensity VI (strong)

Zone III - Intensity VII (very strong)

Zone IV - Intensity VIII (Destructive Zone)

Zone V - Intensity more than VIII (Disastrous, Catastrophic)

Zone I - Comprises some parts of Punjab and Haryana, plains of Uttar Pradesh, Coastal plains of Maharashtra and Kerala.

Zone II - Includes Southern Punjab and Haryana, certain parts of Plains of Uttar Pradesh, Eastern Rajasthan, Coastal areas of Odisha and Tamilnadu. This is the low damage rests zone.

Zone III: Covers Southern and South Eastern parts of Rajasthan, larger parts of Madhya Pradesh, Maharashtra, Karnataka, Jharkhand and Northern and North-Western parts of Orissa.

Zone IV: Covers Jammu and Kashmir, Himachal Pradesh, Northern parts of Punjab, Haryana, Delhi, Eastern Uttar Pradesh, Tasai and Bhabhat regions, the Himalayan areas of Uttaranchal, Bihar and Sikkim

Zone V: Covers certain parts of Jammu and Kashmir, Himachal Pradesh, Uttranchal, Monghy, and Darbhanga districts of Bihar, Northern part of India and Kutchh region of Gujarat.

Some specific areas where the waves are not reported, such zone is called the 'shadow zone'.

MINERALS

Earth is composed of various elements. These are in a solid state in the outer layer of the earth and in molten state in the interior. About 98% of the total crust of the earth is composed of eight elements like oxygen, silicon, aluminum, iron, calcium, sodium, potassium and magnesium. Out of these, silicates carbonates and oxides make up a large group of them. Elements in the earth's crust are combined with other elements

to form various substances. These substances are known as **minerals**. Quartz, for example, has two elements, silicon and oxygen, united together form a compound known as carbonate of lime. The basic source of all minerals is the hot magma in the interior of the earth. When magma cools down, crystals of minerals appear and a systematic series of minerals are formed in sequence so as to form rocks.

ROCKS

The earth is composed of rocks. A rock is an aggregate of one or more minerals. There are three types of rocks on the basis of their formation. (1) Igneous, (2) Sedimentary and (3) Metamorphic rocks.

Igneous Rocks

- Igneous Rock is derived from latin word 'ignis' meaning fire.
- It is formed when magma cools and solidifies either on the surface of the earth or inside the earth.
- Igneous rocks have been forming since the earth was born.
- Out of three kinds of rock – igneous, sedimentary and metamorphic- they were first to be formed and hence they are called **Primary Rocks** and even parent rock.
- It is characterised by hardness which leads to no percolation of water resulting in no chemical weathering. But the grains of igneous rocks are affected by mechanical on physical weathering.
- No fossil is formed and is associated with volcanic activities.
- Igneous rocks are classified into (1) **extrusive** – it is formed when lava gets solidified on reaching the surface of the earth and (2) **intrusive** – it is formed by solidification of magma at moderate depths beneath the earth's surface and its rate of cooling is slow.
- Intrusive rocks are of two types such as plutonic rocks and hypabyssal rocks.
- Plutonic rocks are magmas which cooled deep inside the earth with a slow speed as temperature over there is high. These rocks are of coarse grained like granite.
- Hypabyssal rocks are volcanic magma which cools and solidifies in cracks, pores and hollows beneath the earth's surface, pores, batholiths, lopoliths, phacoliths, laccoliths, sills and dykes are its examples.

Sedimentary Rocks

The word 'sedimentary' is derived from the Latin word sedimentum, meaning settling. These are formed by the sediments brought by rivers, wind and glaciers. Rocks of the earth's surface are exposed to denudational agents, and are broken into smaller fragments. Such fragments are transported by different exogenous agencies and deposited. These deposits due to pressure and compression turn into rocks. Sedimentary rocks are sandstone, shale and conglomerates. These are also formed from organic matter derived from plants or animal remains and thus are called as fossil rocks. Coal and limestone are such sedimentary rocks. Sedimentary rocks are classified into three major groups:

- (i) **Mechanically formed** – Sandstone, Conglomerate, limestone, shale, loess.
- (ii) **Organically formed** – e.g. geyserite, chalk, limestone, coal etc.
- (iii) **Chemically formed** – like chalk, limestone, halite, potash etc.

Metamorphic Rocks

- Metamorphic means 'change of form'. Metamorphism covers all the processes by which rocks are altered in their mineralogy, texture and internal structure owing to external sources of heat, pressure and introduction of new chemical substances. Due to metamorphism, igneous and sedimentary rocks get totally changed in their physical state, chemical composition and crystallization of minerals. Granite is converted into gneiss, clay and shale are transformed into slate, and coal is transformed into graphite, limestone changes into marble. Gneiss, slate, schist and marble are examples of metamorphic rocks.

Four types of metamorphic rocks are recognized on the basis of its composition:

- (1) Contact or thermal metamorphism (due to heat)
- (2) Dynamic and regional metamorphism (due to pressure)
- (3) Hydro-metamorphism (due to hydro-static pressure)
- (4) Hydro-thermal metamorphism (due to water and heat)

LANDFORMS

- Oceans and continents are first order of relief.
- Mountains, plains and plateau are second order of relief.
- When the agent of erosion acts on these second order of relief the third order of relief is formed land forms formed by agents like river, wind and glacier etc.

Mountain

- Mountains are significant relief features of the second order on the earth's surface. A mountain may have several forms.
- **Mountain Ridge** – a system of long, narrow and high hills.
- **Mountain Range** – a system of mountain and hills with several ridges.

- **Mountain Chain** – Which have several parallel, long and narrow mountains belonging to different periods.
- **Mountain Group** – Consists of several unsystematic patterns of different mountain system.
- **Mountain System** – Consists of different mountain ranges belonging to the same period.
- **Classification based on mode of origin**

1. **Structural or Tectonic Mountain:** These mountains come into existence due to tectonic forces. They are of three types.

- (a) **Fold Mountains:** These are created by compressive forces. Fold mountains are of three types:

- (i) **Young Fold mountain** : Some young fold mountain are Alps in Europe, the Rockies of

North America, the Andes of South America, the Himalayas of Asia and Atlas of North Africa. These young fold mountains are still rising under the influence of the earth's tectonic forces.

(ii) **Matured Fold mountain** : The Urals, the Appalachians, the Tien Shan and the Nan Shan were formed during an earlier mountain-building period.

(iii) **Old Fold mountain** : Folded mountains which have originated before tertiary period are called old fold mountain e.g. – Caledonian and Hercynian, Vindhya, Aravalis. These are also called relict fold mountain because of denudation.

(b) **Block Mountains**: Originated by tensile forces leading to the formation of rift valleys e.g. Wasatch range in Utah (USA), Vosges and Black forest mountain (Europe), Salt Range (Pakistan) and Sierra Nevada mountain (USA). They are known as fault block mountains.

(c) **Dome Mountains**: Originated by magmatic intrusions and upwarping of the crustal surface e.g. Normal domes, Lava domes, Laccolithic domes, Salt domes etc.

2. **Mountain of accumulation or volcanic mountain** : Formed due to accumulation of volcanic material. Mount Mauna Loa in Hawaii islands, Mt. Fuji Yama of Japan and Mt. Popa in Central Myanmar are some of examples.

3. **Erosional or Relict Mountain** : These are formed due to the erosion of the earlier mountains. Its examples are the Vindhya range, Aravalis, Satpura, Eastern Ghats, and Western Ghats in India.

Plateau

- Plateaus are extensive area characterized by flat and rough top surface, steep sidewalls which rise above the neighbouring ground surface at least for 300 metres.
- Causes about 33% of the total earth's area.

Classification of plateau

1. Plateau formed by exogenetic processes

- Glacial Plateau**, e.g. Garhwal plateau, Greenland, Antarctica.
- Fluvial Plateau** e.g. Kaimur plateau, Bhandar plateau, Rewa Plateau, Rohtas Plateau.
- Aeolian Plateau** e.g. Potwar Plateau (Pak), Loess plateau (China)

2. Plateau formed by endogenetic processes

- Intermontane Plateau** : The plateaus which are partly or fully enclosed by mountains are known as intermontane plateaus. These are the results of the mountain-building process which was accompanied by a vertical uplift of the adjoining enclosed lands. e.g. Tibetan plateau, Bolivian plateau, Peruvian plateau, Columbian plateau and Mexican plateau.
- Piedmont Plateau** : Situated at the foot of a mountain, they are bounded on the opposite side by a plain or an ocean. These are also called the

plateaus of denudation because areas which were formerly high have now been reduced in elevation by various agents of erosion. e.g. Appalachian plateau, Patagonian plateau (Argentina).

(iii) **Dome Plateau** : Formed when land mass is uplifted e.g. Ozark Plateau (USA), Chotanagpur plateau (Jharkhand)

(iv) **Lava Plateau** : Formed due to accumulation of thick layers of basaltic lava e.g. Columbia plateau (USA), Mahabaleshwar plateau, Panchgani tableland.

(v) **Continental Plateau** : They rise abruptly from the lowlands or from the sea. e.g. Deccan plateau of India, Ranchi plateau, Shillong plateau, Columbia plateau (USA), Mexican plateau etc.

(vi) **Coastal Plateau** : e.g. Coromandel coastal upland of India.

(vii) **Desert Plateau** : Arabian Plateau.

(viii) **Humid Plateau** : e.g. Shillong Plateau, Assam Plateau, Mahabaleshwar Plateau etc.

(ix) **Young Plateau** : e.g. Idaho Plateau (USA), Colorado Plateau (USA), Mahabaleshwar Plateau, Khandala Upland (Maharashtra).

(x) **Mature Plateau** : e.g. Ranchi Plateau, Hazaribagh Plateau (Jharkhand), Appalachian Plateau (USA).

(xi) **Rejuvenated Plateau**: e.g. Missouri Plateau (USA).

Plains

- A plain is a relatively flat and low-lying land surface with least difference between its highest and lowest points. They are formed by endogenetic forces, erosion and exogenetic deposition.

Plains have been divided into:

- (1) Structural Plains
- (2) Erosional Plains
- (3) Depositional Plains

Structural Plains

- These plains are uplift of a part of the sea floor usually bordering a continent, that is, the continental shelf.
- Depressed areas which make up very extensive lowlands on the earth.
- The underlying horizontal beds of rocks are relatively undisturbed by the earth's crustal movements.
- Such plains include the great plains of Russian platform, the great plains of USA and the Central lowlands of Australia.
- The coastal plain lying between the Appalachian Piedmont Plateau and the Atlantic coast of south-eastern United States is an example of an uplifted coastal plain.

Erosional Plains

- These are formed when an elevated tract of land, for instance, a mountain, a hill or a plateau is worn down to a plain by the process of erosion.
- The surface almost smooth is termed as a **peneplain** river, ice and wind eroded regions.

- The plain formed by wind action is called as pediplain.
- Northern Canada, northern Europe and west Siberia are examples of such ice-eroded plains.
- Parts of Sahara in Africa are wind-eroded plain surface.
- The rivers by widening their banks and lowering the higher land between them have eroded parts of the Amazon basin into streameroded type of plains.

Depositional Plains

- The plain formed by deposition of sediments occurs due to the actions of rivers, glaciers, wind, sea wave etc. Different types of depositional plains are:
1. **River Plain** : These are formed by the alluvium brought down by the rivers. The Indo-Gangetic plain in Indian subcontinent, Hwang-ho plain of north China and Povalley in North Italy are some examples of alluvial plain. (1) Food plain (2) Deltaic plain (3) Pediment plain
 2. **Lacustrine plain** : A plain that originally formed in a lacustrine environment. It is a lake basin Valley. Kashmir is its example.
 3. **Lava Plain** : It is formed due to volcanic eruption.
 4. **Loess Plain** : Formed due to wind deposition. Plains of N. China and Russian Turkistan is formed due to it.
 5. **Marine Plain** : Develops near coast of shallow sea coastal areas of Netherlands Germany, Denmark and the Gulf of Mexico in U.S.A.
 6. **Glacial Plain** : Formed due to deposition of glacial and debris.
 - (a) Till plain (b) Morainic plain (c) Out wash plain

Fluvial Landforms

- Running water is the most important agent of denudation. Flow of a river or a stream is very strong. River action includes weathering and mass wasting causes fluvial denudation.

Erosion by rivers It is performed in two ways mechanical erosion and chemical erosion

They are fertile and economic, importance is dependent on the type of sediments brought by agents of denudation.

1. **Solution/Corrosion** : It is a type of chemical weathering in which most of the salts are removed from the bedrocks through the process of carbonation.
2. **Abrasion/Corrosion** : Involve the removal of loosened material of the rock of valley walls and floor by actions of erosional tools.
3. **Attrition** : Wear and tear of transported material when they roll and collide with each other.
4. **Hydraulic Action** : It involves breakdown of rocks of valley sides due to the impact of water currents. It is both mechanical and chemical process.

River Land Forms

- Upper Course landforms** : V-shaped valley, gorges, canyons, rapids and water fall these are mostly erosional plain of river. The river is at youth stage when it is at upper course.
- Middle Course landforms**: Alluvial fan and cone, meanders and oxbow lake are formed in the middle course of a river. The river achieves maturity stage in this course. Depositional features are formed in this course.
- Lower Course landforms** : Delta is formed at the lower course of the river. It is also known as old stage for a river.

Delta

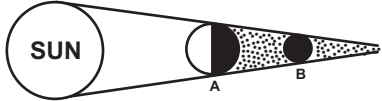
- Depositional feature of triangular shape at the mouth of a river debouching either into a lake or a sea is called a delta.

Types of Delta

- (a) **Bird's foot delta**, Ex-Mississippi delta
- (b) **Arccuate delta**, Ex-Nile, Ganga, Mekong.
- (c) **Estuarine delta** : Amazon, Ob, Vistula, and Tapi.

Their fertility and economic importance is dependent on the type of sediments brought by agents of denudation.

Exercise -1

- In order of their distances from the Sun, which of the following planets lie between Mars and Uranus?
 - Earth and Jupiter
 - Jupiter and Saturn
 - Saturn and Earth
 - Saturn and Neptune
- Which one of the following planets has largest number of natural satellites or moons?
 - Jupiter
 - Mars
 - Saturn
 - Venus
- Which one of the following statements is correct with reference to our solar system?
 - The earth is the densest of all the planets in our solar system
 - The predominant element in the composition of earth is silicon
 - The sun contains 75 percent of the mass of the solar system
 - The diameter of the sun is 190 times that of the earth
- The group of small pieces of rock revolving round the sun between the orbits of Mars and Jupiter are called
 - meteors
 - comets
 - meteorites
 - asteroids
- In the earth's direction of rotation is reversed, what should be the IST when it is noon at the International Date Line?
 - 06.30 hrs
 - 05.30 hrs
 - 18.30 hrs
 - 17.30 hrs
- Which one of the following conditions is most relevant for the presence of life on Mars?
 - Atmospheric composition
 - Thermal conditions
 - Occurrence of ice caps and frozen water
 - Occurrence of ozone
- A meteor is
 - a rapidly moving star
 - a piece of mater which has entered the earth's atmosphere from outer space
 - part of a constellation
 - a comet without a tail
- Diamond ring is a phenomenon observed:
 - at the start of a total solar eclipse
 - at the end of a total solar eclipse
 - only along the peripheral regions of the totality trail
 - only in the central regions of the totality trail
- An instrument to measure the amount of brightness of sunshine received at a point on Earth's surface is
 - Sun-dial
 - Sunshine recorder
 - Campbell - Stoke's recorder
 - All of the above
- Which would you record the maximum angle of incidence of the sun's rays at the North pole ?
 - March 21
 - September 21
 - When the sun's rays fall vertically on the Tropic of Cancer
 - When the sun's rays fall vertically on the Tropic of Capricorn
- In the following diagram, an eclipse is represented. Which statement is correct about it ?
 
 - The diagram represents a lunar eclipse, A is the moon and B is the earth
 - The diagram represents a solar eclipse, A is the earth and B is the moon
 - The diagram represents a lunar eclipse, A is the earth and B is the moon
 - The diagram represents a total solar eclipse
- What can be the maximum duration of totality for a solar eclipse?
 - 12.5 minutes
 - 7 minutes 40 seconds
 - 1 hour 42 minutes
 - There is no maximum duration
- That the earth is a sphere is not proved by
 - the sunrise not being visible from all places at the same time
 - the position of polaris as one travels from equator to North Pole
 - the circular shadow of earth at the time of solar eclipse
 - modern navigation methods
- Which of the following statements with regard to the term 'great circle' is not correct ?
 - Equator is a great circle.
 - A ship can save fuel and time by following the great circle arc between two points.
 - Only one great circle can be drawn on a sphere.
 - A great circle results when a plane passes through the centre of a sphere.
- The solar eclipse achieves totality only in limited geographical regions because
 - The size of the shadow of the moon on the earth is small as compared to the cross section of the earth
 - The earth is not smooth flat surface, but has elevations and depressions
 - The trajectories of the earth around the sun and the moon around the earth are not perfect circles
 - Sun rays can reach most of the peripheral regions of shadow of the moon due to atmospheric refraction
- The axis of rotation of the earth is tilted by 23.5° to the plane of revolution around the sun. The latitude of Mumbai is less than 23.5° whereas the latitude of Delhi is more than 23.5° . Which one of the following statements in this regard is correct?
 - The sun can come overhead at both these places
 - The sun will never come overhead at either of these places
 - At Mumbai the sun can come overhead, but it will never do so at Delhi
 - At Delhi, the sun can come overhead but it will never do so at Mumbai

17. The sidereal month may be defined as
 - (a) the period in which the moon completes an orbit around the Earth.
 - (b) the period in which the moon completes an orbit around the Earth and returns to the same position in the sky.
 - (c) the period of rotation of moon
 - (d) None of these
18. Isostasy is
 - (a) the property of the continental and oceanic crust, that they have the ability to rise and sink.
 - (b) due to which crust floats on top of the mantle like ice cubes in water.
 - (c) the phenomenon when the earth's crust gains weight due to mountain building or glaciation, it deforms and sinks deeper into the mantle.
 - (d) All are correct
19. Magma that is more granitic tends to be very explosive because
 - (a) of relatively higher water content
 - (b) of relatively higher gases content
 - (c) of relatively lower water content
 - (d) of relatively lower gases content
20. Which of the statements does not prove the spherical shape of the earth?
 - (a) If the earth were flat one would come across some sharp edges
 - (b) The sunrise is not visible from all places at the same time
 - (c) The shadow of the earth at the time of the solar eclipse is circular
 - (d) The altitude of the stars from various points on the earth's surface varies
21. One Astronomical Unit is the average distance between
 - (a) Earth and the Sun
 - (b) Earth and the Moon
 - (c) Jupiter and the Sun
 - (d) Pluto and the Sun
22. On the surface of the moon, the
 - (a) mass and weight become lesser
 - (b) mass remains constant and only the weight is lesser
 - (c) only the mass is lesser
 - (d) mass and weight both remain unchanged
23. We always see the same face of the moon because
 - (a) it is smaller than the earth.
 - (b) it revolves on its axis in a direction opposite to that of the earth.
 - (c) it takes equal time for both revolution around the earth and rotation on its own axis.
 - (d) it rotates at the same speed as the earth around the sun.
24. In which region can the phenomenon of midnight sun be observed?
 - (a) In the tropical zone
 - (b) In warm temperate regions
 - (c) In the Arctic and Antarctic regions
 - (d) Anywhere at the time of solar eclipse
25. The theory suggesting that the continents of South America and Africa were once joined together was the
 - (a) continental drift theory
 - (b) tetrahedral theory
 - (c) Kant's theory
 - (d) Ritter's theory
26. When there is noon at I.S.T. meridian people another place of the Earth are taking their 6 O' clock morning tea. The longitude of the place is
 - (a) 17°30' E
 - (b) 7°30' W
 - (c) 172°30' E
 - (d) 90° W
27. When it is 12 Noon at Allahabad, the time at Wellington (New Zealand) would be close to which one of the following?
 - (a) 4:30 pm of the same day
 - (b) 4:30 pm of the previous day
 - (c) 6:30 pm of the same day
 - (d) 6:30 pm of the previous day
28. The correct sequence of the earth's location on its orbit around the sun is
 - (a) autumnal equinox, vernal equinox, summer solstice, winter solstice
 - (b) vernal equinox, summer solstice, autumnal equinox, winter solstice
 - (c) summer solstice, vernal equinox, winter solstice, autumnal equinox
 - (d) winter solstice, summer solstice, autumnal equinox, vernal equinox
29. Beirut time is two hours ahead and Lima time is five hours behind GMT. The longitudes of Beirut and Lima are respectively:
 - (a) 75° West and 30° East
 - (b) 70° East and 35° West
 - (c) 30° East and 75° West
 - (d) 35° West and 70° East
30. If there are four place on the same meridian 500 km apart and the local time at one place is 12.00 noon, what will be the time at the three other places?
 - (a) 12.00 noon
 - (b) 1.00 pm
 - (c) 2.00 pm
 - (d) Different time at different places
31. Which one of the following is the time required for the earth to return to a given point in its orbit with reference to fixed stars called ?
 - (a) Lunar year
 - (b) Solar year
 - (c) Tropical year
 - (d) Sidereal year
32. Which one among the following explains the earthquakes of the eastern margins of Asia?
 - (a) Subduction of Pacific plate under Asiatic plate
 - (b) Subduction of African plate below European plate
 - (c) Subduction of Indian plate under Asiatic plate
 - (d) Subduction of American plate under the Pacific plate
33. Unlike other Meridians International Date Line is drawn zigzag in order to
 - (a) permit certain land areas and groups of islands to have the same calendar day
 - (b) facilitate the sailors to adjust time in their watch

- (c) adjust the day in calendar while sailing from east to west and vice versa
 (d) make 180°E and 180°W coterminous
34. The latitude is the angular distance of a point of the Earth surface, North or South, of the equator as measured from the
 (a) Centre of the Earth
 (b) Equator
 (c) Tropic of Cancer or Capricorn
 (d) Poles
35. Which one of the following pairs is not correctly matched?
- | | Month | Position of Sun |
|-----|--------------|--|
| (a) | June | Midday Sun overhead at Tropic of Cancer |
| (b) | December | Midday Sun overhead at Tropic of Capricorn |
| (c) | March | Midday Sun overhead on Equator |
| (d) | September | Midday Sun overhead at Arctic Circle |
36. The colour of the star is an indication of its
 (a) Distance from the earth
 (b) Distance from the sun
 (c) Temperature
 (d) Luminosity
37. The group of stars arranged in a definite pattern is called
 (a) Milky way (b) Constellation
 (c) Andromeda (d) Solar system
38. Which one of the following is the largest satellite in solar system?
 (a) Ganymede (b) Titan
 (c) Europa (d) Triton
39. The principle of Black hole was enunciated by
 (a) C.V. Raman
 (b) H.J. Bhabha
 (c) S. Chandrasekhar
 (d) H. Khurana
40. The distance of Moon from the Earth is
 (a) 364 thousand kms.
 (b) 300 thousand kms.
 (c) 350 thousand kms.
 (d) 446 thousand kms.
41. Which planet was named after the Roman God Zeus?
 (a) Earth (b) Mars
 (c) Venus (d) Jupiter
42. The approximate diameter of Earth is
 (a) 4200 km (b) 6400 km
 (c) 3400 km (d) 12800 km
43. What is the time taken by the Earth to complete one rotation on its axis?
 (a) 23 hr 52 min 4 sec (b) 23 hr 56 min 4 sec
 (c) 24 hr 53 min 2 sec (d) 24 hr 12 min 6 sec
44. The Blue Moon phenomenon occurs
 (a) when two full moons occur in the same month.
 (b) when two full moons appear in the same month thrice in a calendar year.
 (c) when four full moons appear in two consecutive months of the same calendar year.
 (d) None of the above
45. During the Venus transit, the planet appeared as a tiny black circle moving on the Sun. The black colour on the Sun is because the planet :
 (a) Obstructed all light from the Sun.
 (b) Is black in colour.
 (c) Was invisible due to bright rays from the Sun.
 (d) Behaved as a black hole during its transit.
46. Scientists have determined the temperature near the Earth's centre $1,000^{\circ}\text{C}$ hotter than was reported in an experiment run 20 years ago at
 (a) $6,000^{\circ}$ Celsius
 (b) $5,000^{\circ}$ Celsius
 (c) $4,000^{\circ}$ Celsius
 (d) None of these
47. Space between Earth and Moon is known as
 (a) Cislunar (b) Fulalunar
 (c) Nebula (d) None of these
48. Which of the following stars is known as Fossil star?
 (a) Protostar (b) Dog Star
 (c) Red Giant (d) White Dwarf
49. The energy of sun is produced by
 (a) Nuclear fission (b) Ionisation
 (c) Nuclear fusion (d) Oxidation
50. Which one of the following conditions is most relevant for the presence of life on mars?
 (a) Atmospheric composition
 (b) Thermal conditions
 (c) Occurrence of ice cap and frozen water
 (d) Occurrence of ozone
51. Lunar eclipse occurs
 (a) When moon lies between earth and sun
 (b) When earth lies between sun and moon
 (c) When sun lies between earth and moon
 (d) None of these
52. By how much degree the earth is inclined on its own Axis
 (a) $23^{\frac{1}{2}}$ (b) $66^{\frac{1}{2}}$
 (c) $24^{\frac{1}{2}}$ (d) $69^{\frac{1}{2}}$
53. Which of the following elements occurs the most abundantly in our universe?
 (a) Hydrogen (b) Oxygen
 (c) Nitrogen (d) Helium
54. The black part of the moon is always calm and dark which is called
 (a) Sea of tranquility (b) Ocean of storms
 (c) Area of storms (d) None of these
55. Variations in the length of daytime and nighttime from season to season are due to
 (a) the earth's rotation on its axis
 (b) the earth's revolution round the sun in an elliptical manner
 (c) latitudinal position of the place
 (d) revolution of the earth on a tilted axis.

56. Plate tectonics is a scientific theory that describes the large scale motions of Earth's lithosphere. Which one among the following statements regarding Plate tectonics is not correct?
- (a) Tectonic plates are composed of Oceanic lithosphere and thicker Continental lithosphere
 - (b) Tectonic plates are able to move because the Earth's lithosphere has a higher strength than the underlying asthenosphere
 - (c) The Earth's lithosphere is broken up into Tectonic Plates
 - (d) Along divergent plate boundaries, subduction carries plates into the mantle
57. Albedo effect would be relatively higher in
- (a) early morning and late evening
 - (b) early morning only
 - (c) noon
 - (d) late evening only
58. Which of the following statements regarding the duration of day and night is correct?
- (a) Difference is least near the Equator and progressively increases away from it
 - (b) Difference is maximum at the Equator and progressively decreases away from it
 - (c) Difference is least at the Tropics and progressively increases towards the Equator and Poles
 - (d) Difference is maximum at the Tropics and progressively decreases towards the Equator and Poles
59. Which one of the following is depositional landform ?
- (a) Stalagmite (b) Lapis
 - (c) Sinkhole (d) Cave
60. What explains the eastward flow of the equatorial counter-current? [CSAT 2015 - I]
- (a) The Earth's rotation on its axis
 - (b) Convergence of the two equatorial currents
 - (c) Difference in salinity of water
 - (d) Occurrence of the belt of calm near the equator

Exercise -2

Statement Based MCQ

1. A team of scientists at Brookhaven National Laboratory including those from India created the heaviest anti-matter (anti-helium nucleus). What is/are the implication/implications of the creation of anti-matter?
 1. It will make mineral prospecting and oil exploration easier and cheaper.
 2. It will help probe the possibility of the existence of stars and galaxies made of anti-matter.
 3. It will help understand the evolution of the universe.
 Select the correct answer using the codes given below :

(a) 1 only	(b) 2 and 3
(c) 3 only	(d) 1, 2 and 3
2. About 95 per cent of active volcanoes occur at the
 1. plate subduction zones
 2. transform faults
 3. island arcs
 4. mid-oceanic ridges
 Choose your answer from the following codes.

(a) 1 and 2	(b) 1 only
(c) 1 and 4	(d) All of the above
3. Which of the following is/are cited by the scientists as evidence/evidences for the continued expansion of universe?
 1. Detection of microwaves in space.
 2. Observation of redshift phenomenon in space.
 3. Movement of asteroids in space.
 4. Occurrence of supernova explosions in space.
 Select the correct answer using the codes given below :

(a) 1 and 2	(b) 2 only
(c) 1, 3 and 4	(d) None of the above can be cited as evidence
4. Consider the following statements:
 1. The Axis of the earth's magnetic field is inclined at 23 and half to the geographic axis of the earth.
 2. The earth's magnetic pole in the Northern Hemisphere is located on a Peninsula in Northern Canada.
 3. The earth's magnetic equator passes through Thumba in South India.
 Which of the statements given above is/are correct?

(a) 1, 2 and 3	(b) 2 and 3
(c) 2 only	(d) 3 only
5. Lithosphere covers
 1. Continental crust
 2. Oceanic crust
 3. Upper solid part of mantle
 4. Upper liquid part of mantle
 Select the correct answer from the codes given below.

(a) 2 and 3	(b) 1, 2 and 3
(c) 1, 2 and 4	(d) 3 and 4
6. Earthquakes are important in landform studies because they can trigger catastrophic events like
 1. Landslides and mudflows
 2. Glacier surges
 3. Tsunamis causing damage and shoreline changes
 Select the correct answer from the codes given below.

(a) 1, 2 and 3	(b) 3 only
(c) 1 and 2	(d) 1 and 3
7. Which of the following statements is/are correct?
 1. All stars and planets in the universe shine by the reflected light of the sun.
 2. Sun is believed to be formed of condensation of gases.
 3. The sun has a surface temperature of about 20 million °C.
 4. Sun is about 30,000 times as big as the earth.
 Select the correct answer from the codes given below.

(a) 2 only	(b) 1 and 2
(c) 3 and 4	(d) 2 and 3
8. If the earth's axis had not been inclined,
 1. temperature distribution would have been uniform.
 2. days and nights would not have occurred.
 3. seasons would not have occurred.
 4. poles would not have continuous day.

(a) 1, 3 and 4	(b) 2 and 4
(c) 3 and 4	(d) 2, 3 and 4
9. The permanent tilt of the earth's axis and the revolution of the earth in its orbit together cause
 1. deflection of winds and ocean currents.
 2. differences in time between places on different meridians.
 3. varying lengths of day and night at different times of the year.
 4. changes in the altitude of the mid-day sun at different times of the year.

(a) 2, 3 and 4	(b) 1 and 4
(c) 3 and 4	(d) 1, 2 and 3
10. Which of the following are true with respect to fold mountains?
 1. Their tops are often buried beneath snow and ice.
 2. They form rugged peaks.
 3. They are caused by contraction of the earth's crust.
 4. They contain a core composed of metamorphic and igneous rocks.

(a) 1, 3 and 4	(b) 2, 3 and 4
(c) 1 and 2	(d) 1 and 4
11. On which days do the spring tides occur?
 1. new moon
 2. first quarter of the moon
 3. third quarter of the moon
 4. full moon

(a) 1, 2 and 4	(b) 2, 3 and 4
(c) 1 and 4	(d) 2 and 3
12. Arrange the planets correctly in descending order in terms of number of their satellites.

1. Uranus	2. Jupiter
3. Saturn	4. Mars
(a) 2, 3, 1, 4	(b) 4, 2, 3, 1
(c) 2, 3, 4, 1	(d) 3, 2, 1, 4

13. The standard time of the following countries is ahead of behind Greenwich Mean Time depending on whether they are East or West of the longitude passing through Greenwich.
- | | |
|----------|---------------|
| 1. Cuba | 2. Greece |
| 3. Iraq | 4. Costa Rica |
| 5. Japan | |
- Which one of the following sequential orders gives the correct arrangement of the countries according to their standard time from ahead to behind GMT ?
- (a) 5, 3, 2, 1, 4 (b) 2, 4, 1, 3, 5
(c) 4, 1, 3, 2, 5 (d) 3, 5, 4, 1, 2
14. Consider the following statements :
- As a rock, graphite is formed from coal by thermal metamorphism.
 - Gypsum is an example of sedimentary rock.
- Which of these statements is/are correct ?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
15. Consider the following statement:
- Mechanical weathering takes place only in sedimentary rocks.
 - The rain-water reacts with rocks to form new chemical substances.
- Which of the statements given above is/are correct ?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
16. Which one of the following statements is/are correct with regard to lines of latitude?
- They are concentric circles numbered from 0° to 90° .
 - They are circles on a globe which are parallel to the Equator and which are to the North and South of the Equator.
- Select the correct answer using the code given below:
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
17. Which one of the following statements is/ are correct with regard to Milky Way?
- It is a spiral galaxy.
 - The solar system resides in one of its spiral arms.
- Select the correct answer using the code given below:
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
18. Consider the following statements.
- The tropical year is shorter than the sidereal year.
 - The solar day is longer than the sidereal day.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
19. Which of the following statements regarding the Deccan Traps is/are correct?
- Intense volcanic activity in the form of fissure eruption took place towards the end of Cretaceous period.
 - The volcanic lava spread out in horizontal sheets.
 - The regur soil found here is rich in nitrogen.
- Select the correct answer using the codes given below
- (a) 1 and 2 (b) 1, 2 and 3
(c) 3 only (d) 1 only
20. Which of the following statements is/ are correct?
- The major constituent mineral of granite rock is quartz.
 - The major constituent mineral of sandstone rock is feldspar.
 - The major constituent mineral of limestone rock is dolomite.
- Select the correct answer using the code given below:
- (a) 1, 2 and 3 (b) 3 only
(c) 1 and 2 (d) 2 and 3
21. Which of the following phenomenon is/are the effect of the rotation of the Earth?
- Apparent movement of the Sun, the Moon and the Stars.
 - Flatness of the poles and bulge at the equator.
 - Occurrence of sunrise, noon and sunset.
 - Magnetic field of the Earth.
- Select the correct answer by using the codes given below:
- (a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2, 3 and 4
22. Consider the following statements:
- Our solar system is located in the orion arm of the Milky way galaxy, about two-third of the way out from the centre.
 - The solar system formed from an interstellar cloud of dust and gas or nebulla about 4.6 billion years ago.
- Which of the above statements is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
23. Consider the following pairs:
- Electromagnetic radiation
 - Geothermal energy
 - Gravitational force
 - Plate movements
 - Rotation of the earth
 - Revolution of the earth
- Which of the above are responsible for bringing dynamic changes on the surface of the earth?
- (a) 1, 2, 3 and 4 only
(b) 1, 3, 5 and 6 only
(c) 2, 4, 5 and 6 only
(d) 1, 2, 3, 4, 5 and 6
24. Which of the following phenomena might have influenced the evolution of organisms? (CSAT - 2014)
- Continental drift
 - Glacial cycles
- Select the correct answer using the code given below.
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
25. Which of the following is/are direct source(s) of information about the interior of the Earth?
- Earthquake wave
 - Volcano
 - Gravitational force
 - Earth magnetism
- Select the correct answer using the codes given below
- (a) 1 and 2 (b) Only 2
(c) 3 and 4 (d) All of these
26. Statement I : Sidereal day is shorter than Solar day.
Statement II : The motion of the Earth in its orbit around the Sun is termed as revolution. [CSAT 2015-I]
- (a) Both the statements are individually true and Statement II is the correct explanation of Statement I
(b) Both the statements are individually true but Statement II is not the correct explanation of Statement I

- (c) A - 3 ; B - 2 ; C - 4 ; D - 1
(d) A - 4 ; B - 2 ; C - 3 ; D - 1

30. **List-I** **List-II**
(Volcanoes of the world) **(Countries)**

- (a) A - 3; B - 1; C - 4; D - 2
(b) A - 5; B - 2; C - 3; D - 1
(c) A - 3; B - 2; C - 4; D - 1
(d) A - 5; B - 1; C - 3; D - 2

31. **List-I (Volcano)** **List-II (Country)**

- (a) A - 1; B - 2; C - 3; D - 4
(b) A - 3; B - 4; C - 5; D - 2
(c) A - 1; B - 4; C - 3; D - 2
(d) A - 3; B - 2; C - 5; D - 4

32. Match List I with List II and select the correct answer using the code given below the Lists :

- | List I
(Phenomenon) | List II
(Date) |
|------------------------|-------------------|
| A. Summer solstice | 1. 21st June |
| B. Winter solstice | 2. 22nd December |
| C. Vernal Equinox | 3. 23rd September |
| D. Autumnal
Equinox | 4. 21st March |

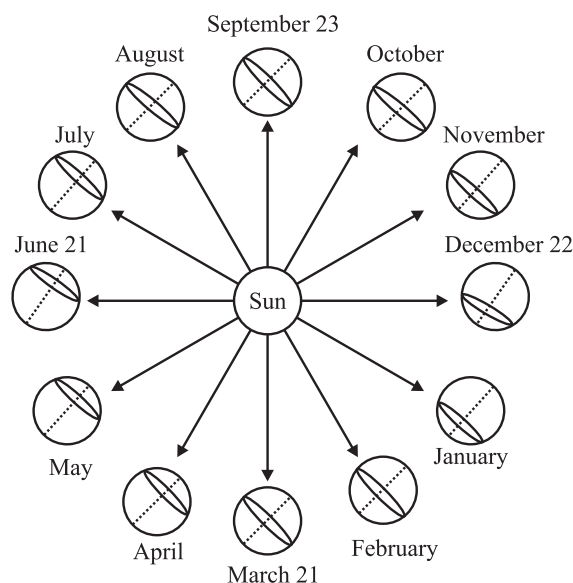
Code :

	A	B	C	D
(a)	1	4	2	3
(b)	1	2	4	3
(c)	3	2	4	1
(d)	3	4	2	1

Hints and Explanations

EXERCISE-1

1. (b) Jupiter and Saturn are located between Mars and Uranus.
2. (a) Jupiter has 63 moons and Saturn has 61.
3. (a) The earth is the densest of all the planet. Density of Earth is 5.52, Venus-5.50, Mercury 5.27, Mars - 3.95, Jupiter and Saturn - 0.69, Uranus - 11.70.
4. (d) The asteroids are group of small pieces of rock revolving round the sun between the orbit of Mars and Jupiter. They are approximately 40,000 in number.
5. (a) India is a large country spanning over 30° of longitude. It is unusual in having a single time zone all over the country, $5\frac{1}{2}$ hours fast.
If the direction of rotation of earth is reversed then Indian Standard Time will be $-5\frac{1}{2}$ hr.
When it is noon the time is 12:00; that will be 6:30 A.M. in place of 17:30 P.M. or 5:30 P.M.
6. (c) Mars is the only planet with similar day time temperatures and an atmosphere similar to earth. The most relevant condition for presence of life on Mars is occurrence of ice caps and frozen water.
7. (b) Meteor is a small celestial body which has entered the earth's atmosphere by the gravitational force of earth, when they pass close to it. Meteors typically occurs in the mesosphere, and most range in altitude from 75 km to 100 km.
8. (c) Diamond ring is observed during solar eclipse, only along the peripheral regions of the totality trail.
9. (c) 10. (c) 11. (c) 12. (b) 13. (c)
14. (c) 15. (a) 16. (c) 17. (b) 18. (d)
19. (a)
20. (c) At the time of the solar eclipse it is not the shadow of the earth that is visible on the surface of the sun. The solar eclipse occurs when the moon comes between the sun and the earth. The shadow of the earth is visible during the lunar eclipse.
21. (a) 22. (b) 23. (c) 24. (c)
25. (a) 26. (b) 27. (c)
28. (b) These sequence occurred in this sequence : 21 March, 21 June, 23 Sept. and 21 December.
29. (a) Beirut is in the east of G.M.T. and Lima is five hours behind the G.M.T.
We know that 1° is equal to four minutes i.e., 2 hours = 30° East and Beirut is 75° West of G.M.T.
30. (a) 31. (d)
32. (a) In 1906, Montessus de Bailore described the earthquakes and their characteristics within a specific region. It generally occurs by subduction of pacific plate under Asiatic plate and triggered a series of devastating Tsunamis.
33. (a) The International Date line is an imaginary line which go through the North pole to the South pole. It is 180° away from the Greenwich Meridian.
34. (b)
35. (d) The parallel at which the sun is overhead at midday is shown in the following diagram :



36. (c) The colour of the star is an indication of its temperature. The glowing is caused by something called Black-Body radiation, which has to do with the heat energy trying to radiate away in more and more energetic wavelengths. Red stars are relatively cool at only a few thousand degrees Celsius, white stars are hot at about ten thousand degrees, and blue stars are the hottest.
37. (b) The group of stars arranged in a definite pattern is called constellation. In modern astronomy, a constellation is an internationally defined area of the celestial sphere. These areas are grouped around asterisms, which are patterns formed by prominent stars within apparent proximity to one another on Earth's night sky.
38. (a) Ganymede is the largest satellite in solar system. Ganymede is a satellite of Jupiter and the largest moon in the Solar System. It is the seventh moon and third Galilean satellite outward from Jupiter. Completing an orbit in roughly seven days, Ganymede participates in a 1:2:4 orbital resonance with the moons Europa and Io, respectively. It has a diameter of 5,268 km (3,273 mi), 8% larger than that of the planet Mercury, but has only 45% of the latter's mass.
39. (c) The principle of Black hole was enunciated by S. Chandrashekhar. A Black Hole is a region of space-time from which gravity prevents anything, including light, from escaping. The theory of general relativity predicts that a sufficiently compact mass will deform spacetime to form a black hole. Around a black hole, there is a mathematically defined surface called an event horizon that marks the point of no return. The hole is called "black" because it absorbs all the light that hits the horizon, reflecting nothing, just like a perfect black body in thermodynamics.

40. (a) The distance of moon from the Earth is 364 thousands kms. The Moon is the only natural satellite of the Earth and the fifth largest moon in the Solar System. It is the largest natural satellite of a planet in the Solar System relative to the size of its primary, having 27% the diameter and 60% the density of Earth, resulting in $1/81$ its mass.
41. (d) Jupiter was named on the Roman God Zeus. Zeus is the "Father of Gods and Men" who rules the Olympians of Mount Olympus as a father rules the family according to the ancient Greek religion. He is the God of sky and thunder in Greek mythology. Zeus is etymologically cognate with and, under Hellenic influence, became particularly closely identified with Roman Jupiter.
42. (d) The approximately diameter of Earth is 12800 km. The rotation of the planet has slightly flattened it out, so it has a larger diameter at the equator than at the poles. The equatorial diameter of Earth is 12,756 km, its polar diameter is 12,713 km, and its average diameter, which is referred to in common usage, is 12,742 km or 7,926 miles.
43. (b) The time taken by the Earth to complete one rotation on its axis is 23 hr-56 min 4 sec.
44. (a) 45. (d) 46. (a)
47. (a) Space between Earth and Moon is known as Cislunar. Pertaining to the space between the earth and the orbit of the moon.
48. (d) White Dwarf is known as Fossil star. They are supported by electron degeneracy pressure. It amplifies the contrast with red giants. They are both very hot and very small. They are the opposite of black holes. They are the end-products of small, low-mass stars.
49. (c) 50. (c) 51. (b) 52. (a) 53. (a)
54. (a) The black part of the moon is called sea of tranquility. Sea of tranquility is not an actual sea but rather the point at which Apollo 11 first landed on when it reached the moon. It is a lunar mare which mainly consists of basalt rock and is located on the Tranquilitatis basin which is on the Moon. The mare has a tint which is slightly blue in colour and stands out from the rest of the moon.
55. (b) Rotation of earth on its axis causes day and night but the revolution of earth in an elliptical manner around the sun causes seasons, equinoxes and solstices.
56. (d) Tectonic plates are composed of oceanic lithosphere and thicker continental lithosphere, each topped by its own kind of crust. Tectonic plates are able to move because the Earth's lithosphere has greater strength than the underlying asthenosphere. The outer shell of the earth, the lithosphere is broken up into tectonic plates. The seven major plates are the African plate, Antarctic plate, Eurasian plate, Indo-Australian plate, North American plate, Pacific plate and South American plate. Along convergent boundaries, subduction carries plates into the mantle.
57. (a) Albedo also varies according to the angle of incidence of the Sun Rays being higher for slanting rays and lower for vertical or nearly vertical rays. From this we can derive that albedo will be relatively higher during early morning and late evening because at both times, Sun rays are slanting.
58. (a) On the equator, the day and night stay approximately the same length all year round.
59. (a) Stalagmite is a onical mineral deposit, usually calcite or aragonite, built up on the floor of a cavern, formed from the dripping of mineral-rich water.
60. (a) The Earth's rotation on its axis explains the eastward flow of the equatorial counter-current. The earth would have been rotating east to west, the piled up water would have come down on the west side. Therefore, essentially it's the earth's rotation that explains the eastward flow of equatorial counter current.

EXERCISE-2

1. (b) It will help probe the possibility of the existence of stars and galaxies made of anti-matter. It will also help in understanding the evolution of the universe.
2. (c)
3. (a) Only 1st and 2nd are correct.
4. (b)
5. (b) 6. (a) 7. (a) 8. (c) 9. (c)
10. (a) 11. (c) 12. (a)
13. (a) Greenwich Mean Time is the standard time for time assessment in different countries of earth. The countries which lies east of the Greenwich line is ahead of Greenwich Mean Time while the countries which lies west of Greenwich has their standard time behind the Greenwich Mean Time. So, the correct sequence of the countries are Japan, Iraq, Greece, Cuba and Costa Rica.
14. (c) 15. (b)
16. (c) The latitude of a place defined as the arc, measured in degree, north or south of the meridian between the place and the equator. The equator is given the value of 0° . The latitudes thus range from 0° at the equator to 90° north or south at the poles.
17. (c) The Milky Way galaxy is the home of our solar system. It is very difficult to study this galaxy because the solar system is located within it. It seems to be a spiral galaxy.
18. (a) 1 Sidereal year = 365.2564 days;
1 Tropical year = 365.2422 days;
So difference is 0.0142 days.
19. (b)
20. (c) Limestone is composed of mineral calcite and aragonite which are different forms of calcium carbonate. Limestones may be formed from skeletons of corals and foraminifera. It is a sedimentary rock easily soluble in water.

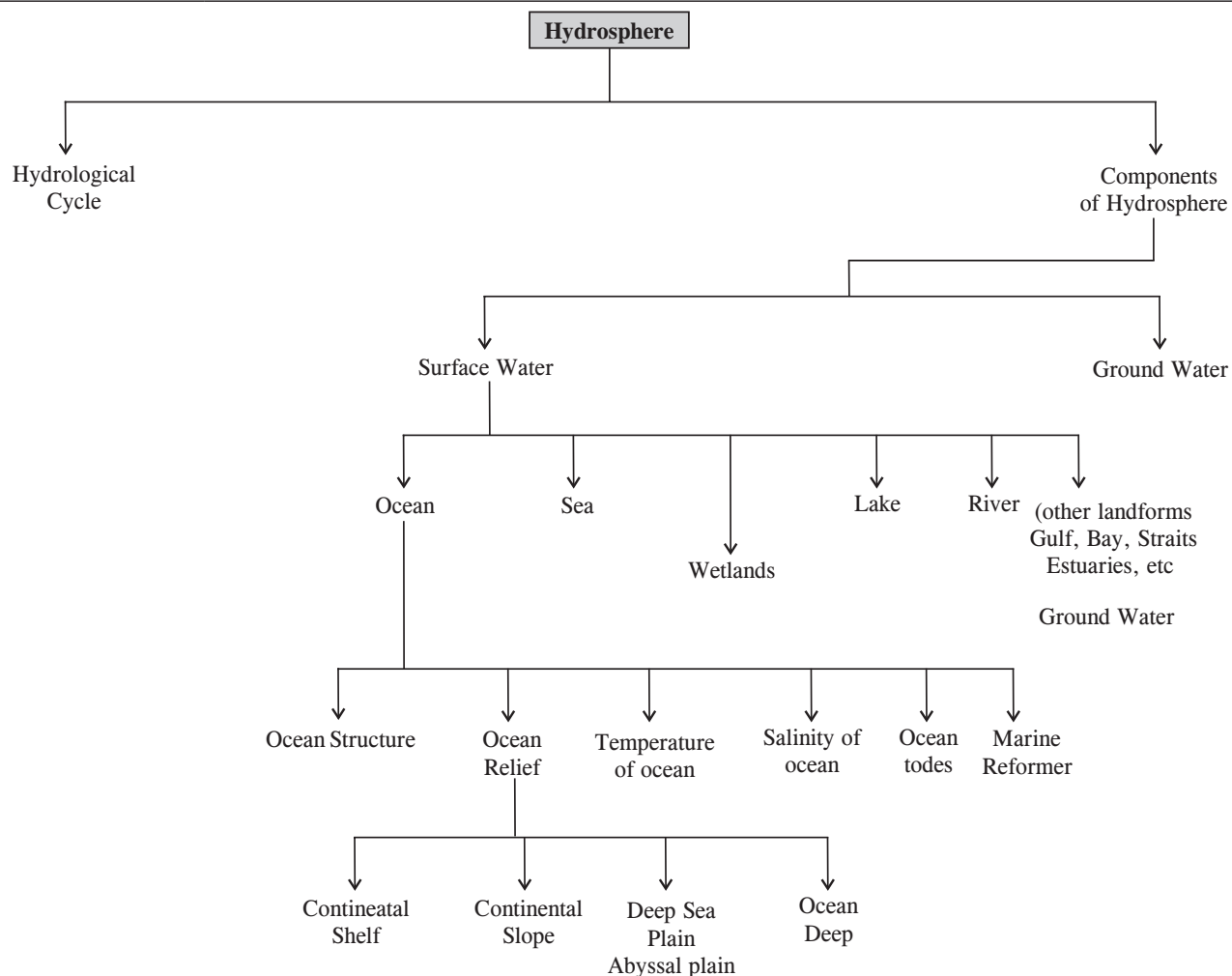
21. (d) All the given phenomena are the effect of the rotation of the earth. The Earth rotates from the west towards the east. As viewed from the North Star or polestar Polaris, the Earth turns counter-clockwise.
22. (c) Our solar system is located in the orion arm of the milky way galaxy, about two-third of the way out from the centre. The sun is about 26,000 light-years from the center of the Milky Way Galaxy, which is about 80,000 to 120,000 light-years across (and less than 7,000 light-years thick). We are located on one of its spiral arms, out towards the edge. It takes the sun (and our solar system) roughly 200-250 million years to orbit once around the Milky Way. In this orbit, we (and the rest of the Solar System) are traveling at a velocity of about 155 miles/sec (250 km/sec).
23. (d) From electromagnetic radiation to revolution of the earth, everything is responsible for bringing dynamic changes on the surface of the earth. For example: Electromagnetic radiation brings changes in the field of microwaves, wavelengths of radio, UV rays, infra red rays, X rays and gamma rays. Geothermal energy is the heat received from the earth's core. This heat continuously flows outward. It transfers to the surrounding layers of rock, the mantle. When temperature and pressure becomes very high some mantle rocks melt becoming magma. It then either comes out as lava or heat up the nearby rocks and water which comes out as hot springs or geysers. Gravitational force is constantly working on all physical bodies. It is giving weights to objects with mass and causes them to fall to the ground when dropped. Plate movement is a dynamic change on the surface of the earth. It explains many aspects of the interrelationship of volcanoes, earthquakes, climate change, and the evolution of life itself. Everything about our planet is related either directly or indirectly to plate tectonic. Rotation causes day and night. Revolution causes seasons, change in the length of day and night.
24. (c) Continental Drift and Glacial Cycles have influenced the evolution of organisms. Continental drift is the movement of the Earth's continents relative to each other by appearing to drift across the ocean bed. A glacial period is an interval of time within an ice age that is marked by glacier advances.
25. (a) Earthquake waves are propagating vibrations that carry energy from the source of the shaking outward in all directions. It is of four types:
P (for primary) S (for secondary)
Love Rayleigh
A volcano is a mountain that opens downward to a pool of molten rock below the surface of the earth. When pressure builds up, eruptions occur.
26. (b) The sidereal day is the time it takes for the Earth to complete one rotation about its axis with respect to the 'fixed' stars. By fixed, we mean that we treat the stars as if they were attached to an imaginary celestial sphere at a very large distance from the Earth. A measurement of the sidereal day is made by noting the time at which a particular star passes the celestial meridian (i.e. directly overhead) on two successive nights. On Earth, a sidereal day lasts for 23 hours 56 minutes 4.091 seconds, which is slightly shorter than the solar day measured from noon to noon. At the same time that the Earth spins on its axis, it also orbits or revolves around the Sun. This movement is called revolution.
27. (b) Falls are possible only when a river is losing height suddenly. Next comes meanders. When a river reaches flatter land, flow is slowed down and meanders are formed through erosion of the river banks and deposition on the inside of bends. Oxbow lake is formed when a wide meander is cut off from the main river, creating a free-standing body of water. Delta is formed at the mouth of a river, where the river flows into an ocean, sea, estuary, lake, or reservoir.
28. (d) A. The smallest planet of the solar system is-Pluto
B. The largest planet of the solar system is-Jupiter
C. The planet second from the Sun in the solar system is-Venus
D. Planet nearest to the Sun is - Mercury.
29. (d) 30. (c)
31. (a) A. Semeru Volcano is situated in Indonesia.
B. Cotopaxi Volcano is situated in Equador.
C. Etma Voplcano is situated in Italy.
D. Kilomanjaro Vaolcano is situated in Kenya.
32. (b) The correct match is as follows:
Summer Solstice- 21st June
Winter solstice- 22nd December
Vernal Equinox- 21st March
Autumnal Equinox- 23rd September
June 21 is called the summer solstice in the Northern Hemisphere. In the Northern Hemisphere, the winter solstice occurs either December 21 or 22. Vernal Equinox is the time when the sun crosses the plane of the earth's equator, making night and day of approximately equal length all over the earth and occurring about March 21. The Autumn Equinox is the first day of the autumn season and occurs when the sun passes the equator moving from the northern to the southern hemisphere. It occurs on 23 September.



Introduction

Hydrosphere refers to the total amount of water present on or under the surface of a planet. Around 70% of the earth surface is covered by water and it is present in all the three state of matter, i.e. solid as glaciers and ice caps, liquid in the form of oceans, seas, lakes and rivers and gas in the form of cloud, and water vapour. Hydrosphere plays significant role in the biosphere as :

- It has the ability to dissolve almost everything.
- It has the potential to store heat for quite a long period of time.
- It nourishes the organism.
- It helps to maintain the flow of elements in the biosphere.



HYDROLOGICAL CYCLE

Commonly known as the water cycle. It is the circular flow of water within earth's atmosphere, involving all the three forms of matter such as solid, liquid and gaseous forms. It also refers to the continuous exchange of water from the one form to another with the application of external energy. Through various process of physical action such as evaporation, condensation, precipitation, interception, infiltration, runoff, surface flow.

Evaporation is the process of transformation of water from the water bodies to water vapour in the atmosphere.

Condensation is the process of changing the state from water vapour to water in the upper air. It generally takes the form of cloud.

Precipitation is the downpour of condensed cloud mostly in the form of rainfall, and sometimes as snow, hails, fog, drop, etc.

Infiltration is the process of penetration of water into underground strata of earth's crust.

Interception is the precipitation trapped by vegetation instead of falling on ground.

Run off is the movement of water on the earth, surface through different channels such as river, streams, etc.

Subsurface Flow is the movement of water within the earth's crust after infiltration which eventually either come up through artesian well or seep to the ocean.

Fishes are the important members of there community. They are further categorised into

Pelagic fish - who leaves neither too close to shore nor to the bottom.

Demersal fish - On the contrary these fishes occupy the bottom area of the sea.

Other than fishes this community is also famous for marine mammals which are again divided into two categories

The mammals who can stay both in water and land for example seal

The second category of mammals spend their entire life in side the sea.

COMPONENTS OF HYDROSPHERE

There are four major components of Hydrosphere.

- Oceans
- Fresh water
- Glacial water
- Atmospheric water vapour
 - (a) Surface water
 - (b) Ground water
- (a) **Surface water:** Oceans, seas, lakes, ponds, rivers and streams are main examples of surface water.
- (b) **Ground water:** Ground water is found in soil pores, rocks, aquifers and springs. The water percolates through pores in rock and soil filling all the available spaces.

Distribution of water through different Sources (in %)

Oceans	97.3	Saline Water
Ice-cap and glaciers	2.05	
Ground water	0.68	
Fresh water lake	0.009	Fresh Water
Inland sea and salt lakes	0.009	
Atmosphere	0.0019	
Rivers	0.0001	

Oceans

As major portion of Earth's Hydrosphere is covered by ocean we are now going to discuss. Ocean and its various features.

An ocean (from Ancient Greek Okeanos,) is a body of saline water that composes much of a planet's hydrosphere. One Earth, an ocean is one of the major conventional divisions of the world Ocean, which covers almost 71% of its surface. These are, in descending order by area, the pacific, Atlantic, Indian, Southern and Arctic Oceans. The word sea is often used interchangeably with "ocean" in American English but, strictly speaking, a sea is a body of saline water (generally a division of the world ocean) partly or fully enclosed by land.

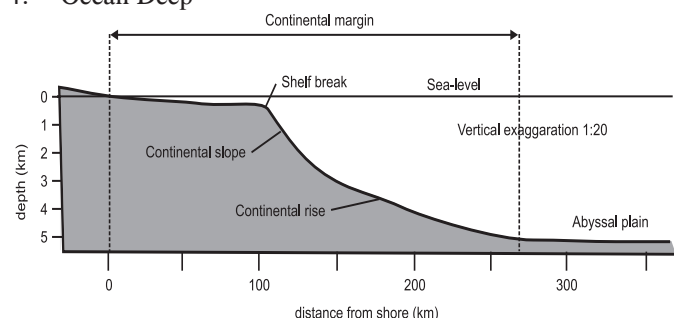
Ocean Structure

- 39.3% of the northern hemisphere comprises of landmass and 60.7% is covered by sea and oceans. This share in the southern hemisphere is of 19.1% and 80.9% between land and water.
- Ocean has been divided into two main groups:
 - (i) the ocean
 - (ii) the sea.
- Ocean contains more than 97% of total water on earth.
- Ocean has the average depth of about 4 km.
- Geographically ocean has been divided into (i) the Pacific (ii) the Atlantic (iii) the Indian (iv) the Arctic.
- Pacific Ocean is the largest and oldest followed by Atlantic (25%), Indian (21%) and Arctic (2.83).
- Arctic is strictly not an ocean and not navigable.

Ocean Relief

Ocean floor is characterized by four relief zone which are as follows:

1. Continental Shelf
2. Continental Slope
3. Deep Sea Plain / Abyssal Plain
4. Ocean Deep



1. Continental Shelf

The continental shelf is margin of continent are submerged under ocean water upto 100 fathoms (600 feet).

- Its slope varies from 1° to 3° .
 - It is determined by the coastal reliefs. Narrow shelf is found where a high mountain runs parallel to coast.
 - In Atlantic Ocean it is 2 km to 80 km wide. The continental shelf in Pacific Ocean varies between 160 km to 1600 km of width.
 - On an average continental shelf of Indian ocean in the west is 640 km wide and in the east near Java and Sumatra is as narrow as 160 km and further narrower along the coast of Antarctic.
 - The point where a mark increase in slope angle occurs is called as shelf break.
 - About 20% of oil and gas production is extracted from them.
 - They support productive fishing ground and are good source of marine food (Dogger bank, Grand bank).

Origin

- Marine waves and currents erode the continental margin continuously.
- Prolong deposition of detritus brought by rivers is found only in the region where sea is calm.
- The compression force acting as thermal convective current formed beneath continental and oceanic plates.
- Generation of parallel faults along the continental plates
- Rise and fall in sea level due to glaciations.
- Submergence and emergence of the continent.

2. Continental Slope

- Continental slope has a steep slope, beyond continental shelf towards ocean.
- Its slope varies from 2° to 5° .
- Depth of water is 200 m to 2,000 m.
- Occupies 8.5% of the total area of ocean basin

Origin

Continental slope is formed due to erosion, tectonic and aggradations (increase in land elevation).

3. Deep Sea Plain / Abyssal Plain

- It is most extensive relief, covering 75.9% of the total area of ocean basin.
- It is flat and rolling submarine.
- Its depth varies between 3000 m to 6000 m.

4. Ocean Deep

- It is made of depression and trenches.
- It runs parallel to coast generally.
- Grouped into two according to size
 - (i) **Deep** – It has great depth and is less extensive depression.
 - (ii) **Trenches** – It is a long and narrow linear depression.

Submarine Canyon

It is long, narrow and very deep valley or trenches. It is located on the continent shelf and slope with vertical walls. It is at maximum 16 km long. Morphogenetic classification:

- (i) Glacially eroded and
- (ii) Non-glacial eroded

Continental Rise

It is a depositional feature made up of thick sequences of continental material that accumulate between the continental slope and the abyssal plain.

Sea Mount

Mountain within ocean does not reaches the water surface is called sea mount. When the top of this mount is eroded by currents making its top flat, it is known as *guyot*.

Ocean Ridge

It is a continuous submarine mountain chain extending to thousands of kilometers. The ridges rise from depths near 5 km (3 miles) to an essentially uniform depth of about 2.6 km (1.6 miles) and are roughly symmetrical in cross section. It is formed across transform faults within fracture zones.

Major Trenches and Ridges in World

Mariana Trench

The cavity located near the Mariana Islands, where the Pacific Plate and the Philippine Plate converge; it is the *world's deepest trench* (about 36,000 feet).

Puerto Rico Trench

It is located off the coast of Puerto Rico, on the boundary between the South American and Caribbean plates; it features the deepest point in the Atlantic Ocean (27,493 feet).

Peru-Chile Trench

The trench (26,460 feet) borders South America and it is the *world's longest trench* (3,700 mi), located on the boundary between the Nazca Plate and the South American Plate.

Pacific-Antarctic Ridge

Mountain range separating the Pacific and Antarctic plates; it joins the eastern Pacific Ridge off the coast of South America.

East Pacific Ridge

Ridge that marks the boundary between the Pacific and Cocos Islands plates to the north, and the Pacific and Nazca plates to the south.

Aleutian Trench

Trench (25,600 feet) extending from Alaska to the Kamchatka Peninsula; it results from the Pacific Plate sliding beneath the North American Plate.

North America

Its area (9.3 million sq. mi) represents about 16% of the world's land; the Central American isthmus is an extension of North America.

Mid-Indian Ridge

Mountain range in the middle of the Indian Ocean that separates the African and Australian-Indian plates.

Southwest Indian Ridge

Ridge separating the African and Antarctic plates; it joins the Mid-Indian and Southeast Indian ridges off the coast of Madagascar.

- **South America**

Represents about 12% of the world's land and is linked to North America by Central America; its features include the Andes in the west and plains and plateaus in east and central regions.

- **Mid-Atlantic Ridge**

Ridge about 7,000 miles long, located in the middle of the Atlantic Ocean; some of its mountains reach the surface, forming islands such as Iceland.

- **Ryukyu Trench**

Trench (24,629 feet) located near the Ryukyu Islands; it marks the boundary between the Philippine Plate and the Eurasian Plate.

- **Japan Trench**

Trench (27,929 feet) located east of Japan, on the boundary between the Pacific Plate and the Eurasian Plate; this zone is marked by intense seismic activity.

- **Kuril Trench**

Trench (34,587 feet) located northeast of Japan; it results from the Pacific Plate sliding beneath the Eurasian Plate.

- **Philippine Trench**

Trench bordering the eastern Philippines, reaching depths of 34,578 feet; it results from the Philippine Plate sinking beneath the Eurasian Plate.

- **Kermadec-Tonga Trench**

Cavity located north of New Zealand, where the Pacific Plate meets the Australian-Indian Plate; it reaches depths of 35,702 feet.

- **Southeast Indian Ridge**

Ridge separating the Antarctic Plate from the Australian-Indian Plate; its topography is more regular than the topography of the Southwest Indian and Mid-Indian ridges.

- **Java Trench**

Trench located south of Indonesia, between the Australian-Indian and the Eurasian Plates; it is the *deepest point in the Indian Ocean* (24,440 ft).

Temperature of Ocean

- Temperature of the ocean is important for both marine organisms and coastal animals as it affects coastal climate.
- Standard thermometer with accuracy upto $\pm 0.02^\circ\text{C}$ is used for its measurement.
- Ocean is divided into three layers according to temperature.
 - (i) **Epilimnion Layer** is the first layer which is upto 500 m from top having temperature of $20^\circ - 25^\circ\text{C}$.
 - (ii) **Thermocline Layer** is below 500 m to 1000 m where temperature decreases at a rapid rate with the increase in depth.
 - (iii) **Hypolimnion Layer** is the third layer, very cold in nature and extend upto deep ocean floor from 1000 m. Only Polar region has this layer starting from surface to deep ocean form.

Daily Range of temperature is the difference of maximum and minimum temperature of a day

- At low latitudes it is usually 0.3°C and 0.2° to 0.3°C at higher latitudes.

- Sky condition, stability or instability of air and stratification of seawater decides daily range of temperature.
- The minimum and maximum temperature is recorded at 5 a.m. and 2 p.m. respectively.

Annual Range of Temperature

- Maximum temperature is recorded in the month of August and minimum in the month of February in northern hemisphere.
- Variation in the temperature is mainly due to insolation, nature of sea, prevailing and location of sea, etc.

Mean Annual Temperature of Water	
Ocean	Mean Temperature (in $^\circ\text{C}$)
Pacific	19.1
Atlantic	16.9
Indian	17.0
Average	17.4

Factors affecting Distribution of Temperature

- Latitude
- Unequal distribution of land and sea.
- Prevailing wind
- Ocean current

Minor factors include: Submarine ridges, local weather, location and shape of sea.

Horizontal distribution

- Average temperature of the ocean is 26.7°C
- Gradual decrease from equator towards poles is 0.5° F per latitude.
- Variation of temperature in the northern and southern hemispheres is mainly due to unequal distribution of land and ocean water.

Vertical Distribution

- Surface of ocean has highest temperature due to insolation and heat transmitted from below through conduction process.
- Solar energy effectively penetrates 20m and seldomly reaches beyond 200 m depth.
- Increase in depth decrease the temperature. Rapid fall in temperature upto 200 m.
- 200 m divides the ocean floor into two vertically-
 - (i) **Photic/ Euphotic zone** is the upper surface upto the depth of 200 m and receives solar radiation.
 - (ii) **Alphatic zone** goes beyond 200 m depth to the bottom and receives no solar ray.

Density of Ocean

- Amount of mass per unit volume of substance, measured in g/cm^3 is called as density of ocean water.
- Density of pure water is 1 g/cm^3 at 4°C and of ocean water is 1.0278 g/cm^3 (2-3% higher than water) at 4°C .
- It increases with lowering of temperature of ocean. Highest density is recorded at -1.3°C .

Factors controlling density

Temperature is inversely proportional to density, whereas pressure and salinity are directly related to it.

Density stratification of Ocean

- Density wise Ocean is a three layered structure
 - Surface layer, ii) Pycnocline layer and iii) Deep layer
- (i) Surface layer**
It has lowest density, having thickness of 100 to 200 m and is called as photic zone. Only 2% of seawater is found there. The zone is important for the growth of phytoplankton through the process of photosynthesis. Phytoplankton is eaten by zooplankton, it initiates food chain process.
- (ii) Pycnocline layer**
It is a transition zone characterized by rapid change of density. It is found between 300 m - 1000 m depth of ocean water. 18% of the total volume of the ocean water lies in this zone.
- (iii) Deep layer**
It has high density. Water mass beyond 1000 m depth to the ocean floor contain 80% of the total volume of ocean water. It denotes high density and low temperature.

Salinity of the Ocean

- Ratio between the weight of dissolve matter and sample sea water (part of salt per thousand part of water) is called as salinity of ocean water.
- Average salinity of ocean water is 35%.
- Salinity of ocean water is affected by marine organism, plant community and physical properties of ocean such as temperature, density, waves, pressure and currents.
- Highest salinity is observed between 20° - 40° N (36%).
- Boiling point of saline water is higher than pure water.
- The instrument used for measurement of salinity of water is called salinometer.
- The line with same salinity is joined by Isohalines.
- Halocline indicates sharp increase in salinity.
- Salts brought by rivers are the main source of salinity. It contains 60% of calcium sulphate, 2% of sodium chloride.

(Salts in Ocean Water)	
Name of Salts	In %
Sodium Chloride	77.8
Magnesium Chloride	10.9
Magnesium Sulphate	04.7
Calcium Sulphate	03.6
Potassium Sulphate	02.5
Calcium Carbonate	00.3
Magnesium Bromide	00.2

Factors controlling salinity

- Evaporation is positively related to the salinity level.
- Precipitation has a negative relation with it.
- Influx of river water has inverse relation with salinity level.
- Atmospheric pressure and wind are of directional help in the redistribution of water salinity.
- Circulation of ocean water is the controlling factor of salinity in a region.

Distribution

- Latitudinal decline in salinity is seen with the increase in distance from equator.
- The equator does not have highest salinity due to heavy rainfall, though temperature and evaporation are high.
- Middle latitudes between 20° to 40° N and 10° to 30° S has high temperature and evaporation but low rainfall.
- The zone between 40° - 60° latitude in both hemisphere record low salinity 31% and 35% respectively in Northern and Southern hemisphere.
- Sub-polar and polar zone has minimum salinity whereas tropics (20°-30°) has maximum.
- Low salinity of the northern part of Caspian is 14% due to the influx of water brought by rivers (Volga, Ural, etc.). On the other hand there is high salinity in the southern part near Gulf of Karabugas (170%). Red sea has 220%, Lake Van has 330% and Dead sea has 238% of salinity.

Ocean Deposits

- Loose materials in the form of sediments derived from various sources, found at ocean floor are called as **ocean deposits**.
- It consists of both organic and inorganic matters.
- Weathering and erosion of continental rocks are transported by river and wind.
- Volcanic eruption, decay and decomposition of marine organisms also contribute to ocean deposits.

Sources and Types of Marine Deposits

On the basis of origin Marine deposits can be classified into the following groups:

- (1) Terrigenous deposits
- (2) Volcanic deposits
- (3) Biotic matter/ Organic deposits
- (4) Abiotic matter/ Inorganic deposits

(1) Terrigenous deposits

- They have continental origin and due to weathering and erosional factors fine to coarse grains are formed.
- They consist of gravel, sand, silt, clay and mud (blue mud, green mud, red mud).
- Gravel range from 2 mm to 256 mm, and mud, clay and silt has the diameter of 1/32 mm to 1/8192 mm. Mud is the finest with 1/ 8192 mm of diametric.

(2) Volcanic deposits

Derived from two sources:

- Volcanic eruption on level (land) and
- Volcanic eruption in the ocean.

(3) Organic deposits

Skeleton of marine organism divided into two:

- Neretic matter*: Skeletons of organism.
- Pelagic matter*: Remains of algae found in liquid mud, generally called as ooze.

Oozes are divided into two

- Calcareous oozes** contain lime and are of two types:
- Pteropod oozes* are formed of floating shells of pteropod molluscs. These shells are conical in shape and ½ inches in diameter. Contain 80% of calcium and found in tropics at depth of 300-1000 fathoms of water and disappears at 2000 fathoms.

- (ii) *Globigerina oozes* are formed from the shells of foraminifera. They contain 64% of calcium, 1.64% of silica and 3.3% of mineral. Usually found in tropics and temperate zone of Atlantic Ocean, west coast of Indian Ocean and in eastern Pacific oceans continental shelves.
- (b) **Siliceous oozes:** Derive from a group of protozoa or radiolarian and benthic animals (sponges). These oozes do not dissolve unlike calcareous oozes. They are found in both warm and cold water at great depth and are of two types such as:
 - (i) *Radiolarian oozes* are formed by shells of radiolarian and foraminifera. Silica dominates and lime content increases with increasing depth. 2000 to 5000 fathoms is the depth between which they are found.
 - (ii) *Diatom oozes* are formed of shells of very microscopic plants rich in silica. They are blue near land and yellow or cream away from land.
- (4) **Inorganic deposits**
 - They come from land by various agencies which include dolomites, amorphous silica, iron, manganese oxide, phosphate, barite, glauconitic, phosphorite, feldspar phillipsite and clay.

Coral Reefs and Atoll

- They are accumulated and compact skeleton of lime secreting organisms known as **coral polyps**, confined between 25°N - 25°S latitude.
- They live on lime and in colony form.
- **Corals are called as rain forests of the oceans.**

Condition for Growth

- High mean annual temperature between 68°F to 70°F (20°C - 21°C) is required for their growth.
- Do not survive in deep water, *i.e.* should not be more than 250 feet (60-77 m).
- Oxygen and sunlight is required for their growth.
- Sediment free water is needed.
- Fresh water is harmful for corals.
- Very high oceanic salinity is injurious for the growth of coral.
- Ocean current and waves are good for the growth of coral as they bring food for them and determine their shape.
- They grow in open sea.
- Extensive submarine platforms are required for the development of coral colonies.
- Human economic activity like deforestation and industrialization, etc. cause global warming which adversely affect corals growth.

Types of Coral Reef

The submarine reefs are classified on the basis of nature, shape and mode of occurrence: (i) fringing reef, (ii) barrier reef, and (iii) atoll.

(i) Fringing Reef

- Develops along the continents in shallow water.
- It has steep slope on seaward side and gentle slope land ward side.

Fringing Reefs are found in *Sakau island* southern florida, mehetia island, etc.

(ii) Barrier Reef

- It is the *largest, highest* and *widest* of all reefs.
 - It runs parallel to coastal platform.
 - It has average slope of 45° but on an average it has 15° to 25° slope.
 - Shallow lagoons are formed in between land and barrier reef.
- Great Barrier reef* located parallel to the east coast of Africa is the *largest* barrier reef.

(iii) Atoll

- A ring of narrow corals growing in a form of rim encircling a lagoon partially or completely.
- It is horseshoe shape.
- Atoll are found in Antilles sea, red sea, China sea Australian sea, Indonesian sea.
- Fun futti Atoll of Ellice Island is the world famous at all

Some Famous Coral Reefs in the World

The Great Barrier Reef

It is the large coral reef in the world. It is located in eastern coast of Queensland in Australia. It has 2,900 individual reefs, and 900 islands stretching for over 2,300 kilometers within an area of approximately 344,400 square kilometers.

The Belize Barrier Reef

The coastal area of Belize consists of the largest barrier reef in the northern hemisphere, offshore atolls, several hundred sand cays, mangrove forests, coastal lagoons and estuaries. It is habitat for threatened species, including marine turtles, manatees and the American marine crocodile.

The Red Sea Reef

It is found on the coast of Red sea and is the northern most reef found in Indian ocean, extending 2,000 km. Fringing reef systems is developed enormously over here.

The Pulley Reef

It is located on the coast of Florida, USA. The corals *Agaricia sp.* is found here in tan-brown shade largely.

Coral Bleaching

Scientifically it is the process of loss of intracellular endosymbionts or zooxanthellae either through expulsion or lost of algae pigmentation. In general it can be viewed as the indicative death of corals with the increase in ocean temperature the coral starts expelling the algae living in their tissue and as a result the corals get completely white.

Causes

Global warming which leads to increase in the temperature of ocean water.

Near shore corals bleach due to storm generated precipitation which carry pollutants and contaminate ocean water.

Over exposure to sunlight leads to the increase in temperature of ocean water and hence result into coral bleaching.

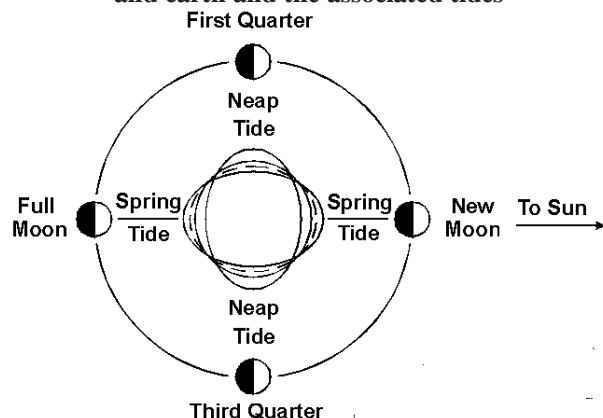
Expocure to air due to extreame lowtides.

Ocean Tide

- The periodic rise and fall of sea water than the average sea level is a **tide**.

- The fall of seawater and movement towards sea is called **ebb**. This generates low water level and called as low tide.
- The difference between high and low tide is called as **tidal range**.
- The variation in the height of both low and high tide can be seen from place to place.
- This variation depends on depth of ocean water
- Configuration of sea coasts and coastlines and openness or closeness of the sea.

Interchanging position of Sun, moon and earth and the associated tides



Origin

- Primarily occurs due to gravitational forces of the sun and moon.
- The revolution of the earth around the sun in an elliptical path and that of moon around earth from west to east. There is change in distance between earth and moon during different time in every month.
- The period of the farthest distance between the moon and the earth (407,000 km) is called as **apogee** and when nearest distance (356,000 km) is called as **perigee**.

The tidal force of moon is maximum during perigee as the distance is less. The resultant tide is known as perigean tide which is 15-20% higher than normal tides.

On the contrary during Apogee, due to maximum distance of moon from earth the tidal force is less hence resulting into Apogean tides 20% lesser than normal tides.

Types of Tide

- Spring Tide
- Neap tide
- Apogean and Pengean tide
- Tropical and Equatorial tide
- Daily and semi diurnal tide
- Equinoctical tide

Apogean & Perigean Tides

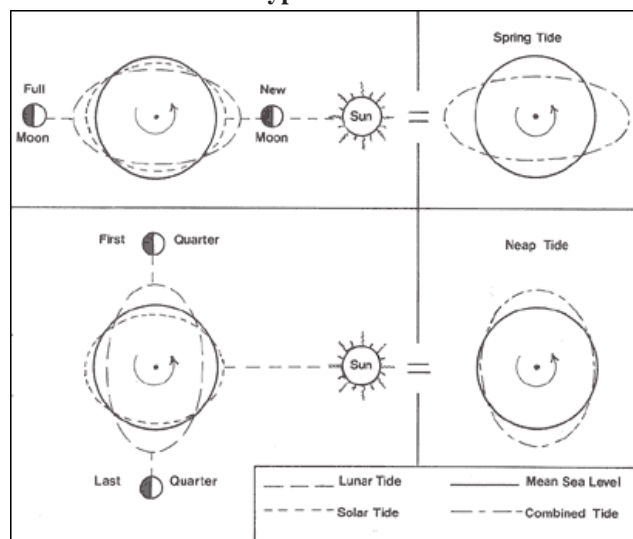
(i) Spring tide

When the sun, the moon and the earth are in the same line, there is formation of spring tide.

- The position when all three are in a straight line is called **syzygy**.
- When the sun, moon and earth are in one side of the earth in a straight line, this position is called **conjunction** (solar eclipse).
- When they are placed at 90°, the position is called **quadrature**.

- When the sun, the moon and the earth are in sequential order in a straight line is called as conjunction. It is when solar eclipse occurs. It occurs on new or full moon day.
- When the earth is in between the moon and the sun are called as opposition. It occurs on new or full moon day.

Types of Tide



(ii) Neap tide

It is a quadrature position between the earth, the sun and the moon on seventh or eighth day of the fortnight. During this time the forces of the sun and the moon acts in opposite direction. It is 20% lower than normal tide.

Tides clear the sediments brought by rivers, retarding the development of delta and clean the coast.

(iii) Perigean and Apogean Tides

When the moon is nearest to the earth in its orbit (at perigee), its tide-producing power is greater than average, resulting in perigean tides. These are 15-20% greater than average. When the moon is farthest from the earth (apogee), the tides are called apogean tide, which are about 15-20% less than average. Coincidence of spring and perigean tides results in an abnormally great tidal range, while when neap and apogean tides coincide the range is abnormally small.

(iv) Tropical and Equatorial Tides

Like the sun moon also undergoes the process of north and southward declination and to complete it. Moon requires 27.5 days of time, which is known as *synodie month*.

During mainnum declination the moon rays are vertical on the tide centre and results into formation of spring tides near Tropic of cancer in northward position and tropic of Capricorn in southward position.

When the moon is vertical on the equator there is no such diurnal change between the neap and spring tides in terms of their height. These types of tides are called as *Equatorial tides*.

(v) Daily & Semi-diurnal Tides

The tides only vary in terms of time duration Daily tides occur in an interval of 24 hrs and 52 min where on semi-diurnal tides occur in 12 hours and 26 min.

(vi) Equinoctical Spring tides

These are forms of spring tides which occurs in a time interval of 6 months.

Ocean Currents

The movement of a mass of oceanic water parallel to the coast is called as ocean current.

Currents are of two types on the basis of temperature.

- (i) Warm current
- (ii) Cold current

On the bases of velocity, dimension and direction.

Drift : It is slow movement of the ocean current under the influence of prevailing wind.

Current : Movement or circulation of ocean water in a definite path having great velocity.

Stream : Movement of ocean water involving large volume in a definite direction with velocity. It is a continuous flow.

Origin

Origin of ocean current occurs due to following factors:

- (i) Rotation of earth
- (ii) Temperature difference in ocean
- (iii) Salinity difference in ocean
- (iv) Density Difference
- (v) Air pressure and wind
- (vi) Rainfall and Evaporation
- (vii) Direction, shape and configuration of coast
- (viii) Bottom relief
- (ix) Seasonal variation

Distribution of Currents

Atlantic Ocean

North Equatorial Current (warm)

South Equatorial Current (warm)

Counter Equatorial Current (warm)

Gulf stream (warm)

Gulf stream

North Atlantic Current

Canary Current (Cold)

Labrador Current (Cold)

Brazil Current (warm)

Falkland Current (Cold)

South Atlantic Drift (Cold)

Benguela Current (Cold)

Pacific Ocean

1. North Equatorial Current (Warm)
2. South Equatorial Current (Warm)
3. Counter Equatorial Current (Warm)
4. Kuroshio System (warm)

- (i) Kuroshio Current
- (ii) Kuroshio extension
- (iii) North Pacific Drift
- (iv) Tsushima Current
- (v) Counter Kuroshio Current

5. Oyashio Current (Cold)

6. California Current (Cold)

7. Peru Current (Cold)

8. El Nino or Counter Current (warm)

9. Eastern Australian Current (warm)

10. West Wind Drift (Cold)

Indian Ocean

1. North-east Monsoon Current (warm)
2. Indian Counter Current (warm)
3. South West Monsoon (warm)
4. Indian Equatorial Current (warm)
5. Mozambique Current (warm)
6. West Wind Drift (cold)

Seas of Ocean

Pacific ocean

Arafura sea, Bering sea, Celebes sea, Coral sea, java sea, Gulf of Carpentria, Yellow sea, Japan sea, Bohol sea, Gulf of Alaska, Molucca sea and South China sea.

Indian ocean

Arabian sea, Andaman sea, Bay of Bengal, Timor sea, Red sea, Laccadive sea, Gulf of Oman, Gulf of Aden, Mozambique Channel.

Atlantic ocean

Caribbean sea, Celtic sea, Labrador sea, Marmara sea, Norwegian sea, Bothnia sea, Gulf of Finland, Bay of Fundy, Barents sea, Beaufort sea, Amunden sea, Baffin Bay, Laptev sea, White sea, Pechora sea, Kara sea, East Siberian sea, Greenland sea, Prince Gustav Adolf sea.

Inland Seas

Caspian sea, Aral sea, Salton sea, Dead sea and Black sea.

Strait is a narrow passage of water connecting two seas or two other large areas of water.

Strait of Malacca is a funnel-shaped narrow waterway of 800 km long that connects the South China & Andaman Sea.

Bass strait is the widest (240 km). The narrowest strait in world used for international navigation, the **Bosporus** connects the Black Sea with the Sea of Marmara.

Major Strait of the World		
Name	Joins	Location
Malacca Strait	Andaman Sea & South China Sea	Indonesia - Malaysia
Palk Strait	Palk Bay & Bay of Bengal	India-Sri Lanka
Sunda Strait	Java Sea & Indian Ocean	Indonesia
Yucatan Strait	Gulf of Mexico and Caribbean Sea	Mexico-Cuba
Messina Strait	Mediterranean Sea	Italy-Sicily
Otranto Strait	Adriatic Sea & Ionian Sea	Italy-Albania

Bab-el-Mandeb Strait	Red Sea & Gulf of Aden	Yemen-Djibouti
Cook Strait	South Pacific Ocean	New Zealand (N & S islands)
Mozambique Strait	Indian Ocean	Mozambique - Malagasy
North Channel	Irish Sea & Atlantic Ocean	Ireland-England
Taurus Strait	Arafura Sea & Gulf of Papua	Papua New Guinea - Australia
Bass strait	Tasman Sea & South Sea	Australia
Bering Strait	Bering Sea & Chukchi Sea	Alaska-Russia
Bonne-Fasio Strait	Mediterranean Sea	Corsika-Sardinia
Bosphorus Strait	Black Sea and Marmara Sea	Turkey
Dardanelle Strait	Marmara Sea and Aegean Sea	Turkey
Davis strait	Baffin Bay & Atlantic Ocean	Greenland-Canada
Denmark strait	North Atlantic and Arctic Ocean	Greenland-Iceland
Dover strait	English Channel & North Sea	England-France
Florida Strait	Gulf of Mexico and Atlantic Ocean	USA-Cuba
Hormuz strait	Gulf of Persia & Gulf of Oman	Oman-Iran
Hudson strait	Gulf of Hudson & Atlantic Ocean	Canada
Gibraltar Strait	Mediterranean Sea & Atlantic Ocean	Spain-Morocco
Magellan strait	Pacific and South Atlantic Ocean	Chile
Makassar Strait	Java Sea & Celebeze Sea	Indonesia
Tsugaru Strait	Japan Sea and Pacific Ocean	Japan (Hokkaido-Honshu island)
Tatar Strait	Japan Sea & Okhotsk Sea	Russia (E Russia-Sakhalin Island)

Gulf

A large area of a sea or ocean partially enclosed by land, especially a long landlocked portion of sea opening through a strait. Gulf and Bay are of economic importance as they serve as excellent harbor in most of the cases. Many important trading centers are located on gulfs. It forms a good fishing ground and oil deposits.

- The Gulf of Mexico, bordering United States, Mexico, and the island nation of Cuba, is the world's largest gulf. It has a coastline of about 5,000 kilometers (3,100 miles).
- The *Persian Gulf* in Arabian Sea borders Iran, Iraq, Kuwait, Saudi Arabia, Qatar, Bahrain, the United Arab Emirates, and Oman. There a vast deposit of petroleum is found.
- The Gulf of Carpentaria, on northeast coast of Australia, is an inlet of the Arafura Sea.

Bay

A bay is a small body of water or a broad inlet that is set off from a larger body of water generally where the land curves inward. Examples of bays include the *Bay of Pigs* (Cuba), *Hudson Bay* (Canada), *Chesapeake Bay* (Maryland and Virginia), and Bay of Bengal (near India).

When a body of water such as a strait is capable of being blocked or even closed in order to control transportation routes, the body is called a “choke point.”

Estuaries

Estuaries are bodies of water and their surrounding coastal habitats typically found where rivers meet the sea. It becomes the home of numerous unique plant and animal communities because their waters are brackish. Brackish is a mixture of fresh water draining from the land and salty seawater.

Coastal Plain

Melting of snow in past rose the sea level as the stream and rivers got flooded with water. The Chesapeake Bay in Maryland and Narragansett Bay in Rhode Island are examples of such estuaries that were once river valleys.

Bar-built

Sandbars or barrier islands built up by ocean currents and waves in coastal areas created a protected area fed by small streams or rivers. The barrier islands off the Atlantic coastline of North Carolina and Massachusetts enclose bar-built estuaries.

Delta system

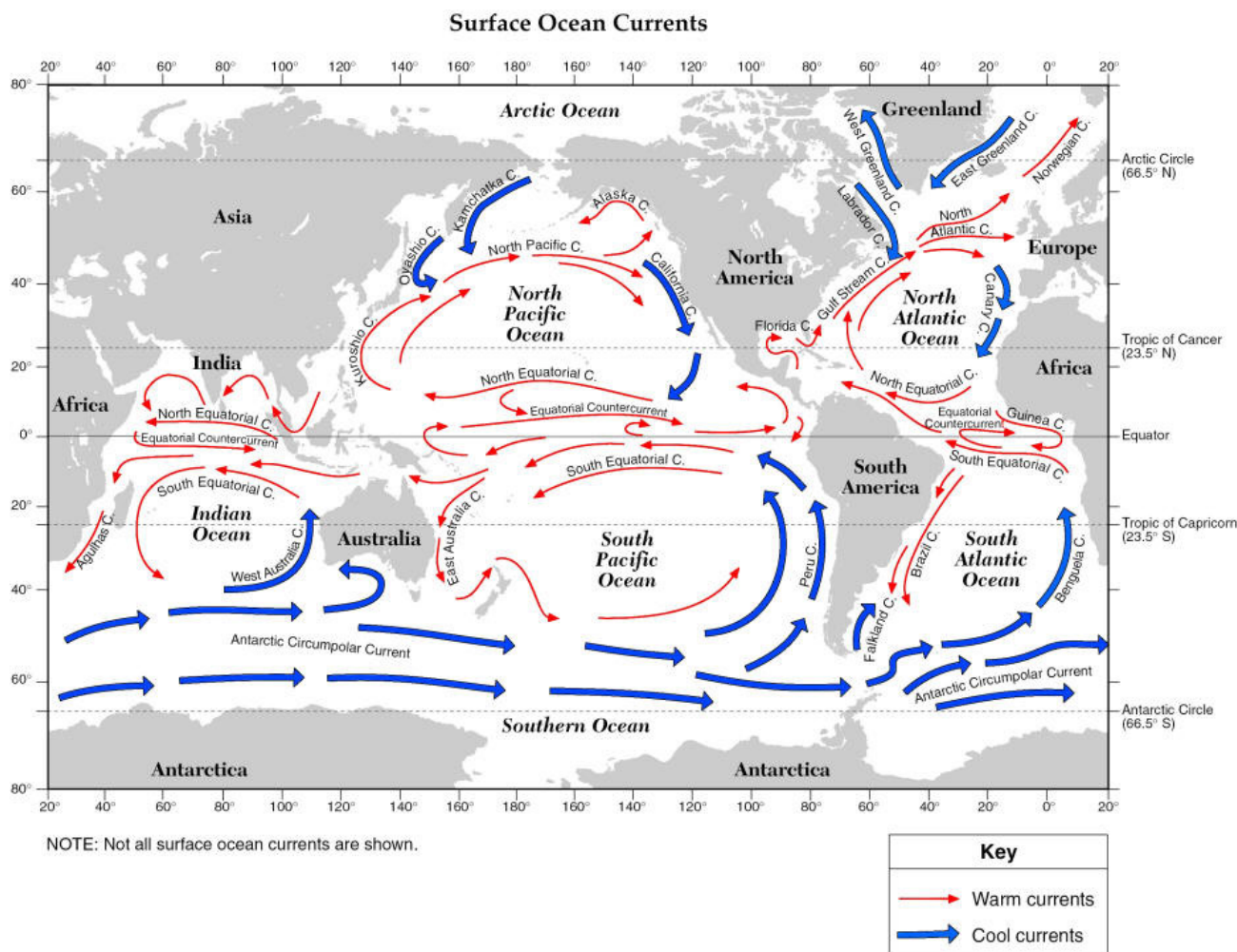
Deltas are formed at the mouths of large rivers from sediment and silt depositing. When the river flow is restricted by the delta, an estuary may form. The Nile River in Egypt and the Mississippi River in Louisiana forms delta systems estuaries.

Tectonic

it is created when a major crack or a large landmass sink, often caused by earthquakes, produced a basin below sea level that fills with water. This type of estuaries usually occurs along fault lines. San Francisco Bay in California is an example of an estuary created by tectonics.

Fjords

Advancing glaciers ground out long, narrow valleys with steep sides. Then when glaciers melted, seawater flooded in. Glacier Bay in Alaska is an example of a fjord.



Major Gulfs and their Location

Gulf of Aden of the southwestern corner of the Arabian Peninsula
Gulf of Alaska in the Pacific Ocean south of the state of Alaska
Amundsen Gulf in the Arctic Ocean northwest of Canada
Gulf of Aqaba in the northern end of the Red Sea, leading to Israel and Jordan
Gulf of Bahrain, part of the Persian Gulf
Gulf of Bothnia, part of the Baltic Sea between Sweden and Finland
Gulf of Cádiz, part of the Atlantic Ocean off the southern border of Spain and Portugal
Gulf of California in the Pacific Ocean in northwestern Mexico
Gulf of Carpentaria a large bay off northern Australia
Gulf of Cazes a large gulf in southern Cuba
Gulf of Corinth, which extends into Greece from the Mediterranean
Davao Gulf in the Philippines
Gulf of the Farallones, westward from the opening of the San Francisco Bay and Drakes Bay to the Farallon Islands
Gulf of Finland, between the southern coast of Finland and the northern coast of Estonia in the Baltic Sea.

Gulf of Genoa inside the Ligurian Sea on the northwestern coast of Italy

Gulf of Guinea in the Atlantic Ocean off the coast of Equatorial Africa

Gulf of İzmir in the Aegean Sea between Turkey and Greece. It was formerly called the Gulf of Smyrna

Gulf of Khambhat in the Arabian Sea, formerly known as the Gulf of Cambay

Gulf of Kutch in the Arabian Sea

Lingayen Gulf off western Luzon, the Philippines, in the South China Sea

Gulf of Lion, a bay on the Mediterranean coastline of Languedoc-Roussillon and Provence in France

Gulf of Maine, off the State of Maine, New Brunswick, and Nova Scotia in the Atlantic Ocean

Gulf of Mannar, between India and Sri Lanka

Gulf of Mexico, between Mexico, the United States, and Cuba

Gulf of Morbihan, a natural harbor on the coast of the Département of Morbihan in the south of Brittany

Gulf of Nicoya, in Costa Rica. Central America.

Gulf of Oman, between the southeastern Arabian Peninsula, Iran, Pakistan and Arabian Sea.

Gulf of Oristano, near Oristano on the Western Sardinian coast

Gulf of Panama in the Pacific Ocean south of Panama
Persian Gulf between Iran and the Arabian Peninsula
Gulf of Roses, the most northeastern bay on the Catalan coast
Gulf of Saint Lawrence, the world's largest estuary and the outlet of the Saint Lawrence River into the Atlantic Ocean
Gulf St Vincent, Separated from Spencer Gulf by the Yorke Peninsula
Gulf of Sidra, just north of Libya in the Mediterranean
Spencer Gulf, near Port Lincoln, South Australia
Gulf of Suez in the northern end of the Red Sea, leading to the Suez Canal
Gulf of Thailand just south of Thailand in the Indian Ocean
Gulf of Tonkin just east of North Vietnam in the Pacific Ocean
Gulf of Tunis in the Mediterranean off the coast of Tunisia

Marine Resources

The biotic and abiotic resources found in the oceanic water and bottoms are called as Marine resources. It includes marine water, inherent energy in ocean water (e.g. wave and tidal energy) biotic life (plants and animals), marine deposits and abiotic elements (minerals, fossils fuels, etc).

Marine Zone

(i) Territorial Sea

It is a region lying between base line and 12 nautical miles towards sea. 12 nautical mile is the seaward limit of territorial sea and called as contiguous zone.

(ii) Exclusive Economic Zone

This zone extends upto a 200 nautical miles from the base line. The coastal state has the right of survey, exploitation, conservation and management of mineral resources of ocean deposits, ocean floor, marine water energy, water and ocean organisms with exclusive economic zone. No other country can venture without the permission of the concern state.

(iii) High Sea

It extends beyond sea limit of the exclusive economic zone and includes the vast ocean area.

Marine Biological Resources

Those marine - related biological resources such as flora, fauna and micro organism whose inter-community behaviour and action affect or get affected by the piece of marine ecosystem they are living in.

Plankton Community

Planktons are floating and drifting micro plants and animals in photic zone. These are divided into phytoplankton (plant planktons) and zooplanktons (animal planktons). Phytoplankton produces food through the process of photosynthesis with the help of sunlight, water and atmosphere. *Algae* and *Diatoms* are most important member of this community. This community grows very fast and covers the large sea surface in no time. This cover of algae and diatom are called as **marine pasture**. The over production of red – gray microscopic plant produces

red tide. They are eaten by zooplanktons. Zooplanktons act as a bridge between marine pasture of phytoplankton of photic zone and large sea animals.

The zooplanktons are divided into three types

- (i) Herbivore zooplanktons
- (ii) Carnivore zooplanktons
- (iii) Detrivore zooplanktons

Nekton Community

It is the collection of actively swimming aquatic organisms in a body of water which are independently moveable without the help of water currents. Most of the animal of this community are vertebrate. Sea fishes are divided into groups:

Fishes are the important members of these community. They are further categorised into –

Pelagic fish - who leaves neither too close to shore nor to the bottom.

Demersal fish - On the contrary these fishes occupy the bottom area of the sea.

Other than fishes of this community is also famous for marine mammals which are again divided into two categories :

- (i) The mammals who can stay both in water and land for example *seal*.
- (ii) The second category of mammals who spend their entire life in the sea.

Benthos Community

It includes the plants and animals which live on the sea bottoms right from littoral marine biomes to the open sea biome. Benthos includes worms, clams, crabs, lobsters, sharks, sponges, and other tiny organisms that live in the bottom sediments. Remineralization of organic matter is an important source of nutrients to the ocean which is done by benthos community as they pump a large amount of water through their body thus distributing the organic matter and sediments of the ocean floor.

Marine food resources

According to the uses marine food resources are divided into two types:

- (i) **Protein rich food resources for the use as food for human being:** Fishes are the rich source of protein and aminoacid along with vitamin B12 and small amount of saturated fat and cholesterol. It helps in reducing high blood pressure and heart diseases.
- (ii) **Animal feed mainly for domesticated animals:** Fishes are used for feeding other animals also.

Fishing Zone of the World

A large share of world's annual income comes from trade and transport of fishes. 90% of world's total fish catches are of Fin fishes and remaining 10% is contributed by whales, lobster, molluscs and several invertebrate animals. There are five major fishing regions in the world according to the degree of development of fishing areas.

North-east Pacific region

This region extends from Aleutians and crossing Russia, China, S. Korea, Japan reaches south Philippines. Most of the catches include salmon, crab, herring, Pollock, halibut and cod.

North-east Atlantic region

This region extends along the European coasts from Iceland to Portugal. Herring and haddock are the important catches from this region.

North-west Atlantic region

It includes the New England region of USA, New Brunswick, Novascotia and Newfoundland of Canada, Grand Bank and George Bank. Cod is the major catch here.

South-east Pacific region

Peruvian and Chilean coastal area are included in this region.

West central Pacific region

this region extends from Philippines to eastern coast of Australia. Major catches here are mackerel, tuna, mullet, redfish, bass, oyster, crab etc.

Conservation of Marine Resources

Marine conservation focuses on limiting human-caused damage to marine ecosystems, and on restoring damaged marine ecosystems. Marine conservation also focuses on preserving vulnerable marine species. Marine resources conservation is the protection and preservation of ecosystems in oceans and seas. It is the need of an hour to lay down step for the exploitation, utilization, conservation and preservation. Strategies and techniques for marine conservation combine theoretical disciplines (population biology with practical conservation strategies) such as setting up protected areas, as with marine protected areas (MPAs) or Voluntary Marine Conservation Areas. National Ocean and Atmospheric Administration (NOAA) created the Damage Assessment, Remediation and Restoration Program (DARRP) to carry out the three steps such as **Preliminary Assessment, Injury Assessment/Restoration Planning and Restoration Implementation**

Restoration actions fall into following categories:

- **Primary restoration** returns the impacted resources to the condition that would have existed if the incident had not occurred.
- **Compensatory restoration** addresses losses from the date of injury until recovery is completed. While the resource is impaired, it is unable to provide services on which other parts of the ecosystem and the public rely (such as fish nursery habitat or recreational use). Restoration projects address the period from injury until recovery.
- **Emergency restoration** includes actions taken prior to the completion of the damage assessment and restoration planning process to prevent or reduce continuing natural resource impacts and prevent potential irreversible loss of natural resources.
- **Early restoration** is a form of compensatory restoration. If the responsible party agrees, it can be implemented prior to the completion of the Natural Resource Damage Assessment (NRDA) process to achieve restoration more quickly.

Marine Mineral Resources

Marine ecosystem is the store house of variety of minerals both metallic and non-metallic which generally found in two form, mineral dissolved in sea water and minerals deposited on different landforms of sea.

Minerals dissolved in Sea water

Minerals such as salt, bromine, gold, zinc, uranium and thorium, etc. are present in sea water in soluble form. Approximately percent of salt dissolved in sea water constituted by sodium and chloride. The salt can be separated from sea water and thus it can be transformed into fresh water like salt all other minerals can be separated through different methods. The commercial value of such minerals depend upon the cost of refinement and real market place.

Minerals of sea Deposits

Location plays a vital role in categorizing the nature of mineral deposits which can be further divided into 3 categories:

- (1) Mineral deposits of continental shelves
 - (2) Mineral deposits of deep sea bottom
 - (3) Sub surface minerals
- (1) **Mineral deposits on continental shelves** are zircon, monazite, magnetite, gold places, diamond, platinum, sulphur, phosphorite and several types of building material.
- (2) **Minerals deposits on deep sea bottom** : Pacific ocean is considered to have the largest deposits of manganese which is present upto a depth of 4000 m. Other than manganese many mineral such as nickel, copper, cobalt, lead, zine, iron, silicon, iron are found in deep sea bottom.
- (3) **Subsurface Deposits**: The mineral stored in oceanic crust of continental shelves are mineral oils and natural gas. They are the most developed mineral resources and contribute around 90% of all marine resources. They provide sample of opportunity to the economic development of any country of shore oil field have developed in the continental shelves of mexican gulf, North sea, North Alaska, Mexico, South California Arctic, Sea, India, Brazil, Australia, Taiwan and Japan.

Marine Energy Resources

Like minerals Ocean is also considered as the supplier of huge amount of energy resources, both *renewable* and *non renewable*. The non renewable sources of energy are petroleum and natural gas whereas the renewable sources are tide, wave and biomass.

The main sources of generation of electricity from ocean tide, sea waves and thermal variation between upper warm surface of sea and lower cold water mass, etc. Tidal energy is produced in two sources firstly by tidal current and the other by tidal range. Sea waves energy is processed by three methods

- (i) vertical displacement methods,
- (ii) salter device, and
- (iii) dam atoll method.

Marine Resources Depletion

Along with other resources marine resources are also in the impact zone of depletion. Which leads to many catastrophic consequences such as extinction of major species, imbalance in marine ecosystems, etc. As oceans are the largest source of water on the earth surface its depletion is also quite noticeable. Increasing human activities play a major role in the polluting marine environment and the possible cares of marine pollution

are spreading of harmful substances such as oil, plastics, industrial and agricultural wastes, etc. The impact of such degradation can be felt through various environmental prospect such as damage of estuaries due to human interventions, negative impacts on coral reef which leads to its continuous decay. These issues are gaining concerns of environmentalists as the loss once done can not be recovered with due course of time.

Causes of Marine Depletion

The threat to marine depletion are the result of variety of factor such as:

Over exploitation of Marine species

The intensity and the magnitude of marine resource exploitation have increased the pressure due to over population. Some marine biological resources (fishes) are over exploited while abiotic (physical) resources (mineral, energy, building material) still await their exploitation. Some of the fishing regions have shown decreases in the Fish population due to over fishing. Most mammals, birds and reptiles in estuaries were depleted by 1900 and declined further by 1950 as a result of the ineresinag demand for food, oil, and luxury items (such as furs, feathers and ivory). Among fish, the highly desirable and easily accessible salmon and sturgeon were depleted first, followed by tuna and sharks, cod and halibut, and herring and sardines. *Oysters* were the first invertebrate resource to degrade because of their value and accessibility as well as destructive harvesting methods.

Accurate estimate and prediction of future demand of fishes for human food and animal feed. Proper arrangement of canning and refrigeration for storage of fishes as it's a perishable good. This would to some extent reduce the catches. In New Zealand, fisheries are managed by a quota system that sets catch limits for commercially important species and aims at sustainable management of our fish stocks.

Pollution

Water bodies have been used as the universal dumping ground since ages be it lake, pond, river, sea or ocean. Land waste brought to ocean including sewage, industrial run-off and chemicals. The accident occurs due to Oil spilling in oceans threatens marine life at all levels. Indirect pollutants brought via storms and rivers also affect them. Some effects may not be immediately obvious, for example, bioaccumulation, and levels of toxic chemicals in organisms increase with successive trophic level in the food web. Marine pollution has the potential to seriously damage marine habitats and life there.

Eutrophication

Eutrophication is the result of a particular type of marine pollution. It is caused by the release of excess nutrients into coastal areas *via* streams and rivers. These nutrients come from fertilizers used in intensive farming practices on land. Additional nutrients in the sea can lead to excessive phytoplankton growth which results in "blooms". Death of these over populated phytoplankton leads to the sharp increase in decomposition of the dead organisms by oxygen-using

bacteria depletes oxygen levels. It may result in the death by oxygen starvation of large numbers of other organisms such as fish.

Ocean acidification

The amount of carbon dioxide in atmosphere has rise dramatically mainly due to human activities. Increase in carbon dioxide in the atmosphere will result in higher levels of dissolved CO₂ in seawater. Recent discovery reveals that even small changes in water pH can have big impacts on marine biology. Ocean acidification is a worldwide issue, but as CO₂ is more soluble in colder water, it is of particular concern in New Zealand's temperate oceans. It is difficult to predict the overall impact on the marine ecosystem but many scientists fear that ocean acidification has the potential to decrease marine biodiversity on a very large scale.

Excessive Human Interference.

The interference of Human being due to coastal tourism and industrial development has been increased to a great extent. It contributes to the degradation of local marine ecosystem in number of countries such as Kenya, Tanzania, etc.

Medical Researches also added to the list of major polluting agent. These kind of researches focuses on the major ingredients of sea for possible cures of various disease of Sponges, raft corals and tunicates are used extensively for producing medicine to some of the rare diseases. The researches are otherwise known as bio- prospecting which has started using coral reef of eastern Africa which is a store house of such species.

Various Ocean Development Programmes in India

Department of Ocean Development (DOD) was created as a nodal and independent department in July 1981 for organising, co-ordinating and promoting ocean development activities. The Ocean Policy Statement is primarily aimed at utilising the marine living and non-living resources for societal benefits in a sustainable manner.

Salient features of the Policy Statement and thrust areas include:

- Exploratory survey, assessment and sustainable utilisation/ harnessing of the ocean resources.
- Technological Development to gear utilisation and preservation. Technology relating to instrumentation, diving systems, position fixing, materials development, oceanic data collecting devices, submersibles, etc.
- Developmental activities related to integrated coastal and marine area management and coastal community development.
- Establishment of an ocean related information system using indigenous and foreign sources; International cooperation in Ocean Science and Technology.
- Development of technologies relating to seabed mining, extractive metallurgy and conducting Environmental Impact Assessment studies.
- Contribution towards basic and applied research in Ocean Science and Technology, Human Resource Management, creation of Centers of Excellence in academic institutions and public awareness on the potential and uses of ocean.

Integrated Coastal and Marine Area Management

United Nations Conference on Environment and Development (UNCED) in 1992, adopted the Agenda 21 which emphasized on the need to adopt the concept of Integrated Coastal Zone Management (ICZM) for sustainable utilisation of coastal and marine resources and prevention of degradation of marine environment. The Ministry of Earth Sciences established it on 2 January, 1998 at Chennai with following objectives:

- Developing capacity towards accomplishing the coastal and ocean related objectives of UNCED and
- To carry out R&D on application of scientific tools and techniques which are helpful in the development of integrated management solutions to address the issues and problems prevalent in the coastal marine areas.

It has established a strong database system related to coastal and marine resources and prevention of degradation of marine environment.

Coastal Ocean Monitoring and Prediction System

Extensive monitoring of marine pollution along the coastal waters was carried out at 70 locations and it has been found that the disposal of untreated sewage from towns, cities and villages cause decrease of dissolved oxygen and increase of nitrate and pathogenic bacteria in the sea close to the shore.

The study also concluded that –

From 120 class I and class A cities toward of coastal areas nearly 6835 ml of waste water being generated out of which on 1492 ml undergoes the process of various level of treatment. The solid waste generated by these 120 coastal cities are not that much harmful as they can be used as dumping material in large dumping ground.

In case of coastal agriculture suspended solid and nutrient contents are causing harmful effect to coast.

Model to predict the movement of oil during oil spills has been developed for the coasts of Mumbai and Chennai. Works to develop similar models for the coasts of Goa, Kerala and Visakhapatnam have been undertaken.

The National Institute of Ocean Technology (NIOT)

The National Institute of Ocean Technology (NIOT) was established in November 1993 as an autonomous society under the Ministry of Earth Sciences, Government of India. The major aim of NIOT is to develop reliable indigenous technology to solve the various engineering problems associated with harvesting of non-living and living resources in the Indian Exclusive Economic Zone (EEZ), which are about two-thirds of the land area of India.

- To develop world class technologies and their applications for sustainable utilization of ocean resources.
- To provide competitive, value added technical services and solutions to organizations working in the oceans.
- To develop a knowledge base and institutional capabilities in India for management of ocean resources and environment.

Indian Antarctic Program

The Indian Antarctic Program is a multi-disciplinary, multi-institutional program under the control of the National Centre

for Antarctic and Ocean Research, Ministry of Earth Sciences, Government of India (1981). The program gained global acceptance with India's signing of the Antarctic Treaty and subsequent construction of the *Dakshin Gangotri Antarctic* research base in 1983. *Maitri* is India's second permanent research station in Antarctica. It was built in 1989. Larsemann hill area became India's third station under the XXX Antarctic Treaty in 2007.

Indian Tsunami Warning System was established in 2006 by National Institute of Ocean Technology (NIOT), Chennai, Ministry of Earth Sciences, Government of India. It comprises of **Data Buoys** with **Bottom Pressure Recorders** in the deep seabed and **Acoustic Tide Gauge Network** in the coastal areas. For Tsunami monitoring applications India uses **Acoustic Tide Gauge**.

Wetlands

- Wetlands are lands which, due to geological or ecological factors, have a natural supply of water – either from tidal flows, flooding rivers, connections with groundwater, or because they are perched above aquifers.
- Wetlands are covered or soaked for at least a part, and often all, of the year and thus are intermediaries between terrestrial and aquatic ecosystems.
- The periodicity of water level fluctuations is termed as *hydroperiod* and it is the key factor that determines the productivity and species composition of the wetland community.
- Generally low lying areas, covered by shallow water and have characteristic soils and water tolerant vegetation.
- Wetlands occupy only 2% of the surface area of earth and they are estimated to contain 10 to 14% of carbon.
- They may be either freshwater or salt water (coastal).
- **Man made wetlands** : paddy fields, fishery ponds, Trapa and Euryale cultivation ponds and other aquaculture habitats.

Significance of Wetlands

- Nutrient rich and have high primary productivity.
- Since they have both aquatic and semi-aquatic environmental conditions to support specialized vegetation and fauna. Often a prime breeding habitat for waterfowl, many migratory birds and other aquatic or semi aquatic vertebrates.



- Helps in controlling flood by holding excess water, and the flood water stored in wetlands then drains slowly back into the rivers, providing a steady flow of water throughout the year.
- Serve as groundwater recharging areas.
- Provide important commercial products, including wild rice and various types of berries (such as black berries, blue berries, etc.).

- Hold sediments and accumulate soils along the shoreline.
- National Wetland Conservation Programme(NWCP) has been initiated for identified wetland which are at present 66 covering 21 states.

Types of Wetland

(i) Marshes

Marshes are defined as palustrine wetlands because they are often or continually inundated with water, characterized by emergent soft-stemmed vegetation adapted to saturated soil conditions. Nutrients are plentiful and the pH is usually neutral to alkaline. We have divided marshes into two primary categories: non-tidal and tidal.

Non-Tidal marshes are highly organic, and mineral rich soils of sand, silt, and clay underlie these wetlands. It provides excellent habitat for waterfowl and other small mammals, such as Red-winged Blackbirds, Great Blue Herons, otters and muskrats. Prairie potholes, playa lakes, vernal pools and wet meadows are all examples of non-tidal marshes.

Tidal marshes are saline. The lower marsh is normally covered and exposed daily by the tide. The saline marsh is characterized by Short Smooth Cord grass, Spike Grass and Salt meadow Rush (*Juncus gerardii*). Saline marshes support a highly specialized set of life adapted for saline conditions. These marshes generally harbour great biological diversity

(ii) Swamp

Swamps are the low lying uncultivated ground which generally collect water. There are many different kinds of swamps, ranging from the forested Red Maple, (*Acer rubrum*), swamps of the Northeast to the extensive bottomland hardwood forests found along the sluggish rivers of the Southeast. Swamps are characterized by saturated soils during the growing season and standing water during certain times of the year. The highly organic soils of swamps form a thick, black, nutrient-rich environment for the growth of water-tolerant trees such as Cypress (*Taxodium* spp.), Atlantic White Cedar (*Chamaecyparis thyoides*), and Tupelo (*Nyssa aquatica*). Some swamps are dominated by shrubs, such as Buttonbush or Smooth Alder. Plants, birds, fish, and invertebrates such as freshwater shrimp, crayfish, and clams require the habitats provided by swamps. Many rare species, such as the endangered American Crocodile, depend on these ecosystems as well. Swamps may be divided into two major classes, depending on the type of vegetation present: *shrub swamps* and *forested swamps*.

(iii) Bogs

Bogs are most distinctive kinds of wetlands. They are characterized by spongy peat deposits, acidic waters and a floor covered by a thick carpet of sphagnum moss. Bogs receive all or most of their water from precipitation rather than from runoff, groundwater or streams. As a result, bogs are low in the nutrients needed for plant growth, a condition that is enhanced by acid forming peat mosses. Bogs develop primarily in two ways: as sphagnum moss grows over a lake or pond and slowly fills it (terrestrialization), or as sphagnum moss blankets dry land and prevents water from leaving the surface (paludification). Over time, many feet

of acidic peat deposits build up in bogs of either origin. The unique and demanding physical and chemical characteristics of bogs result in the presence of plant and animal communities that demonstrate many special adaptations to low nutrient levels, waterlogged conditions, and acidic waters, such as carnivorous plants. Nevertheless, bogs support a number of species of plants in addition to the characteristic Sphagnum Moss, including Cotton Grass, Cranberry, Blueberry, Pine, Labrador Tea and Tamarack. Moose, deer, and lynx are a few of the animals that can be found in northern bogs. The Greater Sand hill Crane, the Sora Rail, and the Great Gray Owl depend on bogs for survival.

(iv) Fens

Fens are *peat-forming wetlands* that receive nutrients from sources other than precipitation, usually from upslope sources through drainage from surrounding mineral soils and from groundwater movement. As fens are less acidic and have higher nutrient levels than bog it supports diverse plant and animal community. These systems are often covered by grasses, sedges, rushes and wildflowers. Some fens are characterized by parallel ridges of vegetation separated by less productive hollows. The ridges of these patterned fens form perpendicular to the down slope direction of water movement. Over time, peat may build up and separate the fen from its groundwater supply. When this happens, the fen receives fewer nutrients and may become a bog.

Distribution of Wetlands in World

Major Wetland Regions of the World

South America

(i) The Orinoco River delta of Venezuela

It is dominated by brackish shoreline mangrove forests.

(ii) The Llanos

It is located on the western part of the Orinoco River found in western Venezuela and northern Colombia. It's one of the largest inland wetland areas of South America. Here winters are wet (yearly flood occurs) and summers are dry. It is dominated by savanna grasslands and scattered palms. It is important wading-bird habitat and is rich with caiman and red piranha.

(iii) The Amazon River

It is 3,000 km long and floods 5-15 meters high every year. It is *world's largest river*, with a flow that results in about one-sixth to one-fifth of all the fresh water in the world. It creates its own weather patterns.

(iv) The Pantanal

One of the largest wetlands in world located in southwestern Brazil. It covers four times the size of the Florida Everglades, with about 131,000 km² of that area flooded annually. It's a haven for 650 to 700 species of birds.

Europe

(i) Mediterranean Sea Deltas

The Rhone River Delta created France's most important wetland, the Camargue. It is highly affected by a hot, dry summer and cool, wet winters. It is home to world's 25 major flamingo nesting sites.

(ii) Coastal Marshes of Northern Europe

Extensive salt marshes and mud flats are found along the Atlantic Ocean and the North Sea coastlines of Europe. Most of the grasses are of type marshes. Rhine River Delta is a major transportation artery in Europe. Much of the Netherlands is on the Rhine River Delta.

(iii) Peatlands

Northern boreal and subarctic are peat land. It is more than half of the world wetlands. Predominately found in Ireland, Scandinavia, Finland, and Russia. It is mostly made up of decomposed sphagnum moss.

Africa

An abundance of wetlands is found in sub-Saharan Africa such as the Congo River Swamps, inner Niger Delta, Sudd of the Upper Nile, and the Okavango Delta.

(i) Northern Africa

Sudd of the Upper Nile: Rainforest where the Blue and White Nile meet in the southern Sudan.

Nile Delta: It is used to be a huge delta; the land has been converted to farm land, no longer affected by floods due to Aswan Dam.

(ii) Okavango Delta

A vast number of rivers, channels, island and lagoons are diverted to the Okavango Delta. Along the coast there are many mangrove forests. Zambezi River system and Etosha Pan in Angola are also included in it.

(iii) Zaire and Congo Rivers and Floodplains of the Senegal, Nile, Lake Chad; in arid Sahel.**Australia**

Wetlands are distinctive due to seasonal dryness from high evaporation rates and low rainfall. Saline wetlands and lakes are common as a result of the high evaporation rates.

New Zealand: One location in North Island has all seven types of wetlands. It has lost 90% of its wetlands. South Island receives 2-10 meters annually of rain, several types of wetlands.

Asia**(i) Northern Asia**

Ob-Irtysh basin, mountains of N. Central Asia (Lake Baikal), Tundra and internally drained seas: Caspian, Aral.

(ii) South and Southeast Asia:

South and Southeast Asia has the biggest wetlands. Some of the major rivers are the Indus, Ganges, Chao Praya and Mekong. The Mekong begins at the Tibet Plateau and runs through China, Laos, Cambodia and Vietnam draining 625,000 km². It is estimated that 20 million people receive their protein from fishing in these areas.

(iii) China

It has approximately 650,000 km² of wetlands, Pearl, and Yangtze River Deltas. Of that 250,000 km² have been reserved, and 400,000 km² have been used for rice paddies, fish ponds, and other functions.

Major Wetland of the world as ranked by World Heritage Site

Country	Wetland Name	Area
Canada	Wood Buffalo National Park	44,807 km ²
USA	Everglades National Park	6,110 km ²
USA	Olympic National Park	3,734 km ²
USA	Yellowstone National Park	8,983 km ²
Bulgaria	Srēbarna Nature Reserve	6 km ²
Romania and Ukraine	Danube Delta	4,152 km ²
Russian Federation	Lake Baikal	31,722 km ²
Russian Federation	Volcanoes of Kamchatka	43781 km ²
Spain	Doñana National Park	543 km ²
Tunisia	Ichkeul National Park	85 km ²
Malawi	Lake Malawi National Park	94 km ²
Senegal	Djoudj National Bird Sanctuary	160 km ²
Dem. Republic of Congo	Virunga National Park	7,800 km ²
India	Kaziranga National Park	430 km ²
India	Keoladeo National Park	28 km ²
India	Manas National Park	950 km ²
Australia	Fraser Island	1,840 km ²
Australia	Kakadu National Park	19,804 km ²
New Zealand	Te Wahipounamu	26000 km ²
Honduras	Río Platano Biosphere Reserve	5250 km ²
Mexico	Sian Ka'an	3,157 km ²
Panama	Darien National Parks	720 km ²
Colombia	Los Katios National Parks	5,970 km ²
PERU	Manu National Park	17,163 km ²

Lakes

Lakes are static bodies of water on the land surface which is surrounded by land on all side. They are always located to land surface. They are variable and changes with time and can be found at heights also (Tso Sekuru lake). Some of the lakes have greater depth (Baykal lake). It can be of varying size.

Types of Lakes

On the basis of salinity there are two types of lakes

- Fresh water lake
- Saline water lake

On the basis of Origin there are following types of lakes:

- Tectonic lake** is formed by uplifting of marine sediments and is often large and shallow such Lake Okeechobee, Lake Baikal, Lake Tanganyika, Lake Tahoe, Pyramid Lake (Nevada), Lake Ohrid (Yugoslavia), Dead Sea, Lake Kinneret (Sea of Galilee), Abert Lake, Lake Kioga, Lake Victoria, Reelfoot Lake.

2. **Volcanic lake** is formed due to volcanic eruption. Crater lake is the most common type of lake found. Crater Lake, Tagus Lake, Eifel Lake District (Black Forest of Germany) and Lake Kivu are the few examples of this type of lake.
3. **Diastrophic lake** is formed by landslides, mudflow and slump of land, e.g. Quake Lake, Yellowstone.
4. **Aeoline lake** formed when the low-lying land among the dunes in the desert is lower than the groundwater level, the water gathers there and forms the aeolian lake, e.g. Ghost lake.
5. **Fluvial lake** results from development of river and river course.
6. **Glacier lake** formed by **glaciers** and ice movement like cirque lake and rock-basin lake. The Great Slave Lake, Laurentian Great Lakes, Finger Lakes and Lake Mendota are examples such lake.
8. **Shoreline lake** is associated with on-shores lakes. It is large lake. It is of two types: **deltaic lake and coastal lake**. **Deltaic lake** is formed by sedimentation as river currents slow when they enter a large lake or the ocean e.g. Wax Lake. **Coastal lake** is related to movement of sand in spits and bars may enclose basins, e.g. Lake Eyre.
9. Lakes formed by falling of **meteor**, e.g. Chubb Crater.
10. **Anthropogenic lake** it is formed by man and even is called as man-made lake, e.g. Kopalinski Lake.

Lakes facts:
Plitvice Lakes (Croatia): Sixteen Lakes interconnected by Spectacular Waterfalls.
Boiling Lake (Dominica): A Flooded Fumarole.
Red Lagoon (Bolivia): Red (algae) + White (borax).
Five-Flower Lake (China): Beautiful Multi-Coloured Lake with Fallen Tree Trunks.
Dead Sea (Israel and Jordan): Lowest Point on Earth.
Lake Baikal (Russia): Deepest and Oldest Lake in the World.
Lake Titicaca (Bolivia and Peru): World's Highest Navigable Lake.
Caspian Sea (Russia): World's Largest Lake.
Crater Lake (USA): Its water is considered one of the World's most clearest.
Lake Karachay (Russia): Most Polluted Spot on the Earth.

Rivers

It's running water which usually empties itself into a sea or ocean, generally flowing with respect to the slope. According to the adjustment to the geological structure the streams or river drainage pattern has been divided into two categories such as:

Generic Classification of Stream

On the basis of the course through which streams generally flow, they can be divided into five basic types.

- **Consequent stream** having a direct consequence of the original slope of the surface on which it is flowing.
- **Subsequent Stream** has a head word erosion along the weak strata of the bed rock. They generally flow along

original streams independent of the original relief of the land.

- **Resequent Streams** flow along side the original stream but the level is lower than the original slope. These streams are generally known as the tributaries of subsequent stream.
- **Obsequent Streams** : Generally flow in the opposite direction of consequent stream.
- **Insequent Stream** : Don't have any fixed pattern hence they form dendrite pattern of drainage. They are generally independent of original slope of the surface its structure, or type of rock.

Types of Rivers

(i) Seasonal Rivers

The rivers which have water in them at a particular period of the year and are rainfed (Godavari, Krishna, Cauveri, Narmada, Tapi and Tungabhadra)

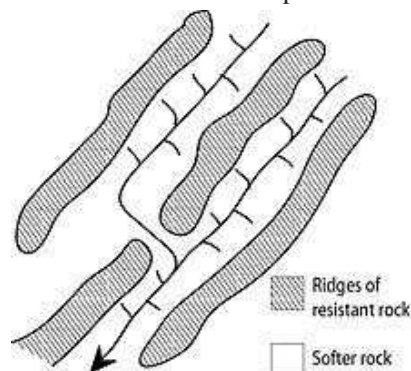
(ii) Perennial Rivers

The rivers which have water throughout the year. It is usually glacier fed (Indus, Sutlej, Ganga and Brahmaputra).

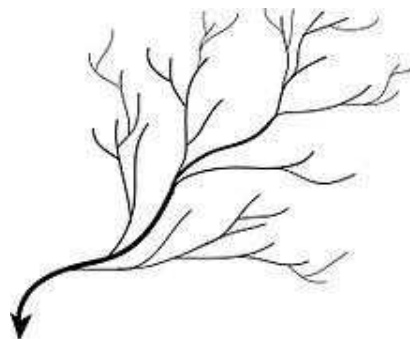
Drainage Pattern of the river

Rivers when flow, form a pattern which are generally divided into following:

- (i) **Trellis Drainage Pattern:** As the river flows along a strike valley, smaller tributaries feed into it from the steep slopes on the sides of mountains. They are formed by consequent stream which follows regional slope and geological structure. Trellis drainage is characteristic of folded mountains, such as the Appalachian Mountains in North America and in the north part of Trinidad.

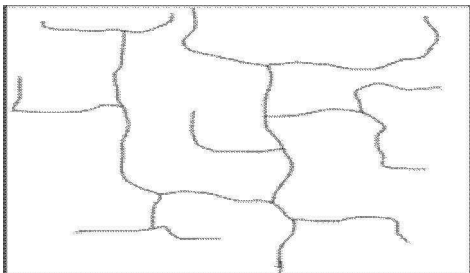


Trellis Drainage Pattern

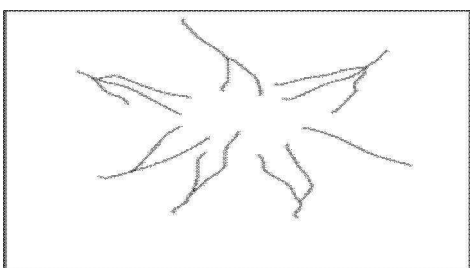


Dendritic Drainage Pattern

- (ii) **Dendritic Drainage Pattern:** The tree like pattern formed by the river as they flow is called as dendritic drainage pattern.
- (iii) **Rectangular Drainage Pattern:** This type of pattern is formed when streams from two directions of joining at approximately right angles. Rectangular drainage develops on rocks that are of approximately uniform resistance to erosion.



Rectangular Drainage Pattern



Radial Drainage Pattern

- (iv) **Radial Drainage Pattern:** When the streams radiate outwards from a central high point they form radial drainage pattern.
- (v) **Centripetal Drainage Pattern:** When the streams flow towards a common depression they form centripetal force. This pattern is opposite of radial pattern.
- (vi) **Annular Drainage Pattern:** The tributaries of consequent stream develop in a form of circle. This pattern is found over a mature and dissected dome mountain characterized by soft and hard rock alternatively. Angles of the streams meeting are both more and less than 90 degrees.
- (vii) **Barbed Drainage Pattern:** It is formed when the tributaries of a river runs in the opposite direction to its master stream.
- (viii) **Herringbone Drainage Pattern:** It is also known as rib pattern. It resembles the rib bones of human being (upper Jhelum, Tamar Kosi, upper Rapti and left bank of Gandak).
- (ix) **Parallel Drainage Pattern:** It follows the regional slope and develops on uniformly sloping landmass. The western coastal plain of India has such type of pattern.

Exercise -1

- Which one of the following factors is responsible for the change in the regular direction of the ocean currents in the Indian Ocean?
 - Indian Ocean is half an ocean
 - Indian Ocean has monsoon drift
 - Indian Ocean is a land-locked ocean
 - Indian Ocean has greater variation in salinity
- Maximum thickness of sediments is found over
 - Abyssal floor
 - Continental shelves and slopes
 - Continental slopes and rises
 - Oceanic ridges
- Which one of the following ocean currents is associated with the 'EL NINO' Phenomenon?
 - Humboldt
 - Benguela
 - Canaries
 - Kurushio
- A submarine mountain or peak rising more than 1000 m above the ocean floor is known as
 - Guyots
 - Seamounts
 - Abyssal hills
 - Trenches
- Dolphin Challenger Ridge is located in
 - Atlantic Ocean
 - Pacific Ocean
 - Arctic Ocean
 - Indian Ocean
- When water freezes in the polar seas the salts
 - form a layer on top
 - remain in the sub-surface water that does not freeze
 - sink to the bottom
 - partly freeze and partly sink to the bottom
- The winter temperatures of north-western Europe are higher than those of eastern Europe in the same latitudinal zone because
 - it is on the western side of the continent
 - it is near the sea
 - it receives only light falls of snow
 - it lies under westerly winds which blow over the Gulf Stream Drift
- Arrange parts of ocean floor according to increasing distance from the coast.
 - continental slope, continental shelf, continental rise, deep sea plain
 - continental shelf, continental rise, deep sea plain, continental slope
 - continental shelf, continental slope, continental rise, deep sea plain
 - continental rise, continental shelf, continental slope, deep sea plain
- What is Gulf Stream?
 - A cold current in the Atlantic Ocean
 - A cold current in the Pacific Ocean
 - A warm current in the Atlantic Ocean
 - A warm current in the Pacific Ocean
- One of the warm currents in the Indian Ocean is the
 - Labrador Current
 - Agulhas Current
 - Curoshio Current
 - Humboldt Current
- The Red Sea is an example of a
 - folded structure
 - faulted structure
 - lava structure
 - residual structure
- What is the triangular landmass that forms at its mouth when a river splits into several branches just before entering the sea?
 - Cape
 - Island
 - Estuary
 - Delta
- From west to east, what are the three main island groups in the Pacific called ?
 - Melanesia, Micronesia and Polynesia
 - Polynesia, Micronesia and Melanesia
 - Melanesia Polynesia and Micronesia
 - Micronesia, Polynesia and Melanesia
- What is the difference between a lake and a tank?
 - A lake is small and a tank is big
 - A lake has flowing water, a tank has still water
 - A lake is a geographical feature, a tank is not
 - Both are land-enclosed water bodies, but the latter is manmade.
- The direction of ocean currents is reversed with season in
 - the Indian Ocean
 - the Pacific Ocean
 - the Atlantic Ocean
 - no area on earth
- The term 'Apogee' refers to which one of the following?
 - A point in the orbit of the moon at which it is farthest from the earth.
 - A point in the orbit of the moon at which it is nearest to the earth.
 - Portion of surface of the moon opposite to the earth.
 - Portion of surface of the moon facing the earth.
- Where is the ocean current called the 'Gulf Stream' found?
 - Atlantic Ocean
 - Indian Ocean
 - North Pacific Ocean
 - South Pacific Ocean
- Sunda Trench lies in
 - Atlantic Ocean
 - Pacific Ocean
 - Indian Ocean
 - Antarctic Ocean
- The exceptionally high and low tides that occur at the time of the new moon or the full moon when the Sun, the Moon and the Earth are approximately aligned is called
 - Spring
 - Fall
 - Neap
 - Diurnal
- The current produced by upwelling of cold water off the coast of Chile and Peru is known as
 - EI Nino
 - Humboldt current
 - Agulhas current
 - Canary current
- On the planet earth, most of the freshwater exists as ice caps and glaciers. Out of the remaining freshwater, the largest proportion
 - is found in atmosphere as moisture and clouds
 - is found in freshwater lakes and rivers
 - exists as groundwater
 - exists as soil moisture

22. Consider the following statements :
1. The Labrador current is a cold current in the North Atlantic Ocean.
 2. The Falkland current is a warm current that flows along the Chile coast of South Pacific Ocean.

Which of the statements given above is/are correct ?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
23. Which one of the following pairs is not correctly matched?
- (a) Kuroshio : Warm ocean current
(b) Labrador : Warm ocean current
(c) Benguela : Cold ocean current
(d) Oyashio : Cold ocean current
24. Which one among the following is a sea without having a coastline?
- (a) North sea (b) Sargasso sea
(c) Baltic sea (d) Bering sea
25. Match List I with List II and select the correct answer using the code given below the Lists :

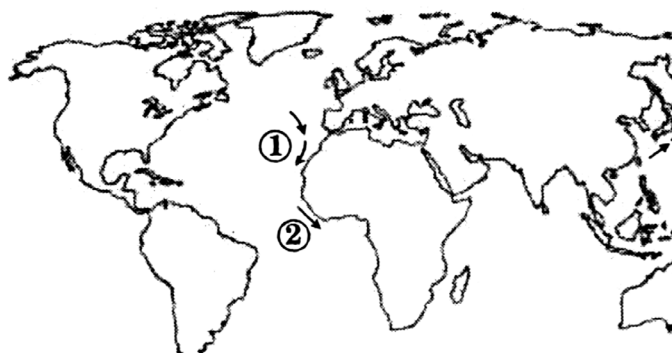
List I

(Ocean current)

- A. Guinea current
B. Oyashio current
C. Canaries current
D. Kuroshio current

List II

(Location in Map)



Code :

- (a) A-4; B-3; C-1; D-2 (b) A-2; B-3; C-1; D-4
(c) A-2; B-1; C-3; D-4 (d) A-4; B-1; C-3; D-2
26. The most important fishing grounds of the world are found in the regions where
- (a) warm and cold atmospheric currents meet
(b) rivers drain out large amounts of freshwater into the sea
(c) warm and cold oceanic currents meet
(d) continental shelf is undulating
27. Read the following statements and select the correct answer from the codes given below:
1. Corals are mainly found in the tropical oceans.
 2. Corals need clean sediment free water.
 3. Corals are mainly found in deeper parts of the ocean.
 4. Rain water promotes the growth of corals.

Codes:

- (a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 2 and 4
28. The horizontal distribution of temperature of ocean water is largely affected by
1. Depth of water in the ocean
 2. Ocean current
 3. Prevailing winds
 4. Latitudes
- Which of the following is correct?
- (a) 1, 2 and 3 (b) 1, 2 and 4
(c) 2, 3 and 4 (d) 1, 2, 3 and 4
29. Which one of the following is correctly matched?
- (a) Gulf of Carpentaria — Italy
(b) Gulf of Sidra — Libya
(c) Gulf of Po hai — Thailand
(d) Gulf of Tonking — Malaysia

Exercise -2

Statement Based MCQ

- Consider the following statements:
 - Annual range of temperature is greater in the Pacific Ocean than that in the Atlantic Ocean.
 - Annual range of temperature is greater in the Northern Hemisphere than that in the Southern Hemisphere.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- A new type of El Nino called El Nino Modoki appeared in the news. In this context, consider the following statements:
 - Normal El Nino forms in the Central Pacific Ocean whereas El Nino Modoki forms in Eastern Pacific Ocean
 - Normal El Nino results in diminished hurricanes in the Atlantic Ocean but El Nino Modoki results in a greater number of hurricanes with greater frequency.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- Consider the following statements:
 - Tides are of great help in navigation and fishing.
 - High tide enables big ships to enter or leave the harbour safely.
 - Tide prevents siltation in the harbours.
 - Kandla and Diamond Harbour are tidal ports.

Which of these statements are correct?

- (a) 1 and 4 (b) 2, 3 and 4
(c) 1, 2 and 3 (d) 1, 2, 3 and 4
- Consider the following statements:
 - Ocean crusts are denser than the continental crust.
 - Oceanic crusts are made up of granitic rocks whose composition is silicon and aluminium.
 - Granitic rocks are lighter in colour than basaltic.
 - Crust extends upto 70 km under mountains, in contrast to oceans which are about 5 km.

Which of the above statements are true?

- (a) 1, 3 and 4 (b) 1, 2 and 4
(c) 2, 3 and 4 (d) 1, 2, 3 and 4
- Which of the following conditions for the survival of corals are correct?
 - The water temperature must not fall below 20°C
 - The depth of water should not exceed 30 fathoms
 - The salinity should be less than 10%
 - The water should be free from sediments

Select the correct answer from the codes given below:

- (a) 1 and 2 (b) 3 and 4
(c) 1, 2 and 3 (d) 1, 2 and 4
- The chief fishing grounds of the world occur where cold and warm currents meet. This is because.
 - cold currents are rich in nitrates and phosphates which feed plankton.

- Such areas cause the water to become well-aerated.
- the warm current brings the right temperature for marine life.

- (a) 1 only (b) 2 only
(c) 1, and 2 (d) 1, 2, and 3

- Which among the following statements about the North Atlantic Drift is/are correct?

- It keeps the west coast of Northern Europe ice free.
- It is responsible for the warm air mass which interacts with the cold air mass from the Polar region and causes rainfall in Western Europe.
- It meets the Labrador current near Vancouver Island and causes dense fog.

Select the correct answer using the code given below

- (a) 1, 2 and 3 (b) 1 and 2
(c) 2 only (d) 1 and 3

- Consider the following statements :

- The Labrador current is a cold current in the North Atlantic Ocean.
- The Falkland current is a warm current that flows along the Chile coast of South Pacific Ocean.

Which of the statements given above is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

Matching Based MCQ

DIRECTIONS (Q. 9 to 10) : Match List-I with List-II and select the correct answer using the codes given below the lists.

- | List-I (Current) | List-II (Ocean) |
|-------------------------|------------------------|
| (A) Agulhas | (1) North Atlantic |
| (B) Aleutian | (2) South Pacific |
| (C) Canaries | (3) South Indian |
| (D) Humboldt | (4) North Pacific |

(a) A - 2; B - 1; C - 4; D - 3
(b) A - 2; B - 4; C - 1; D - 3
(c) A - 3; B - 1; C - 4; D - 2
(d) A - 3; B - 4; C - 1; D - 2
- Match List-I with List-II and select the correct answer using the code given below:

- | List-I (Current) | List-II (Feature) |
|-------------------------|---------------------------------------|
| A. Kuroshio current | 1. Warm current in Atlantic Ocean |
| B. Peru current | 2. Cold current in the Atlantic Ocean |
| C. Labrador current | 3. Warm current in the Pacific Ocean |
| D. Florida current | 4. Cold current in the Pacific Ocean |

Codes:

- | | A | B | C | D |
|-----|---|---|---|---|
| (a) | 3 | 4 | 2 | 1 |
| (b) | 3 | 2 | 4 | 1 |
| (c) | 1 | 4 | 2 | 3 |
| (d) | 1 | 2 | 4 | 3 |

Hints and Explanations

EXERCISE-1

1. (b) Due to the monsoon drift of Indian ocean, its regular direction of the ocean currents changes twice an year.
2. (b) 3. (a) 4. (b) 5. (a) 6. (b)
7. (d) 8. (c) 9. (c) 10. (b) 11. (b)
12. (d) 13. (a) 14. (d) 15. (a) 16. (a)
17. (a) 18. (c)
19. (a) Spring tides occur when the sun and moon are directly in line with the earth and their gravitational pulls reinforce each other, consequently a great flood or rush as of emotion.
20. (a) El Nino is a cold water current which runs along the coast of Peru and Chile. It is also a climate pattern which occurs every five years.
21. (c) The percentage of freshwater that exists on earth is 3% out of which almost 70% is icecaps and glaciers. And the remaining 30% is the ground water. Lakes, rivers and swamps are approximately 10% of that 30% of ground water.
22. (a) The Falkand current is a cold current. It flows along the Argentina's coast in South Atlantic Ocean.
23. (b) The Labrador Current is a cold current in the North Atlantic Ocean.
24. (b) While all other seas in the world are defined at least in part by land boundaries, the Sargasso Sea is defined only by ocean currents. The Sargasso Sea does not have a coastline as it is located in the middle of the Atlantic Ocean.
25. (b) A- 2, B- 3, C- 1, D- 4
The Guinea Current is a slow warm water current that flows to the east along the Guinea coast of West Africa. Oyashio Current is a cold subarctic ocean current that flows south and circulates counter-clockwise in the western North Pacific Ocean. The Canary Current is a wind-driven surface current that is part of the North Atlantic Gyre. The Kuroshio is a north-flowing ocean current on the west side of the North Pacific Ocean.
26. (c) The mixing of warm and cold current in the region where planktons are found, is food for fishes. The temperature is just right for them to survive. The temperature is just right for the growth of first food called planktons.
27. (a) Corals are mainly found in the tropical oceans. Corals need clean sediment free water. A coral polyp has a sack

like body and an opening encircled by stinging tentacles called cnidae. The coral polyp uses calcium carbonate from seawater to build itself a hard skeleton and it is this limestone skeleton that protects the soft coral polyp.

28. (c) The horizontal distribution of temperature of ocean water is largely affected by ocean current, prevailing winds and latitudes.
29. (b) Gulf of Carpentaria – Australia
Gulf of Sidra – Libya
Gulf of Po hai – China
Gulf of Tonking – Vietnam

All these are names of gulfs associated with the following countries.

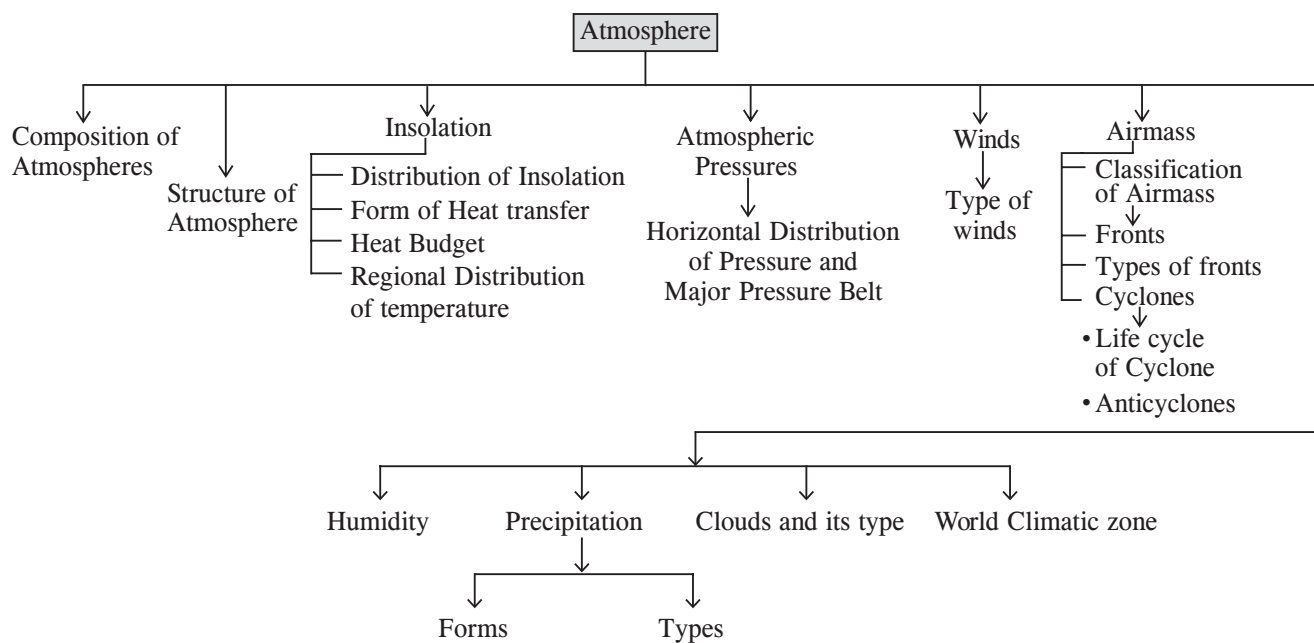
EXERCISE-2

1. (b) Climates in the Southern Hemisphere tends to be milder than those in the Northern Hemisphere. Southern Hemisphere has significantly more ocean and less land. The water heats up and cools down more slowly than land.
2. (b) Normal El Nino forms in south-eastern Pacific whereas as El Nino Modoki forms in central Pacific and causes greater number of hurricanes.
3. (b) Statement '1' is wrong because tides are not good for navigation. Tide may cause sinking of boats and ships. High tide helps ships for enter or leave the harbour safely. Kandla and Diamond harbour are tidal ports, Diamond harbour is located on river Hugli, whereas Kandla port is at the eastern part of Gulf of Kutch, which is a natural port.
4. (a) 5. (d) 6. (c)
7. (b) The North Atlantic current or Drift or sea movement is a powerful warm ocean current that continues the gulf stream northeast, which stretches from Florida to north- western Europe. It moderates the chilled climate of western Europe.
8. (a) The Falkand current is a cold current. It flows along the Argentina's coast in South Atlantic Ocean.
9. (d) There are ocean currents which are found in given ocean. Agulhas is the warm current, Humboldt is cold current and Kanaries is cold current.
10. (a) A. Kuroshio current - warm Pacific current
B. Peru current - cold current in Pacific ocean
C. Labrador current - cold current in Atlantic ocean
D. Florida current - Warm current in Atlantic ocean



Introduction

Atmosphere is a gaseous envelope surrounding the earth extending thousands of kilometers above the earth's surface. Life on earth exists at the bottom of the atmosphere where it meets with the lithosphere and the hydrosphere. The atmosphere directly or indirectly influences the vegetation pattern, soil type and topography of earth. Of the total mass of the atmosphere, 99% is found within 32 km from earth's surface. The atmosphere is held close to earth because of the earth's gravity and is energized by the sun.



COMPOSITION OF THE ATMOSPHERE

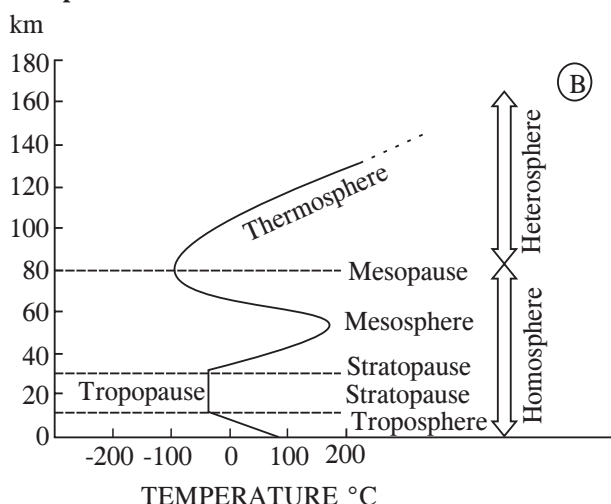
- Atmosphere is a mixture of gases containing huge amount of solid and liquid particles collectively known as aerosols. Pure dry air consists of nitrogen, oxygen, argon, carbon dioxide, hydrogen, helium and ozone. Besides them, water vapour, dust particles, smoke, salts, etc. are also present in the air.
- Nitrogen and oxygen comprise 99% of the total volume of the atmosphere. Nitrogen does not form a chemical union with other substances. It is an agent of dilution and regulates combustion.
- Oxygen combines well with all other in the atmosphere elements. It is easily combustible.
- Carbon dioxide exists in a very small percentage in the atmosphere.
- It absorbs most of radiant energy emitted by earth and reradiates it back to the earth, creating green house effect. This increased the temperature of lower atmosphere.
- It is also called as green house gas effect.

- The percentage of carbon dioxide in atmosphere increased due to burning of fossil fuels (coal, petroleum and natural gas) and deforestation.
- Water vapour and dust particles are crucial to weather and climate. They create condensation and absorb the heat received from the sun and radiated from the earth. Water vapour comprises 3-4% of the total volume of air.
- The amount of water vapour present in the atmosphere decreases from the equator towards the poles. Nearly 90% of the total water vapour lies below 6 km of the atmosphere.

Component	Percent by Volume
Nitrogen	78.08 %
Oxygen	20.94 %
Argon	0.93 %
Carbon dioxide	0.03 %
Neon	0.0018 %
Helium	0.0005 %
Ozone	0.00006 %
Hydrogen	0.00005 %

STRUCTURE OF THE ATMOSPHERE

The atmosphere consists of concentric layers of gases, with varying density and temperature. On the basis of chemical composition, the atmosphere is divided into: **Homosphere** and **Heterosphere**.



1. Homosphere

This extends up to the height of 90 km and is characterized by uniformity in chemical composition. It consists of three thermal layers – troposphere, stratosphere and mesosphere.

(a) Troposphere

Lowest layer of the atmosphere.

- Height of troposphere is 16 km at the equator and 10 km at the poles.
- Temperature decreases with height in this layer roughly at the rate of 1°C per 165 metres, which is known as **normal lapse rate**.
- Upper limit of the troposphere is called **tropopause**.
- All weather phenomena are restricted to troposphere.

(b) Stratosphere

- It lies above tropopause between 25-30 kms on an average over middle latitude. 50 km is considered to be its upper limit on an average.
- At 50 km of altitude the temperature becomes 0°C .
- Stratosphere is more or less devoid of major weather phenomena but there is circulation of feeble winds and cirrus cloud in its lower layer.

- Bacteria is found to be surviving here. This is mainly due to presence of ozone (15-35 km) in lower stratosphere.
- Ozone is blue irritating gas with pungent odour.
- Upper limit of stratosphere is called as stratopause.
- Commercial airlines cruise in lower stratosphere mostly due to low temperature, which results in optimum fuel burn and low density of air reduces parasitic drag on airframe. This layer even provide smooth movement as this layer has very less weather turbulence.

(c) Mesosphere

- It lies above the stratopause.
- Mesosphere extends to 50-80 km.
- The upper limit is called **mesopause**.
- Temperature decreases with height and reaches a minimum of -110°C at 80-90 km in the mesopause.

2. Heterosphere

Proportions of gases are not constant in this layer.

(a) Thermosphere

It lies beyond 80 km above mesopause. Here the temperature increases rapidly with increase in height.

The gases become very light due to extremely low density.

It divides into two layers: inosphere and exosphere.

- Ionosphere** lies between 80 km to 640 km above the sea level. There are number of ionic layers in the sphere, e.g. D layer, E layer, F layer and G layer.
 - D layer disappears with sunset as it is associated with solar radiation.
 - E layer is known as Kennelly-Heaviside, confined to the height between 99 km - 130 km.
 - It reflects the medium and high frequency radiowaves.
 - F_2 layer is called as appleton layer, lying between 150 km-380 km.
 - F_1 and F_2 area the two sub layers of E_2 layer combinedly known as Appleton Layer.
 - G layer is beyond 400 km.
- Exosphere** lies between 640 km above the sea level. Density becomes extremely low here and the atmosphere resembles as nebula because it is highly rarefied.

INSOLATION

- There are three major sources of heat energy through which earth is warmed up, i.e.
 - (i) **Solar radiation**
 - (ii) **Gravity**
 - (iii) **Endogenetic**
- Out of these three sources of heat energy solar radiation is the most significant source of terrestrial heat energy.
- The radiant energy received from the sun is transmitted in the form of short waves. It travels at the rate of 1,86,000 miles per second and is called as (*insolation*) or Incoming Solar radiation
- Solar energy received in the form of solar radiation heats the earth's surface and the atmosphere, resulting in change of pressure gradient. It initiating hydrological cycle, recycle nutrients and chemical elements in biosphere.
- There are two basic laws which govern the nature and flow of radiation such as
 - (i) **Wien's displacement law:** According to this law wave length of the radiation is inversely proportional to the absolute temperature of the emitting body.
 - (ii) **Stefan-Boltzman law:** This law states that flow or influx, of radiation is proportional to the fourth power of the absolute temperature of radiating body.
- Continuous emission of photons from the sun causes bonds of radiations with certain wavelength.
- The radiation is mostly emitted in the form of electromagnetic waves and is known as electromagnetic radiation.
- The electromagnetic radiation radiated from the outer surface of the sun, consisting of four spectrums which are as follows:
 - (i) **The first spectrum** includes, gama rays, hard x-rays, soft x-rays and ultra violet rays. Measured in angstrom (10^{-8} cm) and have short wave lengths.
 - (ii) **The second spectrum** consists of visible rays. It is measured in micron and ranges between 0.4 to 0.07 micron.
 - (iii) **The third spectrum** of electromagnetic waves covers infrared spectrum. It ranges between 0.7 to 300 microns.
 - (iv) **The fourth spectrum** consists of longwaves which include micro-waves, radar waves and radiowaves. These waves are measured in cm and m.
- Polar zone between 66° to 90° latitudes in both the hemisphere receives maximum and minimum insolation once a year only. No direct ray of the sun is received beyond 66° latitude throughout the year.

Factors affecting the Distribution of Insolation

The variation in the amount of insolation received by the earth's surface is mostly due to following factors:

- (i) **Angle of inclination** of the sun's rays depends on the latitude of a place. Higher the latitude less is the angle it makes with the surface of the earth causing slanting rays. The area covered by slanting rays is larger than vertical rays. The energy is distributed over a larger area. The net energy received per unit area decreases. Also slanted rays pass through a greater depth of the atmosphere resulting in more absorption, scattering and diffusion.
- (ii) **Length of the day** : During summer days are longer than nights. The situation is reversed with winter. Longer the day, greater will be the insolation.
- (iii) **Distance between the earth and the sun:** The distance between the earth and the sun is not uniform throughout the year. It is nearest to the sun (147 million km) on 3rd January and farthest (152 million km) on 4th July. The reason behind this is mainly to the orbit of the earth which is elliptical in shape. When the earth is nearest to the sun it is called as *perihelion* and when farthest is called as *aphelion*.
- (iv) **Sunspots:** Sunspots created on the outer surface due to periodic disturbance and explosions. The number of sunspots varies from year to year. Its cycle is completed in 11 years. The energy radiated from the sun increases. When the number of sunspots increases and therefore the amount of insolation received by the earth surface is also increases.
- (v) **Effect of the Atmosphere:** The amount of heat energy absorbed by the atmosphere while on its way to earth's surface. This absorption depends on the composition of gases present in atmosphere. Scattering of solar radiation takes place during sunrise and sunset. The oblique rays pass through the longest path of the atmosphere.

Forms of transfer of heat

Radiation

The process of heat transfer in the form of emitted electromagnetic waves is known as *thermal radiation*. The process of radiation of heat energy back from ground is known as ground radiation.

When part of ground radiation is radiated back to earth after getting absorbed by earth's atmosphere is known as *counter-radiation*.

Convection

The transfer of heat energy from one place to another through a medium of fluids as gaseous substances is known as convection.

Distribution of Insolation

- The amount of insolation received at the earth's surface decreases from equator towards poles.
- There are also temporal variation of insolation received at different latitudes at different period of a year.
- There is very little variation of insolation during winter and summer season from equator upto tropics.
- At middle latitude (23.5° to 66° latitudes), both the hemisphere receive maximum sunlight during summer solstice (in northern hemisphere) and winter solstice (in southern hemisphere).

Conduction

The transfer of heat through molecules of an object is known as conduction. Which can be accomplished via two ways:

1. From one part of the body to another part of the same body.
2. From one body to the other body.

Advection

Advection is the lateral or horizontal transfer of heat. It generally takes place in the ocean in the form of currents.

Heat Budget

- The earth as a unit does not accumulate or loose heat. It maintains its temperature. This is so as the amount of heat received as insolation equals the amount lost through terrestrial radiation. This is the heat budget of the earth.
- It is estimated that out of 100 units of incoming solar energy. The 35 units received at the outer limit of the atmosphere is reflected and scattered back into space by clouds (27 units), dust particles (6 units) and by surface of the earth (4 units) in its original short wave form.
- Remaining 65 units enters atmosphere out of which 51 units are received by the earth's surface and 14 units are absorbed by atmospheric gases alongwith water vapour.
- The 51 units which are received by the earth surface are received through direct radiation (34 unit) and rest 17 units is received as diffuse day light. This is called as heat budget of earth.
- 34 units are returned to atmosphere in the form of outgoing terrestrial radiation (long wave) out of which 6 unit absorbed by the atmosphere, 9 units are spent in convection and turbulence and 19 units are spent through evaporation.
- Heat budget of atmosphere includes absorption of incoming solar radiation (14 units) and heat received from terrestrial radiation (34 units). This is equal to 48 units together.
- The energy sent back to space = 35 units (through direct radiation) + 17 units (through radiation from earth) + 48 units (through radiation from the atmosphere)

Regional Distribution of Temperature

The three major heat zones of the Earth are temperate zone, the torrid zone and the frigid zone. These are based on the distance they have from Equator.

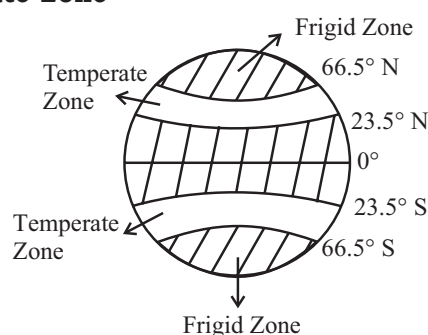
Torrid Zone (Tropical Zone)

This is the hottest zone of the Earth. The region that lies from the Tropic of Cancer (23.5°N), across Equator (0°) to the Tropic of Capricorn (23.5°S) is considered the torrid zone (Tropical Zone). The Sun's ray falls directly at least once a year.

Frigid Zone

This is the coldest zone of the Earth. This area lies to the north of Arctic circle (66.6°N) and to the south of the Antarctic circle (66.5°S) and is permanently frozen. There is no sunlight for most of the months of the year in this zone.

Temperate Zone



- This is the habitable heat zone of the Earth. There are two temperate zones lie in between in both $23\frac{1}{2}^{\circ}$ to $66\frac{1}{2}^{\circ}$ the hemisphere.
- These regions have moderate, tolerable temperature.

Importance of the Heat Zones

This division of the Earth into different heat zones helps in understanding the climate changes and to study weather conditions across the world.

ATMOSPHERIC PRESSURE

Air pressure is defined as total weight of a mass of column of air above per unit area at sea level. Air pressure is maximum at sea level. Pressure exerted by air at a particular point is determined by temperature and density.

There is inverse relation between temperature and pressure.

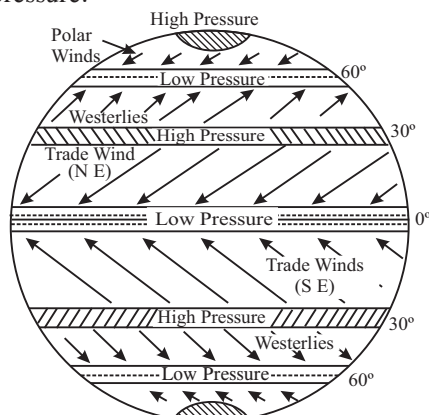
Air pressure is measured as a force per unit area. The unit of air pressure is **millibar or mb** measured by a **barometer**.

The distribution of atmospheric pressure is shown on a map by **isobars**. An isobar is an imaginary line drawn through points of equal atmospheric pressure at the sea level. The spacing of isobars expresses the rate and direction of pressure change and is called pressure gradients.

Major Pressure Belts

- At the equator heated air rises creating a low-pressure area at the surface. This is called the **equatorial low pressure**. It is also known as **doldrums**. Reaching the troposphere the warm air bends towards the pole and descends at $30-35^{\circ}$ latitude or the tropics creating a belt of **sub-tropical high pressure**.
- Subtropical high pressure zone is also known as **horse latitude**. Sub-tropical highs are the source of air moving along the surface towards the poles.
- At $60-65^{\circ}$ latitudes surface air streams coming from polar high pressure and sub-tropical high pressure converge and move up. This upward movement of air creates a low pressure system at the surface which is the **sub-polar low pressure**.

- At the poles, the low temperature makes air contract and creates a zone of high pressure. This thermally created high pressure system is called **polar high pressure**. This dense cold air flows equator-wards away from the area of high pressure.



The planetary System of Pressure and Winds at the Equinoxes

Shifting of Pressure Belts

By late June, sun is overhead at the tropic of cancer. The doldrum (low pressure belt) moves northward from the equator with other belts also shifting in the northern hemisphere. In late December, when the sun is overhead at the tropic of Capricorn, the belts move southwards in the same way.

WIND

Wind is air in motion which can be in any direction. Wind is caused by spatial differences in atmospheric pressure and flows from areas of high pressure to those of low pressure. These differences are caused by uneven absorption of solar radiation at earth's surface. Wind speed is at its greatest during daytime when the greatest spatial extremes in atmospheric temperature and pressure exist. Wind direction is measured as the direction from where a wind comes from.

- Direction is measured by a **wind vane**. Wind speed can be measured using the **Beaufort wind scale**.

Coriolis Force

Rotation of Earth creates Coriolis force. The consequence of Coriolis force is that the moving air changes direction. Instead of wind blowing directly from high to low pressure, rotation of the earth causes wind to deflect. In the northern hemisphere, wind is deflected to wards right, while in the southern hemisphere it is deflected towards left. Magnitude of the Coriolis force varies with the velocity and the latitude of the object. Coriolis force is absent at the equator but its strength increases towards pole gradually.

Geostrophic Wind

Air under the influence of both the pressure gradient and Coriolis force moves parallel to isobars in conditions where friction is low (1000 meters above the surface of the earth) and isobars are straight. Such winds are called **geostrophic winds**.

Gradient Wind

Wind above the earth's surface do not travel in straight lines. The winds blow along the curved isobars of a high (anticyclone) or low (cyclone) pressure center. A wind that blows around curved isobars above the level of friction is called a *gradient wind*.

Types of Winds

- Winds can occur on a large scale.
- On the bases of their area of influence they are divided into two types:
 - Planetary winds:**
 - Trade wind
 - Westerlies
 - Polar wind
 - Periodic winds:**
 - Seasonal wind
 - Local wind
 - Land and sea breeze

Local Winds		
Nature	Nature	Region
Fohn	Warm	Alps
Chinook (snow eater)	Warm	Rockies
Kalbaisakhi	Warm	North India
Berg	Warm	S. Africa
Zonda	Warm	Andes
Loo	Warm	Indian subcontinent
Santa Ana	Warm	Coastal Southern California
Southerly	Cold	New South Wales Burster
Khamsin	Warm	Egypt
Harmattan (Doctor)	Warm	Guinea Coast
Mistral	Cold	S.E. France
Samun	Warm	Iran
Purga	Cold	Russia
Levanter	Cold	France
Pampero	Cold	S. America
Norwester	Warm, dry	New Zealand
Harmattan	Warm, dry	Eastern part of Sahara

1. Planetary Winds

They are a major part of general global circulation of air. They occur due to temperature and pressure variance existing throughout the world. They are of following types :

- Trade winds
- Westerlies
- Polar winds

(i) Trade Winds

Blowing from the subtropical highs or horse latitudes (between 30°N and 30°S) towards the equatorial low pressure are the trade wind.

- In the northern hemisphere, the trade wind blowing from the northeast and is known as the Northeast trade wind in the southern hemisphere, the wind blowing from the southeast and are called the southeast trade wind.

(ii) Westerlies Winds

Prevailing wind in the middle latitudes between 35° and 65° latitude, blowing from high pressure area in the horse latitudes towards the poles are called as Westerlies wind.

- The winds are predominantly from the southwest in the northern hemisphere and from the northwest in the southern hemisphere and bring extra-tropical cyclones with them.
- In southern hemisphere they are stronger and more constant in direction than those of the northern hemisphere because of the vast expanse of water.
- They are best developed between 40° and 65° south latitudes. These latitudes are hence often called Roaring forties, furious fifties and shrieking sixties.

(iii) Polar Wind

Winds blowing in the arctic and the antarctic latitudes are polar winds. They blow from polar high pressure towards sub-polar low pressure belt. In the northern hemisphere, they blow from north-east, and are called the north-east polar winds. In the southern hemisphere, they blow from south-east and are called south-east polar winds. As these winds blow from ice-capped landmass, they are very cold.

2. Periodic Winds

Land and sea breezes local and monsoon winds are periodic winds. They are of the following types:

(i) Seasonal Winds

- **Monsoon Winds:** Monsoons are regional wind systems that predictably change direction with the passing of the seasons. Monsoons occur over distances of thousands of kilometers, and their two dominant patterns of wind flow act over an annual time scale.
- **South-West Monsoon:** During summers, monsoon winds blow from cooler ocean surfaces onto the warmer continents during April to September. The continents become warmer than the oceans due to:
 - (i) Specific heat differences between land and water.
 - (ii) Greater evaporation over water surfaces.
 - (iii) Subsurface mixing in ocean basins, which redistributes heat energy to a deeper layer.

Precipitation is associated with summer monsoons. Onshore winds blowing inland from the warm ocean have very high in humidity. Upon cooling these air masses causes condensation and rain. Some highland in Asia receives more than 10 meters of rain during the summers.

- **North-West Monsoon:** In the month of October and November the wind patterns reverses, as ocean surfaces are now warmer. With little solar energy available,

continents begin cooling as long wave radiation is emitted to space. Ocean surface retains its heat longer as water has high specific heat and subsurface mixing. Winter monsoons bring clear dry weather and winds that flow from land to sea. It brings rain to A.P and T.N as they pick moisture from Ocean.

- **Monsoon Winds of SE Asia:** Asiatic monsoon is caused by a complex climatic interaction between distribution of land, water, topography, and tropical and mid-latitudinal circulation.
 - During summer, a low-pressure forms over northern part of India and Southeast Asia because of high solar insolation.
 - Warm moist air is drawn into the thermal lows from air masses over the Indian Ocean.
 - Summer heat creates a strong latitudinal pressure gradient and development of an easterly jet stream (15 km) at the latitude of 25°N.
 - The jet stream enhances rainfall in Southeast Asia, Arabian Sea, and South Africa. When autumn returns to Asia, thermal extremes between land and ocean decrease. Westerlies of the mid-latitudes move in. Easterly jet stream is replaced with strong westerly winds in the upper atmosphere. Subsidence from an upper atmosphere cold low above the Himalayas produces outflow that creates a surface high-pressure system that dominates the weather in India and Southeast Asia.

Besides, Asian continent, monsoon wind systems exist in Australia, Africa, South America, and North America.

- (ii) Local winds:** Winds flow in comparatively small area and have special characteristics. They are generated by purely local factors (local temperature differences) and their zone of influence is quite limited. They play an important role in the weather of a particular locality.

- (iii) Land and Sea Breezes:** Sometimes local conditions may set air in continuous motion. Even in calm days in summer, heated air rising from land surface may cause strong breezes to move in from over the cooler sea. By night, land cools more rapidly than the sea, cool air may move seawards as land breeze, usually a gentle flow. In some countries, if hot deserts border the seas, strong winds may develop. Sea breeze will be felt miles inland.

Jet Streams: These are strong, rapid and narrow air currents circumpolar westerlies blowing in upper atmosphere or into troposphere.

- Blow from west to east spreading over a few hundred kilometers at the height of 7.5-14 km
- Found between poles and 20° latitude in both the hemisphere.

The minimum velocity is 108 km/hr and the average summer velocity is 130 km/hr and average winter velocity is 65 km/hr. The maximum velocity is 480 km/hr.

AIR MASS, FRONTS, CYCLONE AND ANTI-CYCLONE

Air Mass

An air mass is defined as a large body of air, relatively having similar physical properties (especially temperature and humidity) spreading over hundreds of kilometers.

- In order to acquire the similar physical properties air mass must be stagnate for a longer period of time on the source region.
- The notable centre for the development is anticyclone area which is characterised by high pressure and low pressure.

Classification of Air mass

Generally the air mass is classified into 4 types:

- (1) **Polar Air mass (P)** - It originates in polar region between 60° N and S.
- (2) **Tropical Air mass (T)** - It originates between 25°N and S upto equator.
- (3) **Continental Air mass (C)** - It originates over continents (land mass) and is dry in nature.
- (4) **Marine Air mass (M)** - It originates over the oceans and is moist in nature.

These four air masses are combined with one other to form four principal air masses such as:

- (1) Continental polar mass air (CP) - Cold, dry and stable.
- (2) Maritime polar mass air (MP) - Cold, moist and unstable.
- (3) Continental Tropical air mass (CT) - hot, dry, stable (at height) and at surface unstable.
- (4) Maritime Tropical air mass (MT) - warm, moist and unstable.

Front

It is the transition zone between two air masses with different properties (temperature, humidity, density, pressure and wind direction). An extensive transitional zone between two converging air masses is called as frontal zone on frontal surface. The process associated with the creation of new fronts or the regeneration decaying fronts is called *frontogenesis*. The process of destruction or dying of fronts is called *frontolysis*.

Types of Fronts

Fronts are of four principal types on the basis of their different characteristic features such as

- (1) Warm front
- (2) Cold front
- (3) Occluded front
- (4) Stationary front

(1) Warm front

It has gentle sloping frontal surface along with warm and light air. As they are warm and light they rise up slowly over cold and dense air. This warm air is cooled adiabatically. It is saturated and gets condensed resulting in precipitation over a larger area for several hours. The precipitation varies from moderate to gentle. It is associated with Ci, Cs, As, Ns, St and fog.

(2) Cold front

It can be defined as boundary between warm and cool air masses wherein cold air displaces warm air. It has a steep edge and moves aggressively to invade the warm air territory. As they are denser, they remain at the ground, forcing the warm air to rise. The cold air mass is associated with bad weather-thick clouds, heavy downpour with thunderstorms and lightning. At times it results in snowfall and hailstorms. Clouds associated with this type of front are Ci, Cs and Cb.

(3) Occluded front

When the cold front completely overtakes warm front, displacing the warm air by cold air from the ground system.

(4) Stationary front

It is the stage when the two contrasting air masses become parallel to each other and there is no ascent of air.

Cyclones

Cyclone is a centre of low pressure surrounded by closed isobars. The pressure increases outward and the air moves inward as the centre has low pressure. In the northern hemisphere it has anticlockwise movement and clockwise in the southern hemisphere.

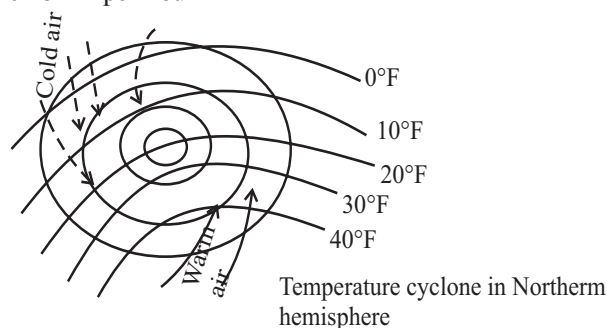
It can be circular, elliptical or 'V' shape.

It is of two types

- (1) Extratropical/Temperature cyclone
- (2) Tropical cyclone

(1) Extratropical/Temperature cyclone

It is a simple depression or atmospheric disturbance having low pressure in the centre and pressure increases outward. It is found in the middle latitude (35°-65°) in both opposing air masses (warm, moist and light tropical air mass, front is created where these two air masses converge and lead to the development of temperate cyclone. The size of cyclones varies from 150 km to 3000 km. On an average vertical extent is 10-12 km. Its average summer velocity is 32 km per hour for winter it is 48 km per hour.



Life Cycle of Cyclone

The period of cyclone from its inception (cyclogenesis) to its termination (cyclolysis) is called as *life cycle of cyclone*. There are six stages in a life cycle of a cyclone

1. **First stage** involves the convergence of two air masses of contrasting physical properties and direction.
2. **Second stage** is a stage when the warm and cold air masses penetrate into the territories of one other, and a wave-life front is formed. It is also called incipient stage.
3. **Third stage** is the mature stage when the cyclone is fully developed and the isobars are almost circular in shape.
4. **Fourth stage** in this stage where warm air mass is narrowed by the cold air mass.
5. **Fifth stage** is related with the starting of occlusion of cyclone. Here, the cold front fully overtake the warm front.
6. **Sixth stage** warm sector completely disappears and occluded front is eliminated and ultimately cyclone dies out.

(2) Tropical Cyclone

The development of the cyclone over the tropical region lying between tropic of cancer to capricorn are called as Tropical Cyclones.

- Its average diameter varies between 80 km to 300 km. At times it is restricted to 50 km or even less in diameter.
- Velocity varies between 32 km/hr to 180 km/hr or more when it is converted into a hurricane.
- Tropical cyclone becomes more vigorous and move with high velocity over ocean but over land it becomes feeble as it reaches interior portion of the continents.
- There are less number of isobars and are more or less in circular shape. This results in rapid rush of wind towards the centre.
- Every year it occurs at a particular period of a year, mainly during summer season.
- It is not generated near equator as the coriolis force is negligible at equator.

Types of Tropical Cyclone

Tropical cyclone are divided into 4 major types:

- (1) Tropical disturbance
- (2) Tropical depression
- (3) Tropical storms
- (4) Hurricane or typhoon

1. Tropical disturbance

It is associated with easterly trade wind. The easterly wave develops between 5° to 20° N latitude in the western part of the oceans. It is associated with large amount of cumulus or cumulonimbus cloud. These clouds bring heavy to moderate rainfall.

2. Tropical Depression

This region is the centre of low pressure and characterised by closed isobars, which are small in size. The wind velocity at the centre is 40-50 km/hr. It is usually influenced by summer weather of India and Australia. Sometimes, it becomes strong and give heavy rainfall, after resulting in floods.

3. Tropical Storm

Tropical storm is related with low pressure centre, closed isobars which rushes towards the centre with the velocity ranging between 40 to 120 km/hr. It develops over Caribbean sea, near Philippines and Bay of Bengal causing devastation of lives and properties.

4. Hurricane or Typhoon

It is a form of massive tropical cyclone surrounded by several closed isobars. It moves with an average speed of 120 km/hr. Hurricane has more symmetrical and circular isobar. The pressure increases sharply from centre towards the outer margin, resulting in pressure gradient. Heavy downpour occurs, which is often uniformly distributed over a larger area as compared to other types of tropical cyclones.

Tropical Cyclones are known by different names :

Hurricanes - Caribbean and Pacific coast of Mexico.

Typhoons - Sea of China and Japan

Cyclones - India / Australia

Willy-Willies - North Australia

Tornado - South and Eastern USA.

Bagguio - Philippines

Comparison between Temperature and Tropical cyclones.

Tropical Cyclone	Extra-tropical cyclone
The tropical cyclone have a thermal origin, exclusively over the tropical seas.	Formed in middle or high latitudes, due to the development of front (35°-65° N and S)
The size of the tropical cyclone is 1/3 of temperate cyclone	It is much extensive size.
Strongest winds of tropical cyclones take place at surface.	Strongest winds of mid-latitude cyclones are higher up in atmosphere.
It is associated with single eye	There is more than one place where wind and rain is active.
Wind velocity is very high	It is low in comparison to tropical cyclone.
Its relation with upper level air is not clear	It has a distinct relation with upper level of air.

Anticyclones

The air mass in which pressure is high at the centre but decreases outwards is anti-cyclone. Winds move in a clockwise out-spiral in the northern hemisphere but in an anticlockwise out-spiral in the southern hemisphere.

- They have a circular shape usually but at times assume V shape.
- They are larger in size than temperate cyclone. 75 times the size of temperate cyclone.
- It has the average velocity of 30-50 km/hour.
- The high pressure at the centre causes the wind to move outward.
- The wind descend from above at the centre and thus weather becomes clear and rainless.
- It has no fronts.

HUMIDITY

It is the amount of water vapour present in the air at a particular period of time and place. Humidity of a place can be expressed in three ways:

Absolute Humidity

The measure of water vapour content of the atmosphere which may be expressed as the actual quantity of water vapour present in a given volume of air is called absolute humidity. This is measured as gms per cubic meter air. Absolute humidity changes with place and time. The capacity of air to hold water vapour depends on temperature. Warm air holds more moisture than cold air.

Specific Humidity

Another way to express humidity as the mass of water vapour per unit weight of air or the proportion of the mass of water vapour to the total mass of air is called the specific humidity. Specific humidity is not affected by changes in pressure or temperature.

Relative Humidity

This is a ratio expressed between actual quantity of water vapour present in the air at a given temperature absolute humidity and the maximum quantity of water vapour that the atmosphere can hold at that temperature. Relative humidity determines the amount and rate of evaporation.

- **Hygrometer** is the instrument used for measuring relative humidity. It comprises of wet and dry bulb thermometer.

$$\text{Relative humidity} = \frac{\text{Absolute humidity}}{\text{Humidity capacity}} \times 100$$

- Temperature and evaporation are directly proportional to humidity.
- The process of transformation of liquid into gaseous form is called as *evaporation*.
- Oceanic and coastal regions record higher humidity capacity of air than the remote continental regions.
- Humidity capacity decreases from equator to polewards as the temperature also decreases.
- The air having moisture content equal to its humidity capacity is called as **saturated air**.

It balances the energy flow of the earth in the form of clouds. The process of evaporation and condensation are major means by which transfer of energy takes place from earth to atmosphere, which is the basis of precipitation.

PRECIPITATION

Condensation of atmospheric water vapour that falls under the gravity is called as precipitation. This could be in the form of rain, snow or hail etc. Its form depend on the temperature at which water vapour condenses.

Forms of Precipitation

Hail

It is a form of solid precipitation consisting of large pellets or spheres of ice balls with the diameter varying between 5 to 50 mm. The falling of hail on the ground surface is called *hailstorm*. It is destructive as it destroys agricultural crops and

claim human and animal lives.

Snowfall

It is the fall of large snowflakes from clouds on the ground surface. The dew point should be below freezing point for receiving snowfall. It is a result of sublimation.

Sleet

It is a mixture of snow and rain. It is a small pellets formed by freezing of raindrops or melting snowflakes.

Rainfall

It is the most common form of precipitation. It is a process wherein warm air ascends, saturates and condenses. Adiabatic cooling takes place when the relative humidity becomes 100 per cent. The condensation of water vapour takes place where large hygroscopic nuclei (salt and dust) is formed. Such droplets are called as cloud droplets. Rainfall occurs when cloud droplets change to raindrops which involves two processes:

1. The warm and moist air ascends to such a height that the process of condensation begins below freezing point. Both the water droplets and ice droplets are formed. The condensation takes place as the water droplets evaporates around ice droplets due to difference in vapour pressure. These ice droplets become large and fall when finally they are not able to be held back in the condensed ice droplets.
2. The suspended cloud droplets in the cloud are of varying sizes. They collide among themselves at different rate as their size varies. They combine to form a large droplet. In this process several cloud droplets are coalesced to form raindrops. When these cloud droplets are large enough that they are unable to hold by ascending air they begin to fall.

Types of Rainfall

Rainfall can be classified into three types:

1. Convictional Rainfall
2. Orographic Rainfall
3. Cyclonic or Frontal Rainfall

1. Convictional Rainfall

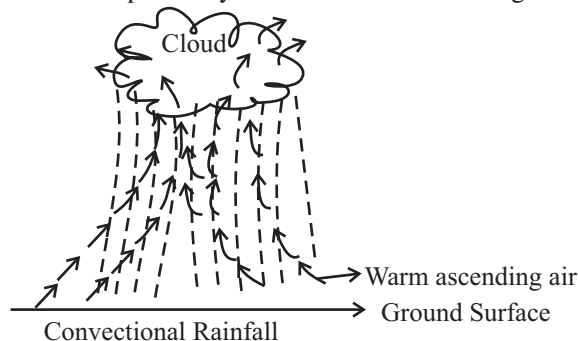
The thermal heating of the ground surface through the process of insolation leads to rise of air as they become warm and light. The process of convictional rainfall mainly depends on two factors.

- (i) The supply of moisture through evaporation should be abundant so that its relative humidity becomes high.
- (ii) There should be intense heating through insolation process.

The process of convictional rainfall involves intense heating of ground surface through solar radiation. As the warm air rises, the vacant space is filled by surrounding air which too warm up when come in contact with already warm air. Ascending warm and moist air cools according to dry adiabatic lapse rate (10°C per 1000 metres) increasing the relative humidity. The moist air becomes saturated soon (relative humidity becomes 100 percent) and further rising of saturated air causes condensation and cloud formation cumulo-nimbus clouds. The air further rises and cools with moist adiabatic lapse rate

(5°C per 1000 metre). When the air reaches the temperature of its surrounding cumulo-nimbus cloud is formed and there is instantaneous heavy rainfall.

- It occurs daily in afternoon in the equatorial regions.
- It is for short duration but point down heavily.
- Occurs through thick dark and extensive cumulo-nimbus clouds.
- It is accompanied by number of thunder and lightning.

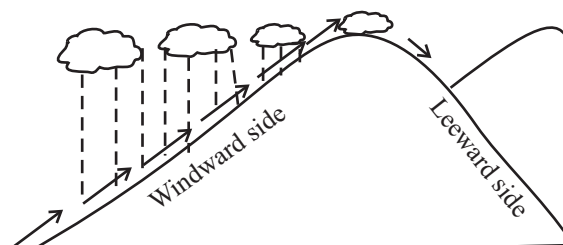


2. Orographic Rainfall

When the warm and moist air is obstructed by any hill or mountain, it starts ascending along the slope of the hill or mountain and get saturated after reaching a height. As a result condenses around hygroscopic nuclei. The presence of hill or mountain triggers the process of ascending of moist air, becoming cool and unstable. The slope of the

mountain facing the wind is called as windward side and the other side of that mountain is known as Leeward side. Windward side receives maximum rain and leeward side receives no or little rain. The leeward side is also known as rainshadow region. This type of rainfall is called as orographic rainfall.

- The mountain should be acting as a barrier across the wind direction.
- Mountain or hill should be running parallel to the coast.
- The height of the mountain too determines



- the amount of orographic rainfall.

3. Cyclone or Frontal Rainfall

Cyclonic or frontal rainfall occurs due to ascending of moist air and adiabatic cooling caused by convergence of two extensive air mass.

CLOUDS

They are mass of very fine water droplets, ice particles or mixture suspended in the atmosphere. Clouds are formed by the adiabatic cooling of air when it is below its dew point. Cooling process is created by upward movement of light and warm moist air which reduces pressure, expands and reaches its dew point. There is further cooling below dew point causing condensation. Adiabatic process involves change in temperature of an ascending or descending air. The air which ascends (due to expansion of volume) results in cooling of air, whereas the air which descends results in warming a air at the rate of 10°C per 1000 meters. This rate of shade change is called as adiabatic lapse rate. 10° per 1000 meters change is only before dew point (condensation level) and after dew point is achieved the rate of change in 5°C per 1000 m.

Types of Cloud

Usually clouds are classified in terms of their vertical arrangement in the atmosphere of high, middle and low clouds.

1. High clouds: 6000 to 12000 meters above sea level.

- **Cirrus** : Wispy, fibrous-looking cloud which indicate fair weather.
- **Cirrocumulus** : A thin cloud, often globular and rippled.
- **Cirrostratus** : Thin white sheet type which gives the sun and moon their haloes.

2. Medium clouds : 2000 to 6000 meter above sea level.

- **Alto cumulus** : Globular, bumpy looking clouds with a flat base covering entire sky..

- **Altostratus** : greyish, watery looking and a wool pack cloud..
- 3. **Low clouds:** below 2000 meters above sea level.
 - **Stratocumulus** : Low rolling, bumpy clouds.
 - **Nimbostratus** : Fog-like low cloud causes dull weather with drizzle. It is also associated with lightning thunder and hailstorm.
 - **Stratus** : These clouds are low, grey and layered, almost fog-like in appearance, bring dull weather and often accompanied by drizzle air or snowfall.
- 4. Clouds of great Vertical extent: 1500 to 9000 m.
 - **Cumulus** : A round-topped and flat based cloud, which form a whitish grey globular mass. It some times becomes thunder cloud.
 - **Cumulonimbus** : Cumulus cloud which reaches up to 9000 meters, often indicates convectional rain, lightning and thunder storm.

Distribution of precipitation across different regions of world.

1. Regions of Heavy Precipitation: Rainfall more than 150 cm per year are:

- Equatorial regions:** Amazon and Congo Basins, Malaysia, Indonesia and New Guinea.

- (ii) **Tropical Monsoon regions:** Parts of India, South-east Asia and South China.
 - (iii) **Mid-latitude West Margin regions:** Coastal regions of British Columbia, North-west Europe, South Chile and South Island of New Zealand.
2. **Moderate rainfall** (100 to 150 cm per year)
- (i) Eastern margins of continents in the trade-wind belt e.g. eastern margin of China, U.S.A., Brazil, South Africa and Australia.

3. **Regions of very low rainfall** (less than 25 cm.)
- (i) Tropical deserts – Western margins of continents in the trade wind belt, Californian desert (USA), Atacama (South America), Kalahari (Southern Africa), Sahara, Arabian Desert and West Australian desert.
 - (ii) Mid-latitude desert – Interiors of large continents like Asia and North America.
 - (iii) Polar Regions – Arctic and Antarctic.

WORLD CLIMATIC ZONES

1. Equatorial Climate or Tropical Rain Forest Climate

Location

- It is found between 5°N to 10°N and S of equator.
 - The equatorial climate is found in the following
- South America :** Largest area is in the Amazon lowlands. It also occurs along the coast of Guianas.
- Africa :** Part of the Congo basin and Guinea coast Africa.
- Southern Asia :** Malaysia, Indonesia, New Guinea and parts of Philippines.

Climate

- It is warm round the year as the sun's rays are always fall vertically. Annual average temperature is uniform at 27°C. The daily range of temperature is in between 10°C and 25°C. The annual range is less than 5°C.
- This region has no dry season. Average annual rainfall is 200 cm to 250 cm. The driest month in this region receives 6 cm of rainfall.
- Thermally included low pressure belt due to the uniform high temperature throughout the year.
- Convection current is formed and results in rainfall in this zone.
- The convergence of trade winds coming from sub-tropical high pressure belt forms intertropical convergence (ITC). ITC is associated with atmospheric distribution (cyclone).

Vegetation

High temperature and year round rain produce most luxuriant vegetation in the region which are tropical rainforest or Selvas in South America.

2. Savanna or Sudan type of climate

Location

This transitional type of climate is bounded between equatorial rainforest and semi-arid and subtropical humid climate. This lies between 5° to 20° north and south of equator.

- **South America:** Columbia and Venezuela.
- Africa :** Sudan, parts of Senegal, Mali, Guinea, Niger, Chad, Ghana, Togo, Kenya, Zimbabwe, Tanzania, Angola and Uganda.
- Australia :** Queensland.

Climate

- This zone has distinct wet and dry season.
- The mean temperature varies between 24°C to 27°C.
- Annual precipitation of 100 cm to 150 cm is received. From December to February there is no rainfall at all.
- 80% to 90% of the rain occurs in rainy season only.

Vegetation

- Seasonal rains allow grass to grow. These are natural grasslands. Rainfall is not sufficient to support tall trees but grass grows well. Elephant grass grows up to 4.5 metres. As we move towards the equator, trees grow along the banks of the streams, are of broad-leaf and umbrella shaped.
- **Fauna:** Elephants, giraffe, zebra, rhinoceros along with varieties of carnivores animals (lion, leopards, tiger, cheeta, hyena etc.) are found.

3. Hot Desert Climate

Location

Arid deserts lie close to the tropic of cancer and tropic of capricorn in the western margins of continents, between 15°-30° in both the hemisphere.

- Sahara, Arabia, Thar, Mohave and Sonoran (U.S.A.), Kalahari and Namib (Africa), Simpson, Gibson, Great Sandy (Australia). These lie in western part of the continents.

Climate

- Climate is dominated by subsidence of air masses and sub tropical anti cyclones.
- Average summer temperature is between 30°C to 35°C. Mid-day temperature is recorded to be 40°C.
- Daily range of temperature is great varying between 22°C to 28°C
- Annual range of temperature varies between 17°C to 22 °C.
- These areas are nearly rainless or receive lowest annual rainfall (Less than 12).

Vegetation

- Normal vegetation is cactus, thorny plants, shrubs and herbs.

4. Steppe (Temperate continental) Climate Underline

Location

These are also called **mid-latitude grasslands**. They are far away from any influence of the sea as they are in the heart of the continents.

Prairies (North America), Pampas (South America), Velds (South Africa), Canterbury (New Zealand), Downs (Australia) and Steppes (Russia).

Climate

- Their climate is continental with extremes of temperature. Summers are very warm and winters are very cold. In northern hemisphere where as in southern hemisphere the temperature variation is mild.
- These are dry lands as they are located in the deep interiors of large land masses away from the oceans. Temperature in summer varies from 18°C to 24°C and in winter from – 4°C to 2°C. Such range of temperature is large. Rain fall occurs in spring and early summer and varies between 25 cm. and 75 cm. Rain is of convectional type and light.

Vegetation

Short grass grows everywhere. Trees are found only on mountain slopes.

5. Mediterranean Climate

Location

- The zone lies between 30°-40°N and S latitudes on the western edge of the continents.
- Mediterranean type of climate is found near the Mediterranean sea in the northern hemisphere (Portugal to Turkey, Morocco, Northern Algeria, Tunisia and Libya) alongwith southern California coast.
- In the southern hemisphere central Chile, Cape town area of South America, South and Southwest coast of Australia.
- Average winter temperature is 5°C to 10°C whereas that of summer it is 20°C to 27°C.
- Mean annual range of temperature is recorded to be between 15°C to 17°C.
- The zone experiences season shift of pressure belt
- Winter rainfall is received through cyclonic storms.
- More than 75% of the rainfall is in winter season.
- The mean annual rainfall varies between 37 cm to 65 cm.
- Summer winds are generally dry and hot.

Vegetation

- Woodland, dwarf forest and scrubs are found in this region.
- The leaves are thick and shiny resisting moisture loss.
- The trees like pine, oak, cedar, madrone, walnut and chestnuts grow here.
- Citrus fruits are grown in this zone.

6. Sub-Tropical Humid Climate/ China type

Location

- It is found in the eastern boundary of the continent between 25°C to 40° N and S latitude.
- It is found in south-east China, PO Basin, Danube Basin, South-east USA; south-east Brazil, Paraguay, Uruguay and north-eastern Argentina and Africa alongwith the east coast of Africa.

Climate

- Mean annual summer temperature lies between 24°C to 26° and during winter it is 6.6°C to 10°C.
- The temperature do not varies spatially.
- Rainfall varies between 75 cm to 150 cm
- Rain decreases as we move inland from coast.
- Tropical cyclone is found and results in rainfall. Even winter cyclones are found here as it is associated with Westerlies.

Vegetation

- Dense evergreen forests are found alongwith deciduous sparse forest and grassland. They have broad leaves.

7. West European type Climate

Location

- The zone lies between 40° and 65° latitude in both the hemisphere along the west coast of the continents.
- North-Western Europe, British Columbia of Canada, Washington and Oregon states of the USA, south-west coast of Chile, south-east coast of Australia and Tasmania and New Zealand.

Climate

- Average temperature during summer ranges between 15°C to 21°C.
- Polar front causes the development of temperate cyclone in this region under the influence of westerlies.
- Annual range of temperature varies between 50 cm to 75 cm.

Vegetation

- Broad-leaf deciduous forest (oak, birch, walnut, maple, elm, chestnut, etc).
- Needle-leaf (coniferous) forest like pine, fir etc are found.
- Mixed forest is also found here.

8. Monsoon Climate

Location

Monsoon region includes the eastern margins of continents which lies between 5° to 30° N and S latitude of equator.

- Eastern Brazil (S. America), Central American countries, Natal coast (S. Africa), Indian subcontinent, South East Asia, Myanmar, Thailand, Vietnam, Philippines, etc. Parts of East Africa including Malagasy, North Australia.

Climate

- Temperature variation results in season formation.
- During the months of summer (March to June) the average temperature is ranging between 27°C to 32°C.
- During the months of winter the average temperature recorded is 10° to 27°C.
- The temperature in this zone is controlled by nearness or remoteness of the sea, latitudinal and altitudinal also influence it.
- The region receives cyclonic rainfall mostly along with orographic rain.
- On an average annual rainfall received is around 150 cm. Rainfall shows temporal variation.
- 80% of the rainfall is received within 3 months (July, August and September).
- Some parts of Indian sub-continent receive winter rainfall (Tamil Nadu and Andhra Pradesh) from north-east monsoons.
- Monsoon rainfall is basically cyclonic in character.
- Dry season is found here

Vegetation

- Trees are mostly deciduous. The forests are open and less luxuriant. Most of the forests yield valuable timber like teak. Other kinds of timber are sal, acacia and eucalyptus.

9. Taiga /Boreal/ Sub-Arctic Climate

Location

This climate type is named after the coniferous forest cover found in the region.

- This region lies between 55° and 70° in northern hemisphere. It forms a continuous belt across southern Canada, northern Europe and Russia.
- South Alaska, Southern Canada, parts of Norway, Sweden, Finland, Northern Russia, Northern Siberia, and Sakhalin Island.

Climate

- Winters are very cold and severe lasting for 6 to 7 months. This region has Verkhoyansk the “cold pole” colder than the arctic region. Summers are short lasting for 3 or 4 months and days are long; at 60°N the sun shines for 18 hours.

- Rainfall varies from 25 to 100 cm. There is more rainfall near the coast.
- Most of the rain comes from cyclonic weather. It falls throughout the year but maximum in summer as frequent showers.
- In winter it is in the form of snow which remains on the ground for 5 to 7 months.

Vegetation

- Vegetation in this climate type is softwood coniferous forests (Spruce, fir, pine).
- Fauna like reindeer, deer, elk, moose and wild cat are found here.

10. Tundra Climate

Location

- The northern most parts of Asia, Europe and North America (include Alaska and Canadian Islands).

Climate

- Average annual temperature is – 12°C.
- Long, bitterly cold and severe winter are experienced.
- Summers are short but cool.
- Precipitation is below 40 cm and as snowfall.

Vegetation

- Very short growing season.
- Dwarf willows and birches grow here.
- Fauna like reindeer, polar bear, fox, musk, ox and arctic hare are commonly found here.

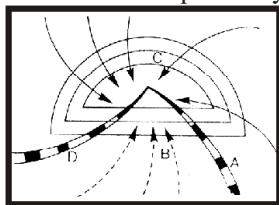
Isolines

It represents the equal value on the map.

- Isobar: A line representing points of equal atmospheric pressure.
- Isobath: A line representing points of equal depths.
- Isobathotherm: A line representing depth of water with equal temperature.
- Isochime: A line representing points of equal mean winter temperature.
- Isochrone: A line representing points of equal time-distance from a point, such as the transportation time from a particular point.

Exercise -1

1. Which one of the areas marked as A, B, C and D in the given figure of the cyclone, witnesses heavy torrential short duration rainfall accompanied by thunderstorms?



- (a) A (b) B
(c) C (d) D
2. Which among the following represents the correct sequence in increasing height, of different types of cloud formation?
- (a) Cirrus, cumulonimbus, stratus, cumulus
(b) Stratus, cumulus, cirrus, cumulonimbus
(c) Cumulonimbus, cumulus, stratus, cirrus
(d) Cirrus, cumulus, cumulonimbus, stratus
3. Which of the following clouds are more commonly formed in the equatorial rainforests?
- (a) Cumulonimbus (b) Stratocumulus
(c) Nimbostratus (d) Alto cumulus
4. The tropical cyclones do not occur close to the equator because of
- (a) Excessive heat
(b) Calm air
(c) Weak coriolis force
(d) The winds are too wet and heavy
5. As per Ferrel's law
- (a) in the northern hemisphere, the north blowing winds are deflected westward
(b) in the southern hemisphere, the north blowing winds are deflected eastward
(c) in the northern hemisphere, the south blowing winds are deflected eastward
(d) in the southern hemisphere the south blowing winds are deflected eastward
6. Spring tides occur when
- (a) the moon, the sun and the earth are in the same line
(b) the moon, the sun and the earth are the right angles, with the earth at the apex
(c) the moon is nearest to the earth
(d) the moon is farthest from the earth
7. Study the following diagram :



The diagram represents

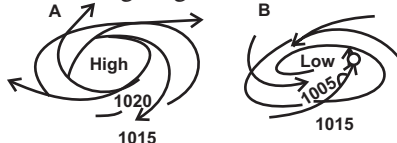
- (a) high tide
(b) spring tide at new moon
(c) spring tide at full moon
(d) neap tide at half moon

8. Which of the following statements is not true of Trade Winds?
- (a) They blow from the Horse latitudes to the Doldrums.
(b) They are deflected to the right to become South-East Trades in the southern hemisphere.
(c) They are constant in strength and direction.
(d) The sometimes contain intense depression.
9. Perpendicular rays are usually more heating than oblique rays. Therefore, south-facing slopes are warmer than north-facing slopes in the summer
- (a) in the northern hemisphere
(b) in the southern hemisphere
(c) everywhere
(d) nowhere
10. Which of the following is wrongly matched?
- (a) Rainfall caused by mountain standing in the way of moisture-laden winds-relief rainfall.
(b) Rain caused by vertical circulation of air currents-orographic rainfall.
(c) Rain caused by a warm moist air mass moving upward over cold air mass- cyclonic rain-fall.
(d) An area sheltered from rain bearing winds by a hill-rain shadow area.
11. Some large lakes in temperate latitudes have a moderating influence upon the climate of adjacent land surfaces. Such an influence may develop when
- (a) the water is shallow
(b) the lake is high above sea level
(c) prevailing outblowing winds cross the lake in water
(d) the lake is a basin of inland drainage
12. What causes snowfall?
- (a) When condensation takes place below 0°C and precipitation occurs in the form of flakes.
(b) When rain freezes into snow flakes while passing through the atmosphere.
(c) When condensation takes place in the form of fog and precipitation occurs as flakes.
(d) When the air temperature is much lower than surface temperature.
13. Heat gain through insolation and heat loss through terrestrial radiation are equal approximately at which latitude ?
- (a) $22\frac{1}{2}^{\circ}$ North and South
(b) $40\frac{1}{2}^{\circ}$ North and South
(c) $42\frac{1}{2}^{\circ}$ North and South
(d) $66\frac{1}{2}^{\circ}$ North and South
14. Which of the following is the annual precipitation of an area?
- (a) All forms of precipitation converted to the quantity of liquid water
(b) The total actual amount of rain, snow, hail and sleet
(c) The total amount of rain per year, irrespective of snow, hail and sleet
(d) All forms of precipitation measure after conversion to equal quantity of ice

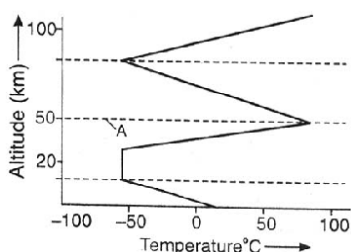
15. Although hot deserts have a very low annual rainfall, occasional heavy rain storms do occur and these sometimes produce steep-sided, flat-floored valleys called
 - (a) gorges (b) wadi
 - (c) canyons (d) barchan
16. Winter temperature ranging from -29°C to -40°C , summer temperature about 10°C , total annual precipitation about 25 cm, and low humidity. These characterise the climate.
 - (a) Taiga (b) Tundra
 - (c) Mountain (d) West European
17. Atmosphere with excess of carbon dioxide as a result of pollution could result in
 - (a) fall in temperature
 - (b) rise in temperature
 - (c) no change of temperature
 - (d) excess of ultraviolet rays reaching the earth
18. Upto a height of 50 km from Earth, the atmosphere is composed of
 - (a) nitrogen, oxygen, argon, minor gases
 - (b) only nitrogen and oxygen
 - (c) only minor gases such as carbon dioxide, hydrogen, neon, helium, methane, etc.
 - (d) None of these
19. The order of the layers in the atmosphere, upwards from below, is
 - (a) stratosphere, troposphere, ionosphere and exosphere.
 - (b) exosphere, ionosphere, troposphere and stratosphere.
 - (c) troposphere, stratosphere, ionosphere and exosphere.
 - (d) troposphere, magnetosphere, stratosphere and exosphere.
20. The most prominent gases in the atmosphere, in terms of volume, are
 - (a) nitrogen and methane
 - (b) nitrogen and oxygen
 - (c) oxygen and carbon dioxide
 - (d) hydrogen and nitrogen
21. What is the name given to winds blowing from subtropical high pressure regions, to the equator?
 - (a) Westerlies (b) Tropical easterlies
 - (c) High latitude easterlies (d) Doldrums
22. 'Horse latitudes' is the term applied to the
 - (a) $0^{\circ} - 5^{\circ}$ N and S latitudes
 - (b) polar circles
 - (c) $30^{\circ} - 40^{\circ}$ N and S latitudes
 - (d) $40^{\circ} - 60^{\circ}$ N and S latitudes
23. The stratosphere is said to be ideal for flying jet aircraft. This is because
 - (a) this layer is rich in ozone which reduces fuel consumption
 - (b) the temperature is constant and ideal for aircraft engine efficiency
 - (c) this layer is out of the firing range of anti-aircraft guns
 - (d) of the absence of clouds and other weather phenomena
24. The velocity of winds is governed by
 - (a) pressure gradient (b) Farrel's Law
 - (c) rotation of the earth (d) temperature
25. Over which region is the temperature the highest near the tropopause?
 - (a) Around the equator
 - (b) Over the Arctic region
 - (c) Near the Tropic of Capricorn
 - (d) Near the Tropic of Cancer
26. Which of the following are not a planetary wind?
 - (a) Easterlies (b) Westerlies
 - (c) Drainage winds (d) Trade winds
27. The water content in the atmosphere
 - (a) is not dependent on temperature
 - (b) increases as temperature increase
 - (c) remains unchanged with change in temperature
 - (d) cannot be measured
28. What will happen if the temperature of water is lowered from 8°C to 3°C ?
 - (a) Water will freeze
 - (b) Water will not change in volume
 - (c) The volume will first increase, then decrease
 - (d) The volume will first decrease, then increase
29. What is the importance of ozone in the atmosphere?
 - (a) It provides protection against ultraviolet radiation.
 - (b) It provides condensation nuclei.
 - (c) It creates the greenhouse effect on earth.
 - (d) It helps scatter blue light.
30. Which one is an anticyclone?
 - (a) Low pressure system with clockwise winds in the northern hemisphere.
 - (b) High pressure system with clockwise winds in the northern hemisphere.
 - (c) Low pressure system with clockwise winds in southern hemisphere.
 - (d) High pressure system with clockwise winds in southern hemisphere.
31. The process of change of state of water from solid directly into vapour is called
 - (a) condensation (b) snow fall
 - (c) sublimation (d) precipitation
32. What is the vapour pressure?
 - (a) The presence of vapour in air
 - (b) Barometric pressure
 - (c) Pressure only due to vapour in the air
 - (d) Pressure in the clouds
33. What is stated in terms of grams of water vapour per kilogram of moist air?
 - (a) Specific humidity (b) Relative humidity
 - (c) Vapour pressure (d) Absolute humidity
34. The temperature at which an air parcel will become saturated with the present amount of water vapour is called
 - (a) critical temperature (b) saturation point
 - (c) dew point (d) condensation point
35. The leeward side of a mountain which does not receive rain is known as the
 - (a) dry zone (b) desert area
 - (c) rain-shadow area (d) adiabatically dry area

36. Dew is caused when
 (a) humid air condenses on cool surface
 (b) the air is colder than the earth's surface
 (c) the sky is overcast at night
 (d) the wind is too dry to cause rainfall
37. Which of the pedogenic regimes is associated with hot, dry climates?
 (a) Podzolisation (b) Gleisation
 (c) Laterisation (d) Calcification
38. With what type of climate is the pedogenic regime of podzolisation associated?
 (a) Hot and dry (b) Humid temperate
 (c) Equatorial (d) Cool and dry
39. Which of the following statement is/are correct?
 (a) Halo is produced by diffraction in ice crystals.
 (b) Corona is produced by refraction by water droplets.
 (c) Halo is an indication of approaching bad weather.
 (d) Though corona and halo is produced by different phenomena, the colour sequence produced in them remains the same.

40. Study the following diagrams, A and B.



- (a) A is cyclone in the southern hemisphere
 (b) A is an anti-cyclone in the southern hemisphere
 (c) B is a cyclone in the northern hemisphere
 (d) B is an anti-cyclone in the northern hemisphere
41. Of the main constituents, which of the following have great influence in earth's climatic conditions?
 (a) Oxygen (b) Nitrogen
 (c) Carbon dioxide (d) Water Vapour
42. Winter rains in North and North-West India are generally associated with the phenomenon of :
 (a) retreating monsoon
 (b) temperate cyclones
 (c) local thunderstorms
 (d) shift in jet stream movement
43. Which one of the following characteristics is NOT found in the Mediterranean climate ?
 (a) Temperatures range from 21° C in summer to 10° C or below in the winter
 (b) Off-shore trade winds blow in the summer
 (c) On-shore westerly winds blow in the winter bringing cyclonic rain
 (d) The annual rainfall ranges from 1500 mm to 2000 mm
44. The layers of the atmosphere are given in the figure. The layer marked 'A' refers to :



- (a) Troposphere (b) Mesosphere
 (c) Stratosphere (d) Ionosphere
45. Which of the following latitudinal extents relate to 'horse latitudes'?
 (a) 20° - 30° North and South
 (b) 30° - 40° North and South
 (c) 40° - 50° North and South
 (d) 50° - 60° North and South
46. Which region of the Earth's surface is called doldrums?
 (a) Equatorial low pressure belt
 (b) Sub-tropical high pressure belt
 (c) Between 10° to 23.5° North and South Latitudes
 (d) Sub-polar low pressure belt
47. Which one of the following is a low cloud?
 (a) Cirrocumulus (b) Cirrostratus
 (c) Altopumulus (d) Nimbostratus
48. Injurious ultraviolet radiations are mostly prevented from reaching the land surface as these are absorbed mostly by which one of the following?
 (a) Stratosphere (b) Troposphere
 (c) Ionosphere (d) Mesosphere
49. The earth's reflectivity of solar radiation, termed albedo, is highest in which one of the following?
 (a) Cropland (b) Forest area
 (c) Sand desert (d) Snow area
50. Which one of the following regions on the surface of Earth has Horse Latitudes?
 (a) Equatorial low pressure belt
 (b) Sub-tropical high pressure belt
 (c) Sub-polar low pressure belt
 (d) Polar high pressure belt
51. Which one of the following is produced by rain water action?
 (a) Gorge (b) Cliff
 (c) Gully (d) Dome
52. What is the general direction of cyclones formed in the Bay of Bengal?
 (a) East to West (b) West to East
 (c) West to South (d) North to South
53. Which one among the following statements relating to an anticyclone is correct ?
 (a) Anticyclone is a wind system with a high pressure centre
 (b) In anticyclone the movement of wind is inward
 (c) The contribution of an anticyclone towards determining weather of an area is quite significant
 (d) The movement of wind is clockwise in an anticyclone of southern hemisphere
54. Doldrums is a
 (a) Tropical wind belt
 (b) Tropical wind deflection belt
 (c) Sub-tropical wind belt
 (d) Tropical no-wind belt
55. The most important fishing grounds of the world are found in the regions where
 (a) warm and cold atmospheric currents meet
 (b) rivers drain out large amounts of freshwater into the sea
 (c) warm and cold oceanic currents meet
 (d) continental shelf is undulating

56. What is the term used to denote the temperature at which the water vapour present in the atmosphere is sufficient to saturate?
 (a) Condensation point (b) Dew point
 (c) Sublimation point (d) Saturation point
57. Tropical cyclone of Philippines is termed as
 (a) Typhoon (b) Willy-willy
 (c) Hurricane (d) Baguio
58. Cloudy nights are warmer than clear nights because of
 (a) greenhouse effect
 (b) depletion of ozone layer
 (c) insolation
 (d) terrestrial radiation
59. As we proceed from equator to poles, the daily range of temperature tends to
 (a) decrease (b) increase
 (c) be constant (d) fluctuate
60. Doldrums are characterized by
 (a) uniform low pressure
 (b) uniform high pressure
 (c) high wind velocity
 (d) low humidity
61. Which of the following statements are correct?
 1. In a cyclone, the area of low pressure is at the centre surrounded by the areas of high pressure
 2. In a cyclone, the areas of low pressure surround the area of high pressure
 3. In an anti-cyclone, the area of high pressure is surrounded by the areas of low pressure
 4. In an anti-cyclone, the area of low pressure is surrounded by the areas of high pressure
 Select the correct answer using the code given below:
 Code:
 (a) 1 and 2 (b) 1 and 3
 (c) 1 and 4 (d) 2 and 4
62. Jet streams are usually found in the
 (a) ozonosphere (b) mesosphere
 (c) tropopause (d) ionosphere
63. What is the general direction of cyclones formed in the Bay of Bengal?
 (a) East to West (b) West to East
 (c) West to South (d) North to South
64. Hot deserts like Sahara, Arabia, etc, receive very negligible amount of rainfall. This is because they
 (a) do not receive moisture-bearing wind from the oceans
 (b) are the most rocky and barren areas of the Earth
 (c) are located on the tropical high pressure belt of the atmosphere
 (d) are not on the path of the monsoons
65. Sirocco is a name used to mean
 (a) a local wind (b) a volcano
 (c) an island (d) an ocean current
66. The intensity of insolation depends on
 (a) Altitude (b) Nature of terrain
 (c) Wind (d) Latitude
67. When the winds blow from all sides to the central low in an anticlockwise direction, then this phenomenon is known as
 (a) anti-tropical cyclones of southern hemisphere
 (b) temperate cyclones of northern hemisphere
 (c) tropical cyclones of northern hemisphere
 (d) tropical cyclones of southern hemisphere
68. Chinook is a
 (a) cold wind in Europe
 (b) tropical desert storm in West Asia
 (c) warm wind in North-America
 (d) depression to South Africa
69. Mist is a result of which one of the following
 (a) Condensation (b) Evaporation
 (c) Sublimation (d) Saturation

Exercise -2

Statement Based MCQ

- Consider the following statements regarding the cumulonimbus clouds
 - This cloud is also referred to as thunder clouds.
 - This brings convectional rain.
 - This is frequently seen in tropical afternoons.
 - These are woolly, bumpy clouds arranged in layers appearing like waves in blue sky.

Which of the above statements is/are not correct?

(a) 2 only (b) 3 only
(c) 1 and 4 (d) 4 only
- Westerlies in southern hemisphere are stronger and persistent than in northern hemisphere. Why?
 - Southern hemisphere has less landmass as compared to northern hemisphere.
 - Coriolis force is higher in southern hemisphere as compared to northern hemisphere.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- The jet aircrafts fly very easily and smoothly in the lower stratosphere. What could be the appropriate explanation?
 - There are no clouds or water vapour in the lower stratosphere.
 - There are no vertical winds in the lower stratosphere.

Which of the statements given above is/are correct in this context?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- Normally, the temperature decreases with the increase in height from the Earth's surface, because
 - the atmosphere can be heated upwards only from the Earth's surface.
 - there is more moisture in the upper atmosphere.
 - the air is less dense in the upper atmosphere.

Select the correct answer using the codes given below:

(a) 1 only (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3
- A layer in the Earth's atmosphere called ionosphere facilitates radio communication. Why?
 - The presence of ozone causes the reflection of radio waves to earth.
 - Radio waves have a very long wavelength.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- Which of the following is/are not correct?
 - The unit of measurement of Sun's radiation is Langley.
 - The incoming solar radiation to earth is $2 \text{ cal/cm}^2\text{-min}$.
 - The mean sea level pressure of atmosphere is 1013.2 bar.
 - We are normally not aware of atmosphere pressure because it acts in all directions.

Select the correct answer from the codes given below:

(a) 1, 2, 3 and 4 (b) 2, 3 and 4
(c) 3 only (d) None of these
- Consider the following statements:
 - Clouds generally reflect some of the sunlight in different directions and the rest is absorbed.
 - Nearly 90% of the water vapour lies below 12 km of the atmosphere.
 - Water vapour does not exceed 3 percent to 4 percent of total volume of air.
 - Nitrogen serves mainly as dilutant and dissolver.

Which of the above statements is/are not correct?

(a) 1 and 2 (b) 2 and 3
(c) 3 only (d) 3 and 4
- Which of the following is/are correct?
 - The Northern Pacific is the deepest part of the Ocean.
 - The South-West Pacific is devoid of marginal seas.
 - The South-East Pacific has broad submarine ridges and plateaus.
 - The ocean floor of the Pacific Ocean is fairly uniform with broad rises and depressions.

Select the correct answer from the codes given below:

(a) 1, 2, 3, 4 (b) 1 and 4
(c) 2 and 3 (d) 1, 3 and 4
- The absence of which among the following causes the temperature of the atmosphere to decrease with ascent along with decreasing density of air?
 - Vegetation
 - Water
 - Carbon dioxide
 - Oxygen
 - Nitrogen

(a) 1 and 2 (b) 3 and 4
(c) 2 and 3 (d) 2 and 5
- Land masses are heated and cooled much faster than water bodies. Identify the true statements in this regard.
 - Land gets heated by conduction
 - Water has a higher specific heat
 - Only the surface layers of water are heated
 - Water gets heated by convection

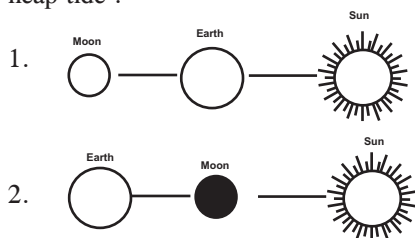
(a) 1, 2 and 3 (b) 2 and 3
(c) 1 and 4 (d) 1, 2 and 4
- Which of the following are true with regard to temperature control of the atmosphere of a given place?
 - The major temperature contrasts on earth's surface are between land masses and oceans.
 - During night, the water surface cools more rapidly than the land surface.
 - Temperature generally decreases with latitude on either side of the equator.
 - Temperature rises to a maximum during summer and falls to a minimum during winter.

(a) 1, 3 and 4 (b) 2, 3 and 4
(c) 1 and 2 (d) 3 and 4

12. Which of the following statements are true with regard to tropical monsoon climate?
1. Temperatures range from 32°C in the hot season to about 15°C in the cool season.
 2. Annual rainfall varies greatly.
 3. The climate is best developed in the southern continents.
 4. It consists of three main seasons, i.e., cool, dry season; hot, dry season; hot, wet season.
- (a) 1 and 4 (b) 2, 3 and 4
(c) 2 and 3 (d) 1, 2 and 4

13. An anti-cyclone
1. is an intense low pressure system.
 2. has winds blowing clockwise in the southern hemisphere.
 3. is accompanied by clear and settled weather conditions.
 4. is more extensive, slower-moving and more persistent than a depression.
- (a) 1, 2 and 4 (b) 2, 3 and 4
(c) 3 and 4 (d) 1 and 2

14. Consider the diagrams below and say which one represents neap tide :



Choose the correct option

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
15. Consider the following climatic characteristics:
1. Temperatures range from 21°C in the summer to 10°C or below in the winter.
 2. Off-shore trade winds blow during the summer.
 3. The summers are hot and dry; and winters are cool and moist.
 4. The annual rainfall ranges from 500 mm to 760 mm.
- Select the correct climatic type which possesses the above characteristics using the codes given below:
- (a) Tropical Monsoon (b) Mediterranean
(c) China type (d) West European type
16. Consider the following statements :
1. Troposphere varies in its thickness between equator and poles.
 2. Stratosphere does not experience Normal Lapse Rate.
 3. Ozone layer occurs in the Mesosphere.
 4. Ionosphere lies between Stratosphere and Mesosphere.
- Which of these statements are correct ?
- (a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1, 2 and 4
17. Consider the following statements :
1. Half the water vapour in the air in atmosphere lies below an altitude of 1000 m.
 2. The amount of precipitable water in the atmosphere increases from the equator to the poles.
- Which of the statements given above is/are correct ?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

18. Consider the following statements:
1. The approach of a cyclone is characterised by a rise in barometric reading.
 2. In the cyclones of the southern hemisphere, the winds circulate in anticlockwise direction.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

19. Consider the following statements:
1. In tropical grassland regions, rainfall mainly occurs in the short summer season with a long dry season.
 2. In Mediterranean region, the winter rainfall is caused by the passage of cyclones in the westerly wind belt which lies over this area.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

20. Consider the following statements:
1. In a cyclone, the direction of wind flow is counter clockwise in the northern hemisphere.
 2. The tropical cyclone fades away when it reaches land because there is no large supply of warm moist air.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

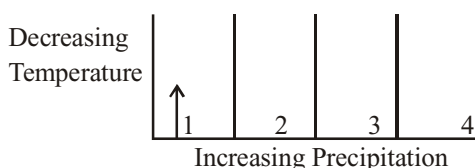
21. The annual range of temperature in the interior of the continents is high as compared to coastal areas. What is/are the reason/reasons?
1. Thermal difference between land and water
 2. Variation in altitude between continents and oceans
 3. Presence of strong winds in the interior
 4. Heavy rains in the interior as compared to coasts
- Select the correct answer using the codes given below.
- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

22. Which among the following statements characterized El Nino?
- I. It occurs at irregular intervals
 - II. It carries warmer water
 - III. It carries less saline water
 - IV. Its atmospheric equivalent is southern oscillation.
- Select the correct, answer using the code given below
- (a) I and II only (b) II and III only
(c) III and IV only (d) I, II, III and IV

23. Which of the following statements is/are correct?
1. Cyclone is a low pressure system.
 2. The wind movement is clockwise in the cyclone of northern hemisphere.
- Select the correct answer using the code given below:
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

24. Which of the following statements regarding hurricanes is/are correct?
1. They develop over the ocean between 8° – 15° N.
 2. They are almost absent in the South Atlantic Ocean.
 3. They do not develop close to the equator.
- Select the correct answer using the code given below.
- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

25. The diagram given below shows the schematic relations of temperature and precipitation of tropical climatic type over land:



Arrange the climatic types in the correct sequence from left to right:

1. Monsoon tropics 2. Wet and dry tropics
3. Arid and semiarid 4. Rainy tropics

Select the correct answer using the code given below.

- (a) 1-4-3-3 (b) 4-3-2-1
(c) 2-1-3-4 (d) 3-2-1-4

Matching Based MCQ

DIRECTIONS (Qs. 26 to 27) : Match List-I with List-II and select the correct answer using the codes given below the lists.

26. **List-I**

- (A) Fog

List-II

- (1) A mass of minute droplets of water or tiny crystals of ice formed by the condensation of water vapour in free air at considerable elevations.

- (B) Cloud (2) Condensation at a dew point which is at or below freezing point.
(C) White Frost (3) Moisture deposited in the form of water droplets on cooler surface of solid objects such as stones, grass blades etc.
(D) Dew (4) A mass of minute droplets of water formed by the condensation of water vapour in free air with its base at or near ground.

- (a) A - 4 ; B - 2 ; C - 1 ; D - 3

- (b) A - 3 ; B - 2 ; C - 1 ; D - 4

- (c) A - 3 ; B - 1 ; C - 2 ; D - 4

- (d) A - 4 ; B - 1 ; C - 2 ; D - 3

27.

List-I

List-II

- (A) Cloudiness and heavy precipitation unfavourable (1) Equatorial region
(B) Westerlies throughout the year and mild winters (2) Mediterranean region
(C) Famous for game park land type of vegetation (3) Savanna region
(D) Dry summers, wet winters and orchard farming (4) West European region

- (a) A - 1 ; B - 4 ; C - 3 ; D - 2

- (b) A - 4 ; B - 1 ; C - 3 ; D - 2

- (c) A - 1 ; B - 4 ; C - 2 ; D - 3

- (d) A - 4 ; B - 1 ; C - 2 ; D - 3

Hints and Explanations

EXERCISE-1

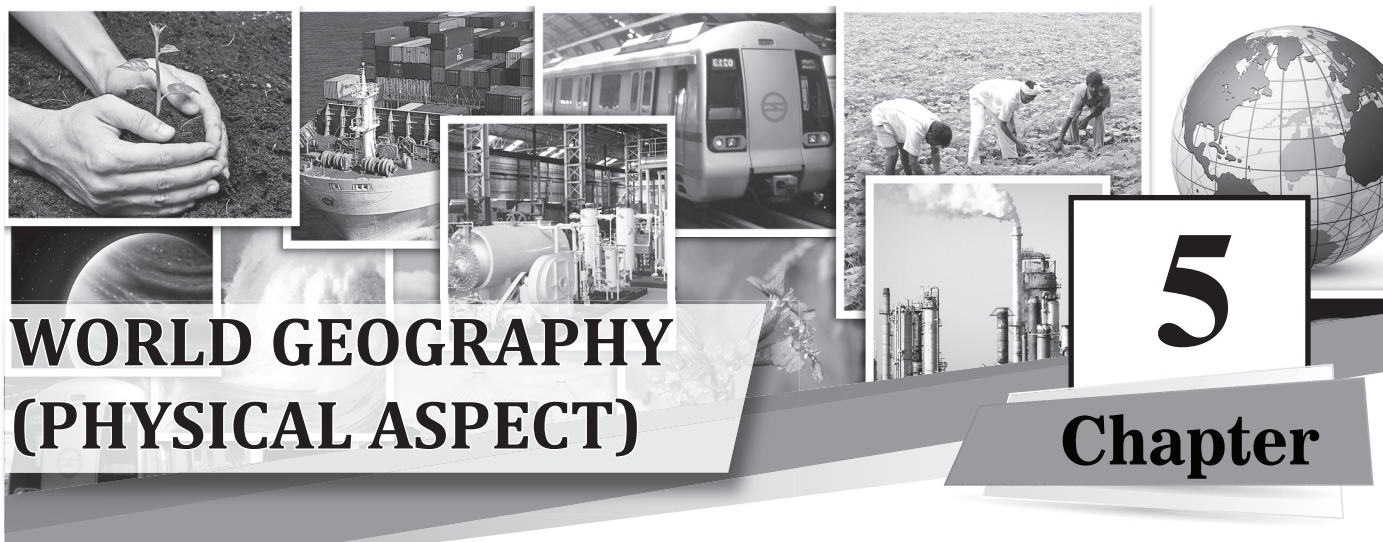
1. (b) In the given figure 'B' lies between two cyclone zone 'A' and 'D'. At the 'B' marked region rainfall would be accompanied by thunder storms.
2. (c) 3. (a) 4. (c)
5. (d) 6. (a) 7. (b)
8. (b) They deflect to the left in the southern hemisphere.
9. (a) 10. (b) 11. (c) 12. (a) 13. (b)
14. (a) 15. (b) 16. (b) 17. (b) 18. (a)
19. (c) 20. (b) 21. (b) 22. (c) 23. (d)
24. (a) 25. (b) 26. (c) 27. (b) 28. (d)
29. (a) 30. (b) 31. (c) 32. (c) 33. (a)
34. (b) 35. (c) 36. (a) 37. (d) 38. (b)
39. (c) 40. (c) 41. (b)
42. (b) In the winter season temperature cyclon occurred in Arabeen sea. This phenomenon led a rain in the North-West India.
43. (d) The option (d) is incorrect. The annual rainfall ranges from 400-600 mm. not 1500-2000 mm.
44. (c) Stratosphere is the second layer of the atmosphere, extends to a height of 50 km above the earth.
45. (b) Two regions, found at latitude 30°-35° north and 35°-30° south of the equator, where there is steady movement of air is known as horse latitudes.
46. (a) The air at the equator is generally rising. For this reason, there is little wind in the region. This region of light shifting winds near the equator is called the doldrums.
47. (d) Low cloud group includes clouds occupying height of up to 2100 metres. This group consists the stratus, the nimbostratus and the stratocumulus types.
48. (a) Injurious ultraviolet radiations are mostly prevented from reaching the land surface as these are mostly absorbed by ozone which are formed in stratosphere.
49. (d) The percentage of radiant energy reflected back by a surface is called the albedo. The total amount of energy lost by scattering and reflection of various kinds and returned to space is called earth's albedo.
50. (b) Horse Latitudes are the subtropical belts of variable winds and columns that is between the latitudes 25° and 35° south and north. They coincide with sub-tropical high-pressure belts.
51. (c)
52. (a) Bay of Bengal is the largest bay in the world. It finds a triangle in a shape and its bordered mostly by the eastern coast of India, Southern west of Bangladesh and Sri Lanka to the west and Burma and the Andaman and Nicobar Islands to the east.
53. (d) An anticyclone is a large scale circulation of wind around a central region of high atmospheric pressure. It is clockwise in northern hemisphere but anticlockwise in the Southern Hemisphere.
54. (d) Doldrums are noted for their calm periods when winds disappear totally. They are a low pressure area around the equator.
55. (c) The temperature is just right for them to survive.
56. (b) Dew point is the temperature at which the air is fully saturated and below which condensation normally occurs, water vapour starts to condense to form water droplets.
57. (a) The tropical cyclone of Philippines is known as Typhoon.
58. (c) Cloudy nights are warmer than clear nights because of insolation.
59. (a) The daily range of temperature decreases on proceeding equator to poles.
60. (a) Doldrums is the name for the equatorial belt of low pressure lying between 5° south and 5° North latitude.
61. (b) In a cyclone, the area of low pressure is at the centre surrounded by the areas of high pressure. In an anti-cyclone, the area of high pressure is surrounded by the areas of high pressure.
62. (c) Jet streams are fast flowing, narrow air current found in the atmospheres of some planets including Earth. The main jet streams are located near the tropopause, the transition between the tropopause, the transition between the troposphere and the stratosphere.
63. (a) Bay of Bengal is the largest bay in the world. It finds a triangle in a shape and its bordered mostly by the eastern coast of India, Southern west of Bangladesh and Sri Lanka to the west and Burma and the Andaman and Nicobar Islands to the east.
64. (a)
65. (a) Sirocco is a local Mediterranean wind that comes from Sahara and can achieve Hurricane speed in North Africa and Europe. Its Arabic name is Quibli.
66. (d) Insolation is the measure of solar radiation energy recieved on a given surface area in a given time. Some of the solar radiation is absorbed and the rest is reflected. Insolation is largest when the surface faces the sun.
67. (d)
68. (c) Chinook is a warm, dry, gusty, westerly wind that blows down the Rocky Mountains in North America.
69. (a) Mist is a thin fog resulting from condensation in the air near the earth's surface.

EXERCISE-2

1. (d)
2. (a) The higher speed and greater persistence of the westerlies in the Southern Hemisphere are caused by the difference in the atmospheric pressure patterns as well as its variation from that of the Northern hemisphere. The landmass in the southern hemisphere is comparatively less and average annual pressure decreases much more rapidly on the pole ward side of the high pressure belt. Statement 1 is correct. As far as coriolis force is concerned,

it is strongest in the polar regions and zero at the equator. At the intermediate levels it varies directly as the sine of the latitude.

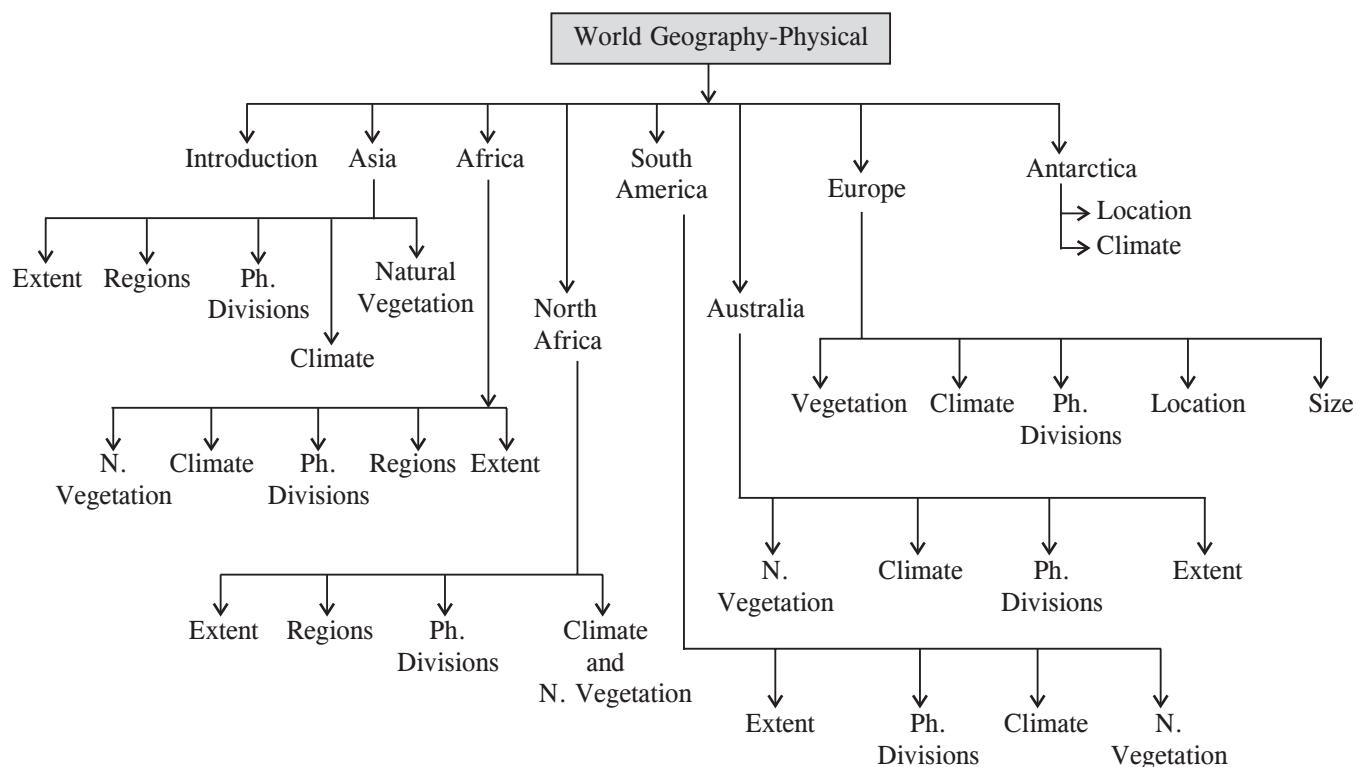
3. (b) The idea is to optimize fuel burn reason is the low air density that reduces parasitic drag on the airframe. It also allows them to stay above any hard weather (extreme turbulence). Air flow in the stratosphere is much lesser turbulent than in the troposphere and this is the reason that jet aircraft like to cruise at stratospheric altitudes for, the flight is less “bumpy.” So statement 2 is correct. Majority of clouds form in the Earth’s troposphere, but there are occasions where clouds in the stratosphere and mesosphere are formed and have been observed. The statement 1 which says that There are no clouds or water vapour in the lower stratosphere, is not a correct statement, to the extent that it does not explain the flight of jets in lower stratosphere.
4. (c) 1. The atmosphere can be heated upwards only from the Earth’s surface.
2. There is more moisture in the upper atmosphere.
3. The air is less dense in the upper atmosphere.
5. (d) 6. (c) 7. (a) 8. (d) 9. (c)
10. (d) 11. (a) 12. (d) 13. (c) 14. (d)
15. (b) 16. (a)
17. (d) Half of the water vapour in the air in atmosphere lies below an altitude of 2 kilometer and 90 per cent of water vapour lies below an altitude of 5 kilometer. Amount of precipitable water in atmosphere increases from the poles to the equator.
18. (c) Cyclone is a mass of air whose isobars form an oval or circular shape, with low pressure at the centre. The air converge at the centre and rises to disposed off. In a dipression, the winds rotate anticlockwise in northern hemisphere. While in the southern hemisphere, the circular movement of winds is in clockwise direction.
19. (c) Tropical grassland located mainly in the continental areas of tropical latitudes where rains fall during the summer season which lasts for about five months. In Mediterranean region, the winter rainfall due to middle latitude fronts and cyclones.
20. (c) Cyclone is the synonym of Hurricane. The air moves in the cyclone inwards in an anti-clock wise direction in the northern hemisphere and clock-wise in the southern hemisphere. Cyclones are divided as -
1. Extra tropical cyclones or temperate cyclones
2. Tropical cyclones
21. (a) The first statement is correct as we all know. One major factor affecting the distribution of the temperature of Earth is distribution of Land and Oceans. Since there is more land in Northern Hemisphere and more waters in Southern hemisphere and there is a big difference between the specific heat of land and water; the loss of heat from the continents is bigger than the oceans. The continents get heated faster and get cooled faster in comparison to the Oceans. This is the reason that the temperatures of the Oceans are moderate while that of continents is extreme. The moderating effect on temperature of the land due to proximity of the seas is called Maritime influence. The increasing effect on temperature of the land at interior of the continents is called Continental Influence.
22. (d) EL Nino, an unusual warming of surface ocean waters in the eastern tropical Pacific is the part of southern oscillation. Scientists do not know exactly how El Nino forms. It is said that El Nino may have contributed to the 1993 Mississippi and 1995 California floods. The average period length of EL Nino is 5 years.
23. (a) Cyclones are moving bodies of air in an enclosed area rotating in the same direction as earth. The air inside the cyclone moves anticlockwise in the Northern Hemisphere.
24. (a) Hurricanes develop over the oceans between 8° - 15° N. The term “tropical” refers to the geographical origin of these systems, which usually form over the tropical oceans. The term “cyclone” refers to their cyclonic nature, with wind blowing counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere. The opposite direction of circulation is due to the Coriolis force. Depending on its location and strength, a tropical cyclone is referred to by names such as hurricane, typhoon, tropical storm, cyclonic storm, tropical depression and simply cyclone.
25. (d) 26. (d) 27. (c)



Introduction

In the previous chapters we have discussed the basic elements of Geography which in general deals with nature and scope of Geography, the Universe, the solar system, and its various branches of Geography. In this special branch of Geography we have also discussed development of earth, its lithosphere, hydrosphere and atmosphere in details.

Moving on to next important branch of Geography, we are now going to discuss Regional Geography which deals with the studies of the world's regions. In general it is called as World Geography which can be further categorised into three important segments such as **physical segment** dealing with unique characteristics of earth's relief, their spatial extent, drainage, climate, natural vegetation; **social segment** that deals with human development, their race and ethnicity and the relationship with their own physical environment; and finally the **economic segment** which deals with the natural resources, livelihood, transport, communications, etc. Let's start with physical segment.



World Geography Physical

On the basis of major physical characteristics the earth can be divided into large continuous land masses known as Continents and the surrounding water bodies known as Ocean. Major continents of the world are **Asia, Africa, Europe, North America, South America, Antarctica, and Australia.**

ASIA

Extent

Asia is the world largest continent, having an area of 44,444,100 sq km. It covers 8.8% of the Earth's total surface area with the population of 4.4 billion which is 60 % of the world's total population. It is a continent of contrast in relief, temperature, vegetation and people also. Asia is to the east of the Suez Canal, the Ural River, and the Ural Mountains, and south of the Caucasus Mountains and the Caspian and Black Seas. It is bounded on the east by the Pacific Ocean, on the south by the Indian Ocean and on the north by the Arctic Ocean.

Regional Divisions

Asia can be divided into six physiographic divisions:

1. **Central Asia:** Kazakhstan, Kyrgyzstan, Tajikistan, Turkmenistan, Uzbekistan
2. **Eastern Asia:** China, Hong Kong, Japan, North Korea, South Korea, Macau, Mongolia, Taiwan
Northern Asia: Russia
3. **South-eastern Asia:** Brunei, Myanmar, Cambodia, Indonesia, Laos, Malaysia, Philippines, Singapore, Thailand, Timor-Leste, Vietnam.
4. **Southern Asia:** Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, Sri Lanka.
5. **Western Asia:** Armenia, Azerbaijan, Bahrain, Cyprus, Georgia, Iran, Iraq, Israel, Jordan, Kuwait, Lebanon, Oman, State of Palestine, Qatar, Saudi Arabia, Syria, Turkey, United Arab Emirates, Yemen.

Major Physical Divisions

The major physical divisions of Asian continent are:

1. The Northern Lowlands
2. The Central Mountains
3. The Central and Southern Plateaus
4. The Peninsulas, the Deserts
5. The Great River Plains, the Island Groups

1. The Northern Lowlands

The Northern Lowlands are the extensive plain areas which comprise of several patch of lowlands of this large continent. The major lowlands are:

Great Siberian plain

It extends between the Ural Mountains in the west and river Lena in the east. It is the largest lowland in the world covering an area of 1,200,000 square miles approx.

Manchurian Plain

It is the area adjoining Amur river and its tributaries of northern part of China with an area of 135,000 square miles approx.

Great Plains of China

It is contributed by two major rivers of China, Hwang Ho and Yangtze river which covers an area of 158,000 square miles approx.

Tigris-Euphrates plains

These are formed by river Ganga 984,000 sq. miles approx.

2. The Central Mountains

These are the prominent and extensive mountain ranges which covers the parts of Central Asia. They consist of Pamir and Tian Shan ranges, and extending across portions of Afghanistan, China, Kazakhstan, Kyrgyzstan, Tajikistan, and Uzbekistan. These mountain ranges are designated as biodiversity hot spot by Conservation International which covers several montane and alpine ecoregions of Central Asia. It encompasses several habitat types, including montane grasslands and shrublands, temperate coniferous forests, and alpine tundra.

A *mountain knot* is a junction of two or more mountain ranges.

The two main mountain knots in Asia are:

- The **Pamir Knot** is the junction of five mountain ranges they are the Sulaiman, the Hindu Kush, the Kunlun, the Karakoram and the Himalayan ranges. Mount Everest, the highest peak in the world in the Himalayan range.
- The **Armenian Knot** is connected to the Pamir Knot by the Elburz and the Zagros Ranges that originate in the Armenian Knot. The Tien Shan and the Altai are other mountain ranges in Asia.

3. The Central and Southern Plateaus

Plateaus are the land areas having a relatively that surface considerably raised above adjoining land on at least one side, and often cut by deep canyon.

Major Plateaus of Asian Continent

Plateau	Location	Types
Ladakh	Between Karakoram and Himalaya mountain ranges	Intermontane
Tibet	Between Kulun and Himalayan Mountain range	Intermontane
Yunan	Situated on the south east of the Tibet Plateau and separated from Szechuan Basin extensive fertile land by the range of Mountains	Piedmont
Pamir	Well connected to the range of mountains such as Himalayas with the Tian Shan, Karakoram, Kunlun, and Hindu Kush ranges on all sides	Intermontane
Armenian	Present in between Caspian and Black Sea	Piedmont
Iranian	Present in between Zagros Mountains, Caspian Sea, Turkmen-Khorasan Mountain Range	Piedmont

Mongolian	Surrounded by the Greater Hinggan Mountains in the east, the Yin Mountains to the south, the Altai Mountains to the west, and the Sayan and Khentii mountains to the north	Intermontane
Shan	Stretched in the Pegu Yoma and Arkan Yoma in the eastern part of Myanmar,	Intermontane
Deccan	Extended in between the Western Ghats in west and the Eastern Ghats in the east of Indian Subcontinent, it almost touches the southern tip of India and in north covered by the Satpura and Vindhya Ranges	Intermontane
Anatolian	Enclosed between Pontic mountain ranges in the South and Taurus in the south west	Volcanic

4. Peninsulas Deserts

A peninsula is a mass of land surrounded by water, but attached to the mainland. The Deccan plateau region is also a peninsula. The major peninsulas of Arabia, India and Malay are in southern Asia. The Kamchatka peninsula lies in north-eastern Asia.

Asia has some big deserts such as the Gobi, the Takla Makan, the Thar, the Kara-kum and the Rub-al-Khali Deserts.

5. (a) Islands of Asia

Asia also has a cluster of islands, also called an archipelago. An archipelago sometimes called an island group or island chain, which are formed close to each other in large clusters. Indonesia, Philippines, Japan, Andaman and Nicobar are some examples of archipelagos.

5. (b) Drainage of Asia

The drainage of Asia consists of mighty oceans, extensive seas, lengthy rivers and their tributaries and distributaries, major lakes, etc.

Oceans: Asian continent is surrounded by three major ocean from three sides such as

- **The Pacific Ocean**
It covers the eastern part of Asia where major rivers of eastern Asia drain, such as Menam Mekong, Xi Jiang, Chang Xiang, Huang Ho and Amur.
- **The Indian Ocean**
It covers the southern part of Asia and the major rivers flow into Indian Ocean are Tigris, Euprates, the Indus, the Ganga, Brahmaputra, Irrawaddy, Salween.
- **The Arctic Ocean**
It covers the North east part of Asia and consists of three major rivers such as Ob, Yenisey and Lena.

Seas

As the continent is covered by sea from its three sides, It has also characterised by long stretch of bay and gulf. Major seas contributing Asian Drainage are Sea of Galilee, Andaman Sea,

Arabian Sea, Banda Sea, Barents Sea, Bering Sea, Black Sea, Caspian Sea, East Siberian Sea, Java Sea, Kara Sea, Laccadive Sea, Sea of Japan, Sea of Okhotsk. South China Sea and Yellow Sea.

Lakes

Major lakes of Asia are Lake Baikal, Onega, Ladoga, and Peipus in Russia; Lake Akan, Mashu, Biwa, Shikotsu in Japan; Qinghai Lake, Lake Khanka in China; Dal Lake, Chilka, Vembanada, Pulicat and Sukhna in India; Lake Matano and Toba in Indonesia, etc.

Climate

As a land of Contrast Asia is characterized by varied climatic type on the basis of the temperature and rainfall condition. The rainfall across the continent is highly influenced by Monsoon winds the Asia can be divided into three major climatic zones such as:

Monsoon Climate

The Monsoon Asia is the zone including south and south east Asia and east Asia where the effect of monsoon is prominent. Hence the climatic condition varies according to monsoonal wind flow. After the onset the wind starts moving in the north west direction, hence causing rain over the eastern coast of Indian subcontinent, and parts of south east and east Asia. Moreover the summer spell in India is very hot and dry which trigger the occurrence of additional heavy precipitation owing to tropical cyclones.

In winters the central land mass of Asia gets cool more rapidly than the surrounding ocean. This climatic phenomenon starts the flow of cold descending air current in the central Asia which results into generation of high pressure in the heart of Asia. The high pressure starts chasing the low pressure zone present over Indian and Pacific Ocean due to comparatively high temperature. This is called as *retreating monsoon or season of winter monsoon*. As a result of these phenomena of both onset and retreat of monsoon there is marked difference in the climate of Northern and Southern part of Asia.

Dry Climate

The Dry Asia consists of South West Asia, Central Asia and Mongolia. Latitudinally it varies from tropical desert of Arabian Peninsular to subtropical steppe in Afghanistan and further to mid latitude steppe and desert of Mongolia and Northern China. As compared to other parts of the continent the rainfall is also very less, i.e. 2.5 cms to 20 cms and it is very unpredictable throughout the region. Moreover a Mediterranean climate is experienced over the coastal region which receives winter rainfall.

Cold Climate

The Cold Asia is experienced in maximum part of Russia as an influence of sub-arctic climate. The summer is comparatively mild and lasts for only for four month. The rainfall is also less as compared to other parts of the continent. The annual rainfall accounts for only 50 cms in the coastal areas whereas towards interior, it decreases up to 25 cms.

Natural Vegetation

There are various types of vegetation found in Asia. For examples:

Tundra

The Tundra extends to 70°N and with further south extensions on high altitudes (Chersk, Verkhoyansk and Kamachotka mountains). The region is covered by cold, treeless plains with permanently frozen subsoil.

The Taiga

The Taiga found in south of tundra is a belt of coniferous forests running across whole of Siberia from west to east reaching Pacific and northern part of Japan. The trees have small leaves, deep roots and thick bark. The species (pine, spruce and fir, etc.) are growing successfully in cold and dry environment.

Temperate Grasslands

Temperate Grasslands

These are elongated, unbroken stretch of the Steppes from Ukraine to Manchuria, which further stretches to several thousand miles in southern Siberia. Region gets low precipitation although cold winters with warm summer. High elevated mountains here are covered with forests.

Mediterranean Scrubland is an area of dry land with small bushes and trees

In this region summers are hot and dry; the winters mild and moist. Thus vegetations grown here are of small size, short leaves, deep roots, and thick barks to retain moisture. It includes countries of Israel, Lebanon, Syria, Iraq, and the plateaus of Turkey and Iran.

Desert Vegetation

Desert Vegetation types are found in the Arabian Peninsula, the deserts of Tibet, Mongolia, and the desert-like steppe-lands bordering the Caspian Sea. The region is sparsely populated by vegetation. Moisture-combating plants, waxy, deep-rooted or thorny shrubs and sporadic stunted trees grow here.

Monsoon Vegetation

Monsoon Region vegetation varies with the amount of annual rainfall each year in this region. The average range of rainfall varies between 40 inches and 80 inches annually. Mostly, tropical deciduous (shedding leaves seasonally) forests, and those which receive less than 40 inches have savanna and steppe-like vegetation are seen. The monsoon lands have been extensively modified by human settlement and put to cultivation, and little trace of the original vegetation survives.

Tropical Rainforest

Tropical Rainforest is the region where evergreen, broad-leaved tall and high-crowned trees are found in this region. Several species having a dense canopy above the floor due to the heavy rainfall received all round the year. The savannas and deciduous trees cover the ground, the subequatorial and the areas that lie in the rain shadow on the leeward slopes.

Malaysia and Indonesia, southern Sri Lanka and Java have such vegetation. Plantation tea, rubber, coffee, cocoa, etc. are found here.

Mountainous Vegetation

Vegetation in the Mountain area is found on southern and eastern Asia. The higher elevated part is snow covered by meadows. Lower parts are covered by broad-leaved deciduous forests, and on higher ground the coniferous trees occur.

Peaks of Asia

- Mount Everest (8848 m), Nepal-Tibet, China border
- K2 (8,611 m), Pakistan-China
- Kangchenjunga (8,586 m), Nepal-Sikkim (India).
- Lhotse (8,516 m), Nepal-Tibet, China
- Makalu (8,462 m), Nepal-Tibet, China
- Cho Oyu (8,201 m), Nepal-Tibet, China
- Dhaulagiri (8,167 m), Nepal

AFRICA

Extent

Africa is the second largest continent in area (30,330,000 sq Km) which covers 6% of Earth's total surface area and 20.4 % of its total land area. *Algeria* is Africa's largest country by area, and *Nigeria* by population. Separated from Europe by the Mediterranean Sea, it is joined with Asia at its northeast extreme end by the *Isthmus of Suez* 163 Km wide. It is bounded by Red Sea along the Sinai Peninsula to the northeast, the Indian Ocean to the southeast, and the Atlantic Ocean to the west. With fully recognized 54 sovereign states, nine territories and two de facto independent states.

Regional Divisions

The physiographic divisions of Africa are into the following six regions:

Northern Africa

It extends from Algeria in the north, through, Canary Islands, Santa Cruz de Tenerife, Ceuta, Egypt, Libya, Madeira, Melilla, Morocco, Sudan and Tunisia, It reaches upto Western Sahara.

Northeast Africa

It is also called the horn of Africa which extends several hundred kilometers into the Arabian Sea and lies along the southern side of the Gulf of Aden. It contains countries such as Djibouti, Eritrea, Ethiopia, and Somalia.

Eastern Africa

The extensive area stretches from Red Sea and horn of Africa to Mozambique including Burundi, Comoros, Kenya, Madagascar, Malawi, Mauritius, Mayotte, Mozambique, Réunion, Rwanda, Seychelles, South Sudan, Tanzania, Uganda, Zambia, Zimbabwe.

Central Africa

It is the large land mass situated exactly in the middle of the continent covering Angola, Cameroon, Central African Republic, Chad, Republic of the Congo, Democratic Republic of the Congo, Equatorial Guinea, Gabon, São Tomé and Príncipe.

Southern Africa

It is the southern most part of the continent and covers the countries such as Botswana, Lesotho, Namibia, South Africa, and Swaziland.

Western Africa

It is situated roughly at 100° E longitude covering countries like Benin, Burkina Faso, Cape Verde, Gambia, Guinea, Guinea-Bissau, Ivory Coast, Liberia, Mali, Mauritania, Niger, Nigeria, Saint Helena, Senegal, Sierra Leone and Togo.

Major Physical Divisions

The major physical divisions of African continent are:

The Plateau

The Fold Mountains

Deserts

Islands

Rivers

The Plateaus

The vast African continents are famous for its saucer shaped and steep edged plateaus which are facing towards the coast and extending from Guinea coast to Somali Land and north Sahara to the Cape Province. These are divided into three groups:

South African plateau

- The South African plateau as far as about 12°S, bounded east, west and south by bands of high ground which fall steeply to the coasts. The South African plateau is connected towards the north-east with the East African plateau.

East African plateau

- The East African plateau, with probably a slightly greater average elevation, and marked by some distinct features. It is formed by a widening out of the eastern axis of high ground, which becomes subdivided into a number of zones running north and south and consisting in turn of ranges, tablelands and depressions.

Ethiopian Highlands

- The third division of the higher region of Africa is formed by the Ethiopian Highlands, a rugged mass of mountains forming the largest continuous area of its altitude in the whole continent.

The Fold Mountains

Africa is famous for its newly formed folded mountains. Prominent mountain ranges with some of the very high raised mountain peaks are the specialty of African Continents. Some of the well known mountain ranges are:

Atlas Mountains

- It is situated on the north western part of the continent stretching over an area of 2400 km towards the south west direction across Morocco, Algeria and Tunisia.
- The range is again sub divided into high, medium and Anti – Atlas group. The *Jebel Toubkal* is among such highlands or mountain whose height is 4165 m from sea level.
- It is a physical separator between the extensive coasts of Mediterranean and Atlantic Sea and the Sahara Desert.

Ruwenzori Mountains

- Stretching over an area of 240 sq. miles the range borders Uganda and Congo (Kinshasa) and thought to be the “Mountains of the Moon.”
- Mount Stanley at Margherita Peak (5,119 m) is the highest pick of this mountain system.
- It is a gigantic horst of six separate glaciated masses which falls steeply westward to the Western Rift Valley.

Mount Elgon:

- It is an extinct volcanic mountain situated at the north east part of lake Victoria on the Uganda – Kenya border.
- The height is about 4,321 m from the mean sea level.
- As a volcanic mountain it has a crater which is 610 m deep and 8 km across.

The Tibesti Mountains

- These are mostly situated in the northern part of **Chad** and spread west into northern Niger and the Southern border area of Libya.
- They have a volcanic origin.
- The highest peak is 3,415 m. from mean sea level.

Ahaggar Mountains

- The Ahaggar Mountains, also known as the Hoggar, are a highland region in central Sahara, or southern Algeria near the Tropic of Cancer. They are located about 1,500 km south of the capital, Algiers. Mount Tahat is the highest peak (2,918 m).
- It has a volcanic formation.

The Drakensberg

These mountains are the highest in Southern Africa rising up at Thabana Ntlenyana at 3,482 m (11,422 ft) in height. They are located in the eastern part of South Africa, running from some 1,000 km. The highest peak is Thabana Ntlenyana at 3,482 m (11,422 ft). It is also the highest peak of Lesotho.

Mount Kenya

Mount Kenya is the highest mountain in Kenya, and the second highest in Africa (after Mount Kilimanjaro). The highest peaks of the mountain are Batian (5,199m - 17,058 ft), Nelion (5,188m - 17,022 ft) and Lenana (4,958 - 16,355 ft). Mount Kenya is located in central Kenya, just south of the equator, around 150 km (95 miles) north-northeast of Nairobi.

Kilimanjaro

Kilimanjaro with its three volcanic cones, Kibo, Mawensi and Shira, is an inactive stratovolcano in north-eastern Tanzania. Kilimanjaro is the tallest free-standing mountain rise in the world rising 4,600 m (15,100 ft) from its base, and includes the highest peak in Africa at 5,895 meters (19,340 ft).

The Deserts

- The Sahara, the largest hot desert in the world, stretches across the entire width of North Africa. It covers an area of approx. 3,320,000 sq. miles. The major countries contributing their lead to Sahara deserts are Libya, Algeria, Egypt, Tunisia, Chad, Morocco, Eritrea, Niger, Mauritania, Mali and Sudan.

- The Kalahari Desert lies in the south and the Namib Desert is along the south-west shore of Africa. It covers an area of 3,50,000 sq. miles and eneroachig parts of Botswana, Namibia, Zambia, Angola and Zimbabwe.
- The Nubian Desert is the eastern region of the Sahara desert, between the Nile and the Red Sea. There is virtually no rainfall here, and there are no oases. It is in Egypt. It covers an area of 1,54,000 sq. miles approx.

Karoo : The **Karoo** is a semi-desert region of South Africa. It has two main sub-regions - The *Great Karoo* in the north and the *Little Karoo* in the south. The Great Karoo has an area of more than 400,000 sq. Km. Currently sheep farming is the economic backbone of the Karoo, with other forms of agriculture established in areas where irrigation is possible. Lately game farms and tourism have also started to make an economic impact. The Little Karoo is the smaller (and more southerly) of the two Karoo sub-regions. Locally, it is usually called the Klein Karoo Geographically it is a fertile valley.

The Islands

- There are very few islands near Africa.
- Madagascar (Malagasay) in the Indian Ocean is the largest island in Africa.
- To the north-west, in the Atlantic Ocean are the Canary Islands.
- West of Africa in the South Atlantic Ocean is the island of Saint Helena where Napoleon died in exile.
- Zanzibar belongs to Tanzania and is closer to the Indian Ocean.

The Rivers of Africa

The most important ones are the Nile, the Congo, the Niger and the Zambezi.

River Nile

- This is the **longest river** in the world.
- It starts from many streams in the equatorial rainforest of the Lake Victoria and Ruwenzori Mountain (the mountains of the moon) region.
- From Lake Albert, it flows as the White Nile.
- At Khartoum, it is joined by the Blue Nile which starts from Lake Tana on the Ethiopian Highlands.
- The Nile flows from 3,000 kilometres through the dry Sahara Desert of Egypt and enters the Mediterranean Sea.
- Egypt is called **the gift of the Nile** because without the river it would have been a desert.

River Congo or Zaire

- The Congo is the **second longest river** of Africa.
- It starts from the south-west of Lake Tanganyika and flows into the Atlantic Ocean.
- The Zaire basin is one of the wettest regions of the Earth and is covered with dense impenetrable jungle.
- The river and its network of tributaries are not navigable because of rapids and waterfalls caused by the descent from the plateau to the coast.

River Niger

- This river is the chief river of West Africa.

- It rises from the Fouta Djllon Mountain quite close to the sea but flows north and then turns south again to form a wide arc.
- Finally it joins the Gulf of Guinea on the Atlantic Ocean.

River Zambezi

- River Zambezi flows into the Indian Ocean.
- The famous Victoria Falls at the head of a long gorge is on this river.

River Limpopo

River Limpopo also flows into the Indian Oceans, which crosses the Tropic of Capricorn twice.

River Orange

This flows from the Drakensberg Mountains into the Atlantic Ocean.

The Lakes of Africa

- **Lake Victoria** is the second largest fresh water lake in the world. It is the largest lake of Africa. It is situated on the block mountain between the two branches of the Great Rift Valley. The Equator passes through it. It is the source of white Nile.
- The lakes of the rift valley. There is a string of lakes in the rift valley. **Lake Tanganyika** and **Lake Nyasa** (Malawi) are the larger ones.
- **Lake Tana** is on the Ethiopian Plateau. It is the source of Blue Nile.
- **Lake Chad** at the southern edge in the Sahara Desert is in a region of inland drainage. Streams start from the surrounding hills and flow into this lake instead of the sea. River Charl is the largest river of this area.
- **Lake Nasser** is on river Nile. It is a man made lake located between Egypt and Sudan
- **Lake Kariba** is situated on the southern most part of Africa an Zambezi river. It is one of the biggest look man-made like and it is the largest producer of hydro electricity in Africa.
- **Lake Assal** is situated in Djibouti and the lowest point in Africa

Famous Lakes of Africa

Lake	Elevation	Lake Type
Victoria	11.33 m	Tropical Lake
Trnganyika	773 m	Rift Valley Lake
Malawi	—	African Rift Valley
Turkana	360 m	Alkaline Lake
Lake Albert	615 m	Western Rift
Lake Edward	912 m	Drain North into Lake Albert
Lake Kivu	1460 m	Ruzizi River

Important Dams and Waterfalls

Dam/Falls	River	Country
Aswan Dam	Nile	Egypt
Kariba Dam	Zambezi	Zimbabwe and Zambia
Kainji Dam	Niger	Nigeria
Cobora Bassa Dam	Zambezi	Mozambique
Victoria Fall	Zambezi	Zambia- Zimbabwe
Boyoma Fall	Zaire	Democratic Republic of Congo

Climate

It is quite obvious that as a large continent Africa experiences a highly variable climate. Keeping in view the prevailing weather conditions such as temperature, rainfall, humidity etc the climatic zones of Africa are:

Tropical Monsoon Climate

- It is associated with high mean annual rain fall and temperature. average annual temperature is 27.05°C and the Annual average rainfall vary from 50-150 cm.
- Mostly the central of Africa and eastern cost of Madagascar experience this type of Climate such as Lagos, Kinshasa, Abidjan, Dar es Salaam and Ibadan.
- The hot Summer is generally due to the nearness of equator as the zone is present 10° to 20° on either side of the equator and it brings good amount of rainfall to this region.

Humid and Sub humid tropical climate

- This type of climate is characterised by hot, usually humid summers and mild to cool winters. Although the temperature is relatively high but the rain fall is evenly distributed.
- The average annual temperature varies between -30°C 180°C in the coolest month whereas it is above 220°C in the warmest month.
- The climate occupies comparatively a large part of African continent like Kwa Zulu-Natal and the Eastern Cape provinces of South Africa, north eastern Zimbabwe, Niassa, Manica and Tete provinces of Mozambique, southern Congo provinces, Malawi, and Zambia, Ethiopian Highlands.

Mediterranean Hot Summer:

- This type of climate is associated with high temperature and they generally get precipitation during autumn, winter and spring.
- The temperature reaches to 48°C in summer and the rainfall is almost nil during this period of 4-6 months.
- The winter is also mild as the rainfall is scanty.
- This type of climate is more common around the Mediterranean Sea, southwestern Australia, southwestern South Africa, sections of Central Asia, the Rogue River Valley region of southwestern Oregon, and in the interior of northern California west of the Sierra Nevada.

Hot Desert Climate:

- It is typically a very hot and dry zone where the temperature ranges between 40°C to 45°C throughout the year.
- They are more common in the deserts of North Africa such as the wide Sahara Desert, the Libyan Desert or the Nubian Desert; deserts of the Horn of Africa such as the Danakil Desert or the Grand Bara Desert; deserts of Southern Africa such as the Namib Desert or the Kalahari Desert.

Tropical Wet/Dry (Savanna) Climate

- This is a typical climate experienced in between 5° -10° and 15°C to 20°C in both the latitude.
- Spatially they are found north and south central part of Africa, Campos of Brazil.
- Diurnal temperature ranges of 10°C to 15°C during winter are not uncommon. The Summers are hot and humid which provide an uncomfortable living environment.

Natural Vegetation

There are extensive areas in Africa where few people live and where natural vegetation and wild animals have not been disrupted by such activities as farming or the raising of livestock. In some parts of the continent large forest reserves have been established, like.

Tropical Rain Forests

Large area of Africa is covered by tropical rain forests, or *selvas* type of vegetation covering less than a tenth of the continent with heavy rain throughout the year. These forests contain several layers of vegetation. The top layer consists of the crowns of trees rising 125 to 250 feet (38 to 76 m) in height; the lower layers are arranged according to their height like shorter trees, shrubs, and vines respectively. Most of the trees are broad-leaved evergreens, along with few conifers ones. They yield pulp, timber, and such cabinet woods as mahogany, ebony, and teak. Oil palms, rubber-producing trees and vines, orchids, and lilies are among the numerous kinds of plants found in these forests.

Tropical Savannas

Savannas, covering perhaps one third of the continent, consist of areas mainly growing grass. There are spots of woodlands, scattered trees, or shrubs, depending on the length of the dry season. Coarse grasses upto 12 feet (3.7 m) high and large woodlands of deciduous trees are found near the border of Tropical forest. They even include many evergreens found in tropical rain forests, such as oil palms, rubber trees, and African ebony trees. There are also shea trees (whose seeds yield an edible fat), baobabs, flat-topped acacias, kapok, and many trees that bear edible fruit. As the dry season becomes pronounced grass grows shorter rarely reaching the height of five feet (1.5 m) high. Palms, baobabs, acacias, and such brightly flowering trees as cassias and erythrinias grow in small clumps or are scattered singly over the grassy areas.

- Away from the Equator to the north and south, rainfall decreases and there is a zone where there is a definite dry season unlike the equatorial region where it rains always.
- This is the **Sudan type of climate** and has tropical Grassland or Savanna vegetation.
- This region continues over the Eastern Highlands and forms a wide area around the equatorial forests.
- The grass is thick and coarse. In some places there is the tall elephant grass.

Tropical Steppes and Deserts

Increased aridity and longer dry seasons are the main features of tropical steppes. These are regions grow short grasses only.

Thorny acacias, euphorbias, dwarf palms, and jujube trees are found here. Steppes bordering on deserts no trees are spotted, rather widely scattered bunches of grasses grow. Rain here brings flowering plants and variety of grasses springs up and thrives. True deserts, of the Sahara and the Namib, are virtually featured with no vegetation except at oases (places watered by springs or wells). Vegetation at oases includes date palms, fig trees, willows, poplars, and tamarisks.

Mediterranean Forests

Mediterranean type of vegetation in Africa is found along the northern and southern coast. Different variety of shrubs and small trees, both deciduous and evergreen are grown. The plants are able to withstand long, dry summers with waxy, leathery leaves and long taproots develop here. The northern region raises cork oak, olive trees, cedars, and pines; in the south, laurels, cedars, and ironwood. Grasses and low flowering plants grow only during the rainy months.

Montane Forests

African vegetation system has localized characters. The Montane vegetation of highlands, particularly in Ethiopia and the mountains of the Great Rift Valley are such example. Depending on elevation, latitude, and direction of the winds vegetation grows. The region under Montane forests yield valuable timber and cabinet woods along with bamboo and wild varieties of coffee and banana. The slopes of mountain are covered with thick evergreen forest. At higher elevations, grasses and colorful, low-growing plants are typical. The High Veld of southern Africa is temperate grassland between 3,500 and 11,000 feet (1,070 and 3,350 m) above sea level.

Mangrove Forests

Mostly found along the African coast, but are most extensive along the Gulf of Guinea. A variety of other trees tailored to life in muddy estuaries and tidal flats are found other than Mangrove. Swamp and marsh also occur along the larger rivers and lakes of western and central Africa. *Papyrus*, tall grasses, and *lotus* are the most common plants. The *Sudd region of the White Nile River is one of the largest marshes in the world.*

NORTH AMERICA

Extent

North America is the **third largest continent** after Asia and Africa. It covers an area of nearly 24 million square kilometers. From south to north, it extends from $7^{\circ}N$ to $85^{\circ}N$ latitude and east to west from $20^{\circ}W$ to $179^{\circ}W$.

In other words, its northern boundary is only about 500 kilometers away from the North Pole and its western boundary only 10 kilometres away from the *International Date Line*.

There are five time zones in North America.

The Tropic of Cancer and the Arctic Circle pass through the continent and the $100^{\circ}W$ longitude cuts through the centre of the continent.

This huge landmass includes three large countries – Canada, United States of America and Mexico, seven small states of Central America and the islands of the West Indies.

The Atlantic, Pacific and Arctic oceans surround North America in the east, west and north respectively.

In the north-west the *Bering Strait* separates it from Asia and in the south-east the *Isthmus of Panama* joins it to South America. North America has a smooth coastline except for the existing in the north-west. Fiord (fiord) is a long, narrow winding inlet from the sea between steep slopes of a mountainous coast. It usually occurs where ocean water flows into valleys near the coast by glaciers.

Regional Divisions

Region wise North America can be classified into 5 parts which are listed below:

Western Region

Young mountains rise in the west. The most familiar of these mountains are probably the *Rockies*, North America's largest chain. They stretch from the province of British Columbia, Canada, to the U.S. state of New Mexico.

Great Plains

In the middle of the continent lies the Great Plain. Deep, rich soil blankets in large areas of the plains in Canada and the United States. Grain grown in this region, called the "*Breadbasket of North America*," feeds a large part of the world. The Great Plains are also home to rich deposits of oil and natural gas.

Canadian Shield

The Canadian Shield is a raised but relatively flat plateau. It extends over eastern, central, and northwestern Canada. The Canadian Shield is characterized by a rocky landscape pocked by an astounding number of lakes.

Eastern Region

This varied region includes the Appalachian Mountains and the Atlantic coastal plain. North America's older mountain ranges, including the Appalachians, rise near the east coast of the United States and Canada. These areas have been mined for rich deposits of coal and other minerals for hundreds of years.

Physical Division

North America can be divided into three physical regions:

1. The Western Cordilleras
2. The Central Lowlands
3. The Eastern Highlands

The Western Cordilleras

- The parallel ranges of young fold mountains run from Alaska and extend into South America as the Andes.
- As they resemble twisted cords they are known as Cordilleras.
- Fold mountains are formed when tectonic plates push the Earth's crust and force it to form ridges and valleys.
- Volcanic rocks form the base of fold mountains.
- The Cordilleras are part of the Pacific Ring of Fire. Mount St. Helena is in the USA.
- The snow-covered Cordilleras act as a barrier to moisture laden winds and cause relief rainfall.
- Some of the rivers flow westwards and some eastwards with the Cordilleras acting as the water divide between them.

- Rocky Mountains, Alaska Range, Cascades, Sierra Nevada and Sierra Madre are the chief ranges of the Western Cordilleras.

The Central Lowlands

- These stretch from around the Arctic Shores and Hudson Bay to the Gulf of Mexico.
- They are hemmed in by the Cordilleras in the west and the highlands in the east.
- In the west, they are known as high plains because of the greater altitudes.
- In the north, they form the Canadian Shield.
- The *Canadian Shield* is a peneplain with a number of lakes. They are large enough to be called seas. They are the five *Great lakes* - Superior, Michigan, Huron, Erie and Ontario.
- Lake Winnipeg, Great Bear Lake and Lake Athabaska are also on the Canadian Shield.
- South of the Canadian Shield, the Central Lowlands are covered with layers of sediment brought by glaciers and rivers. It is a very fertile region.

The Eastern Highlands

- They are old fold mountains that stretch from the valley of River St. Lawrence to Southern USA.
- They are not a high or as continuous as the Cordilleras.
- The highlands are also known as the *Laurentian* highlands in Canada and the *Appalachians* in the USA and are less than 2,000 metres in height.
- Their eastern slopes facing the Atlantic Ocean are very steep causing waterfalls in the streams that flow to the coast.
- The Grand Canyon is a network of deep narrow valley cuts into the dry Colorado Plateau.
- *The Old Faithful* is a natural geyser (a hot waterspout). Once in every 90 minutes, the water from the geyser comes out roaring upto 60 metres high. It is found in Yellowstone National Park.

Gulfs of North America

A *gulf* is a portion of the ocean that penetrates land which is very large in size, shape, and depth. They are generally larger and more deeply indented than bays and often make excellent harbors. Many important trading centers are located on gulfs. Some of the important bays those surrounds North America from all sides are

Gulf of Mexico

It is an important economic site for three countries and surrounded by United States, Mexico, and the island nation of Cuba. As one of the biggest gulf it has the coast line of 5000 kilometers.

Gulf of Alaska

It is situated on the north western part of North America where two types of water run into each other, a light, almost electric blue merging with a darker slate-blue.

Gulf of California

It separates the Baja California Peninsula from the Mexican mainland. It has a coast line of 4000 km (2600 miles). It is considered to be one of the most diversified seas on the planet, and is home to more than 5,000 species of micro-invertebrates.

Gulf of St. Lawrence

It is an water outlet of the North American Great Lakes via Saint Lawrence river. It's a semi enclosed sea which covers 236,000 square kilometres (91,000 sq mi) and containing about 35,000 cubic kilometres (8,400 cu mi) of water, which results in an average depth of 148 meters (486 ft).

Islands

The world famous islands of North America are:

Vancouver Island

It is situated on Canada's Pacific Coast, is known for its mild climate and thriving arts community. It is separated from British Columbia mainland by the Strait of Georgia and Queen Charlotte Strait and from Washington by the Juan De Fuca Strait.

Greenland

It is a massive island situated between Atlantic and Arctic oceans and 80% of its land is covered by ice.

Prince of Wales Island

It is one of the islands of the Alexander Archipelago in the Alaska Panhandle. This ranks four among the island in size.

Hawaii Island

It is otherwise known as the *Big Island* provides a vast canvas of natural environment and it is the largest island of the Hawaiian archipelago in the Central Pacific.

Cuba Islands of Antilles

It is known as the *sugar bowl* of the World and its vast source of metallic resources include cobalt, nickel, iron ore, chromium and copper. Other resources include timber, petroleum, silica, salt, and arable land.

Bermuda Island

It is the territory of British Islands in North Atlantic and famous for its Pink sand beaches such as *Elbow* and *Horseshoe Bay*.

Drainage Pattern

There are many rivers in North America. River of North America can be grouped according to the seas they drain into, like

1. **The rivers draining into the Gulf of Mexico**
These are the Mississippi, Missouri and their tributaries drain the whole of the lower Central Lowlands. They start from the Western Cordilleras. The Ohio and Tennessee. Rivers which are also tributaries of the Mississippi but have their source in the Appalachians are exceptions.
2. **Rivers draining into the Atlantic Ocean**
River St. Lawrence is the large river of this group. In this group the smaller rivers of the Fall Line can also be included.
3. **Arctic Ocean drainage**
River Mackenzie which has many shallow lakes on the Canadian Shield. River Nelson flow into the Hudson Bay.
4. **Pacific Ocean Drainage**
River Yukon in Alaska, Columbia, Fraser and the Colorado along the west coast. The Colorado river cuts across the Colorado plateau and forms the world's most famous and attractive deep gorges, known as **grand canons** having nearly one km depth. Among the other rivers, the Yukon, the Skena, the Fraser, the Snake, the Humboldt, the Sacramento, the San Joaquin, etc. are well known.

Seas in North America

Caribbean Sea

It is a sub-oceanic basin bordered by coasts of Venezuela, Colombia, and Panama; to the west by Costa Rica, Nicaragua, Honduras, Guatemala, Belize, and the Yucatán Peninsula of Mexico; to the north by the Greater Antilles islands of Cuba, Hispaniola, Jamaica, and Puerto Rico; and to the east by the north-south chain of the Lesser Antilles, consisting of the island arc that extends from the Virgin Islands in the north east to Trinidad.

Beaufort Sea

It is situated on the north of Canada and Alaska is known to be the marginal sea of Arctic Ocean covering an area of 184,000 sq. miles and the average depth of 3,239 ft (1,004 m).

Hudson Bay

It is known as the *second largest bay in the world* which encompasses an area of 1,230,000 square kilometer (470,000mi) and large body of Salt water.

Labrador Sea

It is bordered by continental shelves and separates Canada from Green Land.

Bering Sea

It is situated on the extreme North of North America separating the continents of Asia and north America.

Inland Drainage System

- The Great Basin area in the Rocky Mountains (Middle) has rivers which do not reach the coast, but terminate in the land. This is the Inland Drainage System.
- The rivers are small, seasonal and end up in saline lakes.
- The *Humboldt* river ends up in Carson Sink - a saline lake. The area is studded with bolsons.

Lakes in North America

- The Lakes of the Canadian Shield are fresh water bodies.
- The Great Salt Lake between the Rockies and the Sierra Nevada has a high salt content and is an area of 'Inland drainage'.
- Lake Champlain in Quebec, Canada and New York, Vermont in the USA.
- Lake Erie in Ontario, Canada and Michigan, New York, Ohio, Pennsylvania in the USA.
- Lake Huron in Ontario, Canada and Michigan in the USA.
- Lake Memphramagog is a 40-mile long glacial lake that extends from Vermont into Canada.
- Lake Ontario in Ontario, and New York in the USA.
- Lake St Clair in Ontario, and Michigan in the USA.
- Lake Superior in Michigan, Minnesota, Wisconsin in the USA.
- Lake of the Woods in Minnesota, USA; Manitoba, Ontario, Canada.
- Rainy Lake in Minnesota, USA; Ontario, Canada.

Climate and Natural Vegetation

- North America extends from roughly 700 kilometres north of the Equator to about 600 kilometres south of the North Pole, spreading over all the temperature zones.

- It is so wide that the central regions are thousands of kilometers away from oceanic influence.

The climatic regions are:

Tundra or Polar Climate

- The regions within the Arctic Circle along the shores of the Arctic Ocean and the Hudson Bay have extremely cold, long winters. The sun does not rise from many weeks here.
- The winter sky is lit sometimes by the Northern Lights or **Aurora Borealis**.
- In summer, the days are long and cool. The sun shines weakly, is low in the sky and does not set for weeks.
- These are lands of the **mid-night sun**. Rainfall is very low. Precipitation is in the form of snow.
- Its soil called the **permafrost** is permanently frozen and extremely hard.
- In summer, the surface ice melts. Lichens, mosses and low berry bearing bushes appear.
- Along the southern borders, there are stunted willow and birch trees.
- The tundra is the ice desert.

The Taiga or the Coniferous Forest Belt

- This is a broad belt of coniferous forest that stretches from Newfoundland in the east to Alaska in the west and southwards along the slopes of the Cordilleras.
- The winters are long and cold.
- The summers are short and warm. Precipitation is in the form of snow and is more than tundra.
- The trees have many adaptations to withstand the heavy snowfall and lack of water.
- They are cone-shaped, evergreen and *xerophytic* (in nature).
- Pine, fir, larch and spruce are some of the trees.
- In British Columbia and California, there are the giant sequoias, douglas, firs and cedars.

The Temperate East Margin Type

- This stretches from the Great Lakes along the Appalachians and the coastal areas.
- The northern part of this region is influenced by the Cold Labrador Current. It is known as the Laurentian Type or Cool Temperate East Margin Type.
- The southern part known as the Warm Temperate East Margin or China Type lies in the Trade Wind belt.
- This region lies in the track of hurricanes and tornadoes.
- The trees are deciduous.
- They have hardwood and broad leaves which are shed in autumn called "fall" in America.
- Maple, oak, elm and ash are the common trees.
- Many places have mixed forests of evergreen and deciduous trees.
- There is very heavy rainfall in Florida resulting in temperate swamps called 'Everglades'.
- Cypress is the predominant tree and alligator is the common reptile of the swamps.

Cool Temperate West Margin Type or British Type

- This type is found along the west of Canada and North California in USA.

- The Warm Alaska Current makes this region warm and the Westerlies bring rain throughout the year.
- It has warm summers and mild winters. There is heavy rainfall because of Rockies block the Westerlies.
- Vegetation consists of coniferous trees.
- The world's oldest, largest and tallest trees grow here. Douglas fir, redwood and giant sequoias are some of them.

The Prairies Grasslands

- From the south of the taiga, between the Great lakes and Rockies, to the Mississippi-Missouri Basin stretches the largest expanse of temperature grasslands.
- These treesless plains are the Prairies.
- Closer to the Rocky Mountains, where the rainfall is less lies the 'shorter grass prairie'.
- Cattle ranching is carried out here.
- The moderate rainfall is not enough to support tree growth.

The Mediterranean Type of Climate

- This climate is found in Central California on the west coast of North America.
- It lies between 30°N and 40°N latitude.
- The trade winds blow in summer as off-shore land winds.
- The hot dry summers make it difficult for plants to grow.
- The plants have adapted to store water from the winter rains in waxy leaves and bark.
- The original vegetation is scrub like and is called the *chaparral*.
- In wetter parts there are huge trees like cedars, cypress and the giant sequoia or red wood.
- This region is ideal for fruit growing.

Hot Desert

- The Mohave and Sonora are deserts on the southwestern part of North America because the mountains to the east act as a rain shadow for the Trade Winds.
- While the Cool California Current in the west prevents sea winds from bringing rain.
- Only very small parts of the desert are without any type of vegetation.
- Cactii, sage and thorn bushes and coarse grasses are common. The Joshua tree is a taller cactus.

Tropical Rain Forests

- This forest is found in Central America and the West Indies.
- There is heavy rainfall and high temperature
- The equatorial jungles of the Amazon and Orinoco of South America have extended to this region.

Storms

- **Hurricanes** - These are destructive cyclones which blow in from the Gulf of Mexico. It is caused by depressions originating over the sea.
- **Tornadoes** or **twisters** form over land in the southern states of the USA. This is a black cloud which possesses a funnel shaped tail which twists as it rushes past. There is complete destruction wherever the tail passes because of the vacuum effect. Everything in its path is sucked in. It is the most destructive of cyclones.

SOUTH AMERICA

Extent

South America is a long triangular shaped continent. It stretched from 12°N to 55°S latitude. The Equator passes through the northern part of the continent and the Tropic of Capricorn runs roughly through the middle. Because of its tapering shape, a major part of the landmass is in the tropics. South America lies on the west of the *Prime Meridian*. So the time at any place on this continent will be some hours less than or behind the *Greenwich Mean Time*. The 60° meridian divides the continent lengthwise into two halves. It is more to the east compared to North America and is therefore closer to Europe and Africa. South America is the **fourth largest continent** after Asia, Africa and North America. It is two-third the size of Africa and six times the size of India. The coastline of South America is smooth with very few inlets except in the extreme south-west where there are **fiords** and many small islands. Fiords are deep inlets of the sea into mountains land. There are a few large islands off the coast of South America. The Galapagos Islands near the Equator and the Juan Fernandez Islands near Central Chile are in the Pacific Ocean. The Tierra del Fuego is in the Southern Ocean and the Falkland Islands in the South Atlantic Ocean. The island of Trinidad is near Venezuela in the North Atlantic Ocean. *The Andes is the longest mountain range in the world*. South America's three southern countries – Argentina, Chile and Uruguay – constitute a region sometimes referred to as the *Southern Cone* because of its pointed, ice-cream-cone-like shaped.

Physical Divisions

South America can be divided into four physical regions:

1. The Pacific coastal strip

It lies in the west, between the ocean and the Andes. It is the longest coastal plain in the Atlantic world. In most places it is about 80 kilometres wide but in some it is as narrow as 8 metres. The coastline of South America is smooth and regular. At the river mouths there are inlets which are used as harbours. The south-western coast of the continent has fiords or deep inlets of the sea into mountainous land.

2. Mountain Ranges

The Andes stretches through entire continent, running in north-south direction from Isthmus of Panama to Strait of Magellan. They are the continuous range of folded mountain systems which covers the entire western coast of South America. The Andes is the longest mountain range in the world. The highest mountain of this mountain range is Aconcagua, stands at 6,962 meters (22,841 feet) and straddles the Argentina-Chile border.

They form a chain of ranges and knots with enclosed intermontane plateaus namely in Ecuador and Bolivia. Being part of the Pacific Ring of Fire, there are many volcanoes and frequent earthquakes along this region. Mount Cotopaxi and Mount Chimborazo are active volcanic peak, is the highest peak in South America.

Some Mountains of South America

Mountain	Country	Height (In Meters)
Aconcagua	Argentina	6962
Ojos del Sulado	Chile	6891
Monte Pissis	Argentina	6793
Llullaillaco	Argentina/Chile	6738
Cesro Mercedario	Argentina	6720
Neado Sajama	Bolivia	6542
Illinani	Bolivia	6438
Jang'u Uma	Bolivia	6427
Illampu	Bolivia	6368
Cesro El Plomo	Chile	5424

3. The Central Lowlands

The are formed by two great river systems – the Amazon-Orinoco and the Parana-Paraguay. The vegetation of the lowlands are given special names.

The Orinoco Basin has dense tropical forests. The northern part is a plain covered with savanna grass called the **Llanos**. The equatorial jungle of the Amazon Basin is called the **selvas**, a typical tropical rain forest. The rich temperate grasslands around the mouth of the Parana-Paraguay is the **pampas**. At the source of these rivers is a region scrub forest called the **Gran Chaco**.

4. The Eastern Highlands

These are plateaus made up of hard old rocks. The River Amazon separates them into the Guiana Highland to the north and the Brazilian Highland to the south. They have been worn down by wind, rain and rivers. They have steep cliffs along the east coast and slope gently towards the Central Plains. The savanna grasslands of Brazilian Highlands is the **compos**. Towards the Central Lowlands, it is known as the plateau of Matogrosso. Patagonia is a cold, wind swept plateau at the southern end of the tapering continent.

- Guiana Highlands:** It is a geographically stunning part of Planet Earth, over 1,000 miles in length, the Highlands stretch from southern Venezuela across the northern edge of South America to the tip of Brazil. It consists of a vast plateau, one marked by deep gorges, tropical rain forests, numerous rivers and waterfalls. It's famed for the *highest waterfall* in the world (*Angel Falls*) at 3,212 ft (979 m) high. The highest point is Mt. Roraima on the borders of Brazil, Guyana and Venezuela at 2,810 m.
- Brazilian Highlands:** This highlands region is about 800 miles in length and runs through the Brazilian states of Minas Gerais, Goias, Bahia and Sao Paulo is southeastern Brazil. The magnificent landscape includes varied mountain ranges, namely the Serra de Mantiquiera, Serra do Paranapiatiba, Serra Geral, and Serra do Mar.
- Patagonia:** It is located between the Andes and the Atlantic Ocean, and about 1,000 miles in length; Patagonia stretches south from the Rio Negro river in southern Argentina to Tierra del Fuego and the Strait of Magellan. It's mostly rugged, barren land, famed for its beauty and striking scenery.

Deserts of South America

- Patagonian Desert** – the largest desert by area located in Argentina
- La Guajira Desert** – a desert in northern Colombia and some of northwestern Venezuela
- Atacama** – a desert in Chile, the *driest place on Earth*.
- Sechura Desert** – a desert located along a portion of the northwestern coast of South America
- Monte Desert** – in Argentina, a smaller desert above the Patagonian desert.

Drainage System

The drainage system of south America consists of:

- The Amazon Basin
- The Rio de Plata Basin
- The Orinoco Basin
- The Sao Francisco Basin

The Amazon basin

- It is the basin of River Amazon.
- Its length is second to that of the Nile river of Africa.
- It has the largest flow of water in the world.
- The river drains nearly 40 per cent of area of South America.
- The major tributaries of the Amazon river are the Caqueta, the Jurua, the Madeira, the Negro, etc.

The Rio de Plata basin

- This basin is second in size to that of the Amazon.
- The main rivers which form the Basin of Rio de Plata are the river Paraguay, the Parana and the river Uruguay.
- River Parana (4,879 km) rises from Minas Gerais from a water divide Carino de Paranaiba.

The Orinoco basin

- This is considered to be the third largest drainage system of South America.
- It rises in the Southern end of Sierra Parima near Mount Delgado Chalboud at a height of 1000 metres.
- It traverse 2,740 km to meet the Atlantic Ocean.
- The word Orinoco means '*a place to paddle*', i.e. a river where navigation is possible.
- In the North, the Orinoco river passes through a zone called 'Region of Rapids' where there are enormous granite boulders.
- The world's highest waterfall Angel (979 m) is situated on river Churun which is a tributary of river Caroni which is further a tributary of river Orinoco.
- The Orinoco flows through the llanos (savanna grasslands) of Venezuela into the (North Atlantic Ocean).

The Sao Fancisco basin

- The fourth largest river system of South America is the river Sao Francisco which is about 2,914 km in length. It flows within Brazil.
- It originates North-west of the city of Belo Horizonte on the Eastern slope of Sierra da Canastra.

Climate

The climate of South America is influenced by the following factors:

- South America stretches through many degrees of latitudes and through all the temperature zones of the Earth.
- The high Andes cover the full length of the continent. Mountains lower the temperature and form rain shadow regions on their leeward side.
- The Cool Peru current flows along the west coast making the shore very dry.
- The Warm Brazil current on the east coast brings good rainfall.
- The South East Trade Winds bring rain to the east coast in the tropics. Thus beyond the Andes on the west coast, deserts are formed.
- Further south from 30°S latitude to the tip of the continent, westerly winds bring rain.

1. The hot wet equatorial climate

- This is found in the Amazon basin.
- The Equator cuts through this region but on the high Andes rain is less because of the height.
- Quito, the capital, is said to have 'eternal spring', while the Amazon Lowlands are very hot and wet throughout the year.
- Dense impenetrable rainforests called the selvas or Amazon rainforest or amazon are found here.

Amazon Rainforest

It is located within nine nations: Brazil (with 60 percent of the rainforest) Colombia, Peru, Venezuela, Ecuador, Bolivia, Guyana, Suriname and French Guiana.

The Amazon represents over half of the planet's remaining rainforests and comprises the largest and most species-rich tract of tropical rainforest in the world.

- The trees are very tall and head wooded like ebony and mahogany.
- Some valuable trees found here are cinchona from which quinine, the medicine for malaria is made; chicle, the tree which gives chewing gum; the hevea tree which gives rubber; the plant coca from which cocaine, the painkiller is extracted.
- The wildlife consists mostly of tree-top dwellers like monkeys, birds, insects and spiders. There are snakes and giant pythons called anacondas, ant eaters and armadillos; crocodiles and alligators in the marshes.

2. Summer rain climate on the eastern side within the tropics

- These are regions of summer rain brought by the trade winds. The Warm Brazil Current increases the rain.
- The rest of the year is dry.

Campos

- North of the Equator along River Orinoco they are called **llanos** and in the Brazillian highlands south of the Equator they are known as **campos**.

The Gran Chaco

This lies between the Brazilian Highlands and the Andes. Being far inland, the trade winds lose their moisture. There is very little rainfall.

3. The Pacific Coastlands

Between the Andes and the Pacific Ocean lies the long narrow plain which passes through the following regions.

- The northern part of this plain lies in Colombia. There are equatorial forests and swamps.
- In Peru and Northern Chile is the Atacama Desert. The Andes block the trade winds from bringing rain. The Cool Peru Current adds to the dryness of the desert.
- In Central Chile, the plains form a fertile area enjoying the Mediterranean climate. The westerlies bring rain in winter. There is summer drought. Citrus fruits, grapes, almonds and walnuts are cultivated here.
- South of 40°S latitude lies the fiord coast of Southern Chile.

There is rain throughout the year from the westerly winds. The sea flows into the valleys to form a 1,000 kilometres stretch of cold, rainy islands and fiords. Evergreen temperate forests grow here. Some of the conifers like the Chilean pine are found only in South America. Some have leaves instead of needles.

Natural Vegetation

Equatorial Forest

The high temperature and heavy rainfall produce luxuriant vegetation, generally composed of broad-leaved trees is found in wet tropical uplands and lowlands around the Equator. Found in Amazon Basin and coastal lowlands of North-eastern Brazil, coastal Colombia and parts of adjoining Ecuador. Spices of rubber, mahogany, ebony, Cieba and Brazil nuts are commonly found. These forests are also called 'Selvas' lies 0° to 10° north and south.

Temperate Forests

These are characteristics by more or less continuous canopy with wide leaves, big and tall trees along with non seasonal vegetation. Warm temperate forests are found in Brazilian Highlands, Paraguay, Uruguay, Southern Brazil and cool temperate forest are found in upper slopes of Andes and S.W. Chile. Species like Beech, conifers, Parana Pines, Quebracho – source of tannic acid and Yerba mate grows here. This type of biome is found in regions where there is cold winter and warm summer. Regions with such climate are common in the mid-latitudes, far from both the equator and the poles. Temperate forests are almost always made of two types of trees, deciduous and evergreen.

Mediterranean Forests

These are characterized by dry summers and rainy winters, generally composed of broadleaf trees. Evergreen laurels and acacias are found in wetter areas where as thorny shrubs and cacti are found in drier north. They are vulnerable to

degradation by human activities such as logging, overgrazing, conversion to agriculture, urbanization, and introduction of exotic and invasive species as the region is semi-arid, and often have poor soils. Fire, both natural and human-caused, has played a large role in shaping its ecology

Savanna Grasslands

These are widely spaced, scattered trees, tall grass, found in Orinoco basin, Paraguay, northern Argentina and Brazil. Chief species like Acacias, tall coarse grass and patches of scrub are found in drier areas whereas shorter grass in wetter areas. Savanna spreads over Cerra do, Brazil in the lower latitudes. This particular biome falls in between a grassland and a forest region. There are actually two very different seasons in savanna; a very long dry season (winter), and a very wet season (summer).

Pampas

These are large treeless plains with large grasslands areas. It is also known as temperate grasslands. Countries like Argentina (provinces of Buenos Aires, La Pampa, Santa Fe and Córdoba), most of Uruguay and the southern most states of Brazilian cover this region. Small plants such as grasses flourish, and trees are rare. These are fertile South American lowlands where pastoral and farming is done.

Desert vegetation

Desert habitat area have adapted to its dry, hot extremes by using both physical and behavioral mechanisms. They often have few or no leaves. It found in arid areas of Atacama desert, Sertao arid region of N.E. Brazil and N.W. Mexico (Sonora and Baja deserts). Lichens, acacia, scrub and Cacti are vegetation covers mostly found here.

AUSTRALIA AND OCEANIA

Extent

Thousands of islands combine to form Oceania region mostly covering the Central and South Pacific Ocean. The region is dominated by world's biggest island and two other major landmasses, micro-continent of Zealandia (includes New Zealand) and the western half of the island of New Guinea, made up of the nation of Papua New Guinea. Oceania also includes three island regions: Melanesia, Micronesia, and Polynesia (including the U.S. state of Hawaii). It stretches from the Strait of Malacca to the coast of Americas. Tropic of Capricorn divides it into almost two halves.

- Australia is the world's **largest island** and **smallest continent**.
- It is the only nation that completely covers a continent.
- Its total area is nearly double that of India and Pakistan combined.
- It lies entirely in the Southern Hemisphere and is aptly named - *Austral* meaning south.
- It is located between the Indian and Pacific Oceans, stretches west to east from 114°E longitude to 154°E longitude and from 10°S to 40°S latitude.

- The Tropic of Capricorn cuts the continent almost into half.
- Asia is the continent nearest to Australia.
- The nearest point on the mainland of Asia is Singapore
- To the west of Australia, Indian Ocean, to the south, the icy shore of Antarctica to the south-east is New Zealand. To the north-west is the continent of Asia.

Physical Divisions

Australia is the most levelled and lowest of all the continents. There are no high mountains, deep valleys or large rivers. Mount Kosciusko 2,230 metres above sea level, is the highest peak. The coastline is very smooth with no inlets except in the south. So there are very few good harbours. As a smallest continent Australia has no prominent physical division. yet efforts have been made to divide it roughly into four categories, such as:

Eastern Highland

This is a chain of hills and mountains which interrupts the levelled land form of Australia. These are also known as great Dividing Range.

The Great Dividing Range

- It is, also known as the Easter Highlands, is Australia's most substantial mountain range. The range stretches more than 3500 km from the northeastern tip of Queens land, running the entire length of the eastern coastline through New South Wales, then into Victoria and turning water, before finally fading into the central plain at the Grampians in western Victoria.
- The Great Dividing Range does not consist of a single mountain range. It consists of a complex of mountain ranges, plateau, upland areas and escarpments with an ancient and complex geological history.
- The crest of the range is defined by the watershed or boundary between the drainage basins of rivers which drain directly eastward into the Pacific Ocean, and those rivers which drain into the Murray-Darling River system towards the west. In the north, the rivers on the west side of the range drain towards the Gulf of Carpentaria.

The Western Plateau

This eroded plateau has undergone the process of erosion for a quite longer period. It is characterised by several 'sinkhole' a network of underground caves which filled with water. The Western plateau is also called as the home for several deserts and the climate is comparatively dry owing to the cold water current of the western Australia. The series of desert in this region are:

- The ranges were originally home to Australian **Aboriginal** tribes such as the **kulin**.

Desert

- **Gibson Desert** - a central Australian desert
- **Great Sandy Desert** - a northwestern Australian desert

- **Great Victoria Desert** - the sixth largest desert in the world by area, located in south-central Australia.
- **Simpson Desert** - a central Australian desert
- **Little Sandy Desert** - a western Australian desert
- **Strzelecki Desert** - a south-central Australian desert
- **Tanami Desert** - a northern Australian desert
- **Western Desert** - a desert located in western Australia, comprising the Gibson, Great Sandy, and Little Sandy deserts.
- **Rangipo Desert** - a barren light altitude desert on the North Island Volcanic Plateau in New Zealand.

The Central Lowland

The extensive patch of plain or low land has its existence between Eastern Highlands and Western Plateau. The patch is drained by two rivers such as Murray and Darling which also facilitate the land with irrigation facility for farming and other essential activities. The area is also known as the Great Artesian Basin as it is covered by a number of underground pressurized wells from which the water comes out automatically to the surface. Yet they are not suitable for drinking as they are salty by nature.

The Great Barrier Reef

- This is the largest coral reef in the world.
- It lies along the east coast of Queensland, Australia, in the Pacific Ocean.
- It is about 2,000 kilometres long, in some places it is as close as 16 kilometres to the coast while in other places it is 200 kilometres away.
- It is one of the natural wonders of the world.
- It is formed by the tiny coral **polyps**.

The Islands

The islands situated surrounding the younger most continent in size is collectively known as Oceania. Broadly it has been divided into three major island groups Melanesia, Micronesia and Polynesia.

Melanesia is otherwise known as Black Islands and congregated around north and east of Australia. Some of the major islands are East Timor, Fiji, New Caledonia, Papua New Guinea, Solomon Islands, and Vanuatu.

Micronesia which is also called as Little Islands is the agglomeration of islands such as Guam, Kiribati, Marshall Islands, Federated States of Micronesia, Nauru, Northern Mariana Islands, Palau and Wake Island

Polynesia otherwise known as Many Lands which expand from islands of Midway in the north to New Zealand in south and include the series of islands such as American Samoa Cook Islands, French Polynesia, Niue, Pitcairn, Tokelau, Tonga, Tuvalu, Wallis and Futuna Islands.

The islands of Oceania have been formed due to differential activities of volcanoes. The low islands of Micronesia have been formed by building up coral reefs on the rim of volcanic island, hence giving it a shape of a Ring which is known as Atolls. The atolls further encircle the lagoons which are characterised by shallow pools of clear water at a very low altitude of just few feet above sea level.

Drainage System

- Australia has low average rainfall.
- Being a hot dry country the rate of evaporation is high. So there is very little water left to flow as river to the sea.
- As a result of this the total Australian continent is mainly drained by two of the largest drainage basins Murray and Darling and an inland lake is also found which is known as lake Eyre Basin. Which accounts an area for over 1 million square kilometre.
- River Murray starts from the Snowy Mountains of the Great Dividing Range.
- Its tributaries are the Darling, Murrumbidgee and Lachlan. Many dams have been built across these to provide for irrigation and power generation.
- River Swan near Perth is also utilized in the same way.

Climatic zones

There are generally four types of wind prevail over Australian continent throughout the year which affect the climate of the little continent to a large extent.

Seasonal change in the Temperate Zones

The coastal hinterland of New South Wales, much of Victoria, Tasmania, the south-eastern corner of South Australia and the south-west of Western Australia are contributing the temperate zones where the seasonal changes are as follows

- **Summer:** December to February
- **Autumn:** March to May
- **Winter:** June to August
- **Spring:** September to November

The two similarly affected areas to that of temperate zones are:

- **The temperate grassland** that surrounds the arid and semi-arid desert areas in the centre and gradually percolates into the area north of Alice Springs in the Northern Territory.
- **Deserts** are the arid and semi arid areas of the centre of the continent which stretch across the vast amount of South Australia and Western Australia.

Seasonal change in the Tropical Zones

There are three climatic zones in the tropical areas of Australia:

- **Equatorial** – the tip of Cape York and Bathurst and Melville Islands north of Darwin.
- **Tropical** – across northern Australia including Cape York, the Top End of the Northern Territory, land south of the Gulf of Carpentaria, and the Kimberley region.
- **Sub-tropical** – the coastal and inland fringe from Cairns along the Queensland coast and hinterland to the northern areas of New South Wales and the coastal fringe north of Perth to Geraldton in Western Australia.

These above mentioned areas experience two exactly opposite spell of season *i.e.* wet and dry seasons.

The wet season is otherwise called as the monsoon season, which lasts about six months, between November and March. The temperature ranges between 30 and 50 degrees Celsius and comparatively it is hotter than the dry season because of the high humidity during the wet, which is caused by large amounts of water in the air. It is also marked by heavy rainfall which leads to frequent flooding.

The dry season lasts about six months, usually between April and October. Temperatures are lower and the skies are generally clearer during the dry. The average temperature is around 20 degrees Celsius.

Natural Vegetation

Downs

- The temperate grassland of Australia known as **downs** is found in the Murray-Darling basins of South Eastern part.
- They are largely used for pastoral activity.
- Cultivation is profitably done in the down areas where Mitchel grasses grown.
- The Mitchel grasses grow even on the cracking clay soils, basaltic soils, and even on skeletal soils.
- Similarly Canterbury grassland is found in New Zealand.

The tropical Rain forest island regions

These are home to dense forests with coconut palms and mangroves growing nearer the shorelines. High temperatures throughout the year with very little rainfall.

The deciduous forest region

It has a wealth of plant life and includes tall and short trees, shrubs, small plants and mosses. Four distinct seasons with warm summers and cold, wet winters. The trees shed their leaves in autumn.

Savanna

Very high temperatures all year and rain during the summer season only.

The dry desert and desert scrub region

It is home to the eucalyptus as well as those plants that can survive the arid conditions like *cactii*. Warm to high temperatures with very little rainfall. Hot summers and cold winters with above average rainfall.

The varying climate of Australia is largely dependent its size. The temperature ranges below zero in the Snowy Mountains in southern Australia to extreme heat in the Kimberley region in the north-west of the continent. Different climatic zones are characterized by distinct climatic phenomena with prominent seasonal changes

Four prominent seasons such as Summer, Autumn, Winter, and Spring are experienced in the Temperate zone, Desert and the Grassland climatic zones but drastically it has changed to wet and dry seasons in the tropical north of the continent which includes the Equatorial, Tropical and sub-tropical zones.

EUROPE

Size

- Europe is the **second smallest** continent, the smallest being Australia.
- Its area, including the islands around the coast, is about 10 million square kilometers.
- It is roughly three times the size of India and smaller than China.

Location

- A large part lies in the temperate zone as it stretches from 35°N to 80°N latitude.
- Longitudinally, it stretches from 10°W to 60°E
- The Prime Meridian, passes through London.

- In the north, though it stretches into the Arctic Circle, the Warm Gulf Stream keeps the ports ice free.
- The broad continent shelf on its west provides good fishing grounds and there are sheltered harbours along the indented coastline.
- It has the longest coastline in proportion to size.

Boundaries

- To the east, it is separated from Asia by the Ural Mountain, Caspian Sea, Caucasus Mountain and Black Sea.
- To the south is the Mediterranean Sea. The Aegean Sea and Adriatic Sea are two of its branches.
- To the west, is the Strait of Gibraltar separating Europe from Africa and joining the Mediterranean to the Atlantic Ocean. The Bay of Biscay, English Channel and North Sea are parts of the Atlantic Ocean.
- Baltic Sea with two branches - Gulf of Bothnia and Gulf of Finland is an inlet in the north. The Arctic Ocean to the north has a bay called the White Sea.
- The peninsula of Greece, known as the Balkan Peninsula and Italy extend into the Mediterranean Sea.
- In the south-west, is the Iberian Peninsula which is made up of Spain and Portugal.
- In the north-west, is the Scandinavian, Peninsula consisting of Norway and Sweden.

Physical Divisions

The physical features of Europe are divided into four contrasting regions:

Western Upland

It is also known as the Northern Highlands, delineate the western edge of Europe and define the physical landscape of Scandinavia (Norway, Sweden, and Denmark), Finland, Iceland, Scotland, Ireland, the Brittany region of France, Spain, and Portugal. These landforms are result of glaciations of hard rock in the ancient times. Distinct physical features such as marshlands, lakes, and fjords have been emerged with the recession of glaciers form the highland areas. The famous Norwegian Fjords which are Lyse fjord, the Geiranger fjord.

North European Plain

It is the extensive low land spread along the bank of various mighty rivers such as Rhine, Weser, Elbe, Oder and Vistula. These river valleys are favourable for growing seasonal crops. It covers all most half of the Europe. Bordered by Baltican White sea from north and Black and Azov from south the plain is gradually narrowed down towards west. the northern part of the land is characterized by diversified glacial landforms such as Pipet Marshland, Valdai hills of western Russia, glacial lakes etc.

Central Uplands or Plateau

These are the collection of distinctive landscape of summits, steep slopes, valleys and depression which stretches across the central Europe. It extends from Belgium in the East to France in the West and from Czech Republic and south Germany in south to Switzerland and Austria in North. Except some river valleys such as like Rhine, Rhone, Elbe, and Danube river valleys all other areas of this division is sparsely populated.

Alpine Mountain Systems

These are located in south-central Europe, immediately north of the Mediterranean Sea. They extend for almost 700 miles

in a crescent shape from the coastline of southern France (near Monaco) into Switzerland, then through northern Italy and into Austria, and down through Slovenia, Croatia, Bosnia and Herzegovina, Serbia and Montenegro - then ending in Albania on the rugged coastline of the Adriatic Sea.

The highest point is Mont Blanc at 15,771 ft. (4,807m).

Mountains

The Ural Mountains

- These mountains separate Asia from Europe in the east
- From north to south, these are 2,200 km long and 80-120 km broad with many parallel valleys.
- Though their average height is 300 m yet their highest peak is about 1,638 m high.

The Scandinavian Mountains

- The Scandinavia consists of Norway, Sweden and Denmark.
- In fact Scandinavia exists or *fenno-Scandia* which continues into the east through Finland to *Kola Peninsula* in Russia.

The Old Mountain Blocks

- These are *Hercynian* and *Caledonian* mountain chains.
- In the west the *Meseta of Spain*, the *Central Plateau of France*, the *Britanny Peninsula*, the *Rhine Upland*, the *Black Forest*, *Vosges*, *Bohemian Plateau* and *Rhodope Mt* etc, are examples of these old mountains.

The Alpine Mountain Ranges

- The highest peak is (Mount Blanc 5,000 m).
- The mountain range runs in many branches.
- The main ones are the Alps, the Carpathians, the Balkans, the Caucasus etc.
- Another branch is the Apennines (Italy), the Atlas (Africa and the Sierra Nevada Spain).
- Still another branch is the Dinaric and the Pindus mountain (Yugoslavia and Greece) and enters through the Crete island into Asia.

The Pyrenees

- The Pyrenees are half as long and broad as the Alps and separate broadly France from Spain.
- The highest peak is Pice de Aneto (3,404 m).

Islands of Europe

As surrounded by number of seas from all sides, Europe is an island rich continent. British Isles is the largest and the most important group of islands consisting of England, Scotland and Ireland. Some of the major islands of these groups are:

England consists of Isle of Wight, Isle of Sheppey, Hayling Island, Foulness Island, Portsea Island, Canvey Island, Mersea Island, Walney Island, Wallasea Island, Lundy, Isles of Scilly. Ireland is famous for Achill Island, Aran Islands, Rathlin Island.

Scotland has two prominent groups of Islands.

The islands of main archipelago consists of Shetland, Orkney, Outer Hebrides, Inner Hebrides, Islands of the Clyde, Islands of the Forth, Outlying islands and Larger islands are Lewis and Harris, Skye, Mainland, Shetland, Mull, Islay, Jura, Arran, South and North Uist and south Uist.

Deserts

- **Halendi** - a region of Iceland and Europe's largest desert.
- **Bedowska Desert** - a desert located in Lesser Poland Voivodeship, Poland.
- **Deliblatska Peseara** - a desert located in Voivodina and Serbia.
- **Oltenian Sahara** - a desert spanning approximately 80000 hectares or 800 km² in the Romanian historical province of Oltenia.
- **Tabernad Desert** - a desert in Almeria, Spain.

Drainage Pattern

- The rivers of Europe are perennial being fed by melting snow or by the rain brought by the Westerlies.
- Many of them have their origin in the Alps.
- Rivers that flow into the Mediterranean Sea are Rhone (France) and Ebro (Spain).
- River Po of Italy flows into the Adriatic Sea.
- The Danube, Dnieper and Don flows into the Caspian Sea.
- Rivers that flow into the Atlantic Ocean are - Guadalquivir (Spain), Tagus and Douro (Portugal), Loire and Seine (France), The Rhine Weser and Elbe (Germany)
- Many rivers flow into the Baltic Sea.
- Thames, the chief river of England, flows into the English Channel.
- The Rhine and Danube are international rivers because they pass through many countries.
- The Rhine starts from the Alps in Switzerland and flow northwards through Germany and enters the sea through Holland. It passes through heavily industrialized regions and is used for transporting heavy goods. It is the busiest waterway of Europe. Rotterdam, the largest part of Europe, is on its delta.
- The Danube is also an international river. It rises from the Alps in Germany and flows through Austria, Hungary, Serbia and enters the Black Sea in Romania. It is not as important as the Rhine for international trade because the Black Sea in the interior.

Important Rivers of Europe

Rivers	Length in km	Draining into (Tributary of)
Volga	3,692	Caspian Sea
Danube	2,860	Black Sea
Ural	2,428	Caspian Sea
Dnieper	2,290	Black Sea
Pechora	1,809	Barents Sea
Don	1,950	Sea of Azov
Kama	1,805	Caspian Sea (Volga)
Nothern Dvina-Vychegda	1,774	White Sea
Oka	1,500	Caspian Sea (Volga)
Belaya	1,430	Caspian Sea (Kama)
Dniester	1,362	Black Sea
Rhine	1,236	North Sea

Elbe	1,091	North Sea
Donest	1,050	Black Sea (Don)
Vistula	1,047	Baltic Sea
Tagus	1,038	Atlantic Ocean
Daugara	1,020	Baltic Sea
Loire	1,013	Atlantic Ocean
Tisza	976	Black Sea (Danube)
Prut	953	Black Sea (Danube)
Sava	940	Black Sea (Danube)
Neman	937	Baltic Sea
Meuse	925	North Sea
Ebro	910	Mediterranean
Douro	897	Atlantic Ocean
Kuban	870	Sea of Azov
Mezen	857	Barents Sea
Oder	854	Baltic Sea
Rhone	813	Mediterranean
Seine	776	Atlantic Ocean

Gulfs and Bays

These are the parts of large water bodies which are adjacent to a massive land mass. It may be continents or countries which are of economic importance for any human civilization. As Europe is surrounded by number of large water bodies such as Mediterranean Sea, Black Sea, North Sea etc. there are a lot of Gulfs, Bay and straits.

- **The Gulf of Finland** is situated in the eastern most arm of the Baltic Sea and extends between Finland (to the north) and Estonia (to the south) all the way to Saint Petersburg in Russia, where the river Neva drains into it. Other major cities around the gulf include Helsinki and Tallinn. The eastern parts of the Gulf of Finland belong to Russia, and some of Russia's most important oil harbours are located farthest in, near Saint Petersburg.
- **The Gulf of Bothnia** situated at the northern most part of Baltic Sea and bordered by Sweden at its western side and Finland at eastern side.
- **The Gulf of Riga** is a brackish water body which is considered as a sub-basin of the Baltic Sea. The areal extent of the Gulf of Riga is approximately 16,300 km². It is also called as the Bay of Riga which is a very shallow water sea with a maximum depth of 67metres.
- **The Gulf of Lions** extends from the easternmost spurs of Pyrenees and covers various lagoons, the Rhone River delta, limestone hills of Marseille. It's an embayment of the Mediterranean coast line of Languedoc-Roussillon and Provence in France,

Climate

- Europe has a mild climate with moderate rain throughout the year.
- Some places are close to the sea and some are located in the centre of Eurasia.

- There are plains on the windward side and plateaus on the leeward side of mountains.
- The southern parts stretch to the south of the 40°N latitude and face desert winds blowing from the Sahara across the Mediterranean Sea.
- The northern parts lie within the Arctic Circle and are close to the North Pole.
- Temperature decreases from south to north because the south is closer to the Equator and has a lower latitude.
- The Mediterranean lands are warmer than the Arctic lands.

Gulf stream plays a major role in the climatic fluctuation over the Western Europe as it provides warming effects during winter. It is otherwise known as North Atlantic Drift which increases the temperature of coastal Europe.

On the other hand Siberian high brings colder and drier weather from the west.

The Danube region through the Balkan, Ukraine and southern part of Russia experience a continental climate.

Natural Vegetation

- Natural vegetation depends on the temperature and rainfall of that place.
- In Europe, the original vegetation has been cleared to make place for agriculture and development of industries and cities.
- Only in inhospitable places like the tundra and taiga we can see the natural vegetation.

Natural Vegetation are:

1. **Deciduous and Mixed Forest**
 - The trees of the Great European Plain are mostly deciduous.
 - They have hardwood and broad leaves which fall in autumn.
 - Elm oak beech and ash are some of the trees.
 - Some evergreen coniferous trees are found scattered. Hence they are called **mixed forests**.
 - The Black Forest of Germany and the Bavarian Hills are protected areas.
 - The Pyrenees and some highlands still have original forest cover.
2. **Taiga**
 - This is Russian name for evergreen coniferous forests.
 - Like North America, this belt stretches right across Eurasia.
 - The trees are evergreen, can withstand the intense cold and lack of water.
 - Pine, spruce, fir and larch are the chief types.
3. **Tundra**
 - In Lapland to the north of Norway, Sweden and Finland are the ice deserts.
 - Mosses and lichens grow throughout the year and even under the ice in winter.
 - In summer, short grasses start growing.
4. **Steppes**
 - South of the East European mixed forests stretch vast grasslands called the **steppes**.
 - They are temperate grasslands similar to the prairies of North America.

5. Maquis

- The vegetation around the Mediterranean Sea consists of drought resisting evergreens.
- They do not need to shed their small leaves because the waxy coating holds back moisture in summer and the moist winter in the season of limited growth.
- The trees are stunted and grow far apart.
- This sparse woodland is called 'maquis' in Southern France.
- The olive is the most typical tree of the region.
- Cork oak, cypress, walnut; almond and cedar the only conifer, are some of the others.
- Olives, citrus fruit and grapes are of great commercial value.
- This is the 'fruit basket' of the western world.

ANTARCTICA

Location

- To the south of India, beyond the Indian Ocean lies the frozen continent of Antarctica.
- The name means *opposite the Arctic*.
- It is the southernmost continent and lies entirely within the Antarctic Circle spread around the South Pole.
- It separated from the rest of world by the icy waters of the Southern Ocean which comprises of the southern portions of the Indian, Atlantic and Pacific Oceans.
- Its area is over 14 million square kilometers.
- It is the fifth largest continent.
- It is larger than Europe and is twice the size of Australia.
- The continent is a high plateau which is frozen throughout the year.
- There is no coastal plain.
- There are mountain ranges, peaks, a rift valley and volcanoes.
- Two broad inlets, the Weddel Sea and the Ross Sea and the Trans-Antarctic Mountains which cross the entire continent divide the land into **West Antarctica** and **East Antarctica**.
- The former faces the Pacific Ocean. The Antarctic Peninsula points towards south America. It is the continuation of the Andes Mountain range.
- The latter, East Antarctica, faces the Atlantic and Indian Oceans, Mount Erebus, an active volcano, is actually of the Ross Sea.
- It is the only continent that is completely covered by permanent ice and snow hence it is known as the **white continent**.
- In some places its ice cap is 4,000 metres deep.
- The valleys between the mountain ranges are dry, windy, frozen and barren and strangely called **oases**.

Climate

- The climate of Antarctica is frozen cold because of its distance from the Equator and because of the great height of the plateau.

- In the winter months of May, June and July the sun never rises and the temperature at the South Pole falls to minus 90°C.
- In the summer months of December, January and February, the sun never sets and there continuous daylight. The summer temperature is about 0°C.
- Extremely cold and icy winds blow throughout the year.
- There is a marked difference between the summer and winter temperatures.
- There is also a vast difference between the temperatures of the continental interior.
- Most parts of the continent is dry with an average of 5 centimetres of rain annually.
- Antarctica is a cold desert.
- Mosses and lichens which cling to rocky slopes are found along the coast.
- There are scattered clumps of coarse grass and flowering plants in a few places where the climate is mild.

Aurora

- In winter, there is continuous night for 3 months in the polar regions. Curtains of brilliant coloured lights appear on these dark nights. They are caused by magnetic storms in the upper atmosphere. They are called **Aurora Australis** in the south and **Aurora Borealis** in the north.

Minerals

- Scientific studies have shown Antarctica to be rich in gold, platinum, nickel, copper and petroleum.
- But by the international Agreement this continent is to be used only by the scientists to study the climate of the Earth and the origin of its crust.
- About 70 per cent of the Earth's supply of fresh water can be extracted from the ice-caps of Antarctica.

Expeditions of Antarctica

- In 1912, a dramatic contest to reach the South Pole was held by two teams. The British team was led by Captain Robert F. Scott and the Norwegian team by Roald Amundsen.
- The five British men reached the pole only to find the Norwegian flag flying at the South Pole. They had been beaten by 34 days.
- **Indian expedition** to Antarctica had a 21 member team with Dr. S. Z. Quasim as its leader. It left from Goa on the 6th December, 1981, and landed on the frozen continent on 9th January 1982.
- They set up a scientific station called **Dakshin Gangotri** laid plans for a second base called **Maitri** and named a point **Mount Indira**.
- They left automatic weather recorders powered by solar batteries at the stations.
- South Pole is 2,250 kilometres away from Dakshin Gangotri.

Exercise -1

1. Which one of the following straits is nearest to the International Date Line?
(a) Malacca Strait (b) Bering Strait
(c) Strait of Florida (d) Strait of Gibraltar
2. Itaipu Dam built on the river Parana is one of the largest dams in the world. Which of the following two countries have this as a joint project?
(a) Brazil and Peru (b) Paraguay and Ecuador
(c) Brazil and Paraguay (d) Colombia and Paraguay
3. Other than India and China, which one of the following groups of countries border Myanmar?
(a) Bangladesh, Thailand and Vietnam
(b) Cambodia, Laos and Malaysia
(c) Thailand, Vietnam and Malaysia
(d) Thailand, Laos and Bangladesh
4. Which one of the following lakes forms an international boundary between Tanzania and Uganda?
(a) Chained (b) Malawi
(c) Victoria (d) Zambezi
5. Which one of the following countries of Africa finds a place in high human development category of UNDP?
(a) South Africa (b) Kenya
(c) Zimbabwe (d) Mauritius
6. The ports on either end of the Suez canal are
(a) Cairo and Alexandria
(b) Suez and Cairo
(c) Cairo and Port Said
(d) Port Said and Suez
7. Which one of the following town is located in the New England state of the USA?
(a) Boston (b) Chicago
(c) Milwaukee (d) St. Paul
8. Most industrialized country in ASEAN is
(a) Philippines (b) Thailand
(c) Malaysia (d) Singapore
9. Which one of the following is the most industrialized country of Latin America?
(a) Brazil (b) Chile
(c) Colombia (d) Argentina
10. Which of the following is the lowest point of Africa?
(a) Lake Assal (b) Lake Tanganyika
(c) Lake Malawi (d) Lake Chad
11. Which is the shortest route from Moscow to San Francisco?
(a) via Canada (b) over land
(c) over the South Pole (d) over the North Pole
12. Which three continents have the largest areas of coniferous forest ?
(a) Australia, Europe and South America
(b) Asia, North America and Europe
(c) North America, South America and Europe
(d) Asia, Australia and South America
13. Which one the following port cities in Venezuela has been developed as an oil port?
(a) Caracas (b) Maracaibo
(c) Maracay (d) Carupano
14. In which part of the world would you find the Veld ?
(a) Argentina (b) Australia
(c) Central Europe (d) South Africa
15. The satellites of which one of the following countries have helped in the preparation of a detailed and complete map of Antarctica?
(a) Canada (b) France
(c) Russia (d) U.S.A
16. Which one of the following countries is the leading producer of Uranium?
(a) United States of America
(b) Canada
(c) Germany
(d) Zambia
17. Which one of the following countries has replaced Italy as the major importer of bauxite from India?
(a) Canada (b) Greece
(c) Ukraine (d) United Arab Emirates
18. Grand Banks is one of the important producers of
(a) Nitrate (b) Iron ore
(c) Marine fish (d) Bauxite
19. Plantation agriculture is most widespread in
(a) Nile valley (b) Mississippi valley
(c) California (d) Caribbean
20. The largest iron-producing region of USA is
(a) North East Appalachian region
(b) Alabama state
(c) Western region
(d) Lake Superior region
21. Which one of the following cities in USA is a steel manufacturing centre?
(a) New York (b) Rochester
(c) Chicago (d) Dallas
22. Port Stanley is the capital of the
(a) West Indies
(b) Falkland Island
(c) Hawaii Island
(d) Madagascar (Nayalgasey)
23. Which one of the following is the characteristic climate of the Tropical Savannah Region?
(a) Rainfall throughout the year
(b) Rainfall in winter only
(c) An extremely short dry season
(d) A definite dry and wet season
24. In the tropical zone of Africa the sequence of climates on both sides of the equator is
(a) rainy tropical, monsoon tropical, semi-arid tropic, arid tropic
(b) rainy tropical, semi-arid tropic, arid tropic, monsoon tropical
(c) rainy tropical, semi-arid tropic, monsoon tropical, arid tropic
(d) monsoon tropical, rainy tropical, semi-arid tropic, arid tropic

25. Which one of the following countries does not border Mediterranean Sea ?
 (a) Malta (b) Libya
 (c) Italy (d) Bulgaria
26. The wet forest of the Amazon basin are known as which one of the following ?
 (a) Campos (b) Lianos
 (c) Pampas (d) Selvas
27. Which one of the following is a land-locked country?
 (a) Angola (b) Mozambique
 (c) Namibia (d) Zimbabwe
28. In which one of the following regions of the world, are the grasslands called Campos found ?
 (a) Brazil (b) China
 (c) Eurasia (d) North America
29. Savanna natural region is characterized by which one of the following?
 (a) A distinct wet and dry season with annual range of temperature between 3°C – 8°C
 (b) Broad-leaf evergreen forests and grasses
 (c) Uniformly high temperature throughout the year
 (d) No spatial variation in mean annual rainfall
30. Which forests lie in the Amazon Basin in South America?
 (a) Montane forests
 (b) Tropical rain forests
 (c) Wet deciduous forests
 (d) Subtropical mixed forests
31. Which one of the following is the busiest ocean route in the world?
 (a) Mediterranean Suez Route
 (b) South Atlantic Route
 (c) North Atlantic Route
 (d) Pacific Ocean Route
32. The sea of Okhotsk is enclosed by the Kurille Islands, Northern Japan and a long peninsula. What is the name of this peninsula?
 (a) Kamachatka (b) Sakhalin
 (c) Hokaido (d) Korea
33. Which of the following two gulfs are connected by Hormuz Strait?
 (a) Persian Gulf—Gulf of Oman
 (b) Persian Gulf—Gulf of Aden
 (c) Gulf of Aden—Gulf of Oman
 (d) Persian Gulf—Gulf of Aquaba
34. The Kiel canal connects
 (a) Caribbean sea and Pacific Ocean
 (b) Bering sea and Chukchi sea
 (c) Red sea and Mediterranean sea
 (d) Baltic sea and North sea
35. Which of the following is the highest mountain peak of the U.S.A.?
 (a) Albert (b) Kilauea
 (c) Mauna Lao (d) McKinley
36. The water fall 'Victoria' is associated with the river.
 (a) Amazon (b) Missouri
 (c) St. Lawrence (d) Zambezi
37. Which one of the following volcanoes is called the Light House of the Mediterranean?
 (a) Etna (b) Vesuvius
 (c) Kilimanjaro (d) Stromboli
38. Mount Fujiyama, the highest point in Japan is in
 (a) Hokkaido (b) Kyushu
 (c) Shikoku (d) Honshu
39. The correct sequence of languages in descending order in terms of their number of speakers in the world is
 (a) Spanish, English, Chinese, Hindi
 (b) English, Chinese, Hindi, Spanish
 (c) Hindi, Chinese, Spanish, English
 (d) Chinese, English, Hindi, Spanish
40. Which one of the following pairs is not correctly matched?
 (a) Davis strait — Baffin Sea and Atlantic Ocean
 (b) Dover strait — Arctic Sea and N. Atlantic Ocean
 (c) Palk strait — Mannar Gulf and Bay of Bengal
 (d) Sunda strait — Java Sea and Indian Ocean
41. The correct decreasing order of the population of the various continents is
 (a) Asia, Europe, America, Africa, Australia.
 (b) Asia, America, Europe, Africa, Australia.
 (c) Asia, Europe, Africa, America, Australia.
 (d) Asia, America, Africa, Europe, Australia.
42. Which one of the following is not correctly matched?
 (a) Duluth — Lake Superior
 (b) Detroit — Lake Huron
 (c) Chicago — Lake Michigan
 (d) Ottawa — Lake Ontario
43. Which of the following pairs of straits and the countries they separate is wrongly matched?
 (a) Gibraltar strait — Spain and Morocco
 (b) Bering strait — Sumatra and Malaysia
 (c) Magellan strait — Chile and Tierra del fuego
 (d) Bass strait — Australia and Tasmania
44. Currently half of the world's population lives in just six countries. Identify them from the following.
 (a) India, China, Pakistan, Brazil, Bangladesh, Indonesia.
 (b) India, China, Brazil, Pakistan, Bhutan, United State.
 (c) China, India, United State, Indonesia, Brazil, Pakistan.
 (d) China, India, Bangladesh, United State, Pakistan, Brazil.
45. Which of the following is not correctly matched?
 (a) Indonesia — Jakarta
 (b) Maldives — Male
 (c) North Korea — Seoul
 (d) Zimbabwe — Harare
46. Which one of the following 'City River' pairs is not correctly matched?
 (a) Berlin — Rhine
 (b) London — Thames
 (c) New York — Hudson
 (d) Vienna — Danube
47. Which one of the following pairs is correctly matched?
- | Geographical Feature | Region |
|------------------------|------------------------|
| (a) Abyssinian Plateau | : Arabia |
| (b) Atlas Mountains | : North-Western Africa |
| (c) Guiana Highlands | : South-Western Africa |
| (d) Okavango Basin | : Patagonia |

48. "Climate is extreme, rainfall is scanty and the people used to be nomadic herders."
The above statement best describes which of the following regions?
(a) African Savannah
(b) Central Asian Steppe
(c) North American Prairie
(d) Siberian Tundra
49. Which of the following straits is not in Asia?
(a) Malacca Strait (b) Bass Strait
(c) Formosa Strait (d) Molucca Strait
50. Which of the following seas/gulfs are connected by the Suez Canal?
(a) The Mediterranean Sea and the Red Sea
(b) Gulf of Oman and the Red Sea
(c) The Mediterranean Sea and the Gulf of Oman
(d) Persian Gulf and the Arabian Sea
51. The broken hills famous for zinc and lead are located in
(a) Turkey (b) France
(c) Germany (d) Australia
52. Climate change resulting in the rise of temperature may benefit which of the countries/regions?
(a) South Africa
(b) East Indies islands comprising of Java, Sumatra and Borneo
(c) The Western coasts of South America
(d) Russia and Northern Europe
53. Which one among the following is not a landlocked country in Africa?
(a) Botswana (b) Zambia
(c) Lesotho (d) Nigeria
54. Savanna natural region is characterized by which one of the following?
(a) A distinct wet and dry season with annual range of temperature between 3°C – 8°C
(b) Broad-leaf evergreen forests and grasses
(c) Uniformly high temperature throughout the year
(d) No spatial variation in mean annual rainfall
55. Which forests lie in the Amazon Basin in South America?
(a) Montane forests
(b) Tropical rain forests
(c) Wet deciduous forests
(d) Subtropical mixed forests
56. The Victoria Falls in Africa is located on which river?
(a) Zaire (b) Orange
(c) Zambezi (d) Niger
57. Mount Merapi is located on which one of the islands?
(a) Java (b) Sumatra
(c) Borneo (d) Celebes
58. In which one of the following region are the Andes mountains located?
(a) East Europe (b) West Europe
(c) South Africa (d) South America
59. Temperature and rainfall of a meteorological station are given below:

	Temperature (0°C)	Rainfall (cm)
J	9.4	12.2
F	10.6	9.1
M	11.7	7.9

A	12.2	2.5
M	13.3	1.0
J	13.9	0.3
J	13.9	-
A	14.4	-
S	15.6	0.8
	15.0	2.5
N	13.3	6.1
D	10.6	11.7

Average temperature: 12.8°C

Average rainfall: 54.9 cm per annum

Identify the region having the above climatic patterns from amongst the following:

- (a) Mediterranean region
(b) Monsoon region
(c) Steppe region
(d) N.W. European region
60. Claims to the historical Macedonian territory have been a bone of contention between which of the following countries?
(a) Portugal and Spain (b) Bulgaria and Greece
(c) Romania and Bulgaria (d) Portugal and Greece
61. Through which one of the following Straits, does a tunnel connect the United Kingdom and France?
(a) Davis Strait (b) Denmark Strait
(c) Strait of Dover (d) Strait of Gibraltar
62. The largest coral reef in the world is found near the coast of which one of the following countries?
(a) Australia (b) Cuba
(c) Ghana (d) Philippines
63. A class of animals known as Marsupials is a characteristic feature of
(a) Africa (b) Australia
(c) South America (d) South-east Asia
64. The Great Barrier Reef is broken in gaps due to
(a) outpouring of river water into the sea
(b) wave action
(c) oceanic currents
(d) tropical cyclones
65. Outside the Mediterranean region, where else is a Mediterranean climate found ?
(a) The Japanese and Chinese coasts
(b) The southern coast of Australia and parts of the western coast of North and South America
(c) The east coast of North America
(d) The North Atlantic coast of Africa
66. Which one of the following is the continent with the highest mean elevation in the world?
(a) Antarctica (b) North America
(c) Asia (d) South America
67. Which one of the following is the correct sequence of the given towns of Pakistan while moving from the North towards the South?
(a) Islamabad–Gujranwala–Peshawar–Multan
(b) Peshawar–Gujranwala–Multan–Islamabad
(c) Peshawar–Islamabad–Gujranwala–Multan
(d) Islamabad–Multan–Peshawar–Gujranwala

68. Which one of the following pairs is not correctly matched?
 (a) Bahamas : Nassau
 (b) Costa Rica : San Jose
 (c) Nicaragua : Balmopan
 (d) Dominican Republic: Santo Domingo
69. Which among the following has the world's largest reserves of Uranium?
 (a) Australia (b) Canada
 (c) Russian Federation (d) USA
70. Through which one of the following groups of countries does the Equator pass?
 (a) Brazil, Zambia and Malaysia
 (b) Colombia, Kenya and Indonesia
 (c) Brazil, Sudan and Malaysia
 (d) Venezuela, Ethiopia and Indonesia
71. Which one of the following cities is known as the Pittsburg of Japan?
 (a) Kobe (b) Yuwaha
 (c) Tokyo (d) Osaka
72. Intensive subsistence farming is prevalent in
 (a) Canada (b) China
 (c) USA (d) Ukraine
73. Baku is famous as
 (a) Railway junction (b) Industrial city
 (c) Fish port (d) Petroleum extracting centre
74. Ural industrial region is dominated by
 (a) Textiles (b) Metal fabricating
 (c) Electronics (d) Automobiles
75. The best inland waterways system is developed in
 (a) Guinea coast (b) Central Europe
 (c) Amazon basin (d) North America
76. Which one of the following is known as the Manchester of the East?
 (a) Tokyo (b) Shanghai
 (c) Kuala Lumpur (d) Osaka
77. Grozny is famous for the production of
 (a) Petroleum (b) Uranium
 (c) Gold (d) Thorium
78. Which country has the densest railway network in terms of per million people?
 (a) China (b) USA
 (c) India (d) Australia
79. The first three countries in order of iron-steel production in the world are
 (a) USA, China, Germany
 (b) Japan, USA, China
 (c) Germany, China, USA
 (d) Japan, Germany, China
80. Flanders industrial region is renowned for
 (a) Iron & steel industry
 (b) Car manufacturing
 (c) Textile manufacturing
 (d) Electronics
81. The third position in the manufacture of cars in the world is occupied by
 (a) Japan (b) USA
 (c) France (d) Germany
82. Largest producer of silver in the world is
 (a) Mexico (b) USA
 (c) Laos (d) South Africa
83. The main centres of ship building industry in Japan are
 (a) Nagasaki and Yawata
 (b) Yakohama and Wakohama
 (c) Osaka and Kyoto
 (d) Nagita and Issikari
84. The three leading coal producers of the world are:
 (a) USA, Russia, Germany
 (b) England, France, Australia
 (c) France, Poland, Japan
 (d) Germany, India, South Africa
85. The regions with highest and lowest densities are respectively
 (a) Europe and Africa
 (b) Asia and Oceania
 (c) North America and Africa
 (d) Europe and Oceania
86. Which of the following countries are the best examples of highly scientific development and management of pastoral farming?
 (a) France and Spain
 (b) Denmark and Holland
 (c) Czechoslovakia and Poland
 (d) Greenland and Norway
87. "From Aceh in the far north west to Torres Strait in the east 5000 miles, almost as far as from London to Baghdad. The archipelago has 14,000 islands, some mere equatorial rocks, others some of the largest in the world." This description best fits:
 (a) West Indies (b) Japan
 (c) Philippines (d) Indonesia
88. Which one of the following pairs is not correctly matched?
- | Well-known Place | Country |
|------------------|---------------|
| (a) Baikonour | Russia |
| (b) Kourou | French Guiana |
| (c) Borobudur | Indonesia |
| (d) Cannes | France |
89. In which one of the following areas is monsoon climate found?
 (a) Pacific Coast of Columbia
 (b) South-Eastern United States
 (c) Southern Part of South Africa
 (d) Central California
90. Turkey is located between
 (a) Black Sea and Caspian Sea
 (b) Black Sea and Mediterranean Sea
 (c) Gulf of Suez and Mediterranean Sea
 (d) Gulf of Aqaba and Dead Sea
91. Which water body separates Australia from New Zealand?
 (a) Cook Straits (b) Megallan
 (c) Tasman Sea (d) Great Barrier Reef
92. The pacific Islands from new Guinea South East-wards to the Fiji Islands' group is called
 (a) The Polynesia (b) The Mellanesia
 (c) The Micronesia (d) The Autralasia

93. Which one among the following is not a Baltic nation?
 (a) Latvia (b) Slovakia
 (c) Lithuania (d) Estonia
94. Which one of the following mountain ranges separates Europe from Asia?
 (a) Apenine (b) Black Forest
 (c) Ural (d) Sulaiman
95. Which one of the following pairs is not correctly matched?
 (a) Kuroshio : Warm ocean current
 (b) Labrador : Warm ocean current
 (c) Benguela : Cold ocean current
 (d) Oyashio : Cold ocean current
96. Mr X has been invited to participate in a conference to be held at Buenos Aires. He has chosen the following shortest flight route
 Mumbai to Frankfurt (non-stop)
 Frankfurt to Sao Paulo (non-stop)
 Sao Paulo to Buenos Aires (non-stop)
 Which one of the following seas will likely to be flown over by Mr X ?
 (a) Tasman Sea (b) Labrador Sea
 (c) Beaufort Sea (d) Black Sea
97. Consider the following statements :
 1. The Labrador current is a cold current in the North Atlantic Ocean.
 2. The Falkland current is a warm current that flows along the Chile coast of South Pacific Ocean.
 Which of the statements given above is/are correct ?
 (a) Only 1 (b) Only 2
 (c) Both 1 and 2 (d) Neither 1 nor 2
98. What is the name given to an almost circular coral reef inside which there is a lagoon ?
 (a) Fringing reef (b) Barrier reef
 (c) Atoll (d) Isthmus
99. In which one of the following region are the Andes mountains located?
 (a) East Europe (b) West Europe
 (c) South Africa (d) South America
100. The equatorial rain forest is also known as
 (a) Savanna (b) Campos
 (c) Selva (d) Lanose
101. In which one of the following regions of the world, are the grasslands called Campo found ?
 (a) Brazil (b) China
 (c) Eurasia (d) North America
102. Which one of the following pairs of islands is separated from each other by the 'Ten Degree Channel'?
 [CSAT 2014 - I]
 (a) Andaman and Nicobar
 (b) Nicobar and Sumatra
 (c) Maldives and Lakshadweep
 (d) Sumatra and Java
103. Turkey is located between [CSAT 2014 - I]
 (a) Black Sea and Caspian Sea
 (b) Black Sea and Mediterranean Sea
 (c) Gulf of Suez and Mediterranean Sea
 (d) Gulf of Aqaba and Dead Sea
104. Which one of the following countries of South-West Asia does not open out to the Mediterranean Sea?
 [CSAT 2015-I]
 (a) Syria (b) Jordan
 (c) Lebanon (d) Israel

Exercise -2

Statement Based MCQ

1. What could be the main reason/reasons for the formation of African and Eurasian desert belt ?
 1. It is located in the sub-tropical high pressure cells.
 2. It is under the influence of warm ocean currents.
 Which of the statements given above is/are correct in this context ?
 - (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
2. Bermuda Triangle extends up to which of the following places?
 1. Southern Florida
 2. Puerto Rico
 3. Hawaii Islands
 Select the correct answer using the codes given below:
 - (a) 1, 2 and 3
 - (b) 1 and 2
 - (c) 2 and 3
 - (d) 1 and 3
3. Consider the following countries:
 1. Australia
 2. Namibia
 3. Brazil
 4. Chile
 Through which of the above does the Tropic of Capricorn pass?
 - (a) 1 only
 - (b) 2, 3 and 4
 - (c) 1, 2 and 3
 - (d) 1, 2, 3 and 4
4. Consider the following statements :
 1. Tropical deciduous forests are found only in India, Central America and Brazil.
 2. Teak is the most important hardwood tree of tropical evergreen forests.
 Which of the statements given above is/are correct ?
 - (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
5. Consider the following statements :
 1. In Savanna climate, wet summers alternate with dry winters.
 2. In China type of climate, winters are very severe and summers are mild and dry.
 Which of the statements given above is/are correct ?
 - (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
6. Arrange the following continents in the ascending order of their areal size and select the correct answer from the codes given below:
 1. Antarctica
 2. Australia
 3. Europe
 4. South America
 - (a) 1, 2, 3, 4
 - (b) 2, 4, 1, 3
 - (c) 2, 3, 1, 4
 - (d) 1, 3, 4, 2
7. Japan is one of the leading industrial countries in the world because it has:
 1. developed hydel power
 2. large deposits of metallic mineral
 3. high technological capability
 4. insular location
8. Of these statements:
 - (a) 1, 2 and 4 are correct
 - (b) 1, 2 and 3 are correct
 - (c) 1 and 3 are correct
 - (d) 2 and 4 are correct
9. Which of the following are temperate grasslands? Select the correct answer from the codes given below:
 1. Campos
 2. Llanos
 3. Downs
 4. Pampas
 - (a) 1 and 2
 - (b) 2 and 3
 - (c) 3 and 4
 - (d) 1 and 4
10. Arrange the following islands of Japan in ascending order of their areal size and select the correct answer from the codes:
 1. Hokkaido
 2. Honshu
 3. Shikoku
 4. Kyushu
 - (a) 4, 3, 2, 1
 - (b) 1, 2, 3, 4
 - (c) 2, 4, 1, 3
 - (d) 3, 4, 1, 2
11. La Nina is suspected to have caused recent floods in Australia. How is La Nina different from El Nino ?
 1. La Nina is characterized by unusually cold ocean temperature in equatorial Indian Ocean whereas El Nino is characterized by unusually warm ocean temperature in the equatorial Pacific Ocean.
 2. El Nino has adverse effect on south-west monsoon of India, but La Nina has no effect on monsoon climate.
 Which of the statements given above is/are correct?
 - (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
12. Consider the following statements
 1. In Macedonia, ethnic Albanians are a minority.
 2. In Kosovo, Serbians are a majority.
 Which of these statements is/are correct?
 - (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
13. Which of the following items and their leading producers are correctly matched?
 1. Grapes - Italy
 2. Oilseeds - India
 3. Millets - India
 4. Coffee - Ethiopia
 Which of the above statement are correct ?
 - (a) 1, 3 and 4
 - (b) 2, 3 and 4
 - (c) 1, 2 and 3
 - (d) 1, 2, 3 and 4
14. A geographic region has the following distinct characteristics:
 1. Warm and dry climate
 2. Mild and wet winter
 3. Evergreen Oak trees
 The above features are distinct characteristics of which one of the following regions?
 - (a) Mediterranean
 - (b) Eastern China
 - (c) Central Asia
 - (d) Atlantic coast of North America
15. What is the correct sequence of occurrence of the following cities in South-East Asia as one proceeds from south to north?
 1. Bangkok
 2. Hanoi
 3. Jakarta
 4. Singapore

Select the correct answer using the code given below.

- (a) 4-2-1-3 (b) 3-2-4-1
(c) 3-4-1-2 (d) 4-3-2-1

15. Consider the following statements with reference to doldrums:

1. Doldrums comprise the equatorial belt of low atmospheric pressure.
2. The trade winds converge at doldrums.
3. There is strong upward movement of air in doldrums.
4. Doldrums are characterized by turbulent weather.

Which of the statements given above are correct ?

- (a) 1 and 2 only (b) 3 and 4 only
(c) 1, 2 and 3 only (d) 1, 2, 3 and 4

16. Consider the following pairs :

Region Country

1. Chechnya : Russian Federation
2. Darfur : Mali
3. Swat Valley : Iraq

Which of the above pairs is/are correctly matched?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

DIRECTIONS (Qs. 17 to 19) : The following questions consist of two statements, one labelled as the Assertion (A) and the other as 'Reason (R). You are to examine these two statements carefully and select the answers to these items using the codes given below:

17. **Assertion (A) :** The Steppe climate is most widespread in North America and Eurasia.

Reason (R) : The Steppe climate is characterized by wet summers as well as wet winters.

- (a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Major natural regions of the world possess similar topography.

Reason (R): Climate is a major influencing factor in a natural region.

- (a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

19. **Statement I:** Temperatures of countries like United Kingdom, Norway, the Netherlands and Denmark are higher as compared to places located on similar latitudes during the winter.

Statement II: United Kingdom, Norway, the Netherlands and Denmark are located on the coast.

- (a) Both the statements are individually true and Statement II is the correct explanation of Statement I.
(b) Both the statements 'are individually true but Statement II is not the correct explanation of Statement I.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

20. Which among the following statements about the North Atlantic Drift is/are correct?

1. It keeps the west coast of Northern Europe ice free
2. It is responsible for the warm air mass which interacts with the cold air mass from the Polar region and causes rainfall in Western Europe
3. It meets the Labrador current near Vancouver Island and causes dense fog

Select the correct answer using the code given below

- (a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 only (d) 1 and 3 only

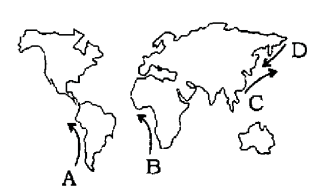
21. What is the correct sequence of occurrence of the following cities in South-East Asia as one proceeds from south to north? [CSAT 2014 - I]

1. Bangkok
2. Hanoi
3. Jakarta
4. Singapore

Select the correct answer using the code given below.

- (a) 4-2-1-3 (b) 3-2-4-1
(c) 3-4-1-2 (d) 4-3-2-1

22. Match **List-I** with **List-II** and select the correct answer using the code given below the Lists :

List-I	List-II
(Map showing ocean current)	(Name of ocean current)
	1. Kuroshio
	2. Humboldt
	3. Benguela
	4. Oyashio

Codes :

- | | | | | | | | |
|-------|---|---|---|-------|---|---|---|
| A | B | C | D | A | B | C | D |
| (a) 2 | 1 | 3 | 4 | (b) 4 | 3 | 1 | 2 |
| (c) 4 | 1 | 3 | 2 | (d) 2 | 3 | 1 | 4 |

Matching Based MCQ

DIRECTIONS (Qs. 23 to 34) : Match List-I with List-II and select the correct answer using the codes given below the lists.

- | List-I (City) | List-II (Industry) |
|----------------|--------------------|
| (A) Osaka | (1) Shipbuilding |
| (B) Hamburg | (2) Textiles |
| (C) Detroit | (3) Iron & Steel |
| (D) Pittsburgh | (4) Automobiles |
- (a) A - 2 ; B - 1 ; C - 4 ; D - 3
(b) A - 1 ; B - 2 ; C - 3 ; D - 4
(c) A - 2 ; B - 3 ; C - 1 ; D - 4
(d) A - 4 ; B - 1 ; C - 2 ; D - 3
- | List-I | List-II |
|------------------|----------------|
| (A) Iron & Steel | (1) Portsmouth |
| (B) Automobile | (2) Dortmund |
| (C) Shipbuilding | (3) Bangalore |
| (D) Aircraft | (4) Detroit |
- (a) A - 1 ; B - 2 ; C - 3 ; D - 4
(b) A - 2 ; B - 1 ; C - 4 ; D - 3
(c) A - 4 ; B - 3 ; C - 2 ; D - 1
(d) A - 2 ; B - 4 ; C - 1 ; D - 3

25. **List-I** **List-II**
(Crop) **(Main Producing area)**
 (A) Coconut (1) Kenya
 (B) Banana (2) Papua New Guinea
 (C) Groundnut (3) Ecuador
 (D) Tea (4) Senegal

26. Which of the following pairs are correctly matched?

(River)		(Countries)
1. Niger	—	Guinea
2. Orange	—	S. Africa
3. Karai	—	Egypt
4. Nile	—	Mali

Select the correct answer using the codes given below:

- (a) 1 and 2 (b) 1 and 3
 (c) 1, 3 and 4 (d) 2, 3 and 4

27. Match List-I and List-II and select the correct answer from the codes given below:

List-I		List-II
(Type of winds)		(Region)
A. Blizzard		1. Prairie Plains
B. Chinook		2. Siberian Plains
C. Bora		3. Greenland
D. Khamsin		4. Egypt

	A	B	C	D
(a)	3	1	4	2
(b)	1	2	3	4
(c)	2	1	3	4
(d)	4	3	1	2

28. Match List-I with List-II and select the correct answer using the code given below:

List-I		List-II
(Current)		(Feature)
A. Kuroshio current		1. Warm current in Atlantic Ocean
B. Peru current		2. Cold current in the Atlantic Ocean
C. Labrador current		3. Warm current in the Pacific Ocean
D. Florida current		4. Cold current in the Pacific Ocean

	A	B	C	D
(a)	3	4	2	1
(b)	3	2	4	1
(c)	1	4	2	3
(d)	1	2	4	3

29. Match List-I with List-II and select the correct answer from the codes given below.

List-I		List-II
(Mountains peak)		(Country)
A. Mt. Everest		1. India
B. K2		2. Nepal
C. Mt. McKinley		3. North America (USA)
D. Mt. Aconcagua		4. South America

Codes:

	A	B	C	D
(a)	2	1	3	4
(b)	1	2	3	4
(c)	4	3	2	1
(d)	3	4	1	2

30. Match List-I with List-II and select the correct answer from the codes given below:

List-I		List-II
(Countries new name)		(Countries old name)
A. Thailand		1. Nippon
B. Ghana		2. Gold Coast
C. Zambia		3. Siam
D. Japan		4. Northern Rhodesia

	A	B	C	D
(a)	3	2	4	1
(b)	1	2	3	4
(c)	2	3	1	4
(d)	4	3	2	1

31. **List-I** **List-II**
(Geographic feature) **(Country)**

(A) Great Victoria Desert	(1) Australia
(B) Grand Canyon	(2) Canada
(C) Lake Winnipeg	(3) New Zealand
(D) Southern Alps	(4) USA

- (a) A - 1 ; B - 2 ; C - 4 ; D - 3
 (b) A - 1 ; B - 4 ; C - 2 ; D - 3
 (c) A - 3 ; B - 2 ; C - 4 ; D - 1
 (d) A - 3 ; B - 4 ; C - 2 ; D - 1

32. **List I (Sea)** **List II (Country)**

(A) Black Sea	(1) Bulgaria
(B) Red Sea	(2) China
(C) Yellow Sea	(3) Eritrea
(D) Caspian Sea	(4) Kazakhstan

- (a) A - 1 ; B - 4 ; C - 2 ; D - 3
 (b) A - 2 ; B - 3 ; C - 1 ; D - 4
 (c) A - 1 ; B - 3 ; C - 2 ; D - 4
 (d) A - 2 ; B - 4 ; C - 1 ; D - 3

33. **List-I (City)** **List-II (River)**

(A) Washington D.C.	(1) River Manzanares
(B) Berlin	(2) River Seine
(C) Paris	(3) River Spree
(D) Madrid	(4) River Potomac

- (a) A - 2 ; B - 3 ; C - 4 ; D - 1
 (b) A - 4 ; B - 1 ; C - 2 ; D - 3
 (c) A - 2 ; B - 1 ; C - 4 ; D - 3
 (d) A - 4 ; B - 3 ; C - 2 ; D - 1

34. Match the following

List I	List II
(Natural Vegetation of India)	(Annual Rainfall Received)
A. Tropical evergreen forests	1. 100-200 cm
B. Tropical deciduous forests	2. Above 200 cm
C. Tropical dryforests	3. Less than 50 cm
D. Arid forests	4. Above 300 cm
	5. 50-100 cm

Codes

	A	B	C	D
(a)	1	2	5	3
(b)	4	3	1	5
(c)	2	1	5	3
(d)	2	1	3	4

Hints and Explanations

EXERCISE-1

1. (b) Bering Strait is nearest to the International Date line, because the international Date line runs equidistant between the American continents, on its East and Asia, Australia, and Europe on its west.
2. (c) Itaipu Dam is built on River Parana is the largest dam in the world. This is a joint project of Brazil – Paraguay.
3. (d) Thailand, Laos and Bangladesh are countries bordering Myanmar.
4. (c) Lake Victoria acts as the international boundary between Tanzania and Uganda. It is the largest lake in Africa and second widest fresh water body in the world.
5. (d) 6. (d) 7. (a) 8. (c) 9. (a)
10. (a) 11. (d) 12. (b)
13. (b) Maracaibo is the oil-port of Venezuela.
14. (d)
15. (a) The satellite launched by Canada has helped in the preparation of a detailed and complete map of Antarctica.
16. (b) Canada is the largest producer of Uranium, accounting for about 22% of world output. Production of uranium in Canada is mainly from the Mc Arthur River mines in northern Saskatchewan province, which is the largest mine in the world.
17. (a) Canada is the largest importer of bauxite from India.
18. (c) 19. (c) 20. (d)
21. (c) 22. (b)
23. (d) The characteristic of the Tropical Savannah Region is a definite dry and wet season.
24. (a)
25. (d) Bulgaria does not border mediterranean sea, shares the boundary with Black-sea.
26. (d) 27. (d) 28. (a)
29. (c) Savanna subtropical grassland are prevailing in both hemisphere from 5° to 20° latitudes. The main countries of this area are South Africa, Sudan, Venezuela and Brazil. Hanoi is the Capital of Vietnam.
30. (b) The largest expanse of tropical rain forest is found in the Amajon basin of Brazil, extending from the Atlantic Coast of the foot hills of the Ande and from the Guiana Highlands in the north to the tropic of capricorn in the South.
31. (c)
32. (a) 33. (a)
34. (d) The Kiel canal connects Baltic sea and North sea. The Kiel Canal, known as the Kaiser-Wilhelm-Kanal until 1948, is a 98-kilometre long canal in the German state of Schleswig-Holstein. The canal links the North Sea at Brunsbüttel to the Baltic Sea at Kiel-Holtenau
35. (d) Mc kinley is the highest mountain peak of the USA. Mount Mc Kinley or Denali is the highest mountain peak in North America, with a summit elevation of 20,237 feet above sea level. At some 18,000 feet, the base to peak rise is considered the largest of any mountain situated entirely above sea level.
36. (d)
37. (d) Stromboli is called the light house of the Mediterranean. Stromboli is a small island in the Tyrrhenian Sea, off the north coast of Sicily, containing one of the three active volcanoes in Italy.
38. (d)
39. (d)
40. (b) Dover strait – English channel and North sea. The Strait of Dover or Dover Strait is the strait at the narrowest part of the English Channel, marking the boundary between the Channel and North Sea, separating Great Britain from continental Europe.
41. (c)
42. (b) Detroit is situated lake Erie. Detroit is the most populous city in the U.S. state of Michigan, and is the seat of Wayne County, the most populous county in the state and the largest city on the United States – Canada border.
43. (b) Bering strait divides Alaska and Russia.
44. (c) China – India – United State – Indonesia – Brazil – Pakistan
45. (c) North Korea – Pyong-Yong (It is the capital of North Korea.)
46. (a)
47. (b) Atlas mountain ranges are situated in the north western part of Africa. They extend almost 2000 km. They pass through Morocco, Algeria and Tunisia.
48. (b) The central Asian steppes run through Kazakhstan, Turkmenistan, Uzbekistan and Mongolia. The climate here is harsh with dust storms, little to no rainfall and temperature ranging from – 4 to 50 degree Celsius.
49. (b) Bass Strait is a sea strait which separates Tasmania from the Australian mainland.
50. (a) The Suez Canal is an artificial sea-level waterway in Egypt, It connects the Mediterranean Sea and the Red Sea.
51. (d) Broken Hills are located in New South Wales, Australia. It is a mining city.
52. (d) Climate change benefits Russia and Northern Europe as they are in colder region.
53. (d) Nigeria is not a landlocked country in Africa. Its coast in the south lies on the Gulf of Guinea in the Atlantic Ocean.
54. (c) Sovanna subtropical grassland are prevailing in both hemisphere from 5° to 20° latitudes. The main countries of this area are South Africa, Sudan, Venezuela and Brazil. Hanoi is the Capital of Vietnam.
55. (b) The largest expanse of tropical rain forest is found in the Amajon basin of Brazil, extending from the

- Atlantic Coast of the foot hills of the Andes and from the Guiana Highlands in the north to the tropic of Capricorn in the South.
56. (c) Lake Victoria is the largest lake in Africa and is the source of River Nile, the longest river in the world.
57. (a)
58. (d) Andes mountains are located in the western part of South America, stretch for about 7200 kilometers from Venezuela in the north to Tierra del Fuego in the south.
59. (c)
60. (b) Macedonian territory is disputed by Bulgaria and Greece.
61. (c) Strait of Dover connects United Kingdom and France. It is about 32 km in length.
62. (a) The largest coral reef is the Great Barrier Reef, located just off the north-eastern coast of Australia. The 1200 mile (1900 km) long reef is protected as a Marine Park.
63. (b) Marsupials are a group of mammals commonly have pouches or pocket of skin, where the mother nurses their young. These marsupials are found in Australia. The common example of marsupials are Kangaroo and Koala.
64. (a) 65. (b)
66. (a) The continent Antarctica is present at the highest mean elevation in the world. This is because it is covered by a huge layer of ice, about 7,100 feet (2,200 m) thick.
67. (c) While moving from the North to the South of Pakistan, the correct sequence of the town is Peshwar – Islamabad – Gujranwala – Multan.
68. (c) Managua is the capital of Nicaragua, not Balmopan.
69. (a) Australia has the world's largest uranium reserves. Approximately 24% of the planet's uranium is present in Australia.
70. (b) Colombia, Kenya & Indonesia are countries through which the equator passes.
71. (b) 72. (b) 73. (d) 74. (b) 75. (b)
76. (d) 77. (a) 78. (d) 79. (b) 80. (c)
81. (d) 82. (a) 83. (a) 84. (a) 85. (d)
86. (b)
87. (d) Indonesia consists of 13,677 islands (nearly 14,000) and this country is 5000 miles from Torres Strait.
88. (a) Baikonour is world's first and largest operational space launch facility. It is located in Kazakhstan, not in Russia.
89. (a) Monsoon type climate is found in Pacific Coast of Columbia.
90. (b) Turkey's smaller part is in Southeastern Europe and its larger part in Western Asia which is located between Black Sea and Mediterranean
91. (c) The Tasman Sea separates Australia from New Zealand.
92. (b) Melanesia region consists of the four countries of Vanuatu, Solomon Islands, Fiji and Guinea. It is a sub-region of Oceania extending from the western end of the Pacific Ocean to the Arafura Sea, and eastward to Fiji. Besides these independent countries, Melanesia also includes New Caledonia, Maluku Islands and West Papua.
93. (b) Baltic Nations are those that have shorelines along the Baltic Sea. The group of countries presently referred Baltic States are Estonia, Latvia, and Lithuania. Slovakia is not a Baltic state.
94. (c) The Ural Mountain range forms the natural boundary between Europe and Asia. The Ural Mountains extend about 2,500 km from the Kara Sea to the Kazakh Steppe along the northern border of Kazakhstan. This range marks the northern part of the border between the continents of Europe and Asia.
95. (b) The Labrador Current is a cold current in the North Atlantic Ocean.
96. (d) Mr X will likely to be flown over Black Sea. Since Black Sea is bounded by Europe, Anatolia and Caucasus, it is ultimately connected to the Atlantic Ocean.
97. (a) The Falkland current is a cold current. It flows along the Argentina's coast in South Atlantic Ocean.
98. (c)
99. (d) Andes mountains are located in the western part of South America, stretch for about 7200 kilometers from Venezuela in the north to Tierra del Fuego in the south.
100. (c) Dense equatorial forest, in the Amazon basin, characterized by tall broad-leaved evergreen trees, epiphytes, lianas, etc.
101. (a)
102. (a) The Ten Degree Channel is a channel that separates the Andaman Islands from the Nicobar Islands in the Bay of Bengal.
103. (b) Turkey's smaller part is in Southeastern Europe and its larger part in Western Asia which is located between Black Sea and Mediterranean
104. (b) Jordan does not open out to the Mediterranean Sea.

EXERCISE-2

1. (a) It is under the influence of cold ocean current. So given second statement is wrong.
2. (b) Bermuda Triangle is a strange and mysterious place in the southern Atlantic Ocean. It is roughly the shape of a triangle. The triangle extends upto south Florida, Puerto Rico and Bermuda Island.
3. (d) The Tropic of Capricorn pass over all the following countries.
4. (c) 5. (a)
6. (c) 2, 3, 1, 4 – Australia – Europe – Antarctica – South America.
7. (c)
8. (c) **Downs** – grassy plains called downs are located in New Zealand's South Island and southeast Australia. **Pampas** – South America's largest grassland, called pampas, which means plain, covers most east-central areas of Argentina.
9. (d) Shikoku – Kyushu – Hokkaido – Honshu
Honshu – Honshu has a total area of 88,017 square miles (227,962 sq km) and it is the world's seventh largest island.

- Hokkaido** – Hokkaido is the second largest island of Japan with a total area of 32,221 square miles (83,453 sq km).
- Kyushu** – It has a total area of 13,761 square miles (35,640 sq km).
- Shikoku** – Shikoku is the smallest of Japan's main islands with a total area of 7,260 square miles (18,800 sq km).
10. (d) La Nina is a coupled ocean-atmosphere phenomenon that is the counterpart of El Nino as part of the broader El Nino-Southern Oscillation climate pattern. During a period of La Nina, the sea surface temperature across the equatorial Eastern Central Pacific Ocean will be lower than normal by 3–5 °C. In the United States. A La Nina, on the other hand, is often beneficial for the monsoon, especially in the latter half.
 11. (b) In Macedonia, ethnic Albanian population is 23%, which is a good number in Macedonia. But in Kosovo 92% Albanians present where Serbians are the minority.
 12. (c)
 13. (a) Mediterranean climate have mild, rainy winter and hot, dry summers and Evergreen, Oak trees.
 14. (c) The correct sequence of occurrence of the following cities in South-East Asia as one proceeds from south to north is Jakarta-Singapore-Bangkok and Hanoi.
 15. (d)
 16. (a) The Chechen Republic is a federal subject (a republic) of Russia. Darfur is a region in western Sudan. Swat is a valley and an administrative district in Pakistan.
 17. (c) The Steppes are areas of comparatively lower temperature and slightly more precipitation. Here annual average temperature is approximately 21°C. Temperate grasses are main vegetation of these regions. Northern America's Prairies and the steppes of Eurasia are the major areas of these grassland.
 18. (d) Major natural regions of the world have not same type of climates. The different climates make different topographies.
 19. (a) Since United Kingdom, Norway, Netherlands and Denmark are located on the coast, their temperature is higher than the places located on similar latitudes during the winter.
 20. (b) The North Atlantic current or Drift or sea movement is a powerful warm ocean current that continues the gulf stream northeast, which stretches from Florida to north-western Europe. It moderates the chilled climate of western Europe.
 21. (c) The correct sequence of occurrence of the following cities in South-East Asia as one proceeds from south to north is Jakarta-Singapore-Bangkok and Hanoi.
 22. (d)

HUMBOLDT	-	South Pacific Cool
BENGUELA	-	South Atlantic Warm/Cool
KUROSHIO	-	North Pacific Warm
OYASHIO	-	North Pacific Cool
 23. (d)
 24. (d)
 25. (a)
 26. (a) The Niger river is the principal river of western Africa, extending about 4,180 km. Its drainage basin is 2,117,700 km² in area. Its source is in the Guinea Highlands in southeastern Guinea. The Orange river, Gariep river, Groote river or Senqu river is the longest river in South Africa. It rises in the Drakensberg mountains in Lesotho, flowing westwards through South Africa to the Atlantic Ocean.
 27. (c) **Blizzard – Siberian Plains.** A blizzard is a severe snowstorm caused by strong sustained winds of at least 56 km/h (35 mph) and lasting for a prolonged period of time – typically three hours or more.
Chinook – Prairie Plains. Chinook winds blow in the interior west of North America, where the Canadian Prairies and Great Plains meet various mountain ranges, although the original usage is in reference to wet, warm coastal winds in the Pacific Northwest of the United States of America.
Bora – Green land. Cold and usually dry katabatic winds, like the Bora, result from the downslope gravity flow of cold, dense air. Katabatic flows slumping down from uplands or mountains may be funneled and strengthened by the landscape and are then known as mountain gap wind.
Khamsin – Egypt. Khamsin can be triggered by depressions that move eastwards along the southern parts of the Mediterranean or along the North African coast from February to June. In Egypt, khamsin usually arrives in April but occasionally occurs between March to May, carrying great quantities of sand and dust from the deserts, with a speed up to 140 kilometres per hour, and a rise of temperatures as much as 20°C in two hours.
 28. (a)

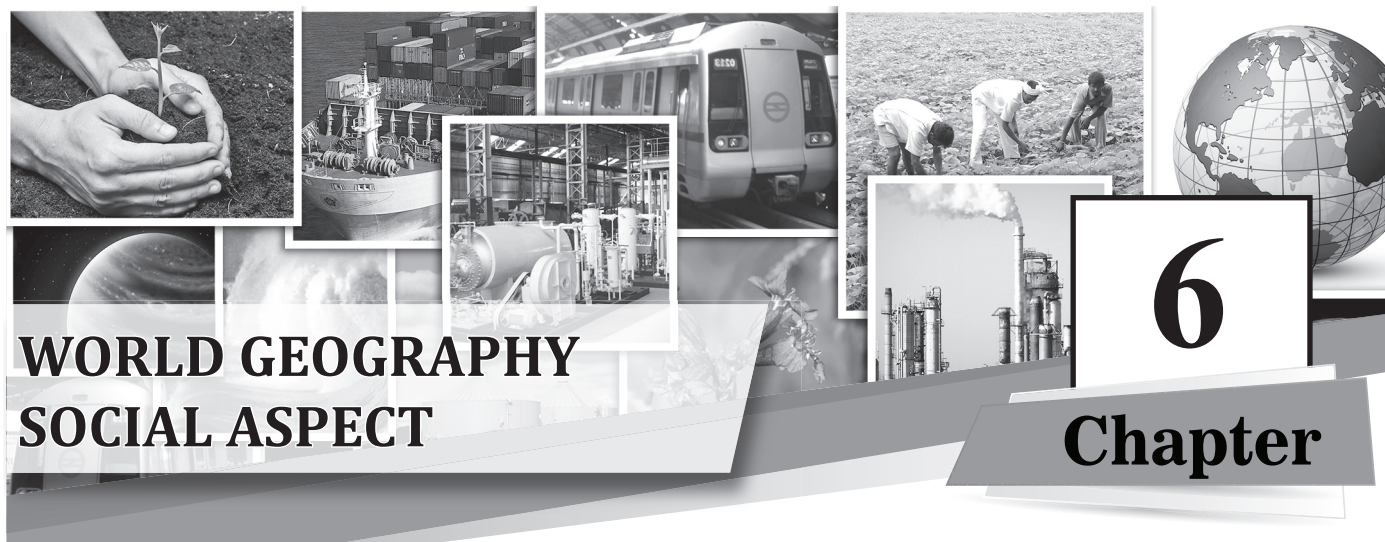
A. Kuroshio current	-	warm Pacific current
B. Peru current	-	cold current in Pacific ocean
C. Labrador current	-	cold current in Atlantic ocean
D. Florida current	-	Warm current in Atlantic ocean
 29. (a)

Mt. Everest	-	Nepal
K2	-	India
Mt. McKinley	-	USA
Mt. Aconcagua	-	South America

 All these are the names of highest mountain peaks located in the countries respectively.
 30. (a)

1. Thailand	-	Siam
2. Ghana	-	Gold Coast
3. Zambia	-	Northern Rhodesia
4. Japan	-	Nippon.
 34. (c)

A. Tropical evergreen forests	-	Above 200 cm
B. Tropical deciduous forests	-	100-200 cm
C. Tropical Dry forest	-	50-100 cm
D. Arid forest	-	Less than 50 cm



WORLD GEOGRAPHY

SOCIAL ASPECT

6

Chapter

As a part of *Human geography*, Cultural geography deals with the socio-cultural aspect of various human groups based upon their geographic location which includes their habitat, clothing, food, shelter, Skills, tools, language, religions, social organization and life style. Basing upon the above criteria each continent has some unique ethnic groups which are otherwise called as *Human Race*, who are confined to a particular region of that continent only.

Race

Race is biologically defined as a geographically isolated population of organisms that differs from other populations of the same species in certain heritable traits. As to clarify it further it is a categorization of humans based on their physical characteristics such as skin colour, stature, head form, face, hair, eye, nose, body type, blood group etc along with regional variations. Accordingly three major human races are identified which are distributed across the world. They are *Caucasoid*, *Mongoloid* and *Negroid* which further sub divided into geographically viable groups. In this chapter we are going to discuss continent wise racial groups and their distribution pattern.

Racial Groups and Their Distribution Pattern

Continents	Major Ethnic Group/Race	Distribution	Remarks
Asia	Arabian	Bahrain, State of Palestine, Lebanon, Comoros, Kuwait, Qatar, Syria, United Arab Emirates, Morocco, Egypt, Jordan, Iraq, Tunisia, Yemen, Djibouti, Sudan, Algeria, Somalia, Saudi Arabia, Oman, Libya, Mauritania	This is a major Pan ethnic Group which is primarily habited in Western Asia. The primary language spoken by Arabian group is <i>Arabic</i> . Besides 15 other languages are spoken among Arabians in different parts of the World
	Indian	Most part of Indian Sub continents (India, Pakistan and Bangladesh)	On the basis of ethno linguistic composition the group is highly diversified, still the majority of population is restricted within two major linguistic groups such as <i>Indo-Aryan</i> and <i>Dravidian</i> .
	Mongoloids	This group is primarily concentrated in Inner Mongolia Autonomous Region and gradually spreading towards Liaoning, Jilin, Heilongjiang, Xinjiang, Qinghai, Gansu, Ningxia, Hebei, Henan, Sichuan, Yunnan and Beijing	They speak mainly the dialect of <i>Atlantic Language</i> family. The major three of them are <i>Inner Mongolian</i> , <i>Barag-Buryat</i> and <i>Uirad</i>
Africa	Pygmies	Rwanda, Burundi, Uganda, the Democratic Republic of the Congo (DRC), the Republic of Congo (ROC), the Central African Republic, Cameroon, the Equatorial Guinea, Gabon, Angola, Botswana, Namibia, Madagascar, and Zambia are populated by these group	A short height tribal group who lives on hunting, gathering fruits, nuts and honey from dense jungles and they are primarily confined to rain forests only.
	Bushmen	They are the huge inhabitant of Kalahari desert. Other than this they spread along Botswana, Namibia, South Africa, Zambia, Zimbabwe and Angola, with loosely related groups in Tanzania.	They are basically yellow-skinned nomads whose primary food intake in plant products collected from jungles which accounts for 70% of their diet and for the rest 20%-30% they depend on hunting.

	Zulus	They are mainly concentrated in KwaZulu-Natal Province of South Africa. Some are also scattered throughout the other provinces. KwaZulu-Natal borders on Mozambique in the north, Eastern Cape in the south, the Indian Ocean in the east, and Lesotho in the west.	They are the descendants of <i>Nguni-speaking</i> people and best known African tribes who were exploited by Britishers during the late 1800s. The most spoken languages are Zulu and English.
	Massai	They are the semi nomadic tribes who originated from the lower Nile valley north of Lake Turkana (Northwest Kenya) and began migrating south around the 15 th century, and finally settled in Kenya around 17 th and 18 th Century	They commonly use Kenya and Tanzania, Swahili and English language for communication as well as education.
	Bantus	The Bantus are primarily found in Rwanda, Angola, Burundi, Zimbabwe, and South Africa, with some among other nations in the Southern part of Africa.	As these tribes are highly resourceful and adaptable their occupancy in the total continent is also the highest. They mainly speak bantu language which gave them their nomenclature. They live on agriculture and metal working which give opportunity to adopt colonization.
North America	Groups of Indigenous peoples	They mostly confined to Canada	The indigenous peoples of the Americas are the descendants of the pre-Columbian inhabitants of North and South America.
	Native Hawaiians	Most Native Hawaiians reside in State of Hawaii and the American Southwest	Native Hawaiians are the indigenous Polynesian people of the Hawaiian Islands or their descendants.
South America	Mestijo	Mainly concentrated in Latin America	They emerged due to extensive intermixing between Europeans and Native Americans early in the colonial period
	Mullatto	Brazil is home to Latin America's largest mulatto population. Other than Brazil they spread in Dominican Republic, Cuba, Panama, Costa Rica, Colombia, Puerto Rico, and Ecuador.	Mulattoes are people of mixture of European and African.
	Zambol	Zambos are the small minorities in the northwestern South American countries such as Colombia, Venezuela, Guyana and Ecuador.	This is a small yet noticeable group resulting from unions of Amerindian women to Afro-Ecuadorian men are not uncommon in major coastal cities of Ecuador.
Europe	Nordic	The Nordics are inhabited in the countries around North and Baltic Seas.	As a sub group of Caucasian race this group is a set of people having distinct physical characteristics such as concurrence of fair, colour, somewhat wavy hair, light eyes, reddish skin, tall stature and a dolichocephalic skull
	Mediterranean	They are mostly concentrated in Afghanistan, Baluchistan, and Hindustan, with perhaps a southward extension into Ceylon.	This is a relatively small, light boned, long skulled race, of brunet color becoming even swarthy in certain portions of its range and possibly emerged during Neolithic times
	Alpine	They acquire all central and eastern Europe, and extending through Asia Minor to the Hindu Kush and the Pamirs.	The Alpines have their special physical characteristic of round skull, medium height and sturdy build, both as to skeleton and muscles. The color of both hair and eyes was originally very dark and still tends strongly in that direction, but many light colored eyes especially gray, are now found in the Alpine populations of western Europe.
Australia	Aborigines and Torres Strait Islanders	They contribute only 3% of the total population though they are the original inhabitant of Australia and the rest 7% are contributed by races of Asiatic origin.	Upto 2010-2012 the average life expectancy of Aboriginal and Torres Strait Islander people was approximately ten years (10.6 years for men and 9.5 years for women) which is much below the race of Asiatic origin, the leading causes of death are being heart disease, diabetes, respiratory disease and cancer.

Population Density

Human being is dependent upon the resource provided by nature for living. Suitable climate, fertile land, favorable environment gives ample of opportunities to grow. Hence the geographical unit which provides all sorts of facilities to human for its growth has a high pressure of population rather than the unfavorable ones. In these areas the density of population is also high in comparison to that of the unfavorable climates.

Population Density refers to the capacity of a piece of land for supporting the population living on it. In other words it refers to the ratio between the numbers of people to the size of land which is generally measured in persons per sq. km.

$$\text{Density of Population} = \text{Population} / \text{Area}$$

Factors affecting Population Distribution

The population on the earth surface is very unevenly distributed as result of so many factors which broadly categorized into three sections.

Geographical Factors

- As nature and its resources provide living aids to human being, factors such as land, water, climate and soil play a major role in influencing the density of population of a particular locality. Smooth terrain, fertile land, suitable climate and proper provision of water for irrigation and other basic uses of human being supports comparatively a large of population that the unfavorable natural conditions:
- For example Indo-Gangetic plain provides favorable living environment than hilly terrain of North eastern part of India.

Economic Factors

- Sometimes the economic prosperity of a region attracts the population of the surrounding areas which leads to migration of population from rural to newly built urban area in search of earning livelihood. The urban agglomeration then gradually increases with further economic growth.
- On the other hand the areas with huge mineral deposits give rise to industrial development which again require skilled and semi-skilled labours to work for the industry hence result into growth of population.
- Canton and Shanghai of China, Tokyo of Japan, Seoul of South Korea, Mumbai and Delhi of India are the brightest example of such economic growth which attracts maximum population of the adjacent rural areas.

Socio-cultural Factors

- Besides geographical and economic factors, social aspects also play a vital role in the growth of population of a particular area. Sometimes regions with religious and cultural background help people find their way in the midst of such places.
- Social and cultural factors in India and China had strongly influence the increase in population as people sometimes attached to the belief of having large families. In the developed world, smaller families are the norm.

Trends in Population Growth

As per the recent statistics of United Nations population Find the world population has reached to 7 billion mark in 2011 which is quite alarming. The rate of growth has increased to seven times to that of early years of economic and social development. The growth was at its peak during 1750. The noticeable change occurred due to the massive change in three major social components of birth, death and migration. With technological advancement during the era of industrial development has helped in reducing the death rate and increasing the fertility rate, decrease in the infant mortality rate, increase in the life expectancy, decrease in the death rate and finally increasing urbanization and migration.

Crude Birth Rate is the number of live births per year per 1000 persons which is generally estimated mid-year.

It can be calculated as::

$$\text{CBR} = (\text{Bi}/\text{P}) \times 1000 \text{ where Bi} = \text{the live births during the year}$$

Crude Death Rate is number of death per 1000 population of a specific geographical location, Like CBR it is also estimated in mid year.

It can be calculated as:

$$\text{CDR} = (\text{D}/\text{P}) \times 1000, \text{ where D} = \text{no. of death per year}$$

$$\text{P} = \text{Estimated mid-year population}$$

Infant Mortality rate (IMR) is the number of deaths of infants under one year old per 1,000 live births. This rate is often used as an indicator of the level of health in a country.

The rate of population growth is the percentage of population change over a period of time and can be divided into:

Natural Growth refers to Birth - Death

Actual Growth refers to birth-death+ In migration-Out migration

Migration

- The third most important component of change in population is *Migration*. It's the movement of people from one place in the world to another for the purpose of taking up permanent or semi permanent residence, usually across a political boundary. The place people move from is known as *Place of Origin* and the place they move to is known as *Place of Destination*.
- Basing upon the duration of stay the migration is of three types permanent, temporary and seasonal.
- There are two sets of factors which greatly influence migration,
 - **Push factor** are the reasons such as unemployment, poor living condition, political turmoil, unsuitable climate, natural disasters and socio-economic backwardness etc compels the dweller of a particular region to leave the place. This is somewhat less attractive. The resultant migration is called *Emigration*.
 - **Pull factors** seem like interesting as they attract the migrants with opportunities like better jobs, living condition, peace of mind and social stability and the resultant migration is known as *Immigration*.

Exercise -1

1. Currently half of the world's population lives in just six countries. Identify them from the following.
 - (a) India, China, Pakistan, Brazil, Bangladesh, Indonesia.
 - (b) India, China, Brazil, Pakistan, Bhutan, United State.
 - (c) China, India, United State, Indonesia, Brazil, Pakistan.
 - (d) China, India, Bangladesh, United State, Pakistan, Brazil.
2. The regions with highest and lowest densities are respectively
 - (a) Europe and Africa
 - (b) Asia and Oceania
 - (c) North America and Africa
 - (d) Europe and Oceania
3. In the Indian context the term 'De-notified tribes' refers to :
 - (a) tribes which are aboriginals
 - (b) nomadic tribes
 - (c) tribes practising shifting cultivation
 - (d) tribes which were earlier classified as criminal tribes
4. ".....They are people of yellow complexion, oblique eyes, high cheek bones, sparse hair and medium height." The reference here is to :
 - (a) Nordic Aryans
 - (b) Austrics
 - (c) Negroids
 - (d) Mongoloids
5. A person of mixed European and Indian blood in Latin America is called a :
 - (a) Mulatto
 - (b) Mestizo
 - (c) Meiji
 - (d) Mau Mau
6. The correct sequence of languages in descending order in terms of their number of speakers in the world is
 - (a) Spanish, English, Chinese, Hindi
 - (b) English, Chinese, Hindi, Spanish
 - (c) Hindi, Chinese, Spanish, English
 - (d) Chinese, English, Hindi, Spanish
7. The Semitic people belong to the?
 - (a) Caucasoid race
 - (b) Mangoloid race
 - (c) Negroid race
 - (d) Australoid race
8. The following Gulf countries have largest Indian communities
 - (a) Saudi Arabia
 - (b) Oman
 - (c) Kuwait
 - (d) Iraq
9. Which one of the following continents has the highest growth of population?
 - (a) Africa
 - (b) Asia
 - (c) South America
 - (d) North America
10. Economic density is ?
 - (a) total number of people per unit of land
 - (b) agricultural population per unit area of cultivated land
 - (c) agricultural population per unit are of cultivated land
 - (d) Ratio between the requirements of population and the resources made available to it.
11. Which of the following pairs are correctly matched?
 - (a) Primary sex ratio: Sex at the time of conception
 - (b) Secondary sex ratio: Sex at the time of enumeration
 - (c) Tertiary sex ratio: Sex at the time of enumeration
 - (d) Iambalanced sex ratio: Sex of workers
 Select the correct answer using the codes given below
 - (a) 1, 2, 3 and 4
 - (b) 1, 3 and 4
 - (c) 2 and 4
 - (d) 1, 2 and 3
12. Population explosion is associated with?
 - (a) high birth rate low death rate
 - (b) high birth rate and high death rate
 - (c) low birth rate and high death rate
 - (d) low birth rate and low death rate
13. Conundrums is the term of skin applied to?
 - (a) Caucasoid Race
 - (b) Negroid Race
 - (c) Mongoloid
 - (d) Australoid Race
14. In Which one the following countries, is Tamil a major language?
 - (a) Myanmar
 - (b) Indonesia
 - (c) Mauritius
 - (d) Singapore
15. Which of the following tribes, belongs to the Kalahari?
 - (a) Hottentots
 - (b) Zulus
 - (c) Kirghis
 - (d) Aborigines
16. Which of the following factors affects the birth rate?
 - (a) Demographic structure
 - (b) Politics
 - (c) Religion
 - (d) All of the above
17. Approximately one-half of the world population lives in?
 - (a) East Asia and Europe
 - (b) Europe and Eastern-North America
 - (c) South and South-East Asia
 - (d) East and South Asia
18. The correct descending order of the following communities of world in terms of their population is
 - (a) Sikhs, Christians, Buddhists, Jains
 - (b) Sikhs, Christians, Jains, Buddhists
 - (c) Christians, Sikhs, Jains, Buddhists
 - (d) Christians, Sikhs, Buddhists, Jains
19. The country which leads in the production of rubber is
 - (a) Australia
 - (b) Indonesia
 - (c) Malaysia
 - (d) Thailand
20. Which one of the following is the correct descending order of the three most populous states of India (2011)?
 - (a) Uttar Pradesh, Maharashtra, Bihar
 - (b) Maharashtra, Bihar, West Bengal
 - (c) Uttar Pradesh, Bihar, West Bengal
 - (d) Uttar Pradesh, West Bengal, Bihar

Exercise -2

1. Consider the following statements
 1. Black Fellow: Australian aborigines.
 2. Yuit: Inhabitants of Siberia and St. Lawrence island of Alaska
 Which of the above statements is/are true?
 - (a) Only 1 (b) Only 2
 - (c) Both 1 and 2 (d) None of these
2. Consider the following statement According to the Malthusian model
 1. Population tends to increase arithmetically.
 2. Positive and preventive checks prevent excessive growth rate of population.
 3. The basis of preventive checks is moral restraint.
 4. Postponement of marriage is to be emphasized.
 Of these statements
 - (a) 1 and 2 are correct
 - (b) 1, 2 and 3 are correct
 - (c) 2, 3 and 4 are correct
 - (d) 1, 3 and 4 are correct
3. Which of the following is/are the stage(s) of demographic transition ?
 1. High death rate and birthrate, low growth rate.
 2. Rapid decline in death rate, continued low birthrate, very low growth rate.
 3. Rapid decline in birthrate, continued decline in death rate
 4. Low death rate and birthrate, low growth rate.
 Select the correct answer using the codes given below
 - (a) Only 1 (b) 1, 2 and 3
 - (c) 3 and 4 (d) 1 and 4
4. Match List-I with List-II. Select the correct answer from the codes given below:

List-I (Tribe)	List-II (Region)
A. Semang	1. Congo Basin
B. Kirghiz	2. Malaysia
C. Bushman	3. Central Asia
D. Pygmies	4. Kalahari desert

Codes:

	A	B	C	D
(a)	4	2	3	1
(b)	1	3	2	4
(c)	2	1	4	3
(d)	2	3	4	1
5. Match List-I with List-II and select the correct answer from the code given below:

List-I (Country)	List-II (Capital)
A. Brunei	1. Bander Seri Begawan
B. Indonesia	2. Phnom Penh
C. Laos	3. Vientiane
D. Combodia	4. Jakarta

Codes:

	A	B	C	D
(a)	1	4	3	2
(b)	2	4	1	3
(c)	4	1	3	2
(d)	3	1	4	2

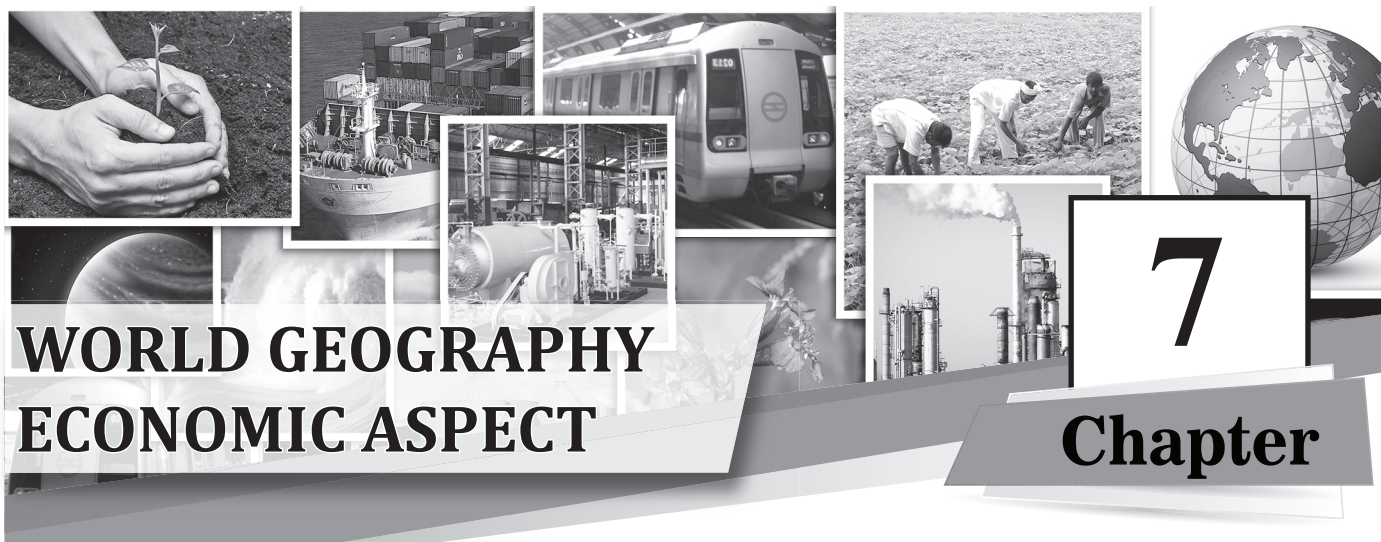
Hints and Explanations

EXERCISE-1

1. (c) China – India – United State – Indonesia – Brazil – Pakistan
2. (d)
3. (d) Denotified tribe in India refers to the tribes who where originally listed under the originally tribble act of 1871. They are also known as Vimukta Jati.
4. (d) Mongloids are inhabitants of northern, eastern and south-east Asia.
Their eyes has a characteristic fold of skin on the upper lid, hair is lank and straight and the height is medium. The group also includes the Chinese, Japanese, Burmese, Thais, Vietnamese and Malays.
5. (b) Mestizo is a type of person with mixed racial ancestry, especially of mixed European and Indian ancestry.
6. (d) 7. (c) 8. (d) 9. (d) 10. (d) 11. (c)
12. (a) 13. (b) 14. (b) 15. (a) 16. (d)
17. (d) 18. (d) 19. (c)
20. (a) Reference-Census 2011
U.P., Maharashtra, Bihar

EXERCISE-2

1. (c) 2. (c)
3. (d) Demographic transition refers to the transition from high death rate and high birth rate to low death and birth rates, as a country develops from a pre-industrial to industrialised economic system.
4. (d) Semang – Malaysia
Kirghiz – Central Asia
Bushman – Kalahari desert
Pygmies – Congo Basin
All these names are of the tribal communities belonging to the countries respectively.
5. (a) Brunei – Bander seri begawan
Indonesia – Jakarta
Laos – Vientiane
Combodia – Phnom Penh
All these are names of south-Asian countries with the names of their capitals.



Economic geography deals with the economic activities of human being basing upon the location, distribution and spatial organization as parts of Human Geography. Human activities meant for earning livelihood are known as *economic activities*. On the basis of resource and techniques used for performing these activities they are divided into *primary*, *secondary* and *tertiary activity*.

Primary activities are the direct interaction of human with environment that refers to the utilisation of earth's resources such as land, water, vegetation, building materials and minerals. Agriculture, pastoral farming, fishing, forestry, mining and quarrying etc. are categorized into primary economic activity.

Secondary Activities refers to the process of the value addition to natural resources by transforming the raw materials into essential finished product. Hence secondary activities are basically manufacturing, processing and construction industries.

Tertiary Activities involve the third level of human activities such as exchange of manufactured products i.e. trade and commerce.

With the advancement of new interventions these categorization has given rise to specialized services which can be termed as Quaternary and Quinary services.

PRIMARY ACTIVITIES

Agriculture

It is the process of producing food, feed and fiber through the cultivation of plants, and rearing livestock and is also known as farming. About 50% of the world's population is engaged in agriculture. In India as the land and climatic conditions are favorable for carrying out agricultural activities more than 2/3rd of the population is dependent on agriculture for their livelihood. The land utilized for agriculture is known as *arable land*.

The system of agriculture varies based on the method of farming, type of crop grown, cropping season etc., and they are as follows.

Subsistence Agriculture is the localized agricultural practice where the agricultural products are taken up by the producers or farmers only. It can be of two types:

Primitive Subsistence Agriculture is the age old practice of farming where people used to burn or slash large vegetative coverage and made them usable for agriculture. After a period of two to three years when the land started losing its fertility they shifted to some other area and usually repeat the same procedure. Hence it is also called as *Shifting Cultivation*. This process was adopted by the nomads or tribes of tropics, mainly in Africa, South and Central America and South East Asia. In different parts of the world it has been named differently such

as *Jhuming* in North Eastern state of India, *Milpa* in Central America and Mexico, *Ladang* in Indonesia and Malaysia.

Intensive Subsistence Agriculture is generally monsoon driven agriculture and mainly practiced in Asia. It is further divided into two categories. They are:

- **Intensive subsistence agriculture dominated by wet paddy** where the dominant crop is rice. This is generally practiced in the areas of comparatively large population and the farm yard manure is used to maintain the fertility of the soil.
- **Intensive subsistence agriculture dominated by crops other than paddy** where the relief, climate and soil along with geographical location plays a major role in crop growth. Hence these factors are all together responsible for growing of crop other than paddy, such as wheat in western India. Wheat, shorgum, barley, soyabean in northern China, Manchuria, North Korea and North Japan.

Plantation Agriculture are the farming which are being practiced in comparatively large estates with the support of huge capital investment, advance managerial and technical aids, scientific methods of cultivation, cheap labour and well connected market. Some of the major crops of plantation agriculture are tea, coffee, cotton, oil palm, sugarcane, banana, and pineapple.

Plantation farming was introduced in European colonies situated over tropics. Later on the French introduced coffee and cocoa in Africa, British set up tea plantation in India and Srilanka, rubber in Malayasia and sugarcane and banana in West Indies, Spanish and Americans introduced coconut and sugarcane plantation in Philippines.

Extensive Commercial Grain Cultivation is practiced for wheat mainly followed by other crops such as corns, barley, oats, rye. The farming is fully mechanized as the farms are comparatively larger than the regular farms. This type of agriculture is profoundly done in the European Steppes, Canadian Prairies, and Argentinean Pampas, the Velds of South Africa and Australian Down and Canterbury Plains of New Zealand.

Mixed Farming is usually done for the farms of medium size. Wheat is again the primary crop followed by barley, oats, rye, maize, fodder and root crops. The crops are being grown in rotation and intercropping plays an important role in maintaining soil fertility. Both crop cultivation and animal husbandry have given equal importance. One of its important characteristics is its high capital expenditure on farm machinery and building and the other is the extensive use of chemical fertilizers. This kind of farming is done usually in the developed countries of the world such as North Western Europe, north eastern Africa, parts of Eurasia and the temperate latitude of southern continents.

Dairy Farming is the advanced level of farming in the categories of milch animals. These are also capital intensive farming mainly which facilitates animal shedding, storage of fodder, feeding and milching machine, cattle breeding health care and veterinary services etc. It is usually practiced near urban and industrial centers, as they provided fresh milk and dairy product. There is no off season for this type of farming. These are extensively performed in north Western Europe, Canada and South Eastern Australia, New Zealand and Tasmania

Mediterranean Agriculture is highly specialized type of agriculture generally practiced in the countries on the either side of the Mediterranean sea in Europe and in north Africa, from Tunisia to Atlantic Coast, South California, Central Chile, south western part of South Africa and Australia. This kind of farming generally emphasizes on the production of Citrus fruits.

Viticulture is otherwise known as the cultivation of Grapes especially in the Mediterranean region. These high quality grapes are used for the production of best quality wines with distinctive flavor and the inferior quality grapes are dried to produce raisins and currants. Olives and figs are also cultivated in these regions. These crops are grown normally in winters and therefore in great demand in European and North America market.

Market Gardening and Horticulture are the specialized crop production of high value where vegetables, fruits and flowers are exclusively grown for urban market. This can be grown in small farms which are well connected with urban centers. The essential factors of this practice are better irrigation facility, High Yield Variety of seeds, fertilizers, insecticides, greenhouses and artificial heating in cold regions. This kind of farming is famous in densely populated industrial district of north western Europe, north eastern United States and America and Mediterranean region. The world famous Tulip is being grown in Netherlands and flown to all major cities of Europe,

The kind of specialized farming being done in the area of vegetable farming is known as Truck Farming as it is named after the distance that a truck can cover over night for supplying vegetables.

Co-operative Farming is meant to pull in the resources of the society more efficiently and use them as the capital for crop growth without hampering the land or the farming practice. These movement were originated a centuries ago and has been implemented in European countries such as Denmark, Belgium, Sweden, Italy.

Mining

Mining is associated with the exploration of new land with enormous of mineral resources. The improvement of mining has begun with period of industrial revolution and since then its importance is continuously increases.

Factors affecting Mining

- The size, grade and mode of occurrence of the mineral
- Demand for the mineral, technologies available and used for extraction, capital to develop infrastructure and finally the labour and transportation cost.

Types of Mining

Depending upon the occurrence of mineral mining is of two types:

- **Surface Mining**, otherwise known as open cast mining which is comparatively easier and cheaper.
- **Under Ground mining** is being done with the help of vertical shaft which has to be sunk underground for the exploration. It is comparatively costlier and risky as at times the explorer may counter with fatal accidents due to poisonous gas, fire and floods.

Mining has lost its importance in developed countries due to high labour costs but in case of developing countries it is still a method of earning livelihood as 50% of the earnings come from mining only. Several countries of Africa and few countries of South America are still in the race.

SECONDARY ACTIVITIES

Manufacturing is the process of transforming natural resources to usable finished product as some of the raw material cannot be used in its crude form.

Classification of Industries

Manufacturing industries can be classified on the basis of their size, input or raw material, output/product and ownership,

1. Industries based on Size

- **Cottage or Household industries** are the smallest manufacturing units producing finished products using localized raw materials and simplest manufacturing tools. The finished products are being sold in the local market only.
- **Small scale industries** are the unit which provide finished product using local raw materials but sometimes these are

manufactured by simple power driven machinery with semi skilled labour. It provides employment opportunities to the local people, hence raising their local purchasing power. These industries are quite famous in countries like India, China, Indonesia and Brazil.

- **Large scale industries** involve large markets, raw material from various sources, specialized workers and hi-tech machinery for production as well as assemblage of various finished products. It has first introduced around 200 years ago in United Kingdom, north-eastern USA and Europe but it has spread to almost all over world.

2. Industries based on Input/Raw Material used

- **Agro based industry** is the industry which uses raw materials from field and farm and turns them into finished product such as sugar, pickle, fruit juice, beverages like tea and coffee, spices and oil, fats and textiles (cotton, jute, silk), rubber, etc.
- **Mineral based industries** use only minerals as their raw materials which further categorized into metal and non-metal minerals. The non- metal mineral industries are cement, pottery industry. But the metallic minerals are further divided into ferrous(which contain iron particle in it) and non ferrous (which does contain iron particle in the metal)
- **Chemical based industry** involves natural chemical minerals such as mineral- oil, petroleum, salt, sulphur, potash industry. At times raw materials are obtained from coal and wood. Some of the finished products are synthetic fiber, plastics, and Petroleum products like Vaseline etc.
- **Forest based industry** uses forest product as their raw materials such as log of woods bamboo, and grasses are used to produce finished product such as paper, wooden furniture etc.
- **Animal based industries** involve raw material extracted from animal either live or dead. Leather and wool are such type of industries.

3. Industries based on Output/Finished Product

- **Basic industries** are those industry whose finished product are being used as the raw material for other industries like iron and steel industries, cotton textile industries etc.
- **Consumer based industries** are the manufacturer of those goods which can be directly consumed by the consumer such as bread, biscuits, tea, soaps, paper, toiletries etc. These are also called non-basic goods.

4. Industries based on Ownership

- **Public Sector Industries** are owned and managed by Government only and also called As Public Sector Undertakings (PSU). In some countries like India both central and state government impose their authority on the industries sometimes individually or jointly.
- **Private Sector Industries** are entirely owned and governed by private companies and hence called as Private Sector Undertaking.
- **Joint Sector Industries** have shares and participation of both Private and Public sectors.

5. Industries based on weight

- **Heavy industries** involves heavy raw materials and its production is also heavy for example iron and steel industry
- **Light industries** use comparatively lighter material as input and the finished product is also light for example electrical industries.

Major industrial centres of the world

Country	Major Industrial Centers	Industries
Britain (Midland is the largest Industrial region centered at Birmingham)	Birmingham Coventry Burton-on-Trent Stoke-on-Trent New Castle Middlesbrough Bradford Halifax Leeds Sheffield (World's largest cutlery town) Manchester (Lancashire region) Liverpool & Birkenhead Along Manchester Canal Glasgow Hamilton Motherwell Coatbridge Pot Glasgow Belfast region (Main industrial region of Ireland)	Iron & Steel, Heavy Machinery Automobile Brewing (largest brewery town of Britain) Pottery (Pottery capital of Britain) Shipbuilding Iron & Steel Worsted textile Garments Cutlery, Iron & Steel Cotton textile Shipbuilding Heavy chemicals Iron & Steel Shipbuilding Shipbuilding & Linen industry
France	Lille Dunkirk St. Etienne Limoges Lyone Marseilles Paris Champagne Lorransar	Textiles Iron & Steel Armaments & Bicycle Pottery Silk making Oil refineries Aircraft & Transport Wine Iron & Steel
Germany (Ruhr-Westphalia region, served by Rhine River, is the largest industrial region of Germany This industrial region is connected to North sea by Dortmund-Ems canal)	Frankfurt Mainz Mannheim Ludwigshafen Hamburg Munich Stuttgart Aachen Leipzig Jena Zeiss Dresden Karl Marx Stadt	Railway engineering Leather, Brewing, Engineering Chemical, electrical engineering Iron & Steel. Shipbuilding Photographic equipment, Musical instrument Automobile Iron & Steel, Textile Optical instrument Photographic equipment Porcelain Textiles
Belgium	Liege Antwerp Ghent	Iron & Steel, Guns, pistols & other firearms Diamond cutting Linen textiles
Luxembourg Netherland	Luxembourg city Rotterdam Amsterdam Arnhem	Engineering Shipbuilding and marine engineering Diamond cutting Tin smelting
Sweden	Goteborg Stockholm	Shipbuilding Shipbuilding

Switzerland	Zurich Basel Baden	Engineering and Textiles Engineering
Denmark	Kopenhagen	Dairy
Italy	Milan (main industrial region) Turin (Detroit of Italy)	Silk textile Motor Car
U.S.A. (Great Lake region) is the most important industrial region	Boston Pittsburg Akron Detroit Pontiac Flint Gary Chicago Toledo Birmingham Troy Buffalo San Francisco (Silicon Valley)	Shipbuilding Iron and Steel (Iron & Steel capital of the world) World's largest synthetic rubber and tyre making centre Motor car and Aeroplane Cars and its spare parts Iron and Steel Automobile Iron and Steel Garment Iron and Steel, Machine (It is also the largest flour milling centre of U.S.A.) Oil refining, Shipbuilding, Computer technology

Canada	Montreal Toronto Ottawa Hamilton Birmingham of Canada) Quebec	Shipbuilding and Aircraft Engineering and Automobile Paper Iron and Steel, Engineering Shipbuilding & Marine Engineering
Russia	Moscow and Gorky Magnitogost Leningrad (St. Petersburg)	Iron and Steel, Chemicals Iron and Steel, Oil refining Textile, Chemical, Paper
Ukraine	Krivoyrog	Iron & Steel and Heavy Machinery
Argentina	Buenos Aires	Shipbuilding
China	Shanghai Wuhan	Textile and Machinery Textile, Machinery, Shipbuilding, Iron and Steel
Japan	Nagoya (Detroit of Japan) Osaka (Manchester of Japan) Kyoto and Kobe Tokyo Nagasaki	Aircraft, Car, Machinery Shipbuilding, Textile, Iron & Steel Shipbuilding, Textile, Iron & Steel Shipbuilding, Engineering, and Textile Shipbuilding, Iron and Steel, Machinery

TERTIARY ACTIVITIES

Tertiary Activities focuses both on production and exchange. Here production involves provision of services which are to be consumed. There is no such involvement of physical processing of raw material rather it's a service meant for exchanging the finished product. Services such as Trade and Transports belong to Tertiary economic activities.

1. Trade and commerce

Trading is essentially involves buying and selling of products intended for earning profit. The process is further divided into retail and wholesale.

2. Transport

It is a service which facilitates the process of trading of people, materials and manufactured products with ease. In other words it is the organized industry to satisfy man's need of mobility. On the basis of the modes of transport it can be of four types:

- **Land Transport** is the most preferred transportation system with reference to the movement of human being, animals and goods from one location to another. The two vital media which facilitate land transport across the world are Roads and Railways.

Road Transport

- Germany was the first country to develop nation wide highway network.
- Usa has the largest highway network in world followed by India.
- Highway 401 in Ontario (Canada) is the busiest highway and is the widest in world.
- The Karakoram highway in Pakistan to China is the highest international highway in world.
- China was the world's largest network of express ways network which extends over 1,23,000 km. USA ranks

second in terms of expressways (7,7017).

- **Transport Canadian Highway** links Victoria with St. John City (7,821 km).
- Alaskan Highway joins the city of Anchorage in Alaska to Edmonlan in Canada. Covering a distance of 2,237 km.
- Pan American Highway is the largest international highway linking USA, Central America and South America. It covers a distance (30,000 km)
- Stuart Highway is the longest highway in Australia starting from Berdum in north Australia it reaches upto Melbourne situated in South Australia.

Road transport is as the most important communication medium as it facilitates door to door services for transporting goods and hence becoming the backbone of world's economic and social infrastructure.

Rail Transport

The history of railway were dated back in 650 BC and it was started with the introduction of wagan ways which were said to be further developed and widely used in 1550 AD in Germany with the invention of steam engine the railway grew many folds over the years. The first steam locomotive was built in 1804 in united kingdom. The first ever mechanised rail transport was introduced by England and ever since railways have remained as an important means of land transport.

- United State (2,50,000 km), China (1,21,000 km), Russia (90,000 km) and India (65,808) rank from first to fourth respectively in world scenario of largest railway network.
- India ranks first in carrying passenger per year by rail. Japan and Germany rank second and third in carrying rail passengers respectively.
- Steam engine was invented by Thomas Newcomen and was improved by James Watt.
- The first full-scale working railway steam locomotive was built in the United Kingdom in 1804 by Richard Trevithick.

- In 1811, John Blenkinsop designed first successful and practical railway locomotives.
- Dieselization was the replacement of steam locomotive with the diesel locomotive. This began in 1930s.
- Electrification of rail was started by Robert Davidson in 1838 in the form of electrical railway car.
- The *Trans-Australian Railway* includes the largest straight stretch of saithack in the world : 301 miles without any curve.
- *Qinghai-Tibet Railway* in China is the world's highest train journey, reaching the height of 5,072m above sea level at the Tanggula Pass.
- World's longest railway tunnel is the Seikan Tunnel, connecting Japan's northern Hokkaido island with Honshu. It is 33.46 miles long.
- The Napier to Gisborne Railway line is unique, as it crosses the main runway of Yesborne Airport.
- Venice Simplon orient express in the train against which all other luxury trains are measured.
- The *Trans-Siberian Railway* is the longest rail line in world. It connects St. Petersburg to Vladivostok via Moscow. Since 1916 its branches stretches into Mongolia, China and North Korea. It runs 9,332 kilometers (5,798.6 miles) and takes full week to cover this distance.

Trans-continental Railways connecting are part of a continent to its other part.

Major Trans-Continental Railways of World

- **Canadian Pacific Rail Route** : It runs from Halifax to Vancouver
- **Candian National Rail Route** : It joins St. John City to Vancouver.
- **Trans-Siberian Rail Route** : It connects St. Petersburg to Vlalivoslok.
- **Trans-caucasus Rail Route** : It starts from Batum, cross Tergana and reaches Krusk.
- **North Trans Continental Rail Route** : It originates at Seattle (USA) New York.
- **Mid Trans Continental Rail Route** : It joins san Francisco to New York.
- **Southern Trans Continental Rail Routes** : It runs from Los Angeles to New York and then reaches New Orleans.
- **Oriental Express Rail Route** : Connects Paris to Kustuntunia (Turkey)
- **Cape-Cairo Rail Route** : It is the longest rail route of Africa running from Cape Town in Southern part of continent and reaches Ciros near the coast of Mediterranean sea situated in the northern part of continent.
- **Trans-Andean Rail Route** : Starting from Valparais (Chile) reaches other end of the continent running west to east to Buenos Aires (Argentina).
- **Trans-Australian Continental Rail Route** : It connects west end of Australia to east end starting from Perth and reaches Sydney.

Transe-Asiatic Railways Line : This is a proposed project of united Nations Economic and Social Commission for Asia and Pacific. It will connect 28 countries including China, Thailand Bangladesh, India, Pakistan, Iran and Turkey.

Water Transport

Water transport is considered to be one of the oldest form of transport. It is cheapest mode of transport. The cost of production alongwith the cost of operation is also low incase of water transport. There are evidences which shows that boats were used in the form dugout canoe (a boat made from a hallowed tree trunk) in ancient period. It is considered that Egyptians probably were the first to use sail boat. The Mesopotamian and Indus valley civilization excavation have shown the presences of habour and ports.

Water transportation includes canals, lakes, rivers, seas and oceans. It is mainly divided into two categories:

Sea Routes

Oceans provide smooth traversable highway in all direction with low maintenance cost. Major sea routes are:

The North Atlantic Sea Route

It is linking two major industrially developed regions of the world such as North-Eastern USA and North-Western Europe.

The Mediterranean-Indian Ocean Sea Route

It is considered to be the most important sea route as it serves more countries and people than this route are port said, Aden, Mumbai, Colombo, and Singapore. The distance covered by this route has drastically reduced with the construction of suex canal.

The cape of Good Hope Sea Route

It connects west European and West African countries with Brazil, Argentina and Urguay in South America.

The North Pacific Sea Route

It is connecting Western European, North American with Australian countries is used for reaching Hongkong, Philippines and Indonesia. Honolulu is the most important sea port of this route.

Inland Water Transport

River, Canals, Lakes and Coastal area the important components of inland water transport as they facilitate transportation of goods and services with in the country. Boats and steam are the important means of transport. The significant waterways of the world are:

The Rhine Waterway

River Rhine flows through Germany and Netherland. It is navigable for 700 km. It serves industrial areas of Switzerland, Germany, France, Belgium and the Netherlands.

The Dawube Waterway

River Danube serves Eastern Europe as it rises from Black forest and flows eastward through many countries. The navigation is possible only upto Taurna Sevenin, this river transport facilitates the export of wheat, maize, timber and machinery.

The Volga Waterways

Volga is one of important water ways of Russia which provides a navigable water route of 11,200 km and it drains into caspian sea.

The Great Lake - St. Lawrence Seaways

Together with canal and welland canal, the great lake of North America Superior, Huron Erie and Ontario forms the Great lakes - St. Lawrence Seaways. Major ports of this routes are, Duluth and Buffalo.

Air Transport

It is the fastest way of transport but at the same time it is the costliest among the all medium of communication, still it is preferred by passenger for travelling across countries or sending cargoes across countries. The growth of Air Transport has reduced the travel time across the world to a great extent. But keeping in view the manufacturing of aircrafts and its associated elaborated infrastructure like hangars, landing, fuelling and maintaining these facilities, the air transport is considered to be the costliest among all transportation system.

Inter-continental Air routes

- These are the distinct air routes of Northern Hemisphere. Denser network of these Air routes are available in Eastern U.S.A, Western Europe and Southeast Asia.
- U.S.A. only accounts for 60% of the total network.
- The major nodal points or airport of this intercontinental air route are New York, London, Paris, Frankfurt, Rome,

Moscow, Karachi, New Delhi, Mumbai, Bangkok, Singapore, Tokyo, San Francisco, Los Angeles and Chicago.

- Africa, Asiatic part of Russia and South America are deprived of any such services.
- There are few air services which are available only between 100 and 350 latitude in the southern hemisphere.

Pipeline Transport

- It is the excellent mode of transporting gaseous or liquid matters from one place to another.
- Cooking gas or L.P.G, oil and natural gas are the items that can be transported extensively through pipelines in many parts of the world.
- In some countries like New Zealand milk is supplied from farm to factories with the help of pipelines.
- In U.S.A, oil is one of the major products which is transported through pipeline from producing areas to consuming areas. Some of the other countries transporting oil through pipelines are Europe, Russia, West Asia and India.
- Big Inch is one of the famous pipelines which connect the oil wells of the gulf of Mexico to the North-eastern States.

Exercise -1

1. The national economy of New Zealand is based on
 - (a) Export oriented agriculture
 - (b) Heavy industries
 - (c) Tourist industry
 - (d) Electronic industry
2. Which one of the following pairs is not correctly matched?
 - (a) Bahamas : Nassau
 - (b) Costa Rica : San Jose
 - (c) Nicaragua : Balmopan
 - (d) Dominican Republic: Santo Domingo
3. Which among the following has the world's largest reserves of Uranium?
 - (a) Australia
 - (b) Canada
 - (c) Russian Federation
 - (d) USA
4. Which one of the following cities is known as the Pittsburg of Japan?
 - (a) Kobe
 - (b) Yuwaha
 - (c) Tokyo
 - (d) Osaka
5. Intensive subsistence farming is prevalent in
 - (a) Canada
 - (b) China
 - (c) USA
 - (d) Ukraine
6. Baku is famous as
 - (a) Railway junction
 - (b) Industrial city
 - (c) Fish port
 - (d) Petroleum extracting centre
7. Ural industrial region is dominated by
 - (a) Textiles
 - (b) Metal fabricating
 - (c) Electronics
 - (d) Automobiles
8. The best inland waterways system is developed in
 - (a) Guinea coast
 - (b) Central Europe
 - (c) Amazon basin
 - (d) North America
9. Which one of the following is known as the Manchester of the East?
 - (a) Tokyo
 - (b) Shanghai
 - (c) Kuala Lumpur
 - (d) Osaka
10. Grozny is famous for the production of
 - (a) Petroleum
 - (b) Uranium
 - (c) Gold
 - (d) Thorium
11. Which country has the densest railway network in terms of per million people?
 - (a) China
 - (b) USA
 - (c) India
 - (d) Australia
12. The first three countries in order of iron-steel production in the world are
 - (a) USA, China, Germany
 - (b) Japan, USA, China
 - (c) Germany, China, USA
 - (d) Japan, Germany, China
13. Flanders industrial region is renowned for
 - (a) Iron & steel industry
 - (b) Car manufacturing
 - (c) Textile manufacturing
 - (d) Electronics
14. The third position in the manufacture of cars in the world is occupied by
 - (a) Japan
 - (b) USA
 - (c) France
 - (d) Germany
15. Largest producer of silver in the world is
 - (a) Mexico
 - (b) USA
 - (c) Laos
 - (d) South Africa
16. The main centres of ship building industry in Japan are
 - (a) Nagasaki and Yawata
 - (b) Yakohama and Wakohama
 - (c) Osaka and Kyoto
 - (d) Nagita and Issikari
17. The three leading coal producers of the world are:
 - (a) USA, Russia, Germany
 - (b) England, France, Australia
 - (c) France, Poland, Japan
 - (d) Germany, India, South Africa
18. Which of the following countries are the best examples of highly scientific development and management of pastoral farming?
 - (a) France and Spain
 - (b) Denmark and Holland
 - (c) Czechoslovakia and Poland
 - (d) Greenland and Norway
19. Which one of the following countries of Africa finds a place in high human development category of UNDP?
 - (a) South Africa
 - (b) Kenya
 - (c) Zimbabwe
 - (d) Mauritius
20. The ports on either end of the Suez canal are
 - (a) Cairo and Alexandria
 - (b) Suez and Cairo
 - (c) Cairo and Port Said
 - (d) Port Said and Suez
21. Most industrialized country in ASEAN is
 - (a) Philippines
 - (b) Thailand
 - (c) Malaysia
 - (d) Singapore
22. Which one of the following is the most industrialized country of Latin America?
 - (a) Brazil
 - (b) Chile
 - (c) Colombia
 - (d) Argentina
23. Which is the shortest route from Moscow to San Francisco?
 - (a) via Canada
 - (b) over land
 - (c) over the South Pole
 - (d) over the North Pole
24. Which one among the following covers the highest percentage of forest area in the world?
 - (a) Temperate coniferous forests
 - (b) Temperate deciduous forests
 - (c) Tropical monsoon forests
 - (d) Tropical rain forests
25. Which one of the following countries has replaced Italy as the major importer of bauxite from India?
 - (a) Canada
 - (b) Greece
 - (c) Ukraine
 - (d) United Arab Emirates
26. Grand Banks is one of the important producers of
 - (a) Nitrate
 - (b) Iron ore
 - (c) Marine fish
 - (d) Bauxite

27. Plantation agriculture is most widespread in
 (a) Nile valley (b) Mississippi valley
 (c) California (d) Caribbean
28. The largest iron-producing region of USA is
 (a) North East Appalachian region
 (b) Alabama state
 (c) Western region
 (d) Lake Superior region
29. Detroit is famous for?
 (a) motor car industry (b) copper industry
 (c) mining industry (d) fishing industry
30. Which one of the following cities in USA is a steel manufacturing centre?
 (a) New York (b) Rochester
 (c) Chicago (d) Dallan
31. The main exporter of copper in the world is
 (a) Ghana (b) Morocco
 (c) Zambia (d) Belgium
32. Starting from the best quality of iron ore, Which one of the following is the correct sequence of the iron ore based on the contents of pure iron?
 (a) Magnetite _ Hematite _ Limonite _ Siderite
 (b) Hematite _ Magnetite _ Limonite _ Siderite
 (c) Magnetite _ Hematite _ Siderite _ Limonite
 (d) Hematite _ Magnetite _ Siderite _ Limonite
33. Which among the following has the world's largest reserves of Uranium?
 (a) Australia
 (b) Canada
 (c) Russian Federation
 (d) USA
34. Which one the following port cities in Venezuela has been developed as an oil port?
 (a) Caracas (b) Maracaibo
 (c) Maracay (d) Carupano

Exercise -2

Statement Based MCQ

1. The main features of the commercial dairy farming are
 1. It is capital intensive.
 2. It is labour intensive.
 3. It is highly productive.
 4. It is highly commercial.
 Select the correct answer from the codes given below:
Codes:
 (a) 1 and 3 only (b) 2 and 3 only
 (c) 1, 3 and 4 only (d) 1, 2, 3 and 4
2. Which of the following are not the examples of shifting cultivation?
 Select the correct answer from the codes given below:
 1. Ladang 2. Hacienda
 3. Fazenda 4. Pondu
Codes:
 (a) 1 and 2 (b) 1 and 3
 (c) 2 and 4 (d) 3 and 4
3. Which of the following are fund resources?
 1. Coal 2. Copper
 3. Petroleum 4. Forest
 Select the correct answer from the codes given below:
Codes:
 (a) 1 and 2 (b) 1 and 3
 (c) 1 and 4 (d) 3 and 4
4. Tank irrigation is practised mainly in Peninsular India because
 1. undulating relief and hard rocks make it difficult to dig canals and wells
 2. rivers are rainfed
 3. of compact nature of population and agricultural field
 Select the correct answer using the codes given below :
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 3 (d) All of these
5. Which of the following are responsible for the decrease of per capita holding of cultivated land in India ?
 1. Low per capita income.
 2. Rapid rate of increase of population
 3. Practice of dividing land equally among the heirs.
 4. Use of traditional techniques of ploughing.
 Select the correct answer using the codes given below :
 (a) 1 and 2 (b) 2 and 3
 (c) 1 and 4 (d) 2, 3 and 4
6. Which of the following items and their leading producers are correctly matched?
 1. Grapes - Italy 2. Oilseeds - India
 3. Millets - India 4. Coffee - Ethiopia
 Which of the above statement are correct ?
 (a) 1, 3 and 4 (b) 2, 3 and 4
 (c) 1, 2 and 3 (d) 1, 2, 3 and 4
7. A geographic region has the following distinct characteristics:
 1. Warm and dry climate 2. Mild and wet winter
 3. Evergreen Oak trees
 The above features are distinct characteristics of which one of the following regions?
 (a) Mediterranean
 (b) Eastern China
 (c) Central Asia
 (d) Atlantic coast of North America
8. Japan is one of the leading industrial countries in the world because it has:
 1. developed hydel power
 2. large deposits of metallic mineral
 3. high technological capability
 4. insular location
 Of these statements:
 (a) 1, 2 and 4 are correct
 (b) 1, 2 and 3 are correct
 (c) 1 and 3 are correct
 (d) 2 and 4 are correct

9. Consider the following statements and select the correct answer from the codes given below:

Assertion (A): The W.T.O. aims to promote free trade.

Reason (R): It does not manage the global economy impartially.

Codes:

- (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true, but R is not the correct explanation of A.
 (c) A is true, but R is false.
 (d) A is false, but R is true.

Matching Based MCQ

DIRECTIONS (Qs. 10-26) : Match List-I with List-II and select the correct answer from the codes given below:

10. **List-I (Towns)** **List-II (Industry)**
 (A) Pittsburg (1) Textile
 (B) Shanghai (2) Iron & Steel
 (C) Nagoya (3) Automobiles
 (D) Moscow (4) Shipbuilding
 (a) A - 1 ; B - 2 ; C - 3 ; D - 4
 (b) A - 2 ; B - 1 ; C - 4 ; D - 3
 (c) A - 3 ; B - 4 ; C - 1 ; D - 2
 (d) A - 4 ; B - 3 ; C - 2 ; D - 1
11. **List-I (City)** **List-II (Industry)**
 (A) Osaka (1) Shipbuilding
 (B) Hamburg (2) Textiles
 (C) Detroit (3) Iron & Steel
 (D) Pittsburg (4) Automobiles
 (a) A - 2 ; B - 1 ; C - 4 ; D - 3
 (b) A - 1 ; B - 2 ; C - 3 ; D - 4
 (c) A - 2 ; B - 3 ; C - 1 ; D - 4
 (d) A - 4 ; B - 1 ; C - 2 ; D - 3
12. **List-I** **List-II**
 (A) Iron & Steel (1) Portsmouth
 (B) Automobile (2) Dortmund
 (C) Shipbuilding (3) Bangalore
 (D) Aircraft (4) Detroit
 (a) A - 1 ; B - 2 ; C - 3 ; D - 4
 (b) A - 2 ; B - 1 ; C - 4 ; D - 3
 (c) A - 4 ; B - 3 ; C - 2 ; D - 1
 (d) A - 2 ; B - 4 ; C - 1 ; D - 3
13. **List-I (Crop)** **List-II (Main Producing area)**
 (A) Coconut (1) Kenya
 (B) Banana (2) Papua New Guinea
 (C) Groundnut (3) Ecuador
 (D) Tea (4) Senegal
14. **List-I (Oil refineries)** **List-II (Country)**
 A. Abadan 1. Saudi Arabia
 B. Haifa 2. Iran
 C. Kirkuk 3. Israel
 D. Ras Tanavra 4. Iraq

Codes:

- A B C D
 (a) 2 1 4 3
 (b) 2 3 4 1
 (c) 1 3 2 1
 (d) 4 2 3 1

15. **List-I (Mineral)**

- A. Coal
 B. Gold
 C. Iron ore
 D. Petroleum

Codes:

- A B C D
 (a) 2 3 1 4
 (b) 2 1 3 4
 (c) 4 1 2 3
 (d) 1 2 3 4

16. **List-I (Mineral)**

- A. Coal
 B. Copper ore
 C. Iron ore
 D. Petroleum

Codes:

- A B C D
 (a) 4 2 3 1
 (b) 4 1 3 2
 (c) 3 1 2 4
 (d) 1 4 3 2

17. **List-I (Country)**

- A. China
 B. Germany
 C. Ukraine
 D. U.S.A.

Codes:

- A B C D
 (a) 4 2 1 3
 (b) 3 2 4 1
 (c) 3 1 4 2
 (d) 4 3 2 1

18. **List-I (Mineral)**

- A. Tin
 B. Thorium
 C. Uranium
 D. Copper

Codes:

- A B C D
 (a) 3 4 2 1
 (b) 1 2 4 3
 (c) 3 2 4 1
 (d) 4 3 2 1

19. **List-I (Country)**

- A. Iran
 B. Iraq
 C. Kuwait
 D. Saudi Arabia

- List-II (Area of production)**

1. Highveld
 2. Karaganda Basin
 3. Krivoi Rog
 4. San Joaquin valley

- List-II (Occurrence)**

1. Bisbee
 2. Baku
 3. Mesabi
 4. Westphalia

- List-II (Coal fields)**

1. Pennsylvania
 2. Saar
 3. Shensi
 4. Donetsk Basin

- List-II (Leading producer)**

1. Zambia
 2. India
 3. Malaysia
 4. Canada

- List-II (Oil field)**

1. Bargan
 2. Damam
 3. Kirkuk
 4. Masjid Sulaiman

Codes:

	A	B	C	D
(a)	4	3	2	1
(b)	3	4	1	2
(c)	2	3	4	1
(d)	4	3	1	2

20.

List-I**(Industrial region)**

- A. Kinki
- B. Lorraine
- C. Midlands
- D. New England

Codes:

	A	B	C	D
(a)	3	2	1	4
(b)	3	1	4	2
(c)	1	2	4	3
(d)	4	1	3	2

21.

List-I**(Industry)**

- A. Iron and steel
- B. Ship building
- C. Electronics
- D. Automobile

Codes:

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	3	4	1	2
(d)	2	4	1	3

22.

List-I**(Iron and steel centre)**

- A. Cleveland
- B. Essen
- C. Hamilton
- D. Tula

Codes:

	A	B	C	D
(a)	1	4	2	3
(b)	3	4	1	2
(c)	3	1	2	3
(d)	4	3	1	2

23.

List-I**(City)**

- A. Shanghai
- B. Chicago
- C. Sheffield
- D. Yokohama

List-II**(Country)**

- 1. France
- 2. U.S.A.
- 3. Japan
- 4. U.K.

List-II**(Centre)**

- 1. Turin
- 2. Taipei
- 3. Montreal
- 4. Pittsburgh

List-II**(Country)**

- 1. Canada
- 2. Russia
- 3. U.S.A.
- 4. Germany

List-II**(Important Industry)**

- 1. Ship building
- 2. Iron and steel
- 3. Cotton textile
- 4. Engineering

Codes:

	A	B	C	D
(a)	3	2	1	4
(b)	1	3	4	2
(c)	4	3	2	1
(d)	3	2	4	1

24.

List-I**(Crops)**

- A. Wheat
- B. Cotton
- C. Sugarcane
- D. Tea

Codes:

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	1	4
(c)	2	4	3	1
(d)	4	1	2	3

25.

List-I

- A. Largest producer of wheat in the world
- B. Largest producer of milk in the world
- C. Largest producer of sugarcane in the world
- D. Largest producer of maize in the world

List-II

- 1. U.S.A.
- 2. China
- 3. India
- 4. Brazil

Codes:

	A	B	C	D
(a)	1	2	4	3
(b)	2	3	4	1
(c)	3	4	2	1
(d)	1	3	2	4

26.

List-I

- A. Iron and steel
- B. Ship building
- C. Automobile
- D. Woollen textile

List-II

- 1. Atlanta
- 2. Bradford
- 3. Cleveland
- 4. Yokohama

Codes:

	A	B	C	D
(a)	3	4	1	2
(b)	1	2	4	3
(c)	2	4	1	3
(d)	3	2	1	4

Hints and Explanations

EXERCISE-1

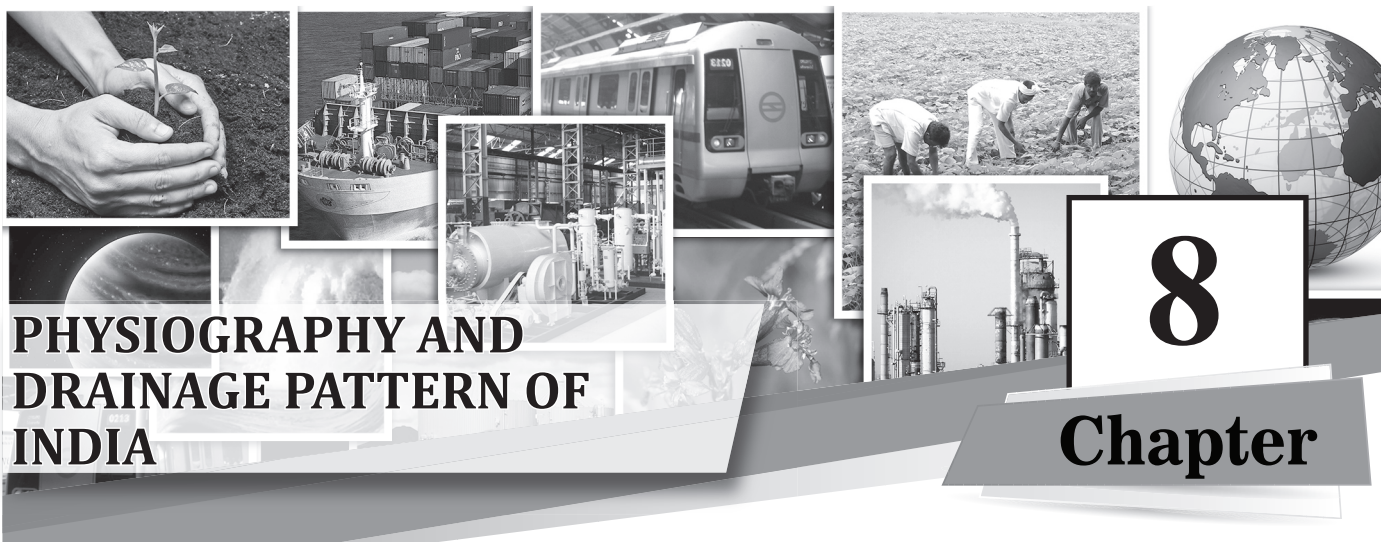
1. (a)
2. (c) Managua is the capital of Nicaragua, not Balmopan.
3. (a) Australia has the world's largest uranium reserves. Approximately 24% of the planet's uranium is present in Australia.
4. (b)
5. (b)
6. (d)
7. (b)
8. (b)
9. (d)
10. (a)
11. (d)
12. (b)
13. (c)
14. (d)
15. (a)
16. (a)
17. (a)
18. (b)
19. (d)
20. (d)
21. (c)
22. (a)
23. (d)
24. (a) Temperature coniferous forests covers the highest percentage of forest area in the world.
25. (a) Canada is the largest importer of bauxite from India.
26. (c)
27. (c)
28. (d)
29. (a)
30. (c)
31. (c)
32. (a)
33. (a) Australia has the world's largest uranium reserves. Approximately 24% of the planet's uranium is present in Australia now 31%.
34. (b) Maracaibo is an oil-port of Venezuela as it has oil rich basis.

EXERCISE-2

1. (d) The main features of the commercial dairy farming are capital intensive, labour intensive, highly productive and highly commercial.
2. (c) Hacienda and Ponda are not the examples of shifting cultivation. Shifting cultivation is known as ladang cultivation in south east Asia. Fazendas (meaning "farms") were plantations found throughout Brazil; during the colonial period (16th - 18th centuries), they were concentrated primarily in the northeastern region, where sugar was produced.
3. (b) Coal and petroleum are the fund resource minerals. Because they are of the highest value in the foreign world market due to demand and also for scarcity.
4. (a) The tank irrigation is practised mainly in the peninsular India due to the following reasons
 1. The undulating relief and hard rocks makes it difficult to dig canals and wells.
 2. There is little percolation of rain water due to hard rock structure and ground water is not available in large quantity.
 3. Most of the rivers of this region are seasonal and dry up in summer season. Therefore, they cannot supply water to canals throughout the year.
 4. The scattered nature of population and agricultural fields also favours tank irrigation.
5. (b) The factors responsible for the decrease of per capita holding of cultivated land in India are
 1. Rapid rate of increase of population.
 2. Practice of dividing land equally among the heirs.
6. (c)
7. (a) Meditterrian climate have mild, rainy winter and hot, dry summers and Evergreen, Oak trees.
8. (c)

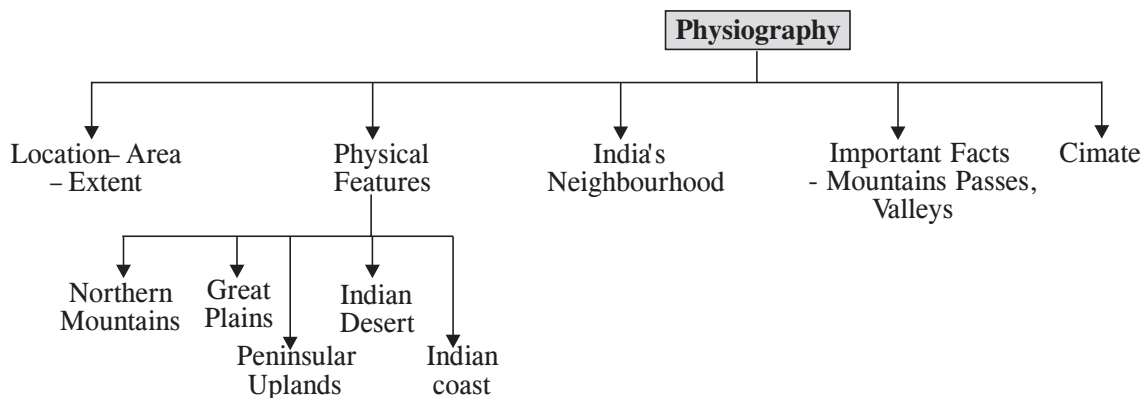
9. (b) The WTO established in 1995. It aims to promote free trade, but it does not manage the global economy impartially. The organization deals with regulation of trade between participating countries; it provides a framework for negotiating and formalizing trade agreements, and a dispute resolution process aimed at enforcing participant's adherence to WTO agreements, which are signed by representatives of member governments and ratified by their parliaments.
10. (b)
11. (d)
12. (d)
13. (a)
14. (b) The oil refineries of Iran are located in Abadan. Abadan is a city in and the capital of Abadan County, Khuzestan province, Iran. It lies on Abadan Island, 53 kilometres from the Persian Gulf, near the Iraq-Iran border. The oil refinery of Israel is located in Haifa. Haifa is the largest city in northern Israel, and the third-largest city in the country, with a population of over 291,000. The oil refinery of Iraq is located in Kirkuk. Kirkuk is a city in Iraq and the capital of Kirkuk Governorate. It is located in the Iraqi governorate of Kirkuk, 236 kilometres north of the capital, Baghdad. The oil refinery of Saudi Arabia is located in Ras Tanura. Ras Tanura is a city in the eastern province of Saudi Arabia located on a peninsula extending into the Persian Gulf.
15. (b) Karaganda Basin is known for coal production. Karagandy, more commonly known by its Russian name Karaganda, is the capital of Karagandy province in Kazakhstan. Highveld is known for gold production. The Highveld is the portion of the South African inland plateau. Krivoi Rog is known for iron ore production. It is a city in central Ukraine. It is situated in Dnipropetrovsk Oblast, to the southwest of the Oblast's administrative centre. San Joaquin valley is known for petroleum exploration. The San Joaquin Valley is the area of the central valley of the U.S. state of California that lies south of the Sacramento - San Joaquin river delta in Stockton.
16. (b) Coal mining occurs in Westphalia. It is a region in Germany. Copper ore occurs in Bisbee. Bisbee is a city in Cochise County, Arizona, United States, 82 miles southeast of Tucson. Iron ore occurs in Mesabi. The Mesabi Iron Range is a vast deposit of iron ore and the largest of four major iron ranges in the region collectively known as the Iron Range of Minnesota. Discovered in 1866, it is the chief deposit of iron ore in the United States. Baku is famous for petroleum exploration. Baku is the capital and largest city of Azerbaijan, as well as the largest city on the Caspian Sea and of the Caucasus region.
17. (b) Shensi is the coal field of China. It is a province of the People's Republic of China, officially part of the northwest China region. Saar is known as

- the coal mining in Germany. The Saarland is one of Germany's sixteen federal states. Its capital is at Saarbrücken. Donetsk Basin is the coal field of Ukraine. It comprises the Donbas Foldbelt, which is the uplifted and compressionally deformed part of the Pripyat–Dniepr–Donets (PDD) Basin. Pennsylvania is known for coal mining in U.S.A. Pennsylvania, officially the Commonwealth of Pennsylvania, is a U.S. state that is located in the northeastern and mid-Atlantic regions of the United States, and the Great Lakes region.
18. (c) Malaysia is the leading producer of tin. India is the leading producer of thorium. Canada is the leading producer of uranium. Zambia is the leading producer of copper.
 19. (d) Masjid-e-Suleiman is the oil field of Iran. Kirkuk is the oil field of Iraq. Barga is the oil field of Kuwait. Daman is the oil field of Saudi Arabia.
 20. (b) Kinki is the industrial region of Japan. The Kansai region or the Kinki region lies in the southern-central region of Japan's main island Honshu. Lorraine is the industrial region of France. Lorraine is one of the 27 regions of France. The administrative region has two cities of equal importance: Metz, the regional prefecture and Nancy. Midlands is the industrial region of U.K. The Midlands is an area comprising central England that broadly corresponds to the early medieval Kingdom of Mercia. It borders southern England, northern England, East Anglia and Wales. New England is the industrial region of U.S.A. New England is a region in the northeastern corner of the United States consisting of the six states of Maine, Massachusetts, New Hampshire, Vermont, Rhode Island, and Connecticut.
 21. (b) Turin is the centre of automobile industry. Turin is a city and an important business and cultural centre in northern Italy, capital of the Piedmont region. Pittsburgh is the centre of iron and steel industry. Pittsburgh is the seat of Allegheny County and with a population of 306,211 is the second-largest city in the U.S. state of Pennsylvania. Montreal is the centre of ship-building. Montreal is a city in the Canadian province of Quebec. It is the largest city in the province. Taipei is the centre of electronics. Taipei, officially known as Taipei City, is the capital of Taiwan. Situated at the northern tip of Taiwan, Taipei is located on the Tamsui river.
 22. (b) Regarding to the Iron and steel centre, Cleveland is located in U.S.A. Essen is located in Germany. Hamilton is located in Canada. Tula is located in Russia.
 23. (d) Cotton textile industry belongs to Shanghai. Iron and steel industry belongs to Chicago. Engineering industry belongs to Sheffield. Ship building industry belongs to Yokohama.
 24. (b) China is the largest producer of wheat. Roughly two-thirds of the total wheat production came from the north China plain and nearly another third from the central provinces. Winter wheat accounts for about 94% of China's total wheat output. U.S.A. is the largest producer of cotton. In its January report, USDA estimated a '13-14 US crop of 13.19 million bales. Upland production was estimated at 12.55 million bales and extra-long staple production at 636,000 bales. Brazil is the largest producer of sugarcane. Brazil's sugarcane industry association UNICA estimates Brazil's sugar cane production in 2012/13 at 531.4 million ton which is 8% up from the 493.2 million ton produced in 2011/12. India is the largest producer of tea. Tea Board of India shows that during January to August, tea production has risen by 6.2% to 705 million kg in 2013.
 25. (b) Largest producer of wheat in the world is China. Largest producer of milk in the world is India. Largest producer of sugarcane in the world is Brazil. Largest producer of maize in the world is U.S.A.
 26. (a) Iron and steel industry belongs to Cleveland. Ship building industry belongs to Yokohama. Automobile industry belongs to Atlanta. Woollen textile industry belongs to Bradford.



Introduction

‘Physiography’ of an area is the outcome of structure, process and the stage of development. The land of India is characterised by great diversity in its physical features. The North has a vast expanse of rugged topography consisting of series of mountain ranges with varied peaks, beautiful valleys and deep gorges. The South consists of stable table land with highly dissected plateaus and denuded rocks. In between these two lies the vast North Indian plain.



LOCATION - AREA - EXTENT

- India lies in the northern and eastern hemispheres of the globe between $8^{\circ} 4' N$ and $37^{\circ} 6' N$ latitudes and $68^{\circ} 7' E$ and $97^{\circ} 25' E$ longitudes.
- The southern most point extent upto $6^{\circ} 45' N$ latitude to cover the last island of the Nicobar group of islands. The southern extreme is called **Pygmalion Point** or **Indira Point**.
- The **tropic of cancer** $\left(23\frac{1}{2}^{\circ} N\right)$ latitudes passes through the middle part of India and crosses the states of Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.
- Its total length of land frontier of 15,200 kilometres passes through marshy lands, desert, level plains, rugged mountains, snow covered areas and thick forests.
- Besides land there is a maritime boundary of 6,100 kilometres along the main land mass which increases to

7,516 kilometres of the coastlines of Andaman-Nicobar and Lakshadweep Islands are added to it.

- India-Afghanistan and Pakistan-Afghanistan international boundary is called the **Durand Line**, determined as a “militarily strategic border between British India and Afghanistan”.
- The India-China boundary (4,225 km) is a natural boundary running along the Himalayan ranges and is based on various treaties. Its eastern part (1,140 km) is called the **Mc Mahon Line**.
- The boundary with Pakistan and Bangladesh (the East Pakistan) was finalized at the time of partition in 1947 through the ‘Redcliffe Award’.
- The **Tin Bigha Corridor** is a strip of land belonging to India is a part of West Bengal and lies adjacent to the Bangladesh border. In 1947, the border between India and Bangladesh was also demarcated by Sir Redcliffe.

- India is the only country which has given its name to an ocean, i.e. *Indian Ocean* encircled by 46 countries (27 littoral including Australia, 7 island states and 12 land locked countries).
- India commands a total geographical area of **32,87,263 sq. km** which is roughly 0.57% of the area of the earth and **2.4%** of the total area of the land hemisphere.
- India has roughly a quadrangular shape. It measures about 3,214 km from north to south and about 2,933 km from east to west, the difference between the two being just 281 km.
- Because of great longitudinal extent, the difference in local time between Eastern and Western extremes of the country is of two hours. In order to avoid the confusion with regard to the time at different places of the country, the local time along 82° 30' East longitudes is taken as the standard time of India, i.e. **India Standard Time (IST)**.
- This meridian is known as the **Standard Meridian of India**. It passes through Mirzapur (UP). The Tropic of Cancer divides India almost into two equal halves. Thus, the Northern half of India is situated entirely in the Northern hemisphere and also belongs to the Eastern hemisphere because of its situation to the East of the Prime Meridian.
- After Russia, China, Canada, USA, Brazil and Australia, India is the **seventh largest country** of the world. Its area is almost equal to the area of Europe (excluding Russia), one-third of Canada, one fifth (1/5) of Russia, eight times of Japan and twelve times of United Kingdom.
- In population-size, India is the second giant country in the world after China.
- Its total population is more than the combined population of USA, Russia, Australia, Canada and Japan.
- The eastern boundaries of India are formed by a complex chain of the Himalayan offshoots consisting of the Mishmi, the Patkai, the Naga hills, the Barail range, the Mizo hills and finally the majestic Arakan Yoma Mountains range.
- The Arakan Yoma is submerged in the Bay of Bengal for sufficiently long stretch and emerges again in the form of Andaman and Nicobar Islands.
- The boundary line between India and Bangladesh crisscrosses the vast Ganga-Brahmaputra delta. This boundary runs is not even a small mount or hill which could be used for demarcating the boundary between the two countries.
- Jammu and Kashmir, Himachal Pradesh, Sikkim and Arunachal Pradesh are the states of India lying completely in the Himalayas, while the state of Uttarakhand lies partly in the Himalayas and partly in the northern plains.
- Madhya Pradesh, Chhattisgarh, Jharkhand, Maharashtra, Odisha, Andhra Pradesh, Karnataka, Kerala and Tamil Nadu, together make the great peninsular plateau.
- Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, Andhra Pradesh, Odisha and West Bengal are states on the coast of India. Among our Union Territories, Daman and Pondicherry have sea boundaries.
- The Indian states on international boundaries (other country/ countries within brackets) are: Gujarat (Pakistan), Rajasthan (Pakistan), Punjab (Pakistan), Jammu and Kashmir (China and Pakistan), Himachal Pradesh (China), Bihar (Nepal), Uttarakhand (China and Nepal), Uttar Pradesh (Nepal), West Bengal (Bhutan, Nepal, and Bangladesh), Sikkim (China, Bhutan and Nepal), Arunachal Pradesh (Bhutan, China and Myanmar), Nagaland (Myanmar), Manipur (Myanmar), Mizoram (Bangladesh and Myanmar), Meghalaya (Bangladesh), Tripura (Bangladesh) and Assam (Bhutan, Bangladesh and Myanmar).
- The states of Haryana, Madhya Pradesh, Chhattisgarh and Jharkhand are the only *land-locked states* which are neither on the coast nor on an international border.

The States having Common Frontiers with Neighbouring Countries

Country	States
Pakistan	Jammu and Kashmir, Punjab, Rajasthan, Gujarat
Afghanistan	Jammu and Kashmir
China	Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim Arunachal Pradesh
Nepal	Uttarakhand, Uttar Pradesh, Bihar, West Bengal Sikkim
Bhutan	Sikkim, West Bengal, Assam, Arunachal Pradesh
Myanmar	Arunachal Pradesh, Nagaland, Manipur, Mizoram
Bangladesh	West Bengal, Meghalaya, Assam, Tripura, Mizoram

Four Ends of India

Easternmost point of India is known as Kibithu; situated on right bank of river Lohit separating India from China-Tibet region. It is a small village with the population at the altitude of 3,350 metre in Arunachal Pradesh. **Westernmost point** is situated in Kuch area of Gujarat called as Ghuar Mota. The region is famous for its harsh climate with 45°C in summer and 20°C in winter. During monsoon season this region looks like tortoise surrounded by seawater. **Northernmost** point of India has been in controversies ever since India's independence. The Siachen Glacier in the state of J&K is the northern boundary of India according to the official division of India during the time of Independences. The **Southernmost** point of the mainland of India is Kanyakumari District in the state of Tamil Nadu. Kanyakumari, formerly was known as Cape Comorin. It is the second largest and urbanized of Tamil Nadu. Indira Point is a village in the Nicobar district of Andaman and Nicobar Islands, India. It is located in the Great Nicobar tehsil. It is the location of the southernmost point of India's territory.

Area Wise Largest States

State	sq km
Rajasthan	342239
Madhya Pradesh	308245
Maharashtra	307713
Andhra Pradesh	160229
Uttar Pradesh	240928

Important Straits

Location	Channel
Indira point-Indonesia	Great Channel
Little Andaman and Nicobar	10° Channel
Minicoy-Lakshadweep	9° Channel
Maldives-Minicoy	8° Channel
India-Sri Lanka	Gulf of Mannar and Palk Strait

PHYSICAL FEATURES

Origin of Landforms

- The oldest landmass, the Indian Peninsula, was a part of the Gondwanaland. The Gondwanaland included India, Australia, Africa and South America as one single land mass. The convectional currents split the crust into a number of pieces, thus leading to the drifting of the Indo-Australian plate after being separated from the Gondwanaland, towards North. The Northward drift resulted in the collision of the Indian plate with the much larger Eurasian Plate.
- Due to this collision, the sedimentary deposit which were accumulated in the geosyncline (a long shallow depression between two large landmasses) known as the **Tethys** were folded to form the mountain system of Western Asia and Himalaya. The Himalayan upliftment out of the Tethys sea and subsidence of the Northern flank of the Peninsular plateau resulted in the formation of a large basin. In due course of time this depression, gradually got filled with deposition of sediments by the rivers flowing from the mountains in the North and the Peninsular plateau in the South.
- A flat land of extensive alluvial deposits led to the formation of the Northern plains of India. Geologically, the Peninsular plateau constitutes one of the ancient landmass on the Earth's surface. It was supposed to be one of the most stable land blocks. *The Himalayas and the Northern plains are the most recent landforms. From the view point of geology, Himalayan mountains form an unstable zone. The whole mountain system of Himalaya represents a very youthful topography with conical peaks, V-shaped valleys and fast flowing rivers. The Northern*

plains are formed of alluvial deposits. The Peninsular plateau is composed of igneous and metamorphic rocks with gently rising hills and wide valleys.

- Out of the total area of the country, about 10.6% is occupied by mountains, 18.5% by hills, 27.7% by plateaus and 43.2% by the plains.
- India may be divided into five major physiographic regions, viz, (I) The Northern Mountains, (II) The Great Plains, (III) The Peninsular Uplands, and (IV) The Indian Coasts and Islands, & (V) The Indian Desert

I. The Northern Mountains

- The region extends all along the northern frontier of the country, for about 2500 km, with a varying width 240 to 320 km and a total area of about 5, 00,000 km².
- Himalayas represent the youngest and the highest folded mountains of the earth, rising to over 8000 m above sea level and consisting of three parallel ranges :
 - Himadri (Greater Himalayas),
 - Himachal (Lesser Himalayas), and
 - The Siwaliks (Outer Himalayas).
- The Himalayas are intersected by numerous valleys like Kashmir valley, the Karewas, the Doon valley, the Kangra and Kullu valley (Himachal Pradesh), Kathmandu valley (Nepal), Bhagirathi valley (near Gangotri) and Mandakini valley (near Kedarnath).

Principal Peaks of India

Peak	Height (Mtrs)	Location
1. Mt. Everest	8848	Nepal-Tibet
2. Mt. K2 (Karakoram)	8611	India
3. Kanchenjunga	8597	Nepal-India
4. Dhaulagiri	8172	Nepal
5. Nanga Parbat	8126	India
6. Annapurna	8078	Nepal
7. Gasherbrum	8068	India
8. Nanda Devi	7817	India
9. Mt. Kamet	7756	India
10. Gurla Mandhata	7728	Tibet

Valleys and its locations

• Araku Valley	: Andhra Pradesh
• Damodar Valley	: Jharkhand and West Bengal
• Darma Valley	: Uttarakhand
• Dzukou Valley	: North-eastern part
• Johar Valley	: Uttarakhand
• Markha Valley	: Ladakh
• Nubra Valley	: Ladakh
• Sangla Valley	: Himachal Pradesh
• Saur Valley	: Uttarakhand
• Suru Valley	: Ladakh
• Tons Valley	: Uttarakhand
• Yumthang Valley	: Sikkim

Divisions of the Himalayas

(a) The sub-Himalayas or Siwaliks :

- The range has a total length of about 2400 km from the Indus gorge to the Brahmaputra valley.
- It is known by various local names, i.e. the Jammu hills (Jammu & Kashmir), the Dundwa range (Uttarakhand), the Churia Muria hills (Nepal), the Daffla, Miri, Abor and Mishmi hills (Arunachal Pradesh).

(b) The Lesser Himalayas or Himachal:

- It is about 80 km wide with average height between 1300 to 5000 m.
- Important ranges include the Dhauladhar, Pirpanjal, Nag Tiba, Mahabharat range and Mussoorie range. The famous hill resorts like Shimla, Chail, Ranikhet, Chakrata, Mussoorie, Nainital, Almora and Darjeeling, etc. are situated over this range.
- Along the slopes are found a number of *small pastures* which are called *Merg in Kashmir* (viz. Gulmerg, Sonmerg, Tanmerg) and *Bugyal and Payar in Uttarakhand*.

Mountain Passes of India

Himalayan passes

- Banihal pass — between Doda and Anantnag (Jawahar Tunnel), J & K.
- Shipki La — River Sutluj enters India from Tibet, Himachal Pradesh.
- Bara Lachan La — between Kyelang and Leh, Himachal Pradesh.
- Rohtang pass — between Kullu and Kyelang, Himachal Pradesh.
- Bomdila pass — between Tezpur and Tawang, Arunachal Pradesh.

Himalayan passes between India and China

- Shipki La — Himachal Pradesh.
- Thaga La and Niti La — Uttarakhand.
- Lipu Lekh La — Tri-junction, India-Nepal-China, Uttarakhand.
- Jelep La — Between India and China (Gangtok-Lhasa Road) Sikkim.
- Nathu La — Between India and China (Entry to Chumbi Valley) Sikkim.

Trans Himalayan passes

- Karakoram pass and Aghil pass — Jammu & Kashmir.

Passes in Western Ghats

- *Palghat* — between Palakkad and Coimbatore.
- *Shenkota* — between Kollam and Madurai.
- *Thalghat* — between Mumbai and Pune.
- *Bhorghat* — between Mumbai and Nasik.

- The best known passes of the Pir Panjal range are the *Pir Panjal Pass* (3480 m), the *Bidil* (4270m), *Golabghar* (9812m) and *Banihal Pass* (235m). The Jammu-Sri Nagar highway uses the Banihal Pass.

(c) The Greater Himalayas or Himadri

- This zone rises abruptly like a wall north of the Lesser Himalayas. It is about 25 km wide with average height above 5000 m.

- The Himadri runs in an arc like shape in a length of 2500 km from *Nanga Parbat* (8126 m) in the west to *Namcha Barwa* (7756 m) in the east.
- This is the northernmost or the innermost of all the Himalayan ranges.
- With an average elevation of 6100 m above sea level this is the loftiest and the most continuous mountain range of the world.
- This mountain range boasts of the tallest peaks of the world, most of which remain under perpetual snow.
- They are, in descending order of altitude, *Mount Everest*, also called *Sagarmatha* or *Chomo Langma* (8848m), *Lhotse 1* (8501m), *Mount Akalu* (8481m), *Kanchenjunga South Peak* (8474m), *Kanchenjunga West Peak* (8420m), *Lhotsa Intermediate Peak* (8410m), *Cho Oyu* (8153m), *Nanga Parbat* (8126m), *Annapurna* (8078m), *Gosainthan* or *Shisha Pangma* (8013m), *Makalu South peak* (8010 m).
- The *Burzil pass* and *Zoji La* in Kashmir, *Bara Lapcha La* and *Shipki La* in Himachal Pradesh, *Thaga La*, *Niti Pass* and *Lipu Lekh Pass* in Uttarakhand and *Nathu La*, *Jelep La* in Sikkim are worthy of mention.
- The *Hindustan - Tibet* road connecting Shimla with *Gartok* in Western Tibet passes through the *Shipki La*.
- Another important trade route connecting *Kalimpong* (near *Darjeeling*) with *Lhasa* in Tibet however passes through *Jelep La* (4386m).
- The Himadri runs in an arc like shape in a length of 2500 km from *Nanga Parbat* (8126 m) in the west to *Namcha Barwa* (7756 m) in the east.

(d) Trans Himalayas

- The Trans-Himalayan Zone with a width of 40 km in its eastern and western ends and a width of 222 km in its central part, it has important ranges such as the *Zaskar Range* and the *Great Karakoram range*. The *Karakoram* extends towards the south-east to form the *Kailash Range* (Tibet). The highest peak in the *Karakoram range* is *K₂* (8,611 m). The longest glacier is *Siachen* in the *Nubra Valley* which is more than 72 km long.

Some important facts about peaks

- Highest Mt. Peak in India: *K₂ or Godwin Austin*
- Highest peak in Aravalli: *Gurushikhar* (in Mt. Abu)
- Highest peak in Satpura - *Dhupgarh* (Mahadeo Hills)
- Highest peak in E. Ghats - *Mahendragiri* (Orissa)
- Highest peak in W. Ghats - *Anaimudi* (Annamalai Hills - Kerala)
- Highest peak in Nilgiris - *Doda Betta*
- Hills in Southern Hill complex - *Nilgiri, Annamalai, Cardamom & Palani*
- Hills in Eastern Ghats: *Shevaroy, Javadi, Palkonda, Nallamalai, Northern Circars*
- Oblique ranges to Western Ghats in Maharashtra: *Ajanta, Satmala, Harishchandra, Balaghat*
- Satpura range from East to West: *Amarkantak - Maikal- Mahadeo - Gawilgarh - Rajpipala*
- Highest peak in Andaman and Nicobar is-lands- *Saddle Peak*
- The highest peak of Naga hills is *Saramati peak*.

(e) The Eastern Hills or the Purvanchal

- After crossing the Dihang gorge, the Himalayas take a sudden southward turn and form a series of comparatively low hills running in the shape of a crescent with its convex side pointing towards the west. These hills are collectively called the Purvanchal because they are located in the eastern part of India.
- The hill ranges running in north-south direction along the Burmese border and passing through Arunachal Pradesh (Tirap division), Nagaland, Manipur and Mizoram are collectively called *Purvanchal*. These are known by various local names, i.e. Patkai Bum (Arunachal Pradesh), Naga hills, Kohima hills, Manipur hills, Mizo hills, Tripura hills and Barail range.
- Extending from Arunachal Pradesh in the north to Mizoram in the south, they form India's boundary with Myanmar.
- In the north is the *Patkai Bum*, which forms the international boundary between Arunachal Pradesh and Myanmar.
- After running for some distance southwards, it merges into Naga Hills where Saramati (3826m) is the highest peak.
- South of Naga Hills are the Manipur hills, which are generally less than 2500 metres in elevation.
- The Barail range separates Naga Hills from Manipur Hills.
- South of the Manipur Hills are the Mizo Hills, which have an elevation of less than 1500 metres. The highest point is the Blue Mountain (2157m) in the South.

Longitudinal divisions of the Himalayas

Longitudinally, the Himalayas can be divided into following sections.

1. The Punjab Himalayas

The 560 km long stretch of the Himalayas between the Indus and the Sutluj rivers is known as the Punjab Himalayas. A large portion of this sector lies in Jammu and Kashmir and Himachal Pradesh as a result of which it is also called the Kashmir and Himachal Himalaya. Karakoram, Ladakh, Pir Panjal, Zaskar and Dhauladhar are the main ranges of this section.

2. Kumaon Himalayas

This section extends from Sutluj to Kali river valleys and is said to have 360 lakes, such as *Naini Tal* and *Bhim Tal*. The Pilgrimage centers (*Badrinath*, Gangotri) located in this section is of particular importance to the Hindus.

3. Nepal Himalayas

This section extends from Kali to Tista and has the distinction of having some of the highest peaks in the world including Mt. Everest.

4. Assam Himalayas

This section extends from Tista to Brahmaputra. The highest peak of this range is Namcha Barwa.

II. The Great Plains

- It is an aggradational plain formed by the alluvial deposits of the Indus, Ganga and the Brahmaputra and

their tributaries. This is the largest alluvial tract of the world, extending for a length of 3200 km and width varies between 150 to 300 km.

- The plain stretches from west (from the banks of the Ravi and Sutluj) to east (the Ganga delta) to a length of 2400 km.
- The plain merges into the Thar desert in the south-west. A low watershed of the Delhi ridge (278 m) along the right bank of the Yamuna river separates the Satluj plains (a part of the Indus plain) from the Ganga plains.

Divisions of the Great Plains

The Great Plains may be divided into a number of smaller units on the basis of the characteristics of the alluvium, surface gradient, drainage channels and regional traits.

Bhabar Plains

It lies all along the foot of the Siwaliks with remarkable continuity from the Indus to the Tista. It is generally 8 to 16 km wide belt consisting of gravel and unassorted sediments deposited by the Himalayan rivers in the foreland zone due to sudden break of slope. The porosity is so high that all streams disappear in the Bhabar tract leaving out only dry channels.

Terai Plains

South of the Bhabar lies a 15-30 km wide marshy tract called terai where streams reappear to the surface.

The Terai is more marked in the eastern part than in the west due to higher amount of rainfall.

It is a zone of excessive dampness, thick forests & rich wild life.

Bangar or Bhangar Plains

The Bhangar represents the uplands (alluvial terrace) formed by the deposition of the older alluvium and lie above the flood-limit of the plains. The main constituent of Bhangar is clay (locally known as kankar) which at places gives way to loam and sandyloam.

Khadar Plains

The younger alluvium of the flood plains of the numerous rivers is called the Khadar or Bet (in Punjab).

Its alluvium is light coloured and poor in calcareous matter consisting of deposits of sand, silt, mud and clay.

Delta Plains

Deltaic plain is an extension of the Khadar plain. It covers about 1.86 lakh sq km of area in the lower reaches of the Ganga river (West Bengal). It mainly consists of old mud, new mud and marsh. Upland area is locally known as *chars* & *marshy* land as '*bills*'. Large part of the coastal deltas is covered by thick impenetrable tidal forests called *Sunderbans*.

On the basis of regional characteristics, the Great Plains may be divided into following sub plains:

The Punjab-Haryana Plains

With a distance of 640 km from north-east to south-west and 300 km from west to east, these flat plains occupy 1.75 lakh sq km. They comprise the *Bist Doab* (between Sutlej and Beas rivers), the *Bari Doab* (between Beas and Ravi rivers), the *Chaj Doab* (between Chenab & Jhelum) & the *Sindh Sagar Doab* (between Jhelum - Chenab & Indus).

The Rajasthan Plains

These comprise the marusthali of Thar and the nearly Bagar areas and occupy an area of 1.75 lakh sq km. Sand dunes are longitudinal (in the west and south) and transverse (in the east). The Luni river flowing towards the south-west is the only river in the region. The Rajasthan desert is sloping towards two directions

- (a) westwards to the Indus Valley in Pakistan, and
- (b) southwards to the Rann of Kutch.

The Ganga Plains

Spreading across the states of Uttar Pradesh, West Bengal and Bihar for 3.57 lakh sq km, the plains comprise the raised bhangar areas and Khadar areas. The Ganga Plain is divided into the

- (a) Ganga-Yamuna Doab,
- (b) Rohilkhand Plain,
- (c) Avadh Plain (covering the eastern half of Uttar Pradesh),
- (d) Bihar Plain, and
- (e) Bengal Plain.

The Ganga delta, which constitutes the Bengal basin, has part of it stretching along the sea and covered with tidal forests (the Sunderbans).

The Brahmaputra Plains

The low-level plains formed by deposits carried by mainly the Brahmaputra river but also the Dibang, the Sibi and the Lohit are bordered by high mountains. Moist soil conditions and thick forests form the northern extreme.

III. The Peninsular Uplands

Rising from the height of 150 m above the river plains up to an elevation of 600-900 m is the irregular triangle known as the *Peninsular Plateau*.

The Peninsular plateau is a tableland composed of the old crystalline, igneous and metamorphic rocks. It was formed due to the breaking and drifting of the Gondwanaland and thus, making it a part of the oldest landmass. This region of the country is surrounded on the three sides by water and thus, is a Peninsular plateau.

The plateau has broad and shallow valleys and rounded hills. Narmada river, which flows into a Rift valley, divided the region into two parts namely, the central highlands in its North and the Deccan plateau in its South.

- Its north-west limit is marked by Aravalli range and its northern extreme has the raised Bundelkhand. At its eastern and western ends are Eastern Ghats and Western Ghats respectively.
- The fault in which the Narmada river flows divides the region into two unequal parts; the smaller one in the north being known as the Central Highlands.
- It is slightly tilted towards north. The southern part has been tilted east with bold heights to the west. This area is popularly known as the Deccan Plateau comprising the Satpuras, Western and Eastern Ghats and a large number of plateaus.

On the basis of prominent relief features, the Peninsular plateau can be divided into three broad groups:

1. The Central Highlands

The Central Highlands are bounded to the West by the Aravalli range. Satpura range demarcates its boundary in the South from Deccan plateau.

An Eastern extension of central high lands is formed by Rajmahal hills.

The general elevation of the central high lands ranges between 700-1000 m above the mean sea level. It slopes towards the North and North-Eastern directions.

The Aravalli Range

It runs North-East to South-West for 800 km from Delhi through Rajasthan to Palanpur in Gujarat. These are the relict mountains representing one of the world's oldest high lands formed as a result of folding process in Archean times. It has a lower elevation between Delhi and Ajmer, where it is characterised by a chain of discontinuous ranges. But it becomes a continuous range South of Ajmer where it rises to 900 m elevation.

The general height of this range varies between 400-1300 m. Gure Skikhar 1722 m) is the highest peak of the range, located in Ubu hills of Rajasthan. Barr, Piplighat, Dewari, Desuri are some of the passes associated with this range.

East Rajasthan Uplands

It is located East of the Aravalli range in North-West India. The upland covers an area of 23,200 sq km and have a general elevation of 350 m. It constitutes the Northern part of Central Highlands.

Madhya Bharat Plateau

It is the Northern of the central highlands. It covers an area of about 22 thousand sq km.

Bundelkhand and Baghelkhand Uplands

It lies to the South of Yamuna river along border region of Uttar Pradesh and Madhya Pradesh. Bundelkhand covers five districts of Uttar Pradesh and four districts of Madhya Pradesh. Baghelkhand lies South-East of the Bundelkhand region and is largely made up of limestones and sandstones. They are represented by rounded Hummocky hills made of granite and sandstone. Streams like Betwa and Ken have carved out steep gorges, rocky banks and waterfalls in these uplands rendering them unfit for cultivation. The region is characterized by 'senile topography'

The Malwa Plateau Largely in Northern Madhya Pradesh, forms a triangular shape and is typical for having two systems of drainage. Rivers like Mahi and Narmada flow through it into the Arabian sea, while rivers like Chambal and Betwa flow through it to join Yamuna and ultimately fall into Bay of Bengal. It is composed of lava flows and is covered with black soils. This plateau is marked in its North by the badlands or ravines formed by Chambal river by exercising gully erosion.

The Vindhya Ranges

It runs parallel to the Narmada Rift valley as an escarpment in an East-West direction from Jabat in Madhya Pradesh to

Sasaram in Bihar for a distance of 1200 km. The general elevation of the ranges is 400-700 m. It consists of horizontal beds of sedimentary rocks.

This range acts as a watershed between North flowing rivers of the Ganga system and Peninsular rivers. It continues Eastwards as Bharner hills and Kaimur hills. Panna hills also lie in these ranges. The Great boundary fault separates the Aravallis from Vindhyan range.

The Chhotanagpur Plateau

It lies East of Baghelkhand in the State of Jharkhand covering some parts of Chhattisgarh and West Bengal. Its average elevation is 700 m above sea level. It is the storehouse of minerals and a large scale mining of iron, manganese, coal, uranium etc is done in this region. This plateau is drained by numerous rivers forming a radial drainage pattern. Damodar river valley is well-known for its coal deposits.

In the North-East of this region lies **Hazaribagh plateau**, while towards East are **Parasnath hills** and towards South-East is **Ranchi plateau**. **Rajmahal hills** form the North-Eastern edge of the Chhotanagpur plateau and are covered by black soil. The plateau is an example of Pat Land.

2. Deccan Plateau

Deccan plateau is bordered by the Western ghats in the West, Eastern ghats in the East and the Satpura, Maikal and Mahadeo hills in North. The Deccan Plateau is higher in the West and slopes gently eastwards. It is higher in South than its North.

The Satpura Ranges

It is a series of seven mountains that run in the East-West direction in between Narmada and Tapi rivers. It is an example of block mountain. Commencing from the Rajpipla hills in the West through the Mahadev hills to Maikal range, it stretches for about 900 km.

Dhupgarh (1325 m) on Mahadev hills near Pachmarhi in Madhya Pradesh is the highest peak of the ranges. Amarkantak is another important peak lying in the Maikal range at Madhya Pradesh-Chhattisgarh border and is the source of river Narmada.

Maharashtra Plateau is a basaltic sheet with a thickness of more than 2000 m. It has been formed by the consolidation of the lava. Maharashtra plateau covers the entire State of Maharashtra except Konkan coast and the Sahyadris.

Karnataka Plateau is situated at the south in Deccan plateau. It covers entire Karnataka except a small portion of North-East. It has the rocks of lava origin.

Telangana Plateau constitutes the North- Eastern part of Deccan plateau. It covers an area of 1.4 lakh sq km. It is located in the Western part of Andhra Pradesh.

The Western Ghats or Sahyadris form the Western edge of the Deccan plateau and lie parallel to the Western coast. They form a continuous water divide. They run continuously for 1600 km from Maharashtra to Kanyakumari and can be crossed through passes only.

Their average width varies from 50 km in the North to about 300 km in the South. It is composed of lava deposits up to 16°

N latitude. Important peaks of the Western Ghats from North to South include - Kalsubai (1646m - highest peak of Maharashtra), Salher (1567 m), Mahabaleshwar (1424 m) in Maharashtra and Kudremukh (1892 m) as the highest peak in Karnataka. The Sivasamudran fall, Gokak fall and Mahatma Gandhi fall are important waterfalls in Western Ghats. Bhore Ghat, Thalghat and Palghat are the important passes facilitating movement between the Western coastal plains and rest of the country. Palghat gap separates Nilgiri hills in the South from Anaimalai hills. It is at the Nilgiris that Western Ghats and Eastern Ghats meet with each other. Anamudi (2695 m) in the Anaimalai hills is the highest peak of whole of Southern India. Another important peak is Dodabetta (2637m) of Nilgiri hills. Cardamom hills are the Southernmost hills and reach upto Kanyakumari and are famous for cultivation of Cardamom and spices.

The Eastern Ghats are discontinuous and irregular and dissected by rivers draining into the Bay of Bengal. These are a series of detached hills of heterogeneous composition which are called by various local names. The Western ghats are higher than the Eastern ghats.

Their average elevation is 900-1600 m as against 600 m of the Eastern Ghats. The Eastern Ghats stretch from the South of Mahanadi valley to the Nilgiris in the South. The Eastern Ghats are comparatively broader and do not form a continuous water divide.

Aroya Konda (1690 m) at Andhra-Odisha border is the highest peak of Eastern Ghats. Other important peaks include Mahendragiri (1501 m) in Odisha, Gali Konda (1643 m) etc. It is a continuous range from south of Mahanadi to Godavari. South of it, it is highly dissected.

They continue South of Krishna river in the form of dissected hills from North to South in Andhra Pradesh and Tamil Nadu as Nallamala hills. Palkonda range, Velikonda range, Javadi hills and Shevaroy hills only to confluence with Western Ghats at Nilgiri hills. Khondalites are predominantly found in Eastern Ghats.

Difference between Eastern Ghats and Western Ghats

Western Ghats	Eastern Ghats
Form a continuous water divide.	Discontinuous and dissected by rivers.
Can be crossed through passes	Series of detached hills
Higher than Eastern Ghats	Lower than Western Ghats
Face Arabian sea and run along the western plateau	Face Bay of Bengal and run along the Eastern Plateau
Comparatively narrow	Comparatively broader
Highest Peak-Anaimudi	Highest Peak-Mahendragiri

3. The North-Eastern Plateau

It is an extension of the main Peninsular plateau. It is believed that due to the force exerted by the North-Eastward movement of the Indian plateau at the time of the Himalayan origion, a huge fault was created between the Rajmahal hills and the Meghalaya plateau. Later this depression got filled up by the deposition activity of the numerous rivers. Today, the Meghalaya and Karbi Anglong plateau stand detached - from the main Peninsular Block. The Meghalaya plateau is further sub- divided into three parts:-

- (i) The Garo hills
- (ii) The Khasi hills
- (iii) The Jaintia hills

These hills are named after the tribal groups inhabiting this region. An extension of this is also seen in Karbi- Anglong hills of Assam. Similar to the Chhotanagpur plateau, the Meghalaya plateau is also rich in mineral resources like coal, iron ore, Sillimanite, limestone and uranium. This area receives maximum rainfall from the South West monsoon. As a result the Meghalaya plateau, especially Cherrapunji, displays a bare rocky surface devoid of any permanent vegetation cover. Highest peak of Meghalaya plateau is Nokrek, which is located in Garo hills.

Wandakaraya Region:- Its Abujhmar hills provide one of the richest iron-ore deposits at bailadilla range.

IV. The Indian Coasts and Islands

The Coastal Plains

The 4,500 km long coastline of India has the Arabian Sea on its west, the Bay of Bengal on its east and the Indian Ocean on its south. It runs from the *Rann of Kutch* in the west to the delta of the Ganga - Brahmaputra in the east. The coastal plain in India has been divided into the following two types:-

Western Coastal Plains

The Western coastal plains are an example of submerged coastal plain. The Western Coastal Plains include plains along Kutch and Kathiawar regions of Gujarat to Konkan plains of Maharashtra-Goa, Karnataka plains and the Southern Malabar plains along Kerala coast.

It is believed that the city of Dwarka which was once a part of the Indian main land situated along the West coast is submerged under water. Because of submergence it is a narrow belt and provides natural conditions for the development of ports and harbours. It consists of three sections:-

- (i) **The Konkan Plains** extend from Daman to Goa for a distance of around 500 km with a width varying between 50-80 km. It consists of features of marine erosion as it is a shore line of submergence with a number islands dotting the coast. Mumbai was also islands until some connecting land was recovered from sea. These plains have fertile stretch of land and some sea water inlets called creeks are also there in between.
- (ii) **Karnataka or Kannada Coastal Plains** stretch from Goa to Mangalore for about 225 km. These are narrow fertile plains. Sharavathi river forms Jog falls (271 m) in this region, which is the highest waterfall of India.
- (iii) **The Malabar Coastal Plains** between Mangalore and Kanyakumari are 500 km long. They are much wider in extent and at places 96 km wide. These are low lying plains with extensive presence of lagoons, back waters, spits etc. The back water locally known as Kayals are the shallow lagoons or the inlets of the sea into the region. The largest among these is the Vembanad lake. Several lagoons have joined to form inland water ways. Every year the famous Nehru Trophy Vallamkali (boat race) is held in Punnamada Kayal in Kerala.

Eastern Coastal Plains

As compared to the Western coastal plain the Eastern coastal plain is broader and is an example of an emergent coast. There are well developed deltas here, formed by the rivers flowing Eastward into the Bay of Bengal. Because of its emergent nature, it has less number of ports and harbours. The plains are divided into two parts:-

- (i) **Northern Circars:** Rivers such as the Mahanadi, the Godavari, the Krishna and the Cauveri have formed extensive deltas on this coast. It comprises the coastal areas of Odisha. Mahanadi forms delta with a thick layer of fine alluvium. Lake Chilka is important here. In South of it, it comprises Andhra coastline upto Krishna delta.
- (ii) **Coromandel Coast:** South of Northern circars coromandel coast lies extending Southwards along Tamil Nadu coast. It covers an area of 23 thousand sq km. These plains constitute a fertile stretch of the cultivation of a range of crop.

The Islands

There are two main groups of islands in the Indian ocean far away from the coast. India has a number of islands (247) both in the Bay of Bengal (204 islands) and the Arabian sea (43 islands). One of them is the Lakshadweep Islands in the Arabian sea and the other is the Andaman and Nicobar islands in the Bay of Bengal.

Lakshadweep Islands

These islands group lies close to the Malabar coast of Kerala. This group of 25 islands is composed of small coral islands. The islands North of 11° N latitude are known as **Amindivi islands** and those South of it are Cannanore islands. They cover small area of 32 sq km.

Most of the islands have low elevation and do not rise more than 5 m above sea level. Shallow lagoons can be seen on their Western sides while sea slope is steeper towards their Eastern coasts. Kavaratti island is the administrative headquarters of Lakshadweep. The Pitli island, which is uninhabited, has a bird sanctuary.

Andaman and Nicobar Islands

Andaman and Nicobar archipelago has been formed by the extension of the tertiary mountain chains of **Arakan Yoma**. These islands lie close to equator and experience equatorial climate and have thick forest cover. Some of the islands are fringed with coral reefs. The entire group of islands is divided into two broad categories: the Andamans in the North and the Nicobars in the South.

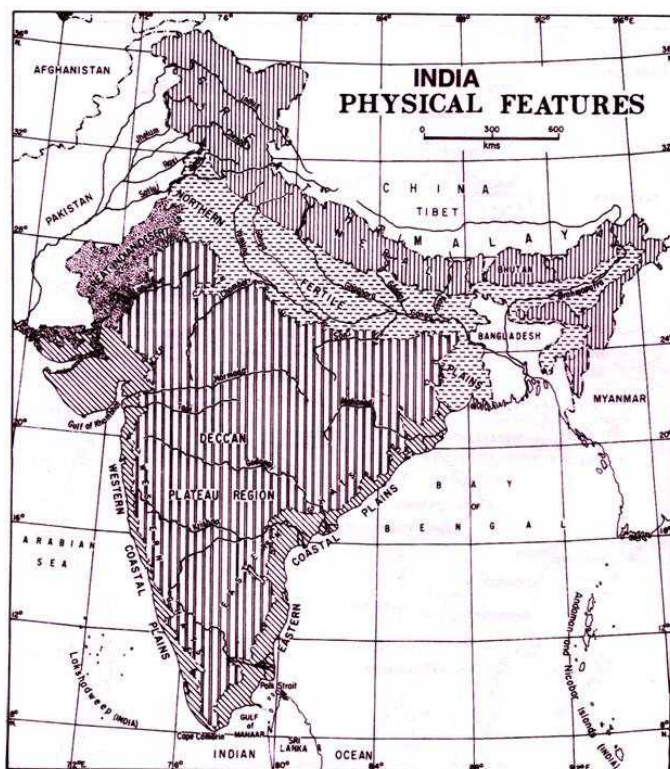
The great Andaman group of islands in the North is separated by the Ten Degree Channel from the Nicobar group in the South. The Andaman islands are sub-divided into four major island groups: North Andaman, Middle Andaman, South Andaman and Little Andaman. Biggest island in Andaman Group is Middle Andaman (1536 sq km) and smallest is Ross island (0.8 km). The capital Port Blair is located on Eastern coast of South Andaman. Barren island and Narcondam island located North of Port Blair are volcanic islands.

Biggest island in Nicobar Group is Great Nicobar (1045 sq km) and smallest is Pilo Milow Island (1.35 sq km). Andaman and Nicobar islands has 4 National Parks and 40 small ports.

The Indian Desert

The Great Indian Desert also called **Thar desert** lies to the North-West of the Aravalli hills. It spreads over four states namely- Punjab, Haryana, Rajasthan, and Gujarat. It is a land of undulating topography dotted with longitudinal dunes and branches. It has arid climate due to very low rainfall (below 150 cm) received by this region.

This region is also called as **Marusthali**. The underlying rock structure of desert is an extension of the peninsular plateau but its surface features have been covered by physical weathering and wind action due to extreme arid condition in this region. The Great Indian Desert can be divided into two parts: the Northern part is sloping towards Sindh and the Southern towards the Rann of Kutch. Most of the rivers in this region are ephemeral. Luni is one of the most important rivers of this region.



Mountain Passes in Peninsular India

Bhor Ghat	Bhor ghat in Western ghat joins Mumbai with Pune.
-----------	---

Rain fall Distribution in India

Amount of Rain fall	Heavy Rainfall (> 200cm)	Moderately Heavy Rainfall (100-200 cm)	Less Rainfall (50-100 cm)	Scanty Rainfall < 50cms
States	West coasts, on the western Ghats, Sub-Himalayan areas in North East and Meghalaya Hills. Assam, West Bengal, Southern slopes of eastern Himalayas.	Southern Parts of Gujarat, East Tamil Nadu, North-eastern Peninsular, Western Ghats, eastern Maharashtra, Madhya Pradesh, Orissa, the middle Ganga valley.	Upper Ganga valley, eastern Rajasthan, Punjab, Southern Plateau of Karnataka, Andhra Pradesh and Tamil Nadu.	Northern part of Kashmir, Western Rajasthan, Punjab and Deccan Plateau

Goran Ghat	In Aravalli hills, joins city of Udaipur with Sirohi in Rajasthan
Haldighati	In Aravalli hills and connects Rajsamand with Pali district in Rajasthan.
Palghat	In Western ghat, joins Coimbatore with Kochi and Kozhikode. It joins Tamil Nadu with Kerala. The river Gayatri flows through it.
Thalghat	Located in Sahyadri range, joins Nashik with Mumbai. The NH-3 and the Bhopal-Indore railway line pass through it.

States with the Longest Coastline

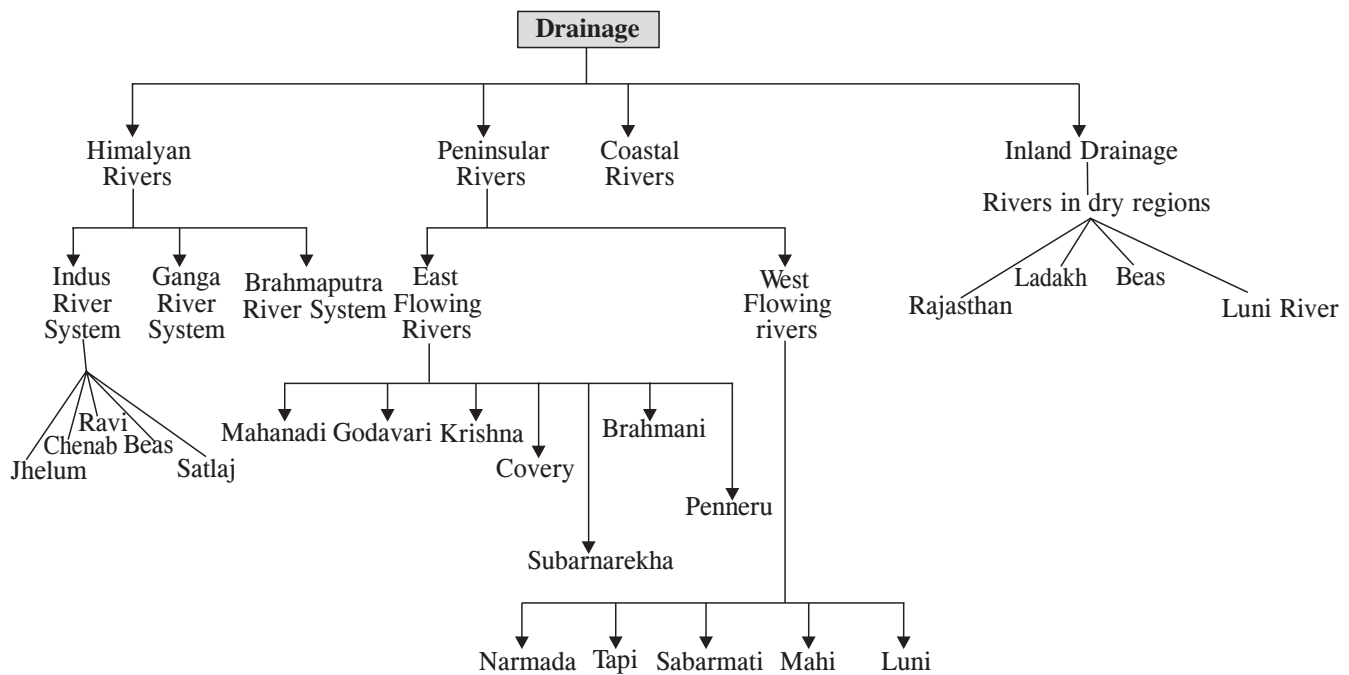
State/UT	Length of Coastline (Km)
1. Andaman & Nicobar Islands	1,962
2. Gujarat	1,215
3. Andhra Pradesh	974
4. Tamil Nadu	907
5. Maharashtra	653

Climate

Although India is basically a tropical country, it experiences wide variation in climatic condition depending upon the altitude, latitude, distance from sea and relief. The variability can be observed in number of factors such as:

Western Rajasthan experiences a high temperature during June where as the areas close to Kashmir are relatively experiencing a much lower temperature. The coastal lands are comparatively having a moderate climate due to the nearness of sea.

The amount of rainfall also varies throughout the country. The rainfall in India is primarily governed by Monsoon wind which generally hits the south west coast of India generally in June and known as onset of Monsoon. The wind then starts circulating via the Bay of Bengal covering the entire eastern, north eastern and parts of central India. The highest rainfall is experienced in Mawsynram in Meghalaya i.e. 1221 cm of annual rainfall every year. On the other hand in the month of October and November the monsoon trough of Low pressure starts receding from Northern Plain results into rain in Southern India. About 50% to 60% of rainfall in Tamil Nadu is caused due to Retreat of Monsoon from North East.



DRAINAGE

The flow of water through well-defined channels is known as Drainage and the network of such channels is called a drainage system. A river drains the water collected from a specific area, is called its catchment Area. An area drained by a river and its tributaries is called a Drainage Basin. The boundary line separating one drainage basin from the other is known as watershed.

India is drained by numerous rivers falling either into the Bay of Bengal or the Arabian Sea. The Ganga, Brahmaputra, Mahanadi, Godavari, Krishna and Cauvery are the major river systems draining into the Bay of Bengal, whereas the major river systems draining into the Arabian Sea are the Indus, Sabarmati, Narmada, Tapi and rivers of the west coast farther south. Indian rivers may be classified into four types depending upon the nature of the river, geographical location, source and drainage area covered: Himalayan, peninsular, coastal and rivers of inland drainage basin.

Himalayan Rivers

- These rivers are *perennial* as they are generally snow-fed and have reasonable flow throughout the year. During the monsoon the Himalayas receive very heavy rainfall and the rivers discharge the maximum quantity of water causing frequent flood. Brahmaputra, Ganga, Jamuna, Ghagra, Gandak, Kosi are the main rivers.
- Some rivers are older than the mountain themselves. Their gorges are the evidence of it. So, they are also examples of antecedent rivers.

The Major river systems of the Himalayas Drainage

Indus River System

- Indus along with its tributaries forms one of the largest drainage systems of the world.
- Also known as Sindhu.
- Indus originates from the northern slope of Kailash range (Tibet) enters India and continues to flow in the north-west direction between the Ladakh and the Zaskar ranges.
- The **Shyok** and **Gilgit** are its important right bank tributaries and the **Zaskar** is left bank tributary.
- Indus river ends its mountainous journey at Attock and is joined by the Kabul river from Afghanistan.
- Just above Mithankot, it receives accumulated waters of five rivers (Panj-nad) Jhelum, Chenab, Ravi, Beas and Sutluj.
- Finally, it empties itself in the Arabian Sea near Karachi, making a big delta.
- It has length of 2900 km from its source to the Arabian sea.
- It is known as Singi Khamban Tibet.

Jhelum

- It rises from a spring at Verinag (in Kashmir).
- Lidar**, **Sind** and **Pohru** are the tributaries of Jhelum in Kashmir.
- At Muzaffarabad, the river takes a sharp hairpin swing southward and the **Kishaganga** joins at on its right bank.
- Thereafter, it forms the India-Pakistan boundary for 170 km and emerges at the Potwar Plateau near Mirpur. It has total length of 724 km.
- It joins the Chenab at Trimmu.

Chenab

- It is the largest tributary of the Indus.

- It originates near the Bara Lacha Pass in the Lahul-Spiti part of the Zaskar Range. The united stream (Chandra and Bhaga) called the *Chandrabhaga* flows in the north-west direction through Himachal Pradesh and enters Jammu & Kashmir as Chenab river.
- It enters the plain area near Akhnur in J&K.
- It cuts a deep gorge near Kishtwar.
- It receives waters of Jhelum and Ravi rivers.

Ravi

- It originates from Kullu hills near the Rohtang Pass in Himachal Pradesh.
- It cuts a deep gorge in the Dhaul Dhar range after crossing Chamba.
- It enters Punjab Plains near Madhopur and later enters Pakistan 26 km below Amritsar.
- It debouches into the Chenab a little above Rangpur in Pakistani Punjab.

Beas

- It also originates near Rohtang Pass, close to the source of the Ravi.
- It crosses the Dhaul Dhar range through a deep gorge from Lorji to Talwara.
- It debouches on the plain near Pong and meets the Satluj river at Harike.
- It lies entirely within the Indian territory.

Satluj

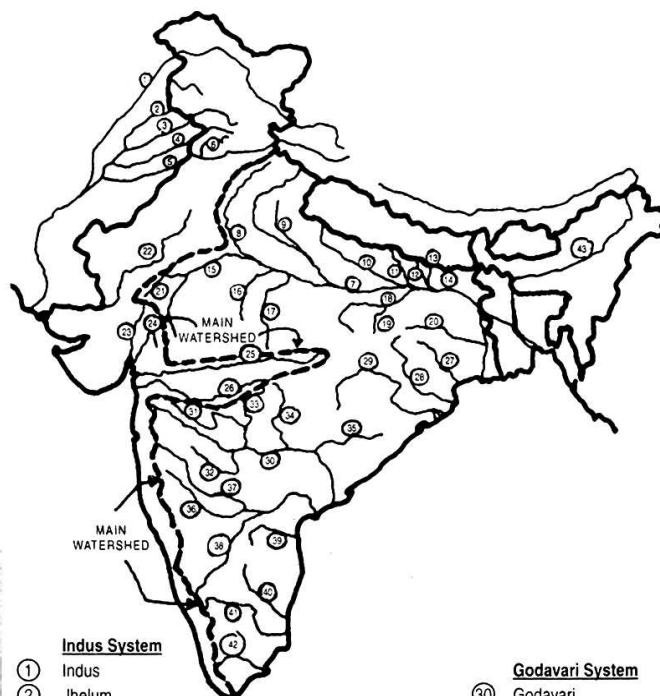
- It rises from the Mansarovar - Rakas Lake near Darma Pass in western Tibet, where it is also known as Langcher Khambab.
- In Nari Khorsan province of Tibet, it has created an extraordinary canyon.
- It is joined by the Spiti river at Namgia near the Shipki La.
- Before entering the Punjab Plain, it cuts a gorge in Naina Devi Dhar (Bhakra Dam has been constructed here).
- It enters the plain at Rupnagar (Ropar).
- It is joined by the Beas at Harike.
- From near Ferozepur to Fazilka, it forms the boundary between India and Pakistan for nearly 120km.
- It joins the Indus a few kilometers above Mithankot.

The Ganga River System

- It is the Largest in India.
- The total area of the Ganga basin in India is 861,404 sq km which accounts for 26.3% of the geographical area of the country.
- The Ganga basin covers over 12,500 sq km in northern India.

This basin is shared by:-

State	(%) share
Uttarakhand & UP	34.2
MP & Chhattisgarh	23.1
Bihar & Jharkhand	16.7
Rajasthan	13.0
West Bengal	8.3



Indus System			Godavari System		
①	Indus	⑩	③①	Godavari	
②	Jhelum	⑪	③②	Penganga	
③	Chenab	⑫	③③	Manjara	
④	Ravi	⑬	③④	Wardha	
⑤	Satluj	⑭	③⑤	Waiganga	
⑥	Beas	⑮	③⑥	Indravati	
Ganga System			Krishna System		
⑦	Ganga	⑮	③⑦	Krishna	
⑧	Yamuna	⑯	③⑧	Bhima	
⑨	Ramganga	⑰	③⑨	Tungabhadra	
⑩	Ghaghra	⑱	④①	North Pennar	
⑪	Gandak	⑲	④②	South Pennar	
⑫	Burhi Gandak	⑳	④③	Cauvery	
⑬	Bagmati	㉑		Waigai	
⑭	Kosi	㉒		Brahmaputra	
⑮	Chambal	㉓			
⑯	Kali Sindh				
⑰	Betwa				
⑱	Son				
㉑	Rihand				
㉒	Damodar				
㉓	Banas				
㉔	Luni				
㉕	Sabarmati				
㉖	Mahi				
㉗	Narmada				
㉘	Tapti				
㉙	Subarnarekha				
㉚	South Koel/Brahmani				
㉛	Mahanadi				

Ganga

- It originates as *Bhagirathi* from the Gangotri glacier.
- Alaknanda* joins it at *Devprayag*. Pindar river joins it at Karan Prayag and *Mandakini* or *Kali Ganga* at Rudra Prayag.
- The combined waters of the Bhagirathi and the Alaknanda flow in the name of the Ganga, below *Devprayag*.
- It debouches on plain from hills in Haridwar.
- It is joined by Yamuna in Allahabad.
- Beyond *Farakka*, it is known as *Padma* in Bangladesh.
- It bifurcates itself into Bhagirathi-Hooghly in West Bengal and Padma-Meghna in Bangladesh.
- The delta formed by the Ganga-Brahmaputra is the largest delta of the world covering an area of 58,752 sq km.
- Sundarbans is a part of the world's largest delta.
- The total length, 2525 km, is distributed among states:
 - Uttar Pradesh – 1140,
 - W. Bengal – 520 km,
 - Bihar – 445 km,
 - Uttarakhand – 310 km.

The Ganga Drainage System			
Name of the river	Source	Length (in km)	Area drained (sq km)
Ganga	Gangotri Glacier at 7,010 m	2,525	861,404

Yamuna	Yamnotri Glacier at 6,330	1,376	366,223
Chambal	Near Mhow (Indore-M.P)	1,050	139,468
Ramganga	Garhwal district at 3,110 m	596	32,493
Ghaghra	Near Gurla Mandhota peak	1,080	127,950
Gandak	South of Manasarovar	425 in India	46,300 (7,620 in India)
Kosi	Tibet-Nepal border at 7,620 Sikkim Nepal- Tibet Himalaya	730 in India	86,900 (21,500 in India)

Difference between Delta and Estuary

Delta	Estuary
1. The triangular deposits made by rivers at their mouth form Delta.	1. The sharp edged mouth of rivers, devoid of any deposits is known as estuary.
2. Deltas are founded in the regions of 100 tides and coastal plains	2. Regions of high tides and rift valleys witness estuaries.
3. Deltas are fertile lands	3. Estuary does not have festive lands.
4. Ganga, Brahmaputra, Krishna, Kaveri and Mahanadi rivers form Delta	4. Narmada and Tapi rivers form estuaries.

Yamuna

- It is the largest and the most important tributary of the Ganga.
- It originates from the Yamunotri glacier on the Bandarpunch Peak in Garhwal in Uttarakhand.
- It enters the plains near *Tajewala*.
- Tons**, a tributary of it, joins it below Kalsi. At this site, the water carried by the Tons is twice the water carried by the Yamuna.
- It takes a southerly course upto Mathura and south easterly in its onward journey upto Allahabad where it unites with the Ganga.

Chambal

- It rises near Mhow in the highlands of Janapao Hills in MP.
- It enters a gorge at Chaurasigarh.
- It joins Yamuna in Etawah district of Uttar Pradesh.
- Banas joins it near Sawai Madhopur.
- Betwa**, rising in Bhopal, joins the Yamuna near *Hamirpur*. Dhasan is an important tributary of Betwa

Son

- It is a large south bank tributary of the Ganga.
- The Son river springs from the Amarkantak Plateau.
- It joins the Ganga near Danapur in Patna district.
- Its catchment area is 71,259 sq km.
- Almost all the tributaries join it on its right bank.
- Tributaries are Johilla, Rihand, Kanhar and North Koel.

Damodar

- It rises in the hills of the Chotanagpur plateau and flows through a rift valley.
- It is also called '**Sorrow of Bengal**'
- It joins the Hugli 48 km below Kolkata.
- The total length of the river is 541 km.
- Its catchment area is 25,820 sq km.

Ramganga

- It rises in the Garhwal district of Uttaranchal.
- It enters the Ganga plain near Kalagarh.
- Its basin covers 32,493 sq km.

Ghaghra

- It originates near the Gurla Mandhota peak, south of Manasarovar in Tibet.
- It is known as the karnali in Western Nepal.
- It joins Ganga a few kilometres downstream of Chapra in Bihar.
- The total catchment area of the river is 127,950 sq km out of which 45% is in India.

Kali

- It rises in high glaciers of snow covered region of trans-Himalayas.
- It forms the boundary between Nepal and Kumaon.
- It is known as the Sarda or Chauka after it reaches the plains near Tanakpur.

Gandak

- It originates near the Tibet-Nepal border.
- Kali Gandak, Mayangadi, Bari and Trishuli are the major tributaries of it.
- Its drainage area is 46,300 sq km out of which 7620 sq km is in India.

Burhi Gandak

- Originating from the western slopes of Sumesar hills near the India-Nepal border, it joins the Ganga opposite Monghyr town.
- Its length is 610 km and drainage area is 12,200 sqkm.

Kosi

- The Kosi river consists of seven streams, namely, Sut Kosi, Tamba Kosi, Talkha, Doodh Kosi, Botia Kosi, Arun and Tamber and is popularly known as Saptkaushiki.
- Seven rivers mingle with each other to form three streams named the Tumar, Arun and Sun Kosi.
- Then all three streams unite at *Triveni* north of the Mahabharat Range to form the Kosi.

The Brahmaputra River System

- The Brahmaputra rises in the great **chemayungdung** glacier in the **Kailas** range of the Himalayas.
- It flows eastward from its source region.
- Mariam La separates the source of the Brahmaputra from the Manasarovar lake.

- With a total length of 2900 km, it is one of the longest rivers of the world & passes through Tibet, India and Bangladesh.
- It is known as Tsangpo (means purifier) in Tibet and Yarlung Zangbo Jiangin in Chinese language.
- It is one of the most remarkable navigable waterways of the world where boats sail at an altitude of about 3000 metre above sea level.
- It emerges as a dynamic river after carving out a deep gorge near Namcha Barwa.
- It emerges from the foothills under the name of Siong or Dihang.
- It enters India west of Sadiya town in Arunachal Pradesh where it receives the Dibang and the Lohit. From here after ward, it is known as the *Brahmaputra*.
- It has a braided channel along most of its length in Assam.
- It is among the four largest rivers of the world in terms of volume of discharge at the mouth.
- The look is like a delta in reverse where Dibang and Lohit rivers meet the Brahmaputra river.
- **Tista**, a tributary of the Brahmaputra, was a tributary of the Ganga prior to the devastating floods of 1787.
- **Majuli** is the river island of the river Brahmaputra (area- 1250 sq km). **Majuli island** is Largest riverine island of world.
- National waterways - 2 is on the Brahmaputra river from Sadiya to Dhubri.

The Peninsular River System

Three main directions of flow:

- Mahanadi, Godavari, Krishna, Cauvery and several smaller rivers draining towards south-east towards into the Bay of Bengal.
- The Narmada and the Tapi flowing towards west as well as several small rivers originating from the Western Ghats flow westwards into the Arabian Sea.
- Tributaries of Ganga and Yamuna such as Chambal, Betwa, Ken, Son and Damodar flow in the north-easterly direction.

The East Flowing Rivers

Mahanadi

- It has its source in Dandakaranya near Sihawa in Raipur district of Chhattisgarh.
- Its upper course lies in the saucer-shaped basin called the 'Chhattisgarh Plains'
- **Hirakud dam** is built on this river.

Godavari

- It is the largest river system of the Peninsular India.
- It is held in reverence as '**Vridha Ganga**' or '**Dakshina Ganga**'
- It has a catchment area of 312,812 sq km which covers about 10% of the area of India.

- It flows in Maharashtra, Andhra Pradesh, Madhya Pradesh, Chhattisgarh, Orissa and Karnataka.
- The source of the river is in the Trimbak Plateau of North Sahyadri near Nasik in Maharashtra and discharges its water into the Bay of Bengal.
- Manjra is the only important right bank tributary which joins the Godavari near Kondalwadi.
- Below Rajahmundry, the river Godavari divides itself into two main streams - the Gautami Godavari on the east and the Vashishta Godavari on the west - and forms a large delta before it pours into the Bay of Bengal.

Krishna

- It is the second largest east flowing river of Peninsula in India.
- It rises in Western Ghats near **Mahabaleshwar**.
- It debouches into the Bay of Bengal, forming a big delta in arcuate shape.
- The Koyna, Tungbhadra and Bhima are its major tributaries.
- *Koyna Dam* is made on the Koyana river, a tributary of the Krishna river.

Cauvery

- It is designated as "the Ganga of the South" or 'Dakshina Ganga'.
- Its source lies at Taal Cauvery on the Brahmagiri range of hills in the Western Ghats.
- Its upper catchment area receives rainfall during summer by the south-west monsoon and the lower catchment area receives rainfall during winter season by the retreating northeast Monsoon.
- It is one of the best regulated rivers and 90 to 95% of its irrigation and power potential already stands harnessed.
- *Sivasamudram waterfalls* is on this river.
- The river divides itself into two distinct channels at Srirangam, the northern channel is called Kollidam and the southern one retains the name Cauvery.
- Cauvery river also forms a big delta in a quadrilateral shape.

Subarnarekha

- It originates from the Ranchi plateau in Jharkhand.
- It forms the boundary between West Bengal and Orissa in its lower course.
- Its total length is 395 km.

Brahmani

- It comes into existence by the confluence of the Koel and the Sankh rivers near Rourkela in Odisha.
- It has a total length of 800 km.
- Its main tributaries are Kura, Sankhad and Tikra.

Penneru

- It springs from the Nandi Durg peak in Karnataka.
- The total length is 597 km.
- The principal tributaries are the Jayamangli, the Kunderu, the Chitravari, the Papagani and the Cheyyeru.

The West Flowing Rivers

Narmada

- It is the largest of all the west flowing rivers of the Peninsula.
- It rises from the Amarkantak plateau in Shahdol district of Madhya Pradesh.
- It flows through a rift valley between the Vindhyan Range on the north and the Satpura range on the south.
- The Dhuandhar (Clouds of Mist) falls is formed by the Narmada river in Jabalpur.
- It makes an estuary studded with several islands. Aliabet is the largest island.
- The **Sardar Sarovar Project** has been constructed on this river.

Tapi (or Tapti)

- It is the second largest west flowing river of the Indian peninsula.
- It is also known as 'the twin' of the Narmada.
- It originates from Multai in Betul district of Madhya Pradesh.

Sabarmati

- This 320 km long river is the name given to the combined streams-the Sabar and the Hathmati.
- It rises from the hills of Mewar in the Aravalli Range. Its tributaries are Hatmati, Sedhi, Wakul, Meshwa, Vatrak etc.

Mahi

- It rises in the Vindhyan range and debouches into the Gulf of Khambhat.
- Its length is 533 km.
- It drains an area of 34,862 sq km.
- The main tributaries are Som, Anas and Panam.
- Mahi river cuts tropic of cancer twice.

Luni (or the Salt River)

- Its water is brackish below Balotra.
- Its source lies to the west of Ajmer (Rajasthan) in the Aravallis.
- The river is known as the Sagarmati in its upper course and from Govindgarh, where Sarsuti joins it, becomes Luni. Finally, it is lost in the Rann of Kachchh.

Inland Drainage

Some rivers of India do not reach upto the sea and constitute inland drainage. These rivers are mostly present in the drier regions of the country like Western Rajasthan, Ladakh and Aksai Chin etc. **Ghaggar river is the most important example of inland drainage.** it is a seasonal stream rising from the lower slopes of Himalayas and is said to flow on the dried bed of ancient river Saraswati. It forms boundary between Punjab and Haryana for much of its length and gets subsumed in Rajasthan desert. Another such river is Luni, which is the largest river of Rajasthan. It originates near Pushkar and flows South-West of Aravalis till it reaches Rann of Kutch.

Famous Cities and River Banks					
City	River	City	River	City	River
Allahabad	At the confluence of the Ganga and Yamuna (Sangam)	Ferozpur	Satluj	Mathura	Yamuna
Agra	Yamuna	Guwahati	Brahmaputra	Nasik	Godavari
Ayodhya	Saryu	Hardwar	Ganga	Patna	Ganga
Ahmedabad	Sabarmati	Hyderabad	Musi	Panjim	Mandavi
Badrinath	Alaknanda	Jabalpur	Narmada	Srinagar	Jhelum
Bareilly	Ram Ganga	Jamshedpur	Swarnarekha	Surat	Tapti
Cuttack	Mahandadi	Jaunpur	Gomti	Sambalpur	Mahanadi
Curnool	Tungabhadra	Kanpur	Ganga	Serirangapatam	Cauvery
Delhi	Yamuna	Kota	Chambal	Tiruchurapalli	Cauvery
Dibrugarh	Brahmaputra	Kolkata	Hooghly	Ujjain	Kshipra
		Lucknow	Gomti	Vijayawada	Krishna
		Ludhiana	Sutlej	Varanasi	Ganga

Annual yield of water	
River	Contribution (%)
Brahmaputra	33.8
Ganga	25.2
Godavari	6.4
Indus	4.3
Mahanadi	3.6
Krishna	3.4
Narmada	2.9

List of the Projects, State, Location and their Purpose

S. No.	Name of the Project	Location	State	Purpose
1	Nagarjuna Sagar multi-purpose Project	River Krishna	Andhra Pradesh	Irrigation, Hydroelectricity
2	Pochampad Project	River Godavari	Andhra Pradesh	Irrigation
3	Lower Sileru Project	River Sileru (Godavari)	Andhra Pradesh	Hydroelectricity
4	Kakrapar Project	River Tapi	Gujarat	Irrigation
5	Kothagudem Project	Singareni Coalfields	Andhra Pradesh	Thermal power
6	Kosi Project	River Kosi	Bihar	Flood Control, Irrigation, Hydroelectricity
7	Gandak Project	River Gandak	Uttar Pradesh. Bihar	Irrigation, Hydroelectricity
8	Dhuvaran Power Station	Kheda District	Gujarat	Thermal Power
9	Sabarigiri (Pamba Kakki) Project	River Pamba-Kakki	Kerala	Hydroelectricity
10	Idukki Project	Rivers Periyar, Cheruthoni, Idukki	Kerala	Hydroelectricity
11	Chambal Project	River Chambal	Rajasthan, Madhya Pradesh	Irrigation, Hydroelectricity
12	Tawa Project	River Tawa (Narmada)	Madhya Pradesh	Irrigation
13	Korba Project	Near Korba Coalfields	Chhattisgarh	Thermal Power
14	Satpura Power Station	Patharkada Coalfields	Madhya Pradesh	Thermal Power
15	Koyna Project	River Koyna	Maharashtra	Hydroelectricity
16	Nagpur Power Station	Koradi, near Nagpur city	Maharashtra	Thermal Power
17	Tungabhadra Multi-purpose Project	River Tungabhadra	Karnataka, Andhra Pradesh	Irrigation, Hydroelectricity
18	Upper Krishna Project	River Krishna	Karnataka	Irrigation
19	Sharavathi Project	River Sharavathi, Near Jog Falls	Karnataka	Hydroelectricity
20	Hirakund multi-purpose Project	River Mahanadi	Odisha	Irrigation, Hydroelectricity
21	Mahanadi Delta Project	River Mahanadi	Odisha	Irrigation
22	Bhakra Nangal Multi-purpose Project	River Sutlej	Himachal Pradesh, Punjab. Haryana	Irrigation, Hydroelectricity
23	Rajasthan Canal Project	River Sutlej in Punjab	Rajasthan, Headworks in Punjab	Irrigation
24	Kundah Project	River Kundah	Tamil Nadu	Hydroelectricity
25	Ramganga Multipurpose Project	Chuisot Stream near Kalagarh	Uttarakhand	Irrigation, Hydroelectricity
26	Matatila Multipurpose Project	River Betwa	Uttar Pradesh. Madhya Pradesh	Irrigation, Hydroelectricity
27	Rihand Scheme	River Rihand	Uttar Pradesh	Hydroelectricity
28	Damodar Valley Project	River Damodar	Jharkhand, Shared with West Bengal	Flood Control, Irrigation
29	Ukai Project	River Tapi	Gujarat	Irrigation
30	Mahi Project	River Mahi	Gujarat	Irrigation
31	Ghataprabha Project	River Ghataprabha	Andhra Pradesh and Karnataka	Irrigation
32	Bhima Project	River Bhima	Maharashtra	Irrigation
33	Sardar Sarovar Project	River Narmada	Gujarat and Madhya Pradesh	Irrigation and Hydroelectricity
34	Bansagar Project	River Son	Madhya Pradesh, Chhattisgarh	Irrigation
35	Dul Hasti Project	River Chenab	Jammu and Kashmir	Hydroelectricity

36	Salal Project	River Chenab	Jammu and Kashmir	Hydroelectricity
37	Their Dam Project	River Ravi	Punjab	Irrigation, Hydroelectricity
38	Malaprabha Project	River Malaprabha	Karnataka	Irrigation
39	Jayakwadi Project	River Godavari	Maharashtra	Irrigation
40	Beas Project	River Beas	Punjab and Haryana	Hydroelectricity
41	Sharda Sahayak	River Ghaghra	Uttar Pradesh	Irrigation
42	Mayurakshi Project	River Mayurkshi	West Bengal	Irrigation, Hydroelectricity
43	Rana Pratap Sagar project	River Chambal	Rajasthan	Hydroelectricity
44	Mettur Project	River Cauvery	Tamil Nadu	Hydroelectricity
45	Pallivasal Project	River Munnar Abuja	Kerala	Hydroelectricity
46	Papanasam Project	River Thamirabarani	Tamil Nadu	Hydroelectricity
47	Loktak Project	Lake Loktak	Manipur	Hydroelectricity
48	Tehri Project	River Bhagirathi (Ganga)	Uttarakhand	Hydroelectricity
49	Farakka Project	Ganga	West Bengal	Irrigation
50	Daman Ganga	River Daman Ganga	Gujarat	Irrigation and River Link
51	Gima Project	River Gima (Tributary of Tapi)	Maharashtra	Irrigation and River Link
52	Pamba Project	River Pamba	Kerala	River Conservation and Irrigation
53	Tapovan-Vishnugarh	River Alaknanda	Uttarakhand	Hydroelectricity
54	Omkareshwar	River Narmada	Madhya Pradesh	Hydropower and Irrigation

Rivers and Disputing States

River	Disputing States
Krishna	Maharashtra, Andhra Pradesh, Karnataka
Godavari	Maharashtra, Andhra Pradesh, Madhya Pradesh, Odisha, Karnataka
Cauvery	Kerala, Karnataka, Tamil Nadu, Puducherry
Narmada	Rajasthan, Madhya Pradesh, Gujarat, Maharashtra
Mahadayi/mandovi	Goa and Karnataka
Vansadhara	Andhra Pradesh and Odisha
Ravi and Beas	Punjab and Haryana
Mullaperiyar	Kerala and Tamil Nadu

Important Falls in India

Water Fall	Height in Meters	River	Location
Kunchikal falls	455	Varahi	Karnataka
Langshiang falls	337	Kynshi	Meghalaya
Nohkalikai falls	335	—	Meghalaya
Nohshgithiang	315	—	Meghalaya
Dudhzagar falls	310	Mandovi	Goa
Kynoem falls	305	—	Meghalaya
Meenmutty falls	300	Karimpuzha	Kerala
Thalaiyar falls	297	Majjalar	Tamil Nadu
Barkana falls	259	Sita	Karnataka
Barchipani falls	399	Budhabalanga	Orissa

Lakes of India			
Wular Lake	Jammu & Kashmir	Bhim Tal	Uttarakhand
Lonar	Maharashtra	Sambhar	Rajasthan
Kolleru	Andhra Pradesh	Sukhna	Chandigarh
Loktak	Manipur	Parashuram Kund	Arunachal Pradesh
Dal Lake	Jammu & Kashmir	Pongong Tso	Jammu & Kashmir
Naini Tal	Uttarakhand	Tso Morari	Jammu & Kashmir
Salt Lake	Kolkata	Nakki Lake	Rajasthan
Chilka	Odisha	Pushkar Lake	Rajasthan
Pulicut	Andhra Pradesh	Udaipur Lake	Rajasthan
Vembanad Kayal	Kerala	Upper & Lower Lake	Bhopal, MP
Ashtamudi	Kerala	Nizam Sagar	Hyderabad

Main Rivers and their Tributaries

Rivers	Left Bank Tributaries	Right Bank Tributaries
Indus	Zaskar, Panjnad, Nubra	Shyok, Gilgit, Kabul
Mahanadi	Ib, Mand, Hasdo, Sheonath	Ong, Jonk, Tel
Godavari	Penganga, Wardha, Wainganga, indravati and Sabari	Manjira
Krishna	Bhima, Doni, Musi, Muneru	Malprabha, Ghatprabha, Tungabhadra
Cauvery	Herangi, Hemavati, Lokpavani, Srimsha and Arkavati	Laksmantirtha, Kabani, Suvarnavati, Bhavani and Amaravati
Narmada	Burhner, Banjar, Sher, Shakkar, Towa and Kundi	Hiran, Barna, Kolar
Tapi	Sipra, Kapra, Khursi, Mona, Girna, Bori, Amaravati	Purna, Betul, Patki, Suki, More, Arunavati, Gomai
Ganga	Gomati, Ghaghara, Gandak, Burhi Gandak and Kosi	Yamuna, Son, Punpun
Yamuna	Tons	Chambal, Sind, Betwa, ken

Difference between the Himalayan & the Peninsular River

S. No.	Aspects	Himalayan River	Peninsular River
1	Place of Origin	Himalayan Mountain covered with glaciers	Peninsular Plateau & Central highland
2	Nature	Perennial	Seasonal
3	Type of Drainage	Antecedent & consequent leading to dendritic pattern in plains	Super imposed, rejuvenated resulting in trellis, radial & rectangular patterns
4	Basins	Very large basins	Relatively smaller basin
5	Depth & Valleys	Rivers form deep vallays & gorges in their source areas	River valleys are often shallow
6	Flow of water	Cause much erosion & have great flow of water	Create much less erosion & also have weaken flow of water.
7	Irrigation	Irrigate the northern plains	Irrigate the Deccan Plateau.
8	Stage	Young & active	Old rivers with graded profile.

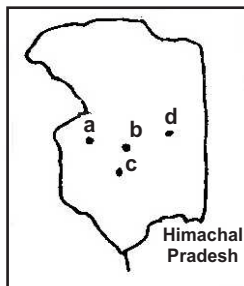
Longest Rivers Flowing in India

River	Length (km)	River	Length (km)
Ganga	2510	Krishna	1400
Godavari	1465	Yamuna	1376
Sutlej	1440	Narmada	1289

Exercise -1

- With reference to, the river Luni, which one of the following statements is correct?
 - It flows into Gulf of Khambhat
 - It flows into Gulf of Kutch
 - It flows into Pakistan and merges with a tributary of Indus
 - It is lost in the marshy land of the Rann of Kutch
- The Brahmaputra, Irrawady and Mekong rivers originate in Tibet and flow through narrow and parallel mountain ranges in their upper reaches. Of these rivers, Brahmaputra makes a "U" turn in its course to flow into India. This "U" turn is due to
 - Uplift of folded Himalayan series
 - Syntaxial bending of geologically young Himalayas
 - Geo-tectonic disturbance in the tertiary folded mountain chains
 - Both (a) and (b)
- The tributary of Indus which does not flow through Pakistan is
 - Ravi
 - Beas
 - Sutlej
 - Chenab
- Which one of the following sections of Himalaya is the longest?
 - Punjab Himalaya
 - Kumaon Himalaya
 - Assam Himalaya
 - Nepal Himalaya
- Which one of the following physiographic units has been created by both exogenic and endogenic forces?
 - The Peninsular plateau
 - The Thar desert
 - The Indo-Gangetic Plain
 - The Himalayas
- Which of the following is the main difference between the Western and Eastern Ghats?
 - Height
 - Continuity
 - Proximity to the coast
 - Vegetation
- Which of the following is not true with regard to the coastal plains of India?
 - The west coast has a narrow alluvial margin interspersed by hilly terrain
 - The eastern coast has a wide plain with well developed deltas of the major rivers
 - The west coast has little indentation except towards the south
 - The west and east coasts have the same alluvial features
- Most of the rivers flowing west-ward from the Western Ghats do not form deltas because
 - of the high gradient
 - they are too slow
 - they are not perennial
 - there is no vegetation
- One of the states through which the Tropic of Cancer passes is
 - Jammu and Kashmir
 - Himachal Pradesh
 - Bihar
 - Jharkhand
- The rocks in the Himalayan system are mainly
 - sedimentary
 - igneous
 - plutonic
 - dyke
- The highest peak in Indian territory is K₂. In which range is it located ?
 - Central Himalayas
 - Trans-Himalayas
 - Karakoram Range
 - Kumaun Himalayas
- The territorial waters of India extend up to
 - 12 nautical miles
 - 6 nautical miles
 - 15 nautical miles
 - 10 nautical miles
- Where is the Maikal range situated ?
 - Rajasthan
 - Jammu and Kashmir
 - Chhattisgarh
 - Tamil Nadu
- Which of the following rivers flows through a rift valley?
 - Ganga
 - Narmada
 - Brahmaputra
 - Krishna
- What is the most important characteristic of the islands (Indian) located in the Arabian Sea?
 - They are all very small in size
 - They are all of coral origin
 - They have a very dry climate
 - They are extended parts of the mainland
- Which area in India has an internal type of drainage ?
 - Central Himalayas
 - Tamil Nadu
 - Chhota Nagpur plateau
 - Western Rajasthan
- Which one is not an important characteristic of the Himalayan rivers ?
 - They have a great capacity for erosion
 - They have a perennial flow
 - They do not form gorges
 - Many of them have their sources in the inner Himalayas
- Most rivers flowing west from the Western Ghats do not form deltas because of
 - lack of eroded material
 - the high gradient
 - lack of vegetation-free area
 - low velocity
- The Thar Desert is believed to be expanding. The most suitable way to check it would be by
 - afforestation
 - artificial rain
 - canal irrigation
 - using the area for cattle-rearing
- Rivers that pass through Himachal Pradesh are :
 - Beas and Chenab only
 - Beas and Ravi only
 - Chenab, Ravi and Sutlej only
 - Beas, Chenab, Ravi, Sutlej and Yamuna

21. Which one of the following is not a lagoon?
 (a) Ashtamudi lake (b) Chilka lake
 (c) Periyar lake (d) Pulicat lake
22. Which of the following industrial towns is located on the Chhotanagpur plateau.
 (a) Bhilai (b) Ranchi
 (c) Asansol (d) Durgapur
23. In the rough outline map of a part of Jammu and Kashmir shown in the figure, places marked a, b, c, d represent respectively :



- (a) Anantnag, Baramula, Srinagar and Kargil
 (b) Baramula, Srinagar, Kargil and Anantnag
 (c) Baramula, Srinagar, Anantnag and Kargil
 (d) Srinagar, Baramula, Kargil and Anantnag
24. Out of the four southern States: Andhra Pradesh, Karnataka, Kerala and Tamil Nadu, which shares boundaries with the maximum number of Indian States?
 (a) Andhra Pradesh only
 (b) Karnataka only
 (c) Each of Andhra Pradesh and Karnataka
 (d) Each of Tamil Nadu and Kerala
25. Which one of the following is the largest canal ?
 (a) Sharda Canal (b) Lower Ganga Canal
 (c) Upper Ganga Canal (d) Yamuna Canal (West)
26. The 'Chilka lake region' lies in between the deltas of:
 (a) Ganga and Mahanadi
 (b) Godavari and Krishna
 (c) Mahanadi and Godavari
 (d) Krishna and Cauvery
27. Which one of the following Indian States does not border Bangladesh?
 (a) Assam (b) Manipur
 (c) Mizoram (d) Tripura
28. Which one of the following pairs is not correctly matched?

River	Tributary
(a) Yamuna	: Betwa
(b) Ganga	: Sone
(c) Mahanadi	: Wardha
(d) Godavari	: Manjra
29. From north to south, which one of the following is the correct sequence of the given hills?
 (a) Kaimur Hills — Mahadeo Hills — Satmala Hills
 (b) Mahadeo Hills — Kaimur Hills — Satmala Hills
 (c) Kaimur Hills — Satmala Hills — Mahadeo Hills
 (d) Mahadeo Hills — Satmala Hills — Kaimur Hills

30. Which one of the following pairs is not correctly matched ?
 (a) Indus : Shyok
 (b) Ganga : Gandak
 (c) Godavari : Musi
 (d) Krishna : Tungabhadra
31. Where is the Nanda Devi peak located ?
 (a) Himachal Pradesh (b) Uttarakhand
 (c) Sikkim (d) Nepal
32. The river Sone is a tributary of which one of the following rivers ?
 (a) Ganga (b) Yamuna
 (c) Narmada (d) Mahanadi
33. Which one of the following is the correct sequence of passes when one travels along the Himalayas from Kashmir to Sikkim?
 (a) Zozila-Nathula-Shipkila
 (b) Nathula-Shipkila-Zozila
 (c) Nathula-Zozila-Shipkila
 (d) Zozila-Shipkila-Nathula
34. From West to East, which one of the following is the correct sequence of Hills?
 (a) Mahadeo Hills—Maikala Hills—Garhjat Hills
 (b) Mahadeo Hills—Garhjat Hills—Maikala Hills
 (c) Maikala Hills—Mahadeo Hills—Garhjat Hills
 (d) Maikala Hills—Garhjat Hills—Mahadeo Hills
35. The Tropic of Cancer passes through which one of the following?
 (a) Assam (b) Manipur
 (c) Mizoram (d) Nagaland
36. Port Blair-the capital of Andaman and Nicobar Islands, is located in which one of the following islands?
 (a) North Andaman (b) Little Andaman
 (c) Middle Andaman (d) South Andaman
37. The Tibetan river 'Tsangpo' enters India through the state of:
 (a) Arunachal Pradesh (b) Assam
 (c) Manipur (d) Nagaland
38. Which one of the following does not characterise the Himalayas?
 (a) Various parallel ranges of the Himalayas form a convex arc
 (b) There exist syntaxial bends at both the terminals of the Himalayas
 (c) Indus, Sutlej and Brahmaputra rivers are examples of antecedent drainage
 (d) The Himalayas are wider in the east than in the west
39. Which among the following statements provides the best evidence that a river is flowing through a rift valley?
 (a) The Chambal valley is marked by bad land topography
 (b) River Tapi does not have Delta but Estuary only
 (c) River Mahanadi flows through a gorge at Satkosia
 (d) River Colorado has the Grand Canyon along its valley

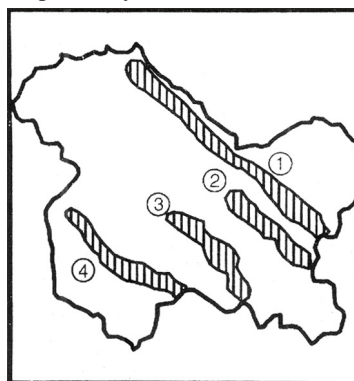
40. The Narmada river in the Peninsular plateau flows westward with a remarkably straight channel. It is because the
- slope gradient in this part controls the river channel pattern
 - river carries a huge amount of water which has created a straight channel course
 - river forms the boundary between the Central Highlands and the Deccan Plateau
 - river flows through the trough of a rift valley inclined westward
41. The reservoir GB Pant Sagar is located on which river?
- Betwa
 - Ghaghara
 - Kosi
 - Rihand
42. River Luni originates near Pushkar and drains into which one of the following?
- Rann of Kachchh
 - Arbian Sea
 - Gulf of Cambay
 - Lake Sambhar
43. Which one of the following states has built the famous Gandhi Sagar across the Chambal River?
- Rajasthan
 - Maharashtra
 - Uttar Pradesh
 - Madhya Pradesh
44. What is 'Operation Kolleru' that was recently in the news?
- A massive river linkage project
 - A project to improve a wetland
 - A project to supply drinking water to a mega city
 - A rural drinking water supply scheme of a southern state
45. In the Cauvery River water dispute, which one of the following groups of states are concerned?
- Kerala and Karnataka only
 - Karnataka, Andhra Pradesh and Maharashtra
 - Kerala, Karnataka, Tamil Nadu and Union Territory of Puducherry
 - Kerala, Goa, Karnataka and Tamil Nadu
46. Which waterway separates India from Sri Lanka?
- 8° Channel
 - Palk Strait
 - 10° Channel
 - Andaman Sea
47. Which one of the following peaks is the highest?
- Nanda Devi
 - Kanchenjunga
 - Godwin Austin
 - Nanga Parbat
48. With which one of the following countries, India shares maximum length of the border?
- Bangladesh
 - Pakistan
 - China
 - Nepal
49. Which of the following tributaries of the Ganga System flows Northwards?
- Kosi
 - Ghaghara
 - Gandak
 - Sone
50. The city of Nasik is situated on the bank of which one of the following rivers?
- Krishna
 - Mandovi
 - Godavari
 - Tapti
51. Rihand Valley Project is located in which one of the following states?
- Odisha
 - Gujarat
 - Himachal Pradesh
 - Uttar Pradesh
52. Which of the following Indian states does not have a common International border with Bangladesh?
- Manipur
 - West Bengal
 - Tripura
 - Assam
53. 28.38° N and 77.12° E are the respective latitude and longitude of which one of the following places?
- Jaipur
 - Delhi
 - Lucknow
 - Allahabad
54. Which one of the following rivers flows into the Arabian Sea?
- Indravati
 - Godavari
 - Cauvery
 - Narmada
55. Madhya Pradesh shares its border with how many States?
- 3
 - 4
 - 5
 - 6
56. Match the following
- | List I
(Multi purpose River Projects) | List II
(Hydel Power Station) |
|--|----------------------------------|
| A. Rihand | 1. Hirakund |
| B. Gandak | 2. Balmikinagar |
| C. Chambal | 3. Pipri |
| D. Mahanadi | 4. Kota |
- Codes :**
- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (a) | 3 | 4 | 2 | 1 |
| (b) | 1 | 2 | 4 | 3 |
| (c) | 3 | 2 | 4 | 1 |
| (d) | 1 | 4 | 2 | 3 |
57. Which one of the following is the oldest mountain range in India?
- Himalayas
 - Aravalli
 - Satpura
 - Nilgiri
58. Which one of the following longitudes determines the Indian standard time?
- 85.5° E
 - 86.5° E
 - 84.5° E
 - 82.5° E
59. The hill range that separates the State of Manipur from the State of Nagaland is known as
- Arakan hills
 - Patkai hills
 - Barail hills
 - Manipur hills
60. The channel separating the Andaman Island from the Nicobar Islands is known as
- Coco channel
 - 10° channel
 - Duncan passage
 - Sombraro channel
61. Which is the correct arrangement of the following rivers from North to South?
- Godavari, Penner, Cauveri, Periyar
 - Penner, Godavari, Periyar, Cauveri
 - Godavari, Cauveri, Penner, Periyar
 - Cauveri, Godavari, Periyar, Penner
62. Which one of the following places is not located on the bank of river Ganga?
- Uttarkashi
 - Kanpur
 - Fatehpur
 - Bhagalpur

63. Which among the following states of India have common borders with Pakistan?
 (a) Jammu and Kashmir, Himachal Pradesh, Punjab and Rajasthan
 (b) Punjab, Jammu and Kashmir, Rajasthan and Gujarat
 (c) Jammu and Kashmir, Punjab, Haryana and Rajasthan
 (d) Punjab, Himachal Pradesh, Gujarat and Rajasthan
64. Which of the following three rivers of the peninsula India have the Amarkantak region as their source?
 (a) Narmada, Krishna Godavari
 (b) Son, Mahanadi, Narmada
 (c) Godavari, Krishna, Cauvery
 (d) Chambal, Betwa, Luni
65. Which one among the following rivers does not flow into the Bay of Bengal?
 (a) Mahanadi (b) Cauveri
 (c) Tapti (d) Godavari
66. The country that shares longest border with India is
 (a) China (b) Bangladesh
 (c) Nepal (d) Pakistan
67. Amarkantak plateau in the Maikal hills marks the origin of the river
 (a) Gandak (b) Chambal
 (c) Narmada (d) Ghaggar
68. Which one among the following states does not form part of the Narmada basin ?
 (a) Madhya Pradesh (b) Gujrat
 (c) Rajasthan (d) Maharashtra
69. Match the following

List I (Hydroelectric power station)	List II (Location in map)
A. Nagarjuna Sagar	
B. Mettur	
C. Hirakund	
D. Sileru	

Codes :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 3 | 4 | 1 | 2 |
| (b) 3 | 1 | 4 | 2 |
| (c) 2 | 1 | 4 | 3 |
| (d) 2 | 4 | 1 | 3 |
70. Which one among the following is nearest to the Tropic of Cancer ?
 (a) Patna (b) Ranchi
 (c) Rourkela (d) Varanasi
71. In which one of the following states is Zojila pass located?
 (a) Arunachal Pradesh (b) Jammu and Kashmir
 (c) Himachal Pradesh (d) Sikkim
72. Which one of the following pairs is not correctly matched?
 (a) Indus : Shyok
 (b) Ganga : Gandak
 (c) Godavari : Musi
 (d) Krishna : Tungabhadra
73. Where is the Nanda Devi peak located ?
 (a) Himachal Pradesh (b) Uttarakhand
 (c) Sikkim (d) Nepal
74. The river Sone is a tributary of which one of the following rivers ?
 (a) Ganga (b) Yamuna
 (c) Narmada (d) Mahanadi
75. Which one of the following is the correct sequence of passes when one travels along the Himalayas from Kashmir to Sikkim?
 (a) Zozila-Nathula-Shipkila
 (b) Nathula-Shipkila-Zozila
 (c) Nathula-Zozila-Shipkila
 (d) Zozila-Shipkila-Nathula
76. Which one among the following peaks is the highest?
 (a) Dhaul Giri (b) Namcha Barwa
 (c) Nanda Devi (d) Nanga Parvat
77. From West to East, which one of the following is the correct sequence of Hills?
 (a) Mahadeo Hills—Maikala Hills—Garhjat Hills
 (b) Mahadeo Hills—Garhjat Hills—Maikala Hills
 (c) Maikala Hills—Mahadeo Hills—Garhjat Hills
 (d) Maikala Hills—Garhjat Hills—Mahadeo Hills
78. The Tropic of Cancer passes through which one of the following?
 (a) Assam (b) Manipur
 (c) Mizoram (d) Nagaland
79. Mahatma Gandhi Hydroelectric project is on which river ?
 (a) Godavari (b) Sharavati
 (c) Cauvery (d) Krishna
80. Indian Standard Time refers to the local time of which one of the following places in India?
 (a) Allahabad (b) Bhopal
 (c) Delhi (d) Lucknow
81. Which one of the following rivers flow between Vindhyan and Satpura ranges?
 (a) Narmada (b) Son
 (c) Mahe (d) Netravati
82. Amravati, Bhavani, Hemavati and Kabini are tributaries of which one of the following rivers?
 (a) Mahanadi (b) Godavari
 (c) Kaveri (d) Krishna
83. The Tropic of Cancer does not pass through
 (a) Orissa (b) Tripura
 (c) Chhattisgarh (d) Rajasthan
84. Which one of the following mountain ranges is spread over only one state in India?
 (a) Aravalli (b) Satpura
 (c) Ajanta (d) Sahyadri
85. Examine the map of Jammu and Kashmir given below:
 The mountains ranges marked 1, 2, 3 and 4 are respectively:



- (a) Ladakh, Zaskar, Karakoram and Pir Panjal
 (b) Karakoram, Ladakh, Zaskar and Pir Panjal
 (c) Karakoram, Zaskar, Pir Panjal and Ladakh
 (d) Ladakh, Pir Panjal, Karakoram and Zaskar